Solomonoff Operator Induction:
Definitions, Versions and Basic Theorems

Kaarel Hänni*

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Abstract

Solomonoff induction is the standard ideal predictor for sequences. We study its analogue for supervised learning, Solomonoff function induction (SFI): a universal Turing machine, run separately on each input, reads the same infinite sequence of fair coin flips from its first bit and outputs a label or diverges; SFI predicts labels by conditioning the law of this random partial function on the observed input–output pairs. This law dominates that of every such machine, universal or not, and does not depend, up to a constant factor, on the choice of universal machine. The shortest consistent program, the program mixture, the mixture of product laws and SFI agree up to the complexity of the set of inputs, but there is no coding theorem for function programs. On every consistent data sequence, SFI's cumulative log loss exceeds that of every computable random labeller by at most the labeller's complexity plus a constant; with i.i.d. inputs and a computable deterministic target f, the averaged predictor has risk $O(K(f)/n)$. SFI cannot model fresh label noise on repeated inputs. Measured on its unnormalised predictive probabilities, SFI's cumulative loss does not depend on the order of the data: on a random cyclic permutation of $\{0,1\}^d$ revealed backwards along its cycle it pays about $(d - \log_2 e)2^d$ bits, while Solomonoff induction on the interleaved sequence, with self-delimiting blocks, pays $O(1)$ bits of label loss in expectation. Hypotheses that see the list of previous inputs give the universal chronological semimeasure. A dominance theorem for computable reveal laws holds pathwise for deterministic laws and, under finiteness assumptions and up to the complexities of the horizon and of the input sets, in expectation for randomised ones. We also treat a coin example of Sam Eisenstat, out-of-distribution behaviour, computable versions and neural-network Bayes.

Contents

1 Introduction **2**
2 Related work **5**
3 Function semimeasures, function machines and SOI **7**
4 Kolmogorov and semimeasure versions **12**
5 Learning theorems: SOI as supervised learning **17**
6 History-aware prediction **22**
7 Worked examples **28**
8 Out-of-distribution behaviour **33**

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9 Computable versions **37**
10 Neural-network Bayes and circuit SOI **41**
11 Steeper simplicity priors (summary of the companion paper) **43**
12 Discussion **44**
Acknowledgements **46**
Code and data **47**
A Proofs for Section 3 (function semimeasures and SOI) **51**
B Proofs for Section 4 (Kolmogorov and semimeasure versions) **58**
C Proofs for Section 5 (learning theorems) **62**
D Proofs for Section 6 (history-aware prediction) **68**
E Proofs for Section 7 (worked examples) **77**
F Proofs for Section 8 (out-of-distribution behaviour) **83**
G Proofs for Section 9 (computable versions) **86**
H Proofs for Section 10 (neural-network Bayes) **91**

1 Introduction

In supervised learning a learner sees input–output pairs $(x_1, y_1), (x_2, y_2), \ldots$ and predicts the output at a new input. For predicting sequences, Solomonoff induction (SI) is the standard ideal reference: a Bayes mixture over all computable sources whose cumulative log loss exceeds that of each computable source by at most a constant that depends on the source (Li and Vitányi, 2008). *Solomonoff function induction* (SFI) is the analogous object for supervised learning. Feed one infinite sequence of fair coin flips to a universal Turing machine; each query x is a separate run on x that reads the sequence from its first bit, as many bits as it needs, and outputs a label or diverges. The same sequence serves every query, so this defines a random partial function; SFI predicts by conditioning its law on the observed pairs.[1] We define SFI precisely, prove its basic properties, compare it with SI and work out examples.

Why SFI. SFI is a candidate for the ideal form of supervised learning. Its guarantees need no assumption on how the inputs are chosen (Theorem A below), and with i.i.d. inputs and a computable deterministic target f its averaged predictor has the Occam rate $O(K(f)/n)$, without a union bound (Theorem 5.5). The analogy has limits, which Remark 5.12 makes precise: the guarantees are for log loss and for averaged predictors, a random function cannot produce fresh label noise on a repeated input, SFI is not computable, and it is blind to the order of the data. SFI is also the natural setting for two further questions. Its hypotheses include samplers that draw private coins at each input, in which a model is kept apart from its noise. On such hypotheses, simplicity priors that charge a bit of program more than a bit of fit become meaningful: under these priors, in Solomonoff induction and among hypotheses with shared randomness, a universal mixture is itself one hypothesis and absorbs all structure, while no hypothesis with private coins is universal. The companion paper studies them for two purposes: splitting data into a model and noise, and not learning a teacher's systematic mistakes (Section 11). And learning a mathematical theory from sentences labelled true or false, by Bayesian inference over axiom systems, is closely

[1] The notion of SFI first appeared in the author's working notes (Hänni, 2026c).

related to the coin-free version of SFI on sentences, restricted to logically consistent labellers; we leave this connection to separate work.[2]

Relation to Solomonoff induction. Up to constant factors, SI is SFI on the inputs $0, 1, \ldots, n-1$ (Proposition 3.24). In general, though, SFI sees the data as a set: each hypothesis answers each query without seeing the other inputs, and SFI's cumulative loss, measured on its unnormalised predictive probabilities, does not depend on the order of the data (Proposition 3.16). SI applied to the interleaved sequence $x_1 y_1 x_2 y_2 \cdots$ sees the order and can do much better (Sections 6 and 7). Giving SFI's hypotheses the list of previous inputs removes the gaps of Sections 6 and 7: the result, $\mathrm{SOI}^{\mathrm{hist}}$, is a universal lower-semicomputable chronological semimeasure (Corollary 6.17), and its loss exceeds SI's joint loss on the interleaved sequence by at most a constant (Theorem 6.18); against SI's loss on the labels alone the comparison involves SI's loss on the inputs (Proposition 6.19). Chronological semimeasures, with actions and percepts in place of inputs and labels, are the setting of Hutter's AIXI (Hutter, 2000, 2005); we define $\mathrm{SOI}^{\mathrm{hist}}$ directly and use no identification with Hutter's construction.

Relation to neural-network Bayes. Bayesian learning with a neural network is function-space Bayes with the prior that the parameter prior induces on functions (Proposition 10.1); the slides of Vaintrob and Hänni (2026b) argue that for small prior scales it behaves like a circuit version of SFI. Sections 9 and 10 prove what holds in general about this link.

1.1 Main results

Two loss conventions. SFI's predictive probabilities are sub-probabilities; the missing mass is the posterior probability that the sampled function is undefined at the query. The *semimeasure convention* scores these raw conditionals, the *normalised convention* rescales them to sum to one at each step. Upper bounds in the first convention hold in both; every lower bound says which convention it is for (Remark 3.18). Logarithms are to base 2, losses are in bits, and c is a constant that depends only on the fixed universal machines and encodings.

Theorem A (informal). A *function machine* is a Turing machine that answers all queries from one random tape, as in the definition of SFI above, but need not be universal. There is a universal function machine whose law M_F, the prior of SFI, satisfies $M_F \geq 2^{-K(\nu)}\nu$ for the law ν of every function machine, including samplers that see only the input and partial computable functions. Changing the universal machine changes M_F by at most a constant factor, and computable injective re-encodings α, β of inputs and labels by at most a factor $2^{K(\alpha,\beta)+O(1)}$. So on every consistent sequence of pairs, however the inputs are chosen, SFI's cumulative log loss is at most that of the random labeller ν plus $K(\nu)$ (Theorems 3.8, 3.20 and 5.1 and Proposition 3.9).

Theorem B (informal). For every $d \geq 2$ there are two inputs $x_1, x_2 \in \{0,1\}^d$ such that every program that outputs 0 on x_1 and 1 on x_2 has length at least $\log d - 2$, while these programs together have algorithmic probability at least $2^{-O(\log\log d)}$ and SFI's loss on the two labels is $O(1)$. So there is no coding theorem for function programs, and no two of the shortest consistent program, the program mixture and SFI agree up to a constant, although these, the mixture of product laws and the universal weak function semimeasure agree up to $K(\mathrm{dom}\, D) + O(1)$ bits on every data set D (Theorems 4.4 and 4.9 and Proposition 4.5).

[2] Axiom induction in this sense first appeared in the author's working notes (Hänni, 2026c).

Theorem C (informal). In the semimeasure convention SFI's cumulative log loss depends only on the set of revealed pairs, not on their order (Proposition 3.16). So SFI cannot use information that only the order carries: if $x_1, x_2, \ldots$ are the consecutive blocks, of lengths $1, 2, 3, \ldots$, of a Martin-Löf random sequence, revealed as $(x_1, \epsilon), (x_2, x_1), \ldots, (x_n, x_{n-1})$, SFI pays $n(n-1)/2 \pm O(\log n)$ bits, and every lower-semicomputable order-independent predictor as much up to its complexity, while a predictor of constant complexity that sees the previous pairs pays nothing (Theorem 6.3 and Proposition 6.5). If SFI's hypotheses see the list of previous inputs, the result, $\mathrm{SOI}^{\mathrm{hist}}$, is a universal lower-semicomputable chronological semimeasure: on every sequence its loss exceeds that of every lower-semicomputable history-aware predictor P by at most $K(P) + O(1)$, and that of SI on the interleaved sequence by at most a constant (Corollary 6.17 and Theorem 6.18).

Theorem D (informal). Let f be a uniformly random cyclic permutation of $\{0,1\}^d$, $N := 2^d$, revealed backwards along its cycle as $(x_1, 0^d), (x_2, x_1), \ldots$ with $x_k := f^{-1}(x_{k-1})$ and $x_0 := 0^d$, so that every label is the previous input. In the semimeasure convention SFI pays at most $\log((N-1)!) + O(1) \approx (d - \log e)N$ bits over the cycle, and with high probability at least $\log((N-1)!) - O(1)$; no order-independent predictor pays less in expectation. In the normalised convention it pays in expectation at least $\delta d - O(\log d)$ bits per step on average over the first $2^{(1-\delta)d}$ steps. SI on the interleaved sequence, with self-delimiting blocks, pays $O(1)$ bits of label loss in expectation, and $\mathrm{SOI}^{\mathrm{hist}}$ $O(1)$ bits on every cycle, uniformly in d (Theorem 7.3 and Propositions 7.5, 7.7, 7.8 and 7.10). In Sam Eisenstat's coin example (each index $i \geq 0$ is labelled 1 independently with probability $\frac{1}{3}$, and each step reveals the smallest unrevealed index labelled 1 with probability $\frac{2}{3}$, else the smallest one labelled 0), SFI's expected loss is $\frac{H(1/3)}{2} n \pm O(\log n)$ bits in the semimeasure convention (and almost surely within $\log n + O(\log \log n)$ bits of the loss of the threshold hypothesis), which is optimal among order-independent predictors up to $O(\log n)$ in expectation, while SI pays at most 2.4 bits plus a constant in expectation (Theorems 7.14, 7.16 and 7.20 and Proposition 7.17).

1.2 Contributions

- *Definitions and versions* (Sections 3 and 4): Theorems A and B; the larger class defined by the one-point inequality alone, whose universal element exceeds M_F by linearly many bits on some data sets (Theorem 3.12); and the comparison of all versions (Table 1).
- *Learning in and out of distribution* (Sections 5 and 8): merging, mistake and online-to-batch bounds (Theorems 5.3 and 5.5 and Corollary 5.4); the failure under fresh label noise on repeated inputs and its repairs (Propositions 5.6 and 5.7); a posterior probability bounded below, whatever the amount of data, for every simple behaviour on an unseen simple property (Proposition 8.1); and a total posterior error of at most $(\ln 2)(K(f) + c)$ on a computable target f over any distinct inputs (Theorem 8.4).
- *History-aware prediction* (Section 6): Theorem C, and the skipped-string example, also normalised (Theorem 6.4 and Proposition D.3). For computable reveal laws Δ_f, which may depend on the target f but see only the history, SFI's loss exceeds that of every computable sampler P that sees the history by at most $K(P) + K(\Delta_f) + c$ on every path for deterministic reveal policies and, for samplers that halt with probability 1 and under finiteness hypotheses, in expectation with an additional $K(n) + K(A)$ term for randomised reveal laws; whether that term is needed is open. A direct simulation argument fails, and the stronger oracle form, with

the complexity of the reveal law measured relative to an oracle for f, is false (Proposition 6.7, Corollary 6.14, Theorem 6.11, and Remark D.7).

- *Worked examples* (Section 7): Theorem D, and positive examples in which SFI is optimal up to lower-order terms (Section 7.4).
- *Computable versions and networks* (Sections 9 and 10): constant regret of time-bounded mixtures against all time-bounded predictors that see only the input (Proposition 9.4 and Corollary 9.5); for circuit Bayes, mistakes polynomially related to circuit size on every ordering (Theorem 9.8), while simplest-consistent learning makes $2^{\Omega(\ell/\log \ell)}$ mistakes on some ordering for the conjunction of d variables, of description length $\ell = \Theta(d \log d)$ (Theorem 9.10); a separation in squared error between Gaussian-process limits and SFI on sparse parities, and a failure of bias-free two-layer ReLU kernels on odd parities (Corollary 10.6 and Remark 10.7).
- *Steeper simplicity priors* (Section 11): a summary of the companion paper (Hänni, 2026a), which gives two reasons for a length penalty steeper than likelihood on hypotheses with private randomness. First, it splits data into a model and noise: with i.i.d. inputs and a sampler that draws fresh coins at every query, every MAP of the steeper objective is eventually a shortest program for the sampler, for fixed factors above 1 at a rate that depends only on the factor, the program's length, a margin and the confidence. Second, with a factor proportional to the sample size, in an explicit window, hypotheses that are near-optimal for the population objective learn a random parity on uniform inputs rather than a teacher's errors, made at a rate below 0.05596 on an error set of product form that is incompressible relative to the parity.

What is not achieved. SFI is not computable, and the constants in our bounds depend on the universal machine and can be large, so the results describe an ideal, not an algorithm. Among the open questions (Section 12.1) are whether M_F dominates every lower-semicomputable law of a random partial function (Question 3.14), several lower bounds in the normalised convention (Question A.8), and a two-sided bridge between NN Bayes and circuit SFI (Question 10.9). We do not prove the main theorems of the slides.

Reader's guide. Section 2 describes related work and Section 3 sets up the definitions used throughout; Section 4, which compares the Kolmogorov and semimeasure versions, can be skipped on a first reading. Each main section states its results with short proof ideas; full proofs are in Appendices A to H.

2 Related work

This section describes the closest work informally; the objects are defined from Section 3 on.

Solomonoff induction and operator induction. Solomonoff induction predicts sequences with a universal prior (Solomonoff, 1978; Li and Vitányi, 2008). Solomonoff's operator induction (Solomonoff, 2008) predicts the second element of a new pair from an unordered set of ordered pairs, the setting of supervised learning. SFI is a universal prior for this setting: the law of a universal machine that reads the same sequence of fair coin flips on every input. The per-query mixture M_{pq} (Section 4) is a mixture in the spirit of operator induction that gives each query fresh randomness; it equals the mixture of product laws on distinct inputs (Remark 4.2) and is the

natural object for fresh label noise (Proposition 5.7). SFI differs from both in sharing randomness across inputs (Proposition 4.7).

Chronological semimeasures. Chronological semimeasures and a universal mixture over them were introduced by Hutter for the AIXI model (Hutter, 2000, 2005), with actions and percepts; here the inputs play the actions and the labels the percepts. We define M_H directly and do not use any identification with Hutter's construction. Hutter (2007) lists supervised learning among the problem classes of the AIXI model. When its hypotheses see the list of previous inputs, SFI becomes $\mathrm{SOI}^{\mathrm{hist}}$, whose law M_H is a universal lower-semicomputable chronological semimeasure (Corollary 6.17). Wood et al. (2013) show that Solomonoff's and Levin's constructions define the same class of priors for sequences; Theorem 6.16 is the corresponding statement for chronological semimeasures. Lattimore et al. (2011) show that normalised Solomonoff induction learns simple recursive sub-patterns of otherwise unstructured sequences, while the unnormalised version can fail; this bears on Solomonoff induction applied to the interleaved sequence $x_1y_1x_2y_2\cdots$ (Proposition D.20).

Universal priors for classification. Lattimore and Hutter (2013) give an off-line classifier inspired by Solomonoff induction and show that random selection of training data can reduce misclassification. Poland and Hutter (2005) show that two-part MDL prediction can have loss bounds exponentially worse than Bayes mixtures; the slides of Vaintrob and Hänni (2026b) compare this with the gap between simplest-consistent learning and weighted majority (Remark G.10). Leike and Hutter (2015a) show that the optimality of AIXI depends on the reference machine; Proposition 8.2 is a supervised analogue. Ryabko (2006) studies classification with deterministic labels and objects that are conditionally independent given the labels.

Learning theory. The guarantees of Section 5 are of the Occam (Blumer et al., 1987), PAC-Bayes (McAllester, 1999), MDL (Barron and Cover, 1991; Grünwald, 2007), halving and weighted-majority (Littlestone, 1988; Littlestone and Warmuth, 1994), expert-advice (Vovk, 1990; Cesa-Bianchi and Lugosi, 2006) and online-to-batch (Littlestone, 1989; Cesa-Bianchi et al., 2004) kinds, with the complexity K in place of $\log|H|$ (Proposition 5.11). Conformal prediction (Vovk et al., 2005) is an exchangeability-based framework for supervised prediction, related to the random splits of Proposition 5.8 (see also Vovk, 1992). Inductive inference of recursive functions (Gold, 1967; Barzdin and Freivalds, 1972; Case and Smith, 1983) is the deterministic ancestor: identification in the limit and prediction of recursive functions, mostly for inputs in the order $0, 1, 2, \ldots$.

Computable versions and networks. Speed priors (Schmidhuber, 2002; Filan et al., 2016) are time-bounded universal priors for sequences; Section 9 gives time-bounded mixtures in the function setting. Dingle et al. (2018) show that input–output maps are strongly biased towards simple outputs, and Valle-Pérez et al. (2019) argue that the parameter–function map of deep networks is biased towards simple functions. The simplicity-bias upper bound holds for every lower-semicomputable prior on functions, while the lower bound needed for a simplicity prior fails (Propositions 10.2, 10.3 and H.1); the slides of Vaintrob and Hänni (2026b) argue the same. Mingard et al. (2021) report that SGD samples functions with probabilities close to the Bayesian posterior, and Mingard et al. (2025) attribute much of the generalisation of deep networks to a prior bias towards simple functions together with the structure of the data. Buzaglo et al. (2024) show that random interpolators under a flat parameter prior generalise when a narrow teacher exists; the slides call this the closest prior work. Goldblum et al. (2024) relate no-free-lunch

theorems to Kolmogorov complexity, and Grau-Moya et al. (2024) train networks on data from universal machines to approximate Solomonoff induction. Steeper simplicity priors, related to λ-weighted MDL, are the subject of the companion paper (Hänni, 2026a), summarised in Section 11.

Novelty. To our knowledge the following are new: the name and the function-machine formulation of SFI; the separations between SFI and $\mathrm{SOI}^{\mathrm{hist}}$ and the comparisons with Solomonoff induction on the interleaved sequence (Sections 6 and 7); the out-of-distribution results Proposition 8.1 (in this form), Theorem 8.3, Propositions 8.5, 8.7, 8.8 and F.4 and Example F.5; the time-bounded mixtures in the function setting (Section 9.1; the speed priors are sequence analogues); the every-ordering form of circuit Bayes with explicit constants (Theorem 9.8); the padding bound for uniform circuit priors (Proposition 10.4); and the statements on Gaussian-process classification and anti-correlation in Remark 10.7(b),(c). Known are operator induction and the universal chronological semimeasure; Solomonoff-inspired classifiers and two-part MDL for classification; Occam, PAC-Bayes and online-to-batch bounds, including the exchangeability argument of Proposition 5.8; the halving bound, its contrast with simplest-consistent selection on the same prior, and the conjunction ladder (Vaintrob and Hänni, 2026b); the Walsh diagonalisation and the dimension bound for kernels and the vanishing odd eigenvalues of bias-free ReLU kernels (Theorem 10.5, Remark 10.7(a)); the fact that $-\log P \approx K$ on P-most objects (Proposition H.1); and the dependence of universal priors on the reference machine.

3 Function semimeasures, function machines and SOI

Solomonoff's operator-induction problem asks, given an unordered set of question–answer pairs, for the probability distribution of the answer to a new question (Solomonoff, 2008); here the questions are inputs and the answers labels. *Solomonoff operator induction* (SOI) is the family of Bayesian solutions to this problem under universal priors. Informally, its *one-tape version* feeds one infinite sequence ω of fair coin flips to a universal Turing machine U. Each query x is a separate run of U on x that reads ω one-way from its first bit, as many bits as it needs, and outputs a label or diverges. The same ω serves every query: the coin flips are shared by all inputs, not drawn afresh at each input. This defines a random partial function, and the one-tape version conditions its law on the observed input–output pairs.[3] Formally, it uses the universal function machine T_U of Theorem 3.8 (Definition 3.15); every universal U gives the same law up to a constant factor. Function machines, the machines of this kind that need not be universal (Definition 3.6), include deterministic programs and samplers with a private random stream per input (Proposition 3.9), and Section 4 compares the versions, among them Solomonoff's own per-query solution. Proofs are in Appendix A.

3.1 Data and conventions

Definition 3.1 (Data)**.** Inputs X, labels Y; by default $X = Y = \{0,1\}^*$, but any infinite decidable $X \subseteq \{0,1\}^*$ and any countable Y with a fixed computable enumeration, $|Y| \geq 2$, will do unless stated. $\perp$ means "undefined", $Y_\perp := Y \cup \{\perp\}$. A *data set* is a finite partial function $D : X \rightharpoonup Y$; $D + (x,y) := D \cup \{(x,y)\}$ for $x \notin \operatorname{dom} D$; $D|_T$ is a restriction. A partial function f *extends* D, $f \supseteq D$, if $f(x) = y$ for all $(x,y) \in D$. A sequence $(x_t, y_t)_{t \leq n}$ is *consistent* if

[3] The author's working notes call this setting function induction (Hänni, 2026c). The private random stream per input, with the resulting dominance (Theorem 5.1(a)), and an informal claim that the input encoding matters little (Theorem 3.20(b)) first appeared there.

$D_n := \{(x_t, y_t) : t \le n\}$ is a partial function; $h_t := (x_1, y_1, \dots, x_t, y_t)$ is the history. For ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, x) := \nu(D + (x,y))/\nu(D)$ for $x \notin \operatorname{dom} D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, x) := \mathbf{1}[y = D(x)]$ for $x \in \operatorname{dom} D$ (the *repeated-input convention*).

Complexity and constants. V is a fixed universal prefix machine, K prefix complexity, φ_e the e-th partial computable function $X \rightharpoonup Y$, and $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \ge 2^{-K(i)}$; "lsc" means lower semicomputable. A *conditional semimeasure* is a map $P(y \mid x) \ge 0$ with $\sum_y P(y \mid x) \le 1$ for every x; the lsc ones have an effective enumeration (P_i) (trim each $P(\cdot \mid x)$ as in the proof of Theorem 3.11). For an object o of a class with a fixed effective enumeration (o_i), such as partial computable functions (φ_e), lsc conditional semimeasures (P_i), function machines (T_i) and their laws (Proposition 3.7), $K(o) := \min\{K(i) : o_i = o\}$. We use the following standard facts (Levin, 1974; Gács, 1974; Chaitin, 1975), (Li and Vitányi, 2008, §4.3): $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor; and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a, z) with $\sum_a P(a \mid z) \le 1$, then $K(a \mid z) \le -\log P(a \mid z) + K(P) + c$. Both relativise. $j(x)$ is the position of x in the length-lexicographic order of $\{0,1\}^*$, starting from $j(\epsilon) = 0$, so $j(x) \le 2^{d+1} - 2$ for $|x| \le d$; $\langle \cdot, \cdot \rangle$ denotes fixed computable bijections $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and $X \times \mathbb{N} \to X$. $\log = \log_2$; losses are in bits; c and $O(1)$ depend only on the fixed machines and codings. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, and $\omega_{1:L}$ the first L tape bits.

Two loss conventions. The predictors below issue sub-probabilities; the missing mass is the probability of "undefined". The *semimeasure convention* scores the raw conditionals, the *normalised convention* divides each by its sum. Statements about losses name their convention; upper bounds in the semimeasure convention hold in both, lower bounds need not (Proposition 3.17 and Remark 3.18).

3.2 Three notions of function semimeasure

Definition 3.2 (Weak function semimeasure)**.** A map ν from data sets to $[0,1]$ with $\nu(\emptyset) \le 1$ and $\nu(D) \ge \sum_{y \in Y} \nu(D + (x,y))$ for every D and $x \notin \operatorname{dom} D$.

Then (W1) ν is monotone, $\nu(E) \le \nu(D)$ for $E \supseteq D$, and (W2) $\sum_{D : \operatorname{dom} D = S} \nu(D) \le \nu(\emptyset)$ for finite S (sum out one input at a time). Call a predictor q (sub-probabilities $q(\cdot \mid D, x)$, with the repeated-input convention) *order-independent* if, for every D, the product $\nu_q(D)$ of its conditionals along a reveal order of D is the same for all orders.

Proposition 3.3 (Predictor reading)**.** *(a) If ν is weak with $\nu(\emptyset) = 1$, its conditionals form an order-independent predictor q_ν with $\nu_{q_\nu} = \nu$. (b) If q is order-independent, ν_q is weak with $\nu_q(\emptyset) = 1$, and $q(y \mid D, x) = \nu_q(D + (x,y))/\nu_q(D)$ when $\nu_q(D) > 0$. So weak semimeasures with $\nu(\emptyset) = 1$ correspond one-to-one to order-independent predictors, identified where $\nu(D) > 0$.*

Definition 3.4 (Random partial function)**.** A *random partial function* is a random element F of $Y_\perp^X$ (product σ-algebra) with law P; its *containment functional* is $\nu_P(D) := P(F \supseteq D)$.

Every containment functional is weak, since the events $\{F \supseteq D + (x,y)\}$, $y \in Y$, are disjoint subsets of $\{F \supseteq D\}$. The converse fails (Example 3.5). Containment functionals are exactly the weak ν with $\nu(\emptyset) = 1$ whose Möbius sums are nonnegative (Proposition A.2 in Appendix A.1), the function-space analogue of the correspondence between completely monotone set functions and random sets (cf. Choquet, 1954; Matheron, 1975; Shafer, 1976).

Example 3.5 (Weak but not a random function)**.** (a) Inputs a, b: $\nu(\emptyset) = \nu(\{(a,0)\}) = \nu(\{(b,0)\}) = 1$, all other values 0. It is weak but not a containment functional: "$F(a) = 0$ surely and $F(b) = 0$ surely, but never both". (b) Inputs a, b, c, labels $\{0,1\}$: $\nu_\triangle(\emptyset) = 1$, and $\nu_\triangle = \frac{1}{2}$ on the six one-point data sets and on $E_1 = \{(a,0),(b,1)\}$, $E_2 = \{(b,0),(c,1)\}$, $E_3 = \{(c,0),(a,1)\}$, and 0 elsewhere. $\nu_\triangle$ is weak, and E_1, E_2, E_3 are pairwise inconsistent with total mass $\frac{3}{2}$; so no law P of a random partial function has $P(F \supseteq \cdot) \geq c\,\nu_\triangle$ with $c > \frac{2}{3}$, and the uniform law on $(F(a), F(b), F(c)) \in \{(0,1,0),(1,0,1),(1,0,0)\}$ attains $c = \frac{2}{3}$.

3.3 Function machines and universality

Definition 3.6 (Function machine)**.** A *function machine* is a c.e. set $T \subseteq X \times \{0,1\}^* \times Y$ of triples that is *consistent*: if $(x,w,y),(x,w',y') \in T$ and w, w' are comparable, then $y = y'$. For a tape ω put $f^T_\omega(x) := y$ if some prefix w of ω has $(x,w,y) \in T$, and $f^T_\omega(x) := \perp$ otherwise. The *machine law* of T is $\nu_T(D) := \mathbb{U}\{\omega :\ f^T_\omega \supseteq D\}$.

Equivalently, T is a Turing machine as in the informal definition, but not necessarily universal: on query x it reads a one-way random tape and either halts with a label after finitely many bits or diverges; all queries read the same tape.

Proposition 3.7 (Machine laws)**.** *Every machine law is the containment functional of a random partial function, with $\nu_T(\emptyset) = 1$, and is lsc uniformly in D and an index of T. Function machines have an effective enumeration (T_i), to which $K(T)$ and $K(\nu)$ for machine laws refer; $K(\nu_T) \leq K(T)$.*

Theorem 3.8 (Universal function machine)**.** $T_U := \{(x,\, pw,\, y) :\ p \in \operatorname{dom} V,\ (x,w,y) \in T_{V(p)}\}$ *is a function machine. Its law $M_F := \nu_{T_U}$ has $M_F(\emptyset) = 1$ and, for every $D \neq \emptyset$,*

$$M_F(D) = \textstyle\sum_i \mathbf{m}(i)\,\nu_{T_i}(D), \qquad \textit{hence} \qquad M_F \ \geq\ 2^{-K(\nu)}\,\nu \quad \textit{for every machine law } \nu. \tag{1}$$

T_U reads the first self-delimiting block p of its tape as a program for $T_{V(p)}$ and the rest as that machine's coin flips: one tape read in two parts, not a second source of randomness. This represents M_F as the Bayes mixture (1) over function machines, with posterior weight $\mathbf{m}(i)\nu_{T_i}(D)/M_F(D)$ on index i after D.

If every conditional of an order-independent predictor q sums to 1, ν_q is the law of a random total function, and M_F dominates every such lsc law (Proposition A.1 and Corollary A.4).

The choice of universal machine. Let U be a Turing machine as in the informal definition. As noted after Definition 3.6, U is a function machine (a halting run that has read w runs identically on every tape extending w, so the triples (x,w,y) with U outputting y on query x after reading exactly w are consistent), and its law ν_U is a machine law. Call U universal if for every function machine T some finite string p_T gives $f^U_{p_T\omega} = f^T_\omega$ for all ω; T_U is universal, with p_T a V-program for an index of T. Every universal U gives M_F up to a constant factor: by (1), $M_F \geq 2^{-K(\nu_U)}\nu_U$; and $\nu_U(D) \geq 2^{-|p_{T_U}|}M_F(D)$, because the tapes $p_{T_U}\omega'$ with $f^{T_U}_{\omega'} \supseteq D$ have measure $2^{-|p_{T_U}|}M_F(D)$ and give U the function $f^{T_U}_{\omega'}$. So ν_U and M_F agree up to a constant factor depending only on U, and so do their posteriors after any D; by Proposition 3.16, the cumulative loss in the semimeasure convention changes with the choice of U by at most an additive constant (compare Theorem 3.20(a)).

Proposition 3.9 (What machine laws include)**.** *(a)* Product laws (samplers that see only the input). *If P is an lsc conditional semimeasure, then $\nu_P(D) := \prod_{(x,y)\in D} P(y \mid x)$ is a machine law with $K(\nu_P) \le K(P) + O(1)$; probabilistic machines that read x and output y (halting or not) define exactly these P. (b)* Deterministic programs. *For partial computable φ, $D \mapsto \mathbf{1}[\varphi \supseteq D]$ is a machine law of complexity at most $K(\varphi)+O(1)$. (c)* Computable laws. *A law of a random partial function whose cylinder probabilities $P(F|_S = v)$, $v \in Y_\perp^S$, are computable uniformly in (S, v) is a machine law; if Y is finite or the function is total, computable ν_P suffices. (d)* Mixtures. *If (ν_i) are machine laws with computable indices and w is an lsc discrete semimeasure, $D \mapsto \sum_i w_i \nu_i(D)$ $(D \neq \emptyset)$ is a machine law.*

In (a) each input samples from a private stream of tape bits: the random bits for different inputs are taken, by dovetailing, from different parts of the universal machine's tape. With Theorem 3.8 this gives dominance over all such samplers (Theorem 5.1(a)).

Remark 3.10 (Per input, not per query). In (a) a repeated query returns the same label; a sampler run afresh at each query would not. This choice fits data that are input–output pairs of a function, and makes ν_P a random function; the per-query reading gives a different object, Solomonoff's per-query solution M_{pq} (Definition 4.1, Remark 4.2, and Proposition 5.7).

Theorem 3.11 (Universal weak semimeasure)**.** *There is an lsc weak semimeasure M^w with $M^w \ge 2^{-K(\nu)}\nu$ for every lsc weak semimeasure ν.*

Theorem 3.12 (Weak versus machine universality)**.** *(i) $M_F \le 2^{O(1)} M^w$. (ii) For every lsc weak ν and every D, $M_F(D) \ge 2^{-K(\nu)-K(\operatorname{dom} D)-O(1)}\nu(D)$; so $M^w(D) \le 2^{K(\operatorname{dom} D)+O(1)} M_F(D)$. (iii) Let $\nu_\triangle^{\otimes k}$ be the product of k copies of $\nu_\triangle$ (Example 3.5(b)) on k disjoint blocks of three inputs, and for $\vec{\imath} \in \{1,2,3\}^k$ let $E_{\vec{\imath}}$ be the union of the copies of $E_{i_1}, \dots, E_{i_k}$ in the blocks: 3^k pairwise inconsistent data sets with $2k$ points each. Some D_k among them has $\log(M^w(D_k)/M_F(D_k)) \ge k \log \frac{3}{2} - K(k) - O(1)$. (iv) The same lower bound holds for the average of $\log(M^w(E_{\vec{\imath}})/M_F(E_{\vec{\imath}}))$ over all $\vec{\imath}$. (v) The factor behind (iii) is exact: the supremum over laws P of random partial functions of $\min_D P(F \supseteq D)/\nu_\triangle^{\otimes k}(D)$, over the D with $\nu_\triangle^{\otimes k}(D) > 0$, is $(2/3)^k$.*

In (ii) the labels of $\operatorname{dom} D$, read in a fixed order, form a semimeasure on strings that a monotone machine realises exactly (Lemma A.6). In (iii) $K(\operatorname{dom} D_k) \le k \log 3 + K(k) + O(1)$, so asymptotically at least a fraction $\log \frac{3}{2} / \log 3 \approx 0.37$ of the term $K(\operatorname{dom} D)$ in (ii) is needed.

Remark 3.13 (Weak and machine universality differ). By analogy with sequences (Zvonkin and Levin, 1970) one might expect lsc function semimeasures and laws of function machines to agree up to a constant factor. For the weak definition this fails: M^w and M_F agree only up to $2^{K(\operatorname{dom} D)+O(1)}$. For laws of random partial functions it is open (Question 3.14).

Question 3.14. Is every lsc law of a random partial function dominated by M_F up to a constant factor? It is for the classes of Proposition 3.9 and for computable filling-in processes (Remark A.5), and up to $2^{K(\operatorname{dom} D)+O(1)}$ always. By a proof sketch in Appendix A.6 it suffices that every such law, up to a constant factor, is the increasing limit of computable finite rational laws P_t with $P_t(U) \le P_{t+1}(U)$ for every up-set U of the extension order; Strassen's theorem (Strassen, 1965) then gives couplings that fill in values.

3.4 The one-tape version of SOI and its conventions

Definition 3.15 (Solomonoff operator induction, one-tape version)**.** The *one-tape version* of SOI is the random partial function $f_\omega := f_\omega^{T_U}$ of one tape $\omega \sim \mathbb{U}$, read by every query, with law

M_F. After data D the *posterior* is the law of f_ω given $f_\omega \supseteq D$, with containment functional $E \mapsto M_F(D \cup E)/M_F(D)$ for E consistent with D (and 0 otherwise). The *predictive conditional* at $x \notin \operatorname{dom} D$ is $M_F(y \mid D, x) = M_F(D + (x, y))/M_F(D)$; its defect $1 - \sum_y M_F(y \mid D, x)$ is the posterior probability that $f_\omega(x) = \perp$; at $x \in \operatorname{dom} D$ it is the stored label with probability 1. The *normaliser* is $\mathcal{Z}(D, x) := \sum_y M_F(D + (x, y))$, and the *normalised conditional* is $\bar{M}_F(y \mid D, x) := M_F(D + (x, y))/\mathcal{Z}(D, x)$. For a weak ν, $L_\nu(D) := -\log \nu(D)$, and the loss of the one-tape version is $L_{\mathrm{SOI}}(D) := L_{M_F}(D)$.

When we say SOI without qualification, we mean the one-tape version M_F: it treats the labelling as one random function, so labels of repeated inputs agree; it is the law of a single universal machine and dominates every function machine (Theorem 3.8); and on distinct inputs it agrees with the other versions of Section 4 up to $K(\operatorname{dom} D) + O(1)$ bits (Theorem 4.4 and Proposition 5.7). $M_F(D) > 0$ for every D (the lookup table of D is a program).

Proposition 3.16 (Order independence)**.** *Semimeasure convention. For every weak ν and consistent $(x_t, y_t)_{t \le n}$, $\sum_{t \le n} -\log \nu(y_t \mid D_{t-1}, x_t) = L_\nu(D_n) + \log \nu(\emptyset) \le L_\nu(D_n)$, which depends only on the set D_n, not on the order or on repetitions. For SOI it is $L_{\mathrm{SOI}}(D_n)$. On an inconsistent sequence it is infinite.*

(A repeated input contributes the factor 1 or 0; the other factors telescope.)

Proposition 3.17 (Normalisation)**.** *For a weak ν and a consistent sequence let $\bar{L}_\nu := \sum_{t \le n} -\log \bar{\nu}(y_t \mid D_{t-1}, x_t)$, where $\bar{\nu}(y \mid D, x) := \nu(D + (x, y))/\sum_{y'} \nu(D + (x, y'))$ at $x \notin \operatorname{dom} D$, and $\bar{L}_{\mathrm{SOI}} := \bar{L}_{M_F}$. (a) $\bar{L}_\nu \le L_\nu(D_n)$; so upper bounds on L_{SOI} hold in both conventions. (b) For laws of random partial functions the normalised loss can depend on the order; for $\nu = M_F$ this is open (Question A.8). (c) For laws of random total functions it equals the semimeasure loss.*

For (b), let $(F(a), F(b))$ be uniform on $\{(0,0), (0,\perp), (1,1), (\perp,\perp)\}$: $D = \{(a,0),(b,0)\}$ has normalised probability $\frac{2}{3}$ if a comes first and $\frac{1}{2}$ if b does, semimeasure probability $\frac{1}{4}$ in both orders.

What does transfer is collected in Proposition A.9: single-point lower bounds, and two general lower bounds, $K^{\emptyset'}(y_{1:n} \mid x_{1:n}) - O(1)$ and the chronological Bayes loss, which are $O(1)$ in the chain, the skipped strings and the cyclic permutation (Sections 6 and 7).

Remark 3.18 (Which lower bounds are semimeasure-only)*.* Lower bounds on $L_{\mathrm{SOI}}(D_n)$, or for every order-free predictor, do not transfer automatically to $\bar{L}_{\mathrm{SOI}}$. In this paper they are proved only in the semimeasure convention for: Theorem 3.21(a) beyond single points; Example 3.22; the $\mathrm{GF}(2^q)$ example (Theorem D.21); the separations of Propositions D.13 and D.14; the cyclic permutation over the full cycle (Theorem 7.3); Eisenstat's example (Theorems 7.14 and 7.16). Whether they hold normalised is part of Question A.8. *Also normalised:* the chain and the skipped strings, up to $O(n \log n)$ and $O(\ell \log \ell)$ bits (Proposition D.3); the cyclic permutation over its first $k \le 2^{(1-\delta)d}$ steps, at $\delta d - O(\log d)$ bits per step on average, in expectation (Proposition 7.7). *In both conventions:* the out-of-distribution lower bounds of Section 8, which bound the posterior mass that is defined and wrong.

Remark 3.19 (Choice of the default version)*.* Besides the reasons given after Definition 3.15, M_F has a posterior over functions (M^w has none), dominates every machine law with constant $2^{-K(\nu)}$, is invariant (Theorem 3.20), and has an order-independent semimeasure-convention loss. The other objects of Section 4, the coin-free version (the program mixture W), the Kolmogorov version K_F, the product mixture M_{prod}, and M^w, agree with it up to $K(\operatorname{dom} D) + O(1)$ bits on data sets, and so does Solomonoff's per-query solution M_{pq} on sequences with distinct inputs, where it equals M_{prod}; they differ by unbounded amounts and should be kept distinct.

3.5 Basic properties

Theorem 3.20 (Invariance)**.** *(a) If M_F, M'_F are built from two universal prefix machines and two enumerations of function machines, $|\log M_F(D) - \log M'_F(D)| \le c$ for all D. (b) For computable injections $\alpha : X \to X'$, $\beta : Y \to Y'$, with $D^{\alpha\beta} := \{(\alpha(x), \beta(y)) : (x, y) \in D\}$ and M'_F the one-tape law over X', Y': $|\log M_F(D) - \log M'_F(D^{\alpha\beta})| \le K(\alpha, \beta) + O(1)$ for all D. So cumulative losses in the semimeasure convention on original and re-encoded data differ by at most $K(\alpha, \beta) + O(1)$ bits, uniformly over data sets and reveal orders. (c) The same holds for the program mixture W, the product mixture M_{prod} (Definition 4.1) and M^w.*

So the encoding of inputs and labels matters little as long as it is simple: a computable re-encoding costs at most $K(\alpha, \beta) + O(1)$ bits. The constant is not small for complex encodings; only the upper bound is proved, and whether such encodings can change losses by that much is not known. Non-computable encodings are not covered.

Theorem 3.21 (Coding-theorem bounds)**.** *For every data set D: (a) $L_{\text{SOI}}(D) \ge K(D|_T \mid T) - O(1)$ for every $T \subseteq \operatorname{dom} D$; in particular $M_F(D) \le 2^{-K(D|\operatorname{dom} D)+O(1)}$, and $M_F(D) \le 2^{-K(y|x)+O(1)}$ for every $(x, y) \in D$; (b) $L_{\text{SOI}}(D) \le \min\{K_F(D), \sum_{(x,y)\in D} K(y \mid x)\} + O(1) \le K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + O(1)$, with $K_F(D)$ the length of the shortest consistent program (Definition 4.1); (c) $L_{\text{SOI}}(\{(x, y)\}) = K(y \mid x) \pm O(1)$.*

Example 3.22 (Swap data)**.** For distinct $x_1, x_2 \in \{0,1\}^d$ let $D := \{(x_1, x_2), (x_2, x_1)\}$. Then $K(D \mid \operatorname{dom} D) = O(1)$, but $L_{\text{SOI}}(D) \ge \max\{K(x_2 \mid x_1), K(x_1 \mid x_2)\} - O(1)$. Conversely $L_{\text{SOI}}(D) \le d + O(1)$ for every pair: the machine $F(x) := x \oplus \omega_{1:|x|}$ has complexity $O(1)$ and extends D exactly when $\omega_{1:d} = x_1 \oplus x_2$, with probability 2^{-d}. For every x_1 some x_2 has $K(x_2 \mid x_1) \ge d$, and then $L_{\text{SOI}}(D) = d \pm O(1)$; also $K_F(D) \le d + 2\log d + O(1)$.

The swap data do *not* separate SOI from history-aware prediction: every lower-semicomputable history-aware predictor ρ shown x_1 first pays at least $K(x_2 \mid x_1) - K(\rho) - O(1)$ bits on the label of x_1 (conditional coding theorem), and M_H pays at least $K(x_2 \mid x_1) - O(1)$. Such separations are Theorems 6.3, 6.4 and 7.3.

Question 3.23. Is there a two-sided characterisation of $L_{\text{SOI}}(D)$ up to $O(1)$ by standard complexities, other than the tautology $L_{\text{SOI}}(D) = \min_\nu[K(\nu) - \log \nu(D)] \pm O(1)$ over machine laws ν?

Proposition 3.24 (Sequence induction is SOI on initial segments)**.** *Let $X = \mathbb{N}$, $\{0,1\} \subseteq Y$, and M Solomonoff's universal monotone semimeasure (Zvonkin and Levin, 1970). For $x \in \{0,1\}^n$ put $D_x := \{(i, x_{i+1}) : i < n\}$. Then $2^{-O(1)} M(x) \le M_F(D_x) \le 2^{O(1)} M(x)$.*

So SOI generalises Solomonoff induction, and is in turn dominated by universal prediction that sees the history (Lemma 6.2 and Corollary 6.17). M_F is lsc but not computable, and its conditionals, raw or normalised, are computable relative to the halting problem but not computable (Proposition A.10); so SOI is an ideal reference, not an algorithm (computable versions: Section 9).

4 Kolmogorov and semimeasure versions

SOI has at least two natural formalisations: a *Kolmogorov version* scores a data set by the length of its shortest consistent program, a *semimeasure version* by a universal mixture such as M_F

or the program mixture W. For strings the two agree up to an additive constant (the coding theorem), so it is natural to ask whether they agree here.[4] They do not, and the versions also differ in where their randomness lives, from Solomonoff's per-query solution M_{pq} (Remark 4.2) to the one-tape version M_F. Table 1 summarises what is equal, what differs and by how much, and Remark 4.12 states what the gaps mean. Proofs are in Appendix B.

4.1 The variants

Definition 4.1 (Variants)**.** Let (P_i) be the effective enumeration of all lsc conditional semimeasures of Section 3.1.

- *Program mixture* (deterministic programs):
  $W(D) := \sum_{p\in\operatorname{dom} V} 2^{-|p|}\,\mathbf{1}[\varphi_{V(p)} \supseteq D] = \sum_e \mathbf{m}(e)\,\mathbf{1}[\varphi_e \supseteq D]$.
- *Kolmogorov version* (single shortest program): $K_F(D) := \min\{|p| : p \in \operatorname{dom} V,\ \varphi_{V(p)} \supseteq D\} = \min\{K(e) : \varphi_e \supseteq D\}$.
- *Product mixture* (private randomness per input, on data sets):
  $M_{\mathrm{prod}}(D) := \sum_i \mathbf{m}(i) \prod_{(x,y)\in D} P_i(y \mid x)$.
- *Per-query mixture*, the per-query version (private randomness per query, on sequences):
  $M_{\mathrm{pq}}(y_{1:n}\|x_{1:n}) := \sum_i \mathbf{m}(i) \prod_{t\le n} P_i(y_t \mid x_t)$, for every finite sequence of pairs, consistent or not.

W is the coin-free version of SOI: the tape only selects a deterministic program, its first self-delimiting block, and no further bits are read. Equivalently, W puts the prior $2^{-|p|}$ on all self-delimiting deterministic programs p and is conditioned on the data like M_F; as p delimits itself without reading the input, W is a countable mixture of partial computable functions with input-independent weights (Remark B.1). Always $W \le 2^{O(1)} M_F$, and $\log(M_F(D)/W(D)) \le \min\{K^{\mathrm{pad}}(L_{1/2}(M_F, D)), K(\operatorname{dom} D)\} + O(1)$, where $L_{1/2}(M_F, D)$ is the least L such that D is produced within the first L tape bits with probability at least $M_F(D)/2$, and $K^{\mathrm{pad}}(N) := \min_{L\ge N} K(L)$ is the *padded complexity*, nondecreasing, unbounded and at most $K(N)$ (Theorem 4.4, Proposition 4.6(a)). The gap is unbounded (Proposition 4.5), but on two-point binary data sets over d-bit inputs it is at most $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ (Proposition 4.6(b)), hence at most $2\log(h(d) + 2) + O_h(1)$ for every computable, nondecreasing, unbounded h (Proposition 4.6(c), Remark B.2).

$W(\emptyset)$, $M_{\mathrm{prod}}(\emptyset)$, $M_{\mathrm{pq}}(\epsilon\|\epsilon)$ are below 1, which changes losses by a constant only. In the language of algorithmic information theory, with $S_D := \{e : \varphi_e \supseteq D\}$, $K_F(D) = \min_{e\in S_D} K(e)$ is the prefix complexity of the constraint "extends D", the length of the shortest deterministic program whose function extends D (not the complexity $K(D)$ of D as a finite object, which is at least $K_F(D) - O(1)$ and can be much larger), and $W(D) = \mathbf{m}(S_D)$ is the algorithmic probability of the constraint with respect to deterministic programs, the total weight $2^{-|p|}$ of the self-delimiting programs $p \in \operatorname{dom} V$ whose function extends D. For a finite string x and the constraint "outputs x" the analogous quantities are $K(x)$ and $\mathbf{m}(x)$, and the coding theorem gives $K(x) = -\log \mathbf{m}(x) + O(1)$ (Li and Vitányi, 2008, §4.3); for the constraint "output begins with x", monotone complexity and $-\log M$ differ by unbounded amounts (Gács, 1983).

Remark 4.2 (Solomonoff's solutions). Solomonoff (2008, §4) gives two solutions to the operator-induction problem. His eq. (9), the joint solution, treats each question–answer pair as one string,

[4] This question first appeared in the author's working notes (Hänni, 2026c).

predicts with his universal mixture over unordered sets of strings, and conditions on the new question; it also models the questions. His eq. (10) takes the probability of the new answer A_{n+1} to be $\sum_j a_0^j \prod_{i\le n+1} O^j(A_i \mid Q_i)$, a mixture over the conditional distributions O^j computed by partial recursive functions, with weights $a_0^j \approx 2^{-l(O^j)}$; with the lsc conditional semimeasures P_i for the O^j and $\mathbf{m}(i)$ for the weights, this is $M_{\mathrm{pq}}(A_{1:n+1}\|Q_{1:n+1})$, and a repeated question gets an independent factor. His eq. (12) is a convergence theorem for it (see after Proposition 5.7). M_{pq} equals $M_{\mathrm{prod}}(D_n)$ on sequences with distinct inputs; with repeats M_{prod} gives a conflicting label probability 0 and M_{pq} allows fresh noise (Proposition 5.7).

Remark 4.3 (Where the randomness lives). There is no randomness in K_F, and none in W beyond the choice of a program; M_{prod} draws fresh coins at each input and Solomonoff's per-query solution M_{pq} at each query; the one-tape version M_F reads the rest of its one tape, shared by all inputs, so one hypothesis may correlate labels across inputs, as $F(x) = \omega_{|x|+1}$ does. Every mixture also correlates labels through its posterior; the classes still differ by more than constants.

4.2 The chain and the gaps

Theorem 4.4 (The chain). *For every data set $D \neq \emptyset$,*

$$2^{-K_F(D)} \le W(D) \le 2^{O(1)} M_{\mathrm{prod}}(D) \le 2^{O(1)} M_F(D) \le 2^{O(1)} M^w(D) \le 2^{K(\mathrm{dom}\, D)+O(1)}\, 2^{-K_F(D)}.$$

So the five quantities agree up to $K(\mathrm{dom}\, D) + O(1)$ bits.

The last step is the conditional coding theorem, $K(D \mid \mathrm{dom}\, D) \le -\log M^w(D) + O(1)$ (Appendix B.1). Let $\Omega_V^{\ge \ell} := \sum\{2^{-|p|} : p \in \mathrm{dom}\, V,\ |p| \ge \ell\}$, which tends to 0.

Proposition 4.5 (Deterministic programs cannot separate inputs cheaply). *(a)* Two points, binary labels *($\{0,1\} \subseteq Y$). For every $\varepsilon > 0$ and all large d there are distinct $x_1, x_2 \in \{0,1\}^d$ with $W(\{(x_1,0),(x_2,1)\}) \le \varepsilon$, while $M_{\mathrm{prod}}(D) \ge 2^{-O(1)}/4$ and $M_F(D) \ge 2^{-O(1)}/4$ for every two-point D with labels in $\{0,1\}$. (b)* One point, long labels *($Y = \{0,1\}^*$). For every d some $y \in \{0,1\}^d$ has $W(\{(0^d, y)\}) \le 2^{1-d}\, \Omega_V^{\ge d-1}$, while $M_{\mathrm{prod}}(\{(0^d,y)\}) \ge 2^{-d-O(1)}$. So M_{prod}/W and M_F/W are unbounded.*

(Pigeonhole: the fewer than 2^ℓ programs shorter than ℓ give the 2^d inputs at most 3^{2^ℓ} output patterns, so when $3^{2^\ell} < 2^d$ some two inputs are separated by none of them.) The largest two-point gap $\log(M_F(D)/W(D))$ (and $\log(M_{\mathrm{prod}}(D)/W(D))$) over d-bit inputs lies between $\log(1/\Omega_V^{\ge \ell_d}) - O(1)$, $\ell_d := \lceil \log(d/\log 3)\rceil - 1$, and $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ (Remark B.2).

Proposition 4.6 (Tape-length bound). *(a) Let T be a function machine with law ν, and for $L \ge 0$ let $\nu^{(L)}(D) := \mathbb{U}\{\omega :$ every $(x,y) \in D$ has a triple $(x,w,y) \in T$ with w a prefix of $\omega_{1:L}\}$. Then $\nu^{(L)}(D) \uparrow \nu(D)$ as $L \to \infty$, and for every $D \neq \emptyset$ and every L,*

$$W(D) \;\ge\; 2^{-K(T)-K^{\mathrm{pad}}(L)-c}\, \nu^{(L)}(D),$$

with c independent of T, D and L. With $L_{1/2}(\nu, D) := \min\{L : \nu^{(L)}(D) \ge \nu(D)/2\}$ this gives $\log\left(M_F(D)/W(D)\right) \le K^{\mathrm{pad}}\big(L_{1/2}(M_F, D)\big) + O(1)$. (b) Fair-coin labels. *For every D with labels in $\{0,1\}$ and $\mathrm{dom}\, D \subseteq \{0,1\}^{\le d}$: $W(D) \ge 2^{-|D|-K^{\mathrm{pad}}(2^{d+1})-c'} \ge 2^{-|D|-K(\lceil \log(d+1)\rceil)-c''}$. (c) $K^{\mathrm{pad}}(2^{2^s}) \le K(s) + O(1)$, and $K^{\mathrm{pad}}(N) \le 2\log(g(N)+2) + c_g$ for every computable, nondecreasing, unbounded $g : \mathbb{N} \to \mathbb{N}$.*

The $2^{L'}$ programs "simulate T with stored tape prefix w", $|w| = L' \geq L$, $K(L') = K^{\mathrm{pad}}(L)$, carry weight $2^{-K(T)-K^{\mathrm{pad}}(L)-O(1)}\nu^{(L')}(D)$. So W imitates any shared-tape law, paying only for a self-delimiting bound on the number of tape bits used: a deterministic program can carry the random bits it needs. (b) applies (a) to $F(x) := \omega_{j(x)+1}$.

Proposition 4.7 (Shared randomness is not a mixture of private randomness)**.** *Let $\{0,1\} \subseteq Y$ and let ν_{len} be the law of $F(x) := \omega_{|x|+1}$: one fair coin per input length, shared by all inputs of that length. ν_{len} is a machine law of complexity $O(1)$, but $\sup_D \nu_{\mathrm{len}}(D)/M_{\mathrm{prod}}(D) = \infty$. Hence M_F/M_{prod} is unbounded.*

(A product hypothesis labels 0^n and 1^n independently; a compactness argument shows that M_{prod} covers only a small part of the non-constant patterns of k shared coins.)

Remark 4.8 (Size of this gap). The witnesses of Proposition 4.7 are not explicit, and their gap is small: on data with inputs of length at most d, $M_{\mathrm{prod}}(D) \geq 2^{-O(1)}W(D) \geq 2^{-K^{\mathrm{pad}}(d+1)-O(1)}\nu_{\mathrm{len}}(D)$ by Proposition 4.6. A lower bound on the rate is open (Question 4.13).

4.3 No coding theorem for function programs

Unlike for strings, shortest description and algorithmic probability of data sets do not agree up to a constant.

Theorem 4.9 (No coding theorem for function programs)**.** *Let $\{0,1\} \subseteq Y$ and $d \geq 2$. There are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(x_1, 0), (x_2, 1)\}$ satisfies*

(i) $K_F(D) \geq \log d - 2$; *(ii)* $-\log W(D) \leq 2 + K(\lceil \log(d+1) \rceil) + c$; *(iii)* $L_{\mathrm{SOI}}(D) \leq 2 + c$,

with c independent of d. Hence $K_F(D) + \log W(D) \geq \log d - K(\lceil \log(d+1) \rceil) - c - 4 \to \infty$, and on these data sets $-\log W(D) \leq \log(K_F(D) + 4) + 2\log\log(K_F(D) + 4) + O(1)$.

(i) is the colouring argument with $\ell = \ell_d$, (ii) is Proposition 4.6(b), (iii) the fair-coin product law. So a c.e. set of programs, $\{e : \varphi_e \supseteq D\}$, can have algorithmic probability $2^{-O(\log\log d)}$ while its simplest member has complexity $\log d - O(1)$ (related literature: Appendix B.5).

Corollary 4.10 (Two-point rate)**.** *Let $\rho_K(d)$ be the largest value of $K_F(D) - L_{\mathrm{SOI}}(D)$ over two-point data sets D with binary labels and d-bit inputs. Then $\log d - O(1) \leq \rho_K(d) \leq \log d + K(\lceil \log(d+2) \rceil) + O(1) = \log d + O(\log\log d)$.*

Proposition 4.11 (Covering bound for the single shortest program)**.** *For every D with labels in $\{0,1\}$, $|D| = k$ and $\operatorname{dom} D \subseteq \{0,1\}^{\leq d}$,*

$$K_F(D) \;\leq\; k + \log\big(k(d+2)\big) + K\big(\lceil \log(d+2) \rceil, k\big) + O(1).$$

Consequently, with $\delta_D := k - K(D \mid \operatorname{dom} D)$ (the deficiency of randomness of the labels given the inputs), $K_F(D) \leq L_{\mathrm{SOI}}(D) + \delta_D + \log(k(d+2)) + K(\lceil \log(d+2) \rceil, k) + O(1)$.

(About $2^k k(d+2)\ln 2$ random truth tables cover all such data sets; a program indexes the first covering family.) A logarithmic cost in d is real for the single shortest program already on two points (Theorem 4.9); the covering argument shows that it is at most logarithmic in $k(d+2)$. W, compared with the fair-coin law, pays no factor depending on $|D|$ (Proposition 4.6(b)).

Large gaps have no computable witnesses: along a computable family $k \mapsto D_k$ the five quantities of Theorem 4.4 agree up to $K(k)+O(1)$ bits (Remark B.4; whether some gap tends to infinity along such a family is Question B.5).

comparison	equal up to $O(1)$?	gap: lower bound (witness)	gap: upper bound	open
2^{-K_F} vs W	no (Thm. 4.9)	$\log d - K(\lceil \log(d+1) \rceil) - O(1)$ on some two-point binary D (non-computable witness)	$K(\operatorname{dom} D) + O(1)$ (Thm. 4.4); binary labels, k points: $\delta_D + \log(k(d+2)) + K(\lceil \log(d+2) \rceil, k) + O(1)$ (Prop. 4.11; a bound on the gap to M_F, hence to W)	rate in general
2^{-K_F} vs M_F	no	$\log d - O(1)$ on two points (Thm. 4.9)	two points: $\log d + O(\log \log d)$ (Cor. 4.10); general: $K(\operatorname{dom} D) + O(1)$	general rate
W vs M_{prod} (and vs M_F)	no (Prop. 4.5)	two points: $\log(1/\Omega_V^{\geq \ell_d}) - O(1)$; one point, $Y = \{0,1\}^*$: $\log(1/\Omega_V^{\geq d-1}) - O(1)$ (non-computable witnesses)	two points, binary: $K^{\mathrm{pad}}(2^{d+1}) + O(1)$; against a machine law ν: $K(\nu) + K^{\mathrm{pad}}(L_{1/2}(\nu, D)) + O(1)$ (Prop. 4.6); general: $K(\operatorname{dom} D) + O(1)$	Q. 4.13; rates
M_{prod} vs M_F	no (Prop. 4.7)	unbounded, no rate (compactness witnesses)	against ν_{len}, inputs of length $\leq d$: $K^{\mathrm{pad}}(d+1) + O(1)$ (Rem. 4.8); general: $\min\{K^{\mathrm{pad}}(L_{1/2}(M_F, D)), K(\operatorname{dom} D)\} + O(1)$	lower rate
M_F vs M^w	no (Thm. 3.12)	$k \log \frac{3}{2} - K(k) - O(1)$ on some $2k$-point D; on average over an explicit family	$K(\operatorname{dom} D) + O(1)$	how close to $K(\operatorname{dom} D)$
M_F vs every lsc random-function law	unknown	—	$K(\operatorname{dom} D) + O(1)$	Q. 3.14
M_{prod} vs M_{pq}	equal on sequences with distinct inputs	with repeats $M_{\mathrm{pq}} \leq M_{\mathrm{prod}}$; $M_{\mathrm{pq}} > 0 = M_{\mathrm{prod}}$ on inconsistent sequences (Prop. 5.7)	—	—

Table 1: The Kolmogorov and semimeasure versions compared. Data sets D with inputs of length at most d; gaps in bits (log of the ratio of the two quantities, or K_F plus log of the other). Every entry is proved where cited. By Rem. B.4, along a computable family $k \mapsto D_k$ of data sets no gap between the five quantities of Thm. 4.4 exceeds $K(k) + O(1)$; the last row compares objects on sequences, where M_{pq} and M_{prod} can differ completely.

4.4 Summary and meaning for prediction

Remark 4.12 (Summary: the versions are not equivalent, and what the gaps mean). No two of K_F, W and M_F agree up to a constant; the five quantities of Theorem 4.4 agree up to $K(\operatorname{dom} D)+O(1)$ bits, and there is no coding theorem for function programs (Theorem 4.9; for $\mathrm{SOI}^{\mathrm{hist}}$, Remark B.7). For prediction: (1) *Gaps are cumulative losses:* by Proposition 3.16, a gap of g bits between two of $W, M_{\mathrm{prod}}, M_F, M^w$ on D is a difference of g bits in total log loss on D in every order (semimeasure convention, up to constants). (2) *Coin flips or not:* the coin-free version W predicts as well as the one-tape version up to $K^{\mathrm{pad}}(L_{1/2}(M_F, D)) + O(1)$ bits in total (Proposition 4.6). On binary labels its loss exceeds the fair-coin loss $|D|$ by at most $K(\lceil \log(d+1) \rceil) + O(1)$ bits however many labels are predicted, hence exceeds the loss of M_F by at most $\delta_D + K(\lceil \log(d+1) \rceil) + O(1)$ with

$\delta_D = |D| - K(D \mid \operatorname{dom} D)$ (Theorem 3.21(a)). So coin flips beyond the program matter little unless the explanation of the data by M_F needs tape bits with no simple bound, and likewise for private versus shared randomness ($W \le 2^{O(1)} M_{\mathrm{prod}} \le 2^{O(1)} M_F$). (3) *The shortest program:* $K_F(D)$ is a description length (MDL with one hypothesis), not a predictor's loss. It can exceed the loss of M_F by $\log d - O(1)$ bits on two points (Theorem 4.9), and on k binary labels with randomness deficiency δ_D by at most $\delta_D + \log(k(d+2)) + K(\lceil \log(d+2) \rceil, k) + O(1)$ (Proposition 4.11); whether gaps of order $\log k$ occur is open (Question 4.13). For deterministic targets it has the learning rate of M_F (Proposition 5.11). (4) M^w can exceed M_F by linearly many bits on some data sets (Theorem 3.12(iii)), because it dominates the weak semimeasure $\nu_{\triangle}^{\otimes k}$, which no law of a random function dominates by a factor better than $(2/3)^k$; it pays for this by having no posterior over functions. (5) For a deterministic computable f all five quantities are at least $2^{-K(f)-O(1)}$ on its data, so the basic learning bounds (Theorem 5.1(b), Corollary 5.4, Theorem 5.5(a)) hold for W, M_{prod}, M_F and M^w; for K_F the corresponding statement is the Occam bound Proposition 5.11. (6) Fresh label noise on repeated inputs separates Solomonoff's per-query solution M_{pq} from all set-based objects (Propositions 5.6 and 5.7). (7) On a single query the one-tape version is the conditional universal distribution, $2^{-K(y|x) \pm O(1)}$, which commits to where the output ends (Proposition B.6).

Question 4.13. Is $\log(M_F(D)/W(D)) = O(K(d, |D|))$ for inputs of length at most d, or even K^{pad} of a computable function of $(d, |D|)$? Proved: $\min\{K^{\mathrm{pad}}(L_{1/2}(M_F, D)), K(\operatorname{dom} D)\} + O(1)$, and $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ against fair coins. Also open: the exact rate of the two-point gap of W (between $\log(1/\Omega_V^{\ge \ell_d})$ and $K^{\mathrm{pad}}(2^{d+1})$); whether the single shortest program has gaps of order $\log k$ on k points; a lower rate for M_F/M_{prod}; and how close $\log(M^w/M_F)$ comes to $K(\operatorname{dom} D)$.

5 Learning theorems: SOI as supervised learning

SOI can be viewed as ideal supervised learning; this section states the guarantees behind that view and where they stop. Throughout, $D_n = \{(x_t, y_t) : t \le n\}$; unless stated otherwise the results concern the one-tape version, and $L_{\mathrm{SOI}}(D_n) = -\log M_F(D_n)$ is its cumulative log loss in the semimeasure convention, independent of order and repetitions and infinite on inconsistent sequences (Proposition 3.16); upper bounds in it also hold normalised (Proposition 3.17). We write μ for the input distribution, c for a machine constant, and $L_\nu(D) := -\log \nu(D)$. Complexities $K(P)$ of lsc conditional semimeasures and $K(\varphi)$ of partial computable functions refer to the enumerations of Section 3.1.

5.1 Dominance

Theorem 5.1 (dominance)**.** *Let ν be a machine law (Definition 3.6). Then $M_F(D) \ge 2^{-K(\nu)} \nu(D)$ for every data set D. Hence for every consistent sequence, however its inputs are chosen (in advance, adversarially, adaptively, non-computably),*

$$L_{\mathrm{SOI}}(D_n) \;\le\; -\log \nu(D_n) + K(\nu). \tag{2}$$

In particular: (a) x-only samplers: *for every lsc conditional semimeasure P, $L_{\mathrm{SOI}}(D_n) \le \sum_{t \le n} -\log P(y_t \mid x_t) + K(P) + c$, where the right side counts a repeated input each time it appears and the left side once; (b)* deterministic programs: *if $y_t = \varphi(x_t)$ for all t, with φ partial computable, then $L_{\mathrm{SOI}}(D_n) \le K(\varphi) + c$; (c)* weak comparators: *for every lsc weak semimeasure ν (Definition 3.2), $L_{\mathrm{SOI}}(D_n) \le -\log \nu(D_n) + K(\nu) + K(\operatorname{dom} D_n) + c$. The*

term $K(\operatorname{dom} D_n)$ cannot be replaced by a constant (semimeasure convention): on the $2k$-point data sets of Theorem 3.12(iii), L_{SOI} exceeds $-\log M^w$ by at least $k \log \frac{3}{2} - K(k) - c$, while $K(\operatorname{dom} D) \le k \log 3 + K(k) + c$.

Proof idea (Appendix C.1). Dominance is Theorem 3.8, and Proposition 3.16 turns it into a pathwise bound; (a) and (b) use the machine laws of Proposition 3.9(a),(b), and (c) is Theorem 3.12(ii),(iii).

Taking expectations gives the bound under every data distribution. Part (a) is dominance over samplers that see only the input (samplers that also see the history are treated in Section 6.3): every computably samplable predictor of the label from the input loses to M_F at most a constant in log loss, depending on the predictor but not on the data distribution. It holds for data from a function (consistent sequences), with randomness read once per input. For the one-tape version, with randomness shared by all inputs, this theorem and the bounds below that rest on it are ours in this form; they follow by the standard dominance argument, which also underlies Solomonoff's convergence theorem for his per-query solution (Proposition 5.7).

Remark 5.2 (scope of dominance). (i) The comparators include all computable laws of random total functions, for every countable Y (Proposition 3.9 and Lemma A.3). (ii) The label sequence must be consistent: if an input reappears with another label, $L_{\mathrm{SOI}} = \infty$, while a sampler with fresh randomness per *query* has finite loss. Randomness per input is the right reading for data from a function (Remark 3.10); per query it gives Solomonoff's per-query solution M_{pq} (Proposition 5.7). (iii) For general lsc laws of random partial functions the statement is open (Question 3.14).

5.2 Merging and mistakes

Theorem 5.3 (merging)**.** *Let ν be a machine law of a random total function, equivalently a computable law of a random total function (Lemma A.3), and $F \sim \nu$. Let the inputs follow any measurable rule $x_t = s_t(h_{t-1}, \xi_t)$, with h_{t-1} the history, (ξ_t) independent of F, and s_t arbitrary (possibly non-computable); $y_t := F(x_t)$. With the one-step conditionals of Definition 3.1 let*

$$d_t := \mathrm{KL}\left(\nu(\cdot \mid D_{t-1}, x_t) \,\|\, M_F(\cdot \mid D_{t-1}, x_t)\right) \text{ bits} \qquad (d_t = 0 \text{ at a repeated input}).$$

Then $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} \log(\nu(D_n)/M_F(D_n)) \le K(\nu)$ for every n. Hence $\sum_{t \ge 1} \mathbb{E} d_t \le K(\nu)$, $d_t \to 0$ almost surely, and $\sum_{t \ge 1} \mathbb{E} \|\nu(\cdot \mid D_{t-1}, x_t) - M_F(\cdot \mid D_{t-1}, x_t)\|_1^2 \le 2 \ln 2 \cdot K(\nu)$.

Proof idea. Given the past, y_t has law $\nu(\cdot \mid D_{t-1}, x_t)$, so the expected log ratio at step t is d_t; the ratios telescope to $\log(\nu(D_n)/M_F(D_n)) \le K(\nu)$ (Theorem 5.1; integrability in Appendix C.2).

This is the one-tape analogue of Solomonoff's convergence theorems (cf. Solomonoff, 1978; for the per-query solution, Proposition 5.7). If the rule may see F, e.g. picks inputs by their label, the expected bound can fail: in a finite analogue with a bound of 1 bit, a direct computation (code in the accompanying repository) gives a summed KL divergence of 12.5 bits. The pathwise bound (2) still holds, and Theorem 6.10 bounds the excess by directed information.

Corollary 5.4 (mistake bound; cf. the halving argument of Barzdin and Freivalds, 1972; Littlestone, 1988)**.** *Let φ be partial computable, $x_1, x_2, \ldots \in \operatorname{dom} \varphi$ any inputs, and $y_t = \varphi(x_t)$. At most $K(\varphi) + c$ steps have $M_F(\varphi(x_t) \mid D_{t-1}, x_t) \le \frac{1}{2}$. So $\hat{y}_t := \arg\max_y M_F(y \mid D_{t-1}, x_t)$ (the maximum exists because the conditionals are positive and sum to at most 1; ties are broken arbitrarily) makes at most $K(\varphi) + c$ mistakes on every input sequence.*

Proof. Each such step adds at least one bit to $L_{\mathrm{SOI}}(D_n) \le K(\varphi) + c$ (Theorem 5.1(b)); above $\frac{1}{2}$, $\varphi(x_t)$ is the unique maximiser. □

This is the halving bound with $K(\varphi)$ in place of $\log|H|$; the predictor is not computable (Proposition A.10); examples: Proposition 7.24.

5.3 Online-to-batch and noise

Theorem 5.5 (online-to-batch)**.** *Let $x_1, x_2, \ldots$ be i.i.d. from any distribution μ on X, computable or not.*

(a) Deterministic labels. *Let φ be partial computable, $\mu(\operatorname{dom}\varphi) = 1$, $y_t = \varphi(x_t)$. Put $Q_t(\cdot \mid x) := M_F(\cdot \mid D_t, x)$, $\bar{Q}_n := \frac{1}{n}\sum_{t<n} Q_t$, $R(Q) := \mathbb{E}_{x\sim\mu}[-\log Q(\varphi(x) \mid x)]$. Then $\sum_{t<n}\mathbb{E}R(Q_t) \le K(\varphi) + c$, $\mathbb{E}R(\bar{Q}_n) \le (K(\varphi)+c)/n$, and the expected 0–1 risk $\mathbb{E}\,\mu\{x : \arg\max_y \bar{Q}_n(y \mid x) \ne \varphi(x)\}$ is at most $(K(\varphi)+c)/n$.*

(b) Noisy labels, stamped inputs. *Let $y_t \sim P(\cdot \mid x_t)$ independently given the inputs, for any conditional probability P (possibly non-computable). Run SOI on $\tilde{x}_t := \langle x_t, t\rangle$, for a computable bijection $\langle\cdot,\cdot\rangle : X \times \mathbb{N} \to X$, with data sets $\tilde{D}_t$; put $q_t(\cdot \mid x) := M_F(\cdot \mid \tilde{D}_{t-1}, \langle x, t\rangle)$ and $\bar{q}_n := \frac{1}{n}\sum_{t\le n} q_t$. Then for every lsc conditional semimeasure P',*

$$\mathbb{E}\,\mathbb{E}_{x\sim\mu}\,\mathrm{KL}\left(P(\cdot \mid x) \,\|\, \bar{q}_n(\cdot \mid x)\right) \le \tfrac{1}{n}\big(K(P') + c\big) + \mathbb{E}_{x\sim\mu}\,\mathrm{KL}\left(P(\cdot \mid x) \,\|\, P'(\cdot \mid x)\right).$$

For lsc (hence computable) P this is $(K(P)+c)/n$. The expected excess 0–1 risk of $\arg\max_y \bar{q}_n(y \mid x)$ over the Bayes classifier is at most $\sqrt{2\ln 2}$ times the square root of the right side.

(c) Frozen noise. *If F has the product law of a computable conditional probability P (one draw per input) and $y_t = F(x_t)$, then $\sum_{t\le n}\mathbb{E}d_t \le K(P) + c$, with d_t as in Theorem 5.3, for every input rule that sees only the past.*

Part (a) is online-to-batch conversion (Littlestone, 1989; Cesa-Bianchi et al., 2004): as x_{t+1} is independent of D_t, $\mathbb{E}R(Q_t)$ is the expected online loss at step $t+1$, Theorem 5.1(b) bounds the sum, and convexity moves the average inside. Averaging over a random order of a finite set instead gives Proposition 5.8 (cf. Haussler et al., 1994). In (b) the stamps make inputs distinct; (c) is Theorem 5.3 for a product law (proofs: Appendix C.3). The bounds are for the Cesàro average $\bar{Q}_n$, not for Q_n (Remark C.2).

Proposition 5.6 (the one-tape version cannot model fresh label noise)**.** *Let $\mu(x_0) = \alpha > 0$, let $P(\cdot \mid x_0)$ give positive probability to two labels, and $p_{\max} := \max_y P(y \mid x_0) < 1$. Draw the pairs (x_t, y_t) i.i.d. from $\mu \times P$, with a fresh label at every draw, and predict with $\nu(\cdot \mid D, x)$ for a weak semimeasure ν, raw or normalised, with the repeated-input convention of Definition 3.1. This covers M_F, W, M_{prod} and M^w. (a) For every $n \ge 2$, $\mathbb{P}[D_n$ inconsistent$] > 0$, so the expected cumulative log loss is infinite, in both conventions. (b) $\mathbb{P}[D_n$ consistent$] \le (1-\alpha)^n + p_{\max}^{-1}(1-\alpha+\alpha p_{\max})^n \to 0$. (c) The natural strengthening "for every input distribution and every computable conditional probability P, the expected cumulative log loss of SOI exceeds that of P by at most $K(P) + O(1)$" fails for the one-tape version M_F. For P with finite expected log loss it holds for all n exactly when $P(\cdot \mid x)$ is a point mass for every x with $\mu(x) > 0$; then it is Theorem 5.5(a).*

(Proof: Appendix C.3; for P with infinite expected log loss the inequality in (c) is trivial.) The one-tape version models the label as a function of the input; a fresh-noise source is not one. With 10 equally likely inputs and fair-coin labels, a direct computation (code in the accompanying repository) gives the probability that the data stay consistent as 0.95, 0.113 and $5 \cdot 10^{-4}$ at $n = 2, 10, 20$. The repairs are stamps and frozen noise (Theorem 5.5(b),(c); example: Proposition 7.29) and Solomonoff's per-query solution:

Proposition 5.7 (Solomonoff's per-query solution)**.** *Let* $M_{\mathrm{pq}}(y_{1:n}\|x_{1:n}) = \sum_i \mathbf{m}(i)\prod_{t\le n} P_i(y_t \mid x_t)$ *on all finite sequences of pairs (Definition 4.1). (a)* M_{pq} *is an lsc chronological semimeasure:* $M_{\mathrm{pq}}(\epsilon\|\epsilon) \le 1$ *and* $\sum_y M_{\mathrm{pq}}(y_{1:n}y\|x_{1:n}x) \le M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$*. It depends only on the multiset of pairs; it equals* $M_{\mathrm{prod}}(D_n)$ *on sequences with distinct inputs, is at most* $M_{\mathrm{prod}}(D_n)$ *on consistent sequences with repeats (often strictly), and is positive on inconsistent sequences. (b) On every sequence, consistent or not,* $-\log M_{\mathrm{pq}}(y_{1:n}\|x_{1:n}) \le \sum_{t\le n} -\log P(y_t \mid x_t) + K(P)$ *for every lsc conditional semimeasure* P*. (c) Let* (x_t, y_t) *be i.i.d. from* $\mu\times P$*,* μ *arbitrary and* P *any conditional probability,* $q_t(\cdot \mid x) := M_{\mathrm{pq}}(y_{<t}\,\cdot\,\|x_{<t}x)/M_{\mathrm{pq}}(y_{<t}\|x_{<t})$ *and* $\bar{q}_n := \frac{1}{n}\sum_{t\le n} q_t$*. For every lsc conditional semimeasure* P'*,* $\sum_{t\le n}\mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t) \le K(P') + n\,\mathbb{E}_\mu \mathrm{KL}(P\|P')$*, so* $\mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|\bar{q}_n) \le K(P')/n + \mathbb{E}_\mu \mathrm{KL}(P\|P')$*. (d) With the stamps of Theorem 5.5(b),* $M_F(\tilde{D}_n) \ge 2^{-c} M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$*.*

M_{pq} is Solomonoff's second solution, his eq. (10) (Remark 4.2). Part (b) is the pathwise log-loss form of its dominance; Solomonoff (2008, eq. (12)) proves the expected squared-error form: for labels drawn from $P(y \mid x)$ computed by a probabilistic algorithm describable in k bits, and any given sequence of inputs, the expected sum of the squared errors of the predicted symbol probabilities, over all symbols of all labels (in binary, with separators), is less than $k \ln 2$; he notes that the proofs of the underlying sequence-prediction theorems also give the bound for Kullback–Leibler loss (his Section 2). Part (c) is the standard consequence for i.i.d. data. So the regret bound holds under fresh noise for M_{pq}, with no stamps (proofs: Appendix C.4); $\mathrm{SOI}^{\mathrm{hist}}$ dominates it (Proposition D.15); Proposition 9.4 is a time-bounded analogue.

5.4 Learning curves and in-distribution error

The expected risk of a Bayes predictor need not decrease with the sample size (Proposition C.1), and whether the last iterate Q_n of Theorem 5.5(a) attains the Cesàro rate is a leave-one-out question (Remark C.2); these results, the leave-one-out bound (Proposition C.3) and the corresponding conjecture for M_F (Conjecture C.4) are in Appendix C.5. In the main text we treat random splits and per-step errors.

Errors. For $\nu \in \{W, M_F\}$, a target f, training inputs A with $D = f|_A$, and a test input x^*: $e(x^*) := 1 - \nu(f(x^*) \mid D, x^*)$ is the posterior mass wrong or undefined at x^*; $e^{\mathrm{wr}}(x^*) := \sum_{y\ne f(x^*)} \nu(D + (x^*, y))/\nu(D)$ is the mass defined and wrong; $e_{\mathrm{norm}}(x^*)$ is the error of the normalised predictor (all three errors are 0 if $x^* \in A$, by the repeated-input convention). Since the normaliser is at most $\nu(D)$, $e^{\mathrm{wr}} \le e_{\mathrm{norm}} \le e$: lower bounds on e^{wr} and upper bounds on e hold in both conventions.

Proposition 5.8 (random splits)**.** *Let* $\nu \in \{W, M_F\}$*,* B *any finite input set (possibly adversarial),* f *any labelling of* B*. Train on the first* $k-1$ *inputs of a uniformly random order of* B *and test on the* k*-th, with* k *uniform on* $\{1, \dots, |B|\}$ *and independent. The expected test loss is* $-\log(\nu(f|_B)/\nu(\emptyset))/|B| \le L_\nu(f|_B)/|B| \le (K_F(f|_B) + c)/|B|$ *(semimeasure convention; in the normalised convention the expected test loss is at most this), with* $c = 0$ *for* W*, and the expected test error is at most* $(\ln 2) L_\nu(f|_B)/|B|$*.*

Proof idea (Appendix C.5). In every order the step losses sum to the order-free total; the exchangeability argument is standard (Cesa-Bianchi et al., 2004).

Proposition 5.9 (separating properties are typical even in distribution)**.** *Let* $f \equiv 0$ *on* $\{0,1\}^d$*,* $x_1, \dots, x_n$ *i.i.d. uniform training inputs and* x^* *an independent uniform test input,* $d \ge \ell := \lceil \log n \rceil + 1$*. Let* G_u *be the set of strings that start with the* ℓ*-bit prefix* u *of* x^**. Then (1)* G_u *separates* x^* *from the training inputs with probability* $(1-2^{-\ell})^n \ge \frac{1}{2}$*; (2)* $K(G_u) \le \ell + 2\log(\ell+1) + c$*;*

(3) for $\nu \in \{W, M_F\}$, $\mathbb{E}\, e^{\mathrm{wr}}(x^*) \geq \frac{1}{2} \cdot 2^{-\ell - 2\log(\ell+1) - c} \geq 2^{-c}/(8n(\log n + 3)^2)$, in both conventions. For computable binary f the same argument gives $\mathbb{E}\, e^{\mathrm{wr}}(x^*) \geq 2^{-K(f)-c}/(8n(\log n + 3)^2)$.

Proof idea. On the event of (1), answering 1 on G_u and running a training-consistent program elsewhere costs $K(G_u) + c$ extra bits (a case of Proposition 8.1; Appendix C.6). So the natural condition for generalisation that no simple property separates the test input from the training inputs (Section 8.4) typically holds only at scales of about $\log n$ bits. This per-step lower bound is not comparable with the Cesàro bound of Theorem 5.5(a) (Remark C.7). Per step:

Corollary 5.10 (per-step in-distribution error)**.** *Let μ be any distribution on X, f partial computable with $\mu(\mathrm{dom}\, f) = 1$, $x_1, \ldots, x_n, x^*$ i.i.d. from μ, $\nu \in \{W, M_F\}$, and e the semimeasure-convention error (so the bounds hold normalised too). (a) For every $\varepsilon \in (0,1)$, $\mathbb{E}e(x^*) \leq \varepsilon + 2^{K(f)+c}\exp(-\varepsilon n)$; hence $\mathbb{E}e(x^*) \leq ((K(f)+c)\ln 2 + \ln n + 1)/n$. With probability at least $1-\delta$ over the training sample, $\mathbb{E}_{x^*}[e(x^*) \mid D_n] \leq \varepsilon + 2^{K(f)+c}\exp(-\varepsilon n)/\delta$. The posterior-majority predictor has expected 0–1 risk at most $2\mathbb{E}e(x^*)$. (b) $\mathbb{E}e(x^*) \leq \mathbb{E}[\frac{\nu(\emptyset) - \nu(f|_S)}{(n+1)\nu(f|_S)}] \leq 2^{K(f)+c}/(n+1)$, where S is the set of the $n+1$ sampled values. For $f \equiv 0$ and uniform inputs on $\{0,1\}^d$, $d \geq \lceil \log n \rceil + 1$: $2^{-c}/(8n(\log n + 3)^2) \leq \mathbb{E}e(x_{n+1}) \leq 2^c/(n+1)$ in both conventions, so the per-step error is $\Theta(1/n)$ up to the $\log^2 n$ factor.*

Proof idea (Appendix C.6). (a) A hypothesis with error above ε survives with probability at most $\exp(-\varepsilon n)$; the posterior denominator $\nu(D_n)$ is at least $2^{-K(f)-c}$. (b) By exchangeability the error is an average of leave-one-out errors. Part (a) is the Markov half of the relevance-based bound in the slides of Vaintrob and Hänni (2026b), for W and M_F (Remark C.8).

5.5 Comparison, and the sense in which SOI is ideal

Proposition 5.11 (Occam bound in description-length form; standard, Blumer et al., 1987)**.** *Let x_t be i.i.d. from μ, φ partial computable with $\mu(\mathrm{dom}\, \varphi) = 1$, $y_t = \varphi(x_t)$. With probability at least $1 - \delta$, every program $p \in \mathrm{dom}\, V$ consistent with D_n has $\mu\{x : \varphi_{V(p)}(x) \neq \varphi(x)\} \leq (|p| \ln 2 + \ln(1/\delta))/n$, where "$\neq$" includes "undefined". So the shortest consistent program, of length $K_F(D_n) \leq K(\varphi)$, has 0–1 risk at most $(K(\varphi)\ln 2 + \ln(1/\delta))/n$.*

We include the union-bound proof for completeness (Appendix C.7). So for deterministic targets the one-tape version (Theorem 5.5(a)) and the Kolmogorov version learn at the same rate, though their code lengths can differ (Theorem 4.9). Appendix C.8 compares with PAC-Bayes and resolvability bounds (McAllester, 1999; Barron and Cover, 1991).

Remark 5.12 (in what sense SOI is ideal supervised learning)*.* Established: (1) The one-tape version is the Bayes predictor for the law M_F of T_U, a universal machine on one random tape; with the tape read as a program followed by its coin flips, M_F is the mixture of all machine laws with weights $\mathbf{m}(i) \geq 2^{-K(i)}$ on nonempty data sets (Theorem 3.8), and it is universal for that class pathwise (Theorem 5.1). (2) With i.i.d. inputs from any distribution and computable labels, the averaged predictor has expected log and 0–1 risk at most $(K(\varphi) + c)/n$; for computable P, with frozen noise, or fresh noise on stamped inputs or for M_{pq}, the cumulative bound $K(P) + c$ holds, and for non-computable P the resolvability bound against lsc comparators (Theorem 5.5 and Proposition 5.7). This is the Occam rate with K for $\log |H|$, without a union bound or confidence parameter. (3) Inputs are never predicted, so the input distribution may be arbitrary at no cost (chosen inputs: Proposition C.5; positive examples: Remark C.9). Limits: *loss*: the guarantees are for log loss on labels, and 0–1 bounds follow (K/n without noise, $\sqrt{K/n}$ excess with

noise); *averaged predictor*: Proposition C.1 and Remark C.2; *function semantics*: Proposition 5.6; *computability*: Proposition A.10, and Section 9 for computable versions; *constants*: machine constants, and $K(\alpha, \beta)$ for encodings (Theorem 3.20); *distribution shift*: Section 8; *comparator class*: simple history-aware predictors have loss 0 where the loss of SOI is unbounded (Theorems 6.3, 6.4 and 7.3; the swap data of Example 3.22 do not show this), and $\mathrm{SOI}^{\mathrm{hist}}$ removes this limit (Corollary 6.17).

6 History-aware prediction

SOI sees the past only as a set: its hypotheses answer each query without seeing the other inputs, and its loss $L_{\mathrm{SOI}}(D_n)$ in the semimeasure convention ignores the order of the data (Proposition 3.16). We give two examples in which seeing the history helps without bound,[5] determine in which formulations a dominance statement for simple reveal laws holds (some cases remain open), and show in what sense the history-aware version of SOI, whose hypotheses see the list of previous inputs, is as good as SI (Section 6.4). Lower bounds on L_{SOI} are in the semimeasure convention unless stated; c_ω depends on the machines and on a Martin-Löf random ω. Proofs and some secondary statements are in Appendix D.

6.1 History-aware predictors

Definition 6.1 (History-aware predictors)**.** A *history-aware predictor* P gives sub-probabilities $P(y \mid h, x)$; its loss is $L_P := -\log P(y_{1:n} \| x_{1:n})$, $P(y_{1:n} \| x_{1:n}) := \prod_{i \le n} P(y_i \mid h_{i-1}, x_i)$. A *computable sampler* is a probabilistic machine that, given (h, x), reads random bits and halts with a label or diverges (*defective* if with positive probability); $P(y \mid h, x)$ is its probability of output y, lsc in the arguments. A sampler may keep memory between calls; it then predicts through its sequence law $P(y_{1:n} \| x_{1:n})$ and the conditionals this law induces (Lemma D.4). An *input sampler* (reveal law) Δ outputs an input given h. A *chronological semimeasure* is a map $\rho(y_{1:n} \| x_{1:n}) \ge 0$ with $\rho(\epsilon \| \epsilon) \le 1$ and $\sum_y \rho(y_{1:n} y \| x_{1:n} x) \le \rho(y_{1:n} \| x_{1:n})$ for all x; $L_\rho := -\log \rho(y_{1:n} \| x_{1:n})$.

Chronological semimeasures go back to Hutter (see the paragraph after Corollary 6.17). The dominance statements of Section 6.3 concern computable samplers; one that halts with probability 1 has computable probabilities ($1 - \sum_{y' \ne y} P(y' \mid h, x)$ is upper semicomputable).

Lemma 6.2 (SOI is a chronological semimeasure)**.** *$\rho_F(y_{1:n} \| x_{1:n}) := \mathbb{U}\{\omega : f_\omega(x_i) = y_i \ \forall i \le n\}$ is an lsc chronological semimeasure, equal to $M_F(D_n)$ on consistent sequences and 0 otherwise. Hence the loss L_H of $\mathrm{SOI}^{\mathrm{hist}}$ (Definition 6.15) satisfies $L_H \le L_{\mathrm{SOI}}(D_n) + c$ in every order. The lsc chronological semimeasures that depend only on D_n and vanish on inconsistent sequences are exactly the lsc weak function semimeasures.*

6.2 Counterexamples for order-free prediction

In two examples a simple predictor that sees the history beats SOI without bound. In the first, a deterministic sequence of increasingly long random strings is revealed as a chain, each label being the previous input; in the second, every string of a given length but one is labelled 0, and the remaining one, revealed last, is labelled 1.

[5] The examples of Section 6.2 and of Theorem 6.11, an informal dominance statement for simple reveal laws, the idea of proving it by simulating the reveal law inside a function machine (Theorem 6.8) and the question behind Conjecture D.10 first appeared in the author's working notes (Hänni, 2026c).

Theorem 6.3 (The chain example)**.** *Let $x_1, \dots, x_n$ ($n \geq 2$) be distinct nonempty strings, $D_n := \{(x_1, \epsilon)\} \cup \{(x_{i+1}, x_i) : i < n\}$, revealed as $(x_1, \epsilon), (x_2, x_1), \dots$. (a) $K(x_1, \dots, x_{n-1} \mid x_n) - c \leq L_{\mathrm{SOI}}(D_n) \leq K(x_1, \dots, x_{n-1}) + c$ (semimeasure convention). (b) If $1 \leq \ell_1 < \ell_2 < \cdots$ are computable, x_i is the i-th block, of length ℓ_i, of a Martin-Löf random ω, and $\bar{\ell}_k := \sum_{i \leq k} \ell_i$, then $\bar{\ell}_{n-1} - K(\ell_n) - c_\omega \leq L_{\mathrm{SOI}}(D_n) \leq \bar{\ell}_{n-1} + K(n) + c$ (semimeasure convention); for $\ell_i = i$, $L_{\mathrm{SOI}}(D_n) = n(n-1)/2 \pm (2 \log n + c_\omega)$. (c) The predictor "$\epsilon$ on the empty history; otherwise the unique input of the history that is not one of its labels" has loss 0; it is deterministic, set-based and of complexity $O(1)$.*

Proof idea. Iterating f_ω from x_n recovers $x_{n-1}, \dots, x_1$, so the chain events are disjoint (Lemma D.1); apply the conditional coding theorem.

For the second example fix $s_\ell \in \{0,1\}^\ell$ for each ℓ and let $f_s(x) := \mathbf{1}[x = s_{|x|}]$. Block ℓ reveals the other strings of length ℓ in lexicographic order, then s_ℓ; $D^{(\ell)}$ is the data after blocks $1, \dots, \ell$.

Theorem 6.4 (Skipped strings)**.** *(a) "Predict 1 at x iff every other string of length $|x|$ is already an input" has loss 0; it is deterministic, set-based and of complexity $O(1)$. (b) $K(s_1, \dots, s_\ell \mid \ell) - c \leq L_{\mathrm{SOI}}(D^{(\ell)}) \leq \ell(\ell+1)/2 + c$ (semimeasure convention). (c) If $s_1 s_2 \cdots$ are the consecutive blocks of a Martin-Löf random ω, then $L_{\mathrm{SOI}}(D^{(\ell)}) \geq \ell(\ell+1)/2 - K(\ell) - c_\omega$; for each ℓ, all but a 2^{-j} fraction of tuples $(s_1, \dots, s_\ell)$ give $L_{\mathrm{SOI}}(D^{(\ell)}) \geq \ell(\ell+1)/2 - j - c$ (semimeasure convention).*

The lower bound is the domain bound of Theorem 3.21(a). In the length-lexicographic order $\mathrm{SOI}^{\mathrm{hist}}$ pays as much as SOI (Corollary 6.22), in the reveal order above, $O(1)$: the order carries the information.

Proposition 6.5 (Every order-free predictor pays)**.** *Let ν be a weak function semimeasure (Definition 3.2), e.g. M^w, M_{prod} or W. (a) For every x, $\sum_z \nu(D(x,z)) \leq 1$ over all chains $D(x,z) := \{(x, z_1), (z_1, z_2), \dots, (z_k, \epsilon)\}$ ($k \geq 0$) of distinct nonempty strings $z_i \neq x$. (b) If ν is lsc, the lower bounds of Theorems 6.3 and 6.4 hold for $-\log \nu$ with $K(\nu)$ subtracted. (c) For every ν, with independent uniform blocks, $\mathbb{E}[-\log \nu(D_n)] \geq \bar{\ell}_{n-1}$ and $\mathbb{E}[-\log \nu(D^{(\ell)})] \geq \ell(\ell+1)/2$.*

So every predictor that sees the past as a set *and* is order-independent pays the cost (the predictors of Theorem 6.3(c) and Theorem 6.4(a) are not order-independent). The normalised predictor $\bar{M}_F$ is not order-free, but on both examples (with $\ell_i = i$ in the chain) it pays the same quadratic loss up to $O(n \log n)$, respectively $O(\ell \log \ell)$ bits (Proposition D.3): a total *broom* that names the past blocks and sends every other input to the last of them keeps its normaliser large (Lemma D.2).

6.3 A dominance theorem for computable reveal laws

In these examples the reveal sequence is complex. If instead the inputs are drawn from a reveal law Δ that is simple to sample from given f, one may hope that every computable sampler P that sees the history beats SOI in expected log loss by at most the complexity C_Δ of sampling from Δ plus a constant depending on P. How C_Δ is measured matters, since the randomness of the chain example can be moved into the reveal law (Theorem 6.11).

Definition 6.6 (Three formulations)**.** Let $y_i = f(x_i)$. *Formulation A*: one reveal law $\Delta(x_i \mid h_{i-1})$ that does not see f (private randomness and memory allowed), f arbitrary. *Formulation B* (the oracle form): $x_i \sim \Delta(\cdot \mid f, h_{i-1})$, with C_Δ the complexity of Δ relative to an oracle for f. *Formulation C*: for each f a reveal law $\Delta_f(x \mid h)$ that sees only the history, with $C_\Delta := K(\Delta_f)$ as a standalone sampler; A is the case $\Delta_f = \Delta$.

Formulation C is the natural one for the one-tape version of SOI: a function machine can simulate a reveal law that sees only the history, not one that sees f. The randomised chain has $C_\Delta = O(1)$ in Formulation B but is excluded in Formulation C (Theorem 6.11). Let $I(f \to x_{1:n}) := \sum_{i \le n} I(f; x_i \mid h_{i-1})$ (*directed information*). A reveal law that does not see f may keep private memory; it enters only through its *reveal-sequence law* $\Delta(x_{1:n} \| y_{1:n-1})$, and $I(f \to x) = 0$ (Lemma D.4). A natural proof simulates Δ and P alternately inside a function machine until the query appears, outputs its label, and applies Jensen's inequality. With separate randomness per input, this direct simulation fails:

Proposition 6.7 (A direct simulation fails)**.** *Let $X = \{a, b\}$, $f(a) = f(b) = \beta$ for a uniform bit β, and let Δ show a, then b. The predictor with y_1 uniform and $y_2 = y_1$ has loss 1 bit; the direct simulator, run with separate randomness for each input, has loss 2 bits.*

A simulator with private randomness per input loses the cross-input correlations that P uses, and the Jensen step averages over *simulated* histories, whose labels come from P, not f. (The one-tape version itself pays at most $1 + c$ bits here.) One simulation shared by all inputs works, at the price of the reveal sequence's log loss:

Theorem 6.8 (Shared-randomness simulator)**.** *For computable samplers Δ (inputs) and P (labels), possibly defective and with private memory, and every consistent sequence, $M_F(D_n) \ge 2^{-K(\Delta,P)-c}\Delta(x_{1:n}\|y_{1:n-1})P(y_{1:n}\|x_{1:n})$; so $L_{\mathrm{SOI}}(D_n) \le L_P + L_\Delta + K(\Delta) + K(P) + c$ with $L_\Delta := -\log \Delta(x_{1:n}\|y_{1:n-1})$. For distinct inputs the product may be summed over all orders of D_n.*

Theorem 6.9 (Bayes form)**.** *Let $f \sim \nu$, a computable random function (a machine law of a random total function, Lemma A.3), and let the inputs see f only through revealed labels: $x_i = \mathrm{sel}_i(\Gamma, h_{i-1})$, arbitrary maps, Γ independent of f. Then the Bayes predictor $\nu(y \mid D_h, x)$ (D_h the data of h; Definition 3.1) is the true conditional law, with loss $-\log \nu(D_n)$, and $L_{\mathrm{SOI}}(D_n) \le -\log \nu(D_n) + K(\nu) + c$ on every path. So $\mathbb{E}L_{\mathrm{SOI}}(D_n) \le \mathbb{E}L_P + K(\nu) + c$ for every predictor P, computable or not.*

Theorem 6.10 (Directed information)**.** *Let $f \sim \nu$, let the inputs have any joint law with f, $y_i = f(x_i)$, finite alphabets up to time n, and $L_{\mathbb{P}} := -\sum_{i \le n} \log \mathbb{P}(y_i \mid h_{i-1}, x_i)$. (a) For the law ν of f (computable or not), $\mathbb{E}[-\log \nu(D_n)] - \mathbb{E}L_{\mathbb{P}} = I(f \to x_{1:n}) - \mathbb{E}\,\mathrm{KL}(\mathbb{P}(f \in \cdot \mid h_n) \| \nu(f \in \cdot \mid f \supseteq D_n)) \le I(f \to x_{1:n})$. (b) For computable ν and every P, $\mathbb{E}L_{\mathrm{SOI}}(D_n) \le \mathbb{E}L_P + I(f \to x_{1:n}) + K(\nu) + c$.*

Theorem 6.11 (The randomised chain; the oracle form fails)**.** *Let x_i be independent and uniform on $\{0,1\}^i$, $f(x_{i+1}) := x_i$ and $f := \epsilon$ elsewhere; reveal $x_1, x_2, \ldots$. Then f is a computable random function with $K(\nu) = O(1)$; an $O(1)$ program computes the reveal sequence from an oracle for f (x_i is the unique string of length $i \ge 2$ with $f(x_i) \ne \epsilon$, and $x_1 = f(x_2)$); the chain predictor has loss 0; $I(f \to x_{1:n}) = n(n+1)/2$; and $\mathbb{E}L_{\mathrm{SOI}}(D_n) \ge n(n-1)/2$, $\mathbb{E}L_{\mathrm{SOI}} \ge n(n-1)/2 - n(2\log n + c)$.*

This refutes only Formulation B: in Formulation C, $K(\Delta_f^{(n)}) \ge n(n+1)/2 - j - c$ for all but a 2^{-j} fraction of chains, where $\Delta_f^{(n)}$ is the restriction of Δ_f to n steps. One may ask for the statement with the algorithmic mutual information of Δ and f in place of the complexity of Δ; in Formulation B this also fails.

For arbitrary f assume (A1) Y is finite and (A2) a computable map $j \mapsto A_j$ to finite, listed subsets of X, of complexity $K(A)$, has all inputs up to step n in A_n. Let $C_{\mathrm{dir}}(\Delta, n) := \sup_\nu I_\nu(f \to x_{1:n})$ over laws ν on Y^{A_n} (0 if Δ does not see f).

Theorem 6.12 (Minimax theorem)**.** *Fix n, assume (A1), (A2), and let the reveal law see the history (with private memory) and possibly $f|_{A_n}$ in case (a), and only the history in case (b). Assume that $P(y \mid h, x)$ and the reveal-sequence law have (a) rational values computable from their arguments, or (b) computable real values (this includes all computable samplers that halt with probability 1; Lemma D.4). Then for every $f : A_n \to Y$, and on average over any law of f independent of the reveal law's private randomness, $\mathbb{E}_{x|f} L_{\mathrm{SOI}}(D_n) \le \mathbb{E}_{x|f} L_P + C_{\mathrm{dir}}(\Delta, n) + K(P) + K(\Delta) + K(n) + K(A) + c$.*

Proof idea. Sion's theorem (Sion, 1958) and Theorem 6.10(a) give a law on Y^{A_n} within $C_{\mathrm{dir}}+2$ bits of P for every f, found by a search computable from (P, Δ, n, A).

Remark 6.13 (Reveal laws that see f, case (b))*.* For computable real values and reveal laws that may see $f|_{A_n}$, the argument of case (b) goes through if $C_{\mathrm{dir}}(\Delta, n)$ can be computed to within one bit from (P, Δ, n, A). *Proof sketch.* C_{dir} is the maximum over a simplex of finitely many terms $u \log(u/v)$, with u, v polynomial in the law of f and in computable values of Δ, and should be computable to within one bit by a modulus of continuity of entropy on a finite alphabet; with such a value the certified search in the proof of Theorem 6.12 halts as before. We do not carry this out.

Corollary 6.14 (A dominance theorem for computable reveal laws)**.** *(a) Under (A1), (A2), with P and every Δ_f of computable real values (e.g. samplers halting with probability 1, private memory allowed): $\mathbb{E}_{x|f} L_{\mathrm{SOI}}(D_n) \le \mathbb{E}_{x|f} L_P + K(P) + K(\Delta_f) + K(n) + K(A) + c$ for every f; for random f, with $\mathbb{E}_f K(\Delta_f)$. (b) If $x_i = \delta_f(h_{i-1})$ with δ_f computable: for every f, sampler P (defective allowed) and n, $L_{\mathrm{SOI}}(D_n) \le L_P + K(\delta_f) + K(P) + c$ on every path (Corollary D.5). (c) For all samplers P (defective allowed) and all reveal samplers Δ_f that halt with probability 1, with no finiteness assumption, $\mathbb{E}_{x|f} L_{\mathrm{SOI}} \le \mathbb{E}_{x|f} L_P + H(x_{1:n} \mid f) + K(\Delta_f) + K(P) + c$, where $H(x_{1:n} \mid f)$ is the entropy of the reveal sequence given f.*

So in Formulation C the dominance statement holds pathwise for deterministic reveal policies (b); in expectation, for samplers that halt almost surely and finite alphabets up to the horizon, it holds with the additional term $K(n) + K(A)$ (a); and without finiteness assumptions it holds with the reveal entropy added (c). Defective samplers, infinite alphabets and the removal of $K(n) + K(A)$ are open (Remark D.7); a direct simulation fails (Proposition 6.7). No pathwise version with $K(\Delta)$ alone holds (Proposition D.8). The Shannon form of the mutual-information variant holds: if the reveal law is indexed by a random Θ and does not see f, then $I(f \to x) \le I(f; \Theta)$, with equality in the randomised chain (Proposition D.9). The algorithmic form, with the algorithmic mutual information of Δ_f and f in place of $K(\Delta_f)$, is open (Conjecture D.10).

6.4 SOI with access to the previous inputs

Between the versions of SOI considered so far and SI lies a natural intermediate, the history-aware version: the hypotheses see the list of previous inputs but not the previous labels, which they can recompute. We show in what sense it is as good as SI.

Definition 6.15 ($\mathrm{SOI}^{\mathrm{hist}}$)**.** Let X_H be the set of nonempty finite input lists. A *history-oracle function machine* is a function machine T on X_H (Definition 3.6), queried at time i at $x_{1:i}$; it never sees labels but can recompute its own. Its law $M_T^{\mathrm{hist}}(y_{1:n} \| x_{1:n}) := \mathbb{U}\{\omega : T^\omega(x_{1:i}) = y_i \ \forall i \le n\}$ ($n \ge 1$; 1 at the root) is an lsc chronological semimeasure (Lemma D.11). $\mathrm{SOI}^{\mathrm{hist}}$, the history-aware version of SOI, is the one-tape version over X_H: its law is $M_H := M_{T_U^H}^{\mathrm{hist}}$, T_U^H the universal function machine for X_H (Theorem 3.8), and its loss is $L_H := -\log M_H(y_{1:n} \| x_{1:n})$. We *define* the universal chronological semimeasure as M_H.

Theorem 6.16 (Representation)**.** *For every lsc chronological semimeasure ρ there is a history-oracle function machine T_ρ with $M^{\mathrm{hist}}_{T_\rho} = \rho$ at every non-root node and $K(T_\rho) \le K(\rho) + c$.*

The proof allocates dyadic intervals as Zvonkin and Levin (1970) do for sequences; for sequences, Wood et al. (2013) show that machine-based and semimeasure-based universal priors form the same class.

Corollary 6.17 (Universality of $\mathrm{SOI}^{\mathrm{hist}}$)**.** *(a) $M_H \ge 2^{-K(\rho)-c}\rho$ for every lsc chronological semimeasure ρ; so M_H is a universal element of this class and agrees with every other one up to constant factors. (b) For every lsc history-aware predictor P (e.g. every computable sampler, defective or not) and* every *sequence, consistent or not: $L_H \le L_P + K(P) + c$. (c) $L_H \le L_{\mathrm{SOI}}(D_n) + c$ on every consistent sequence, in every order.*

In this sense $\mathrm{SOI}^{\mathrm{hist}}$ is as good as SI. A trimming construction as in Theorem 3.11 gives the same law up to constants. Chronological semimeasures and a universal mixture over them were introduced by Hutter for the AIXI model (Hutter, 2000, 2005), with actions and percepts; here the inputs play the actions and the labels the percepts. We define M_H directly and do not use any identification with Hutter's construction. Hutter (2007) lists supervised learning among the problem classes of the AIXI model. Oracles that see the previous pairs, the time i, or the set of previous inputs define $\mathrm{SOI}^{\mathrm{pairs}}$, $\mathrm{SOI}^{\mathrm{count}}$ (SOI on time-stamped inputs) and $\mathrm{SOI}^{\mathrm{set}}$, with laws $M_{\mathrm{pairs}}, M_{\mathrm{count}}, M_{\mathrm{set}}$ and losses $L_{\mathrm{pairs}}, L_{\mathrm{count}}, L_{\mathrm{set}}$; on distinct inputs $\mathrm{SOI} \preceq \mathrm{SOI}^{\mathrm{count}} \preceq \mathrm{SOI}^{\mathrm{set}} \preceq \mathrm{SOI}^{\mathrm{hist}}$ up to constant factors, and M_{pairs} agrees with M_H up to constant factors (Proposition D.12). Each inclusion is strict, with gaps at least linear in n on suitable orders and labels and also along single infinite sequences (semimeasure convention; Propositions D.13 and D.14); so the oracle must give the list of previous inputs, not their set (also Proposition 7.9).

Solomonoff's per-query solution M_{pq} (Remark 4.2) ignores the history: $\mathrm{SOI}^{\mathrm{hist}}$ dominates it, and on distinct inputs it pays the lower bounds of Theorems 6.3 and 6.4 (Proposition D.15).

For deterministic programs on X_H, the Kolmogorov and mixture versions differ unboundedly, as for SOI (Theorem 4.9): against fair-coin labels the program mixture pays at most $K^{\mathrm{tail}}(n) + c$ extra bits, which grows more slowly than every computable unbounded function, while for the worst labelling, along every input sequence, the single shortest program pays more than $\log n + \log\log n$ extra bits infinitely often (Proposition D.16).

6.5 As good as Solomonoff induction?

Let M be the universal monotone semimeasure (Proposition 3.24), $\mathrm{code}(h_n) := \bar{x}_1\bar{y}_1 \cdots \bar{x}_n\bar{y}_n$ with $\bar{s} := 1^{|s|}0s$, and $L^{\mathrm{joint}}_M := -\log M(\mathrm{code}(h_n))$; SOI's loss is at most $L^{\mathrm{joint}}_M + c$ (Corollary D.6). Let L^x_M be the sum over i of $-\log$ of M's conditional probability of $\bar{x}_i$ given $\mathrm{code}(h_{i-1})$ (the loss on the inputs), and $L^y_M := L^{\mathrm{joint}}_M - L^x_M$, Solomonoff induction's unnormalised conditional label loss. Let M^{in}_H be the law of the universal function machine on finite label lists (the empty list included) that answers with the next input; by Theorem 6.16 with the roles exchanged (same proof), $M^{\mathrm{in}}_H \ge 2^{-K(\rho')-c}\rho'$ for every lsc chronological semimeasure $\rho'(x_{1:n}\|y_{1:n-1})$ for inputs given previous labels. Let $L^x_{\mathrm{in}} := -\log M^{\mathrm{in}}_H(x_{1:n}\|y_{1:n-1})$.

Theorem 6.18 (Solomonoff induction on the joint sequence)**.** *On every sequence: (a) $L_H \le L^{\mathrm{joint}}_M + c$ and $L^x_{\mathrm{in}} \le L^{\mathrm{joint}}_M + c$; (b) $L^{\mathrm{joint}}_M \le L_H + L^x_{\mathrm{in}} + c$; (c) $L_H - L^y_M \le L^x_M + c$ and $L^y_M - L_H \le L^x_{\mathrm{in}} - L^x_M + c$; (d) if $x_i = \delta(h_{i-1})$ for a partial computable δ, then $L^{\mathrm{joint}}_M \le L_H + K(\delta) + c$. This holds for every prefix-free block code with nonempty codewords.*

Proposition 6.19 (Input-loss identity)**.** *For a partial computable δ and every sequence with $x_i = \delta(h_{i-1})$ for all i: $L^x_M - K(\delta) - c \le L_H - L^y_M \le L^x_M + c$.*

So for deterministic input policies, "$L_H \le L^y_M + O(1)$" holds iff M predicts the inputs with bounded loss. Under two assumptions on unnormalised M that are suggested by work of Lattimore, Hutter and Gavane (Lattimore et al., 2011) and are not proved here (Assumptions D.18 and D.19), neither $L^y_M \le L_H + O(1)$ nor $L_H \le L^y_M + O(1)$ holds on all sequences of binary inputs and labels under the identity code (Proposition D.20); this does not contradict Corollary 6.17, as M's conditionals need not be lsc. For normalised M both directions are open.

Remark 6.20 (Summary: hypotheses that see the previous inputs). With the list of previous inputs SOI becomes the universal chronological semimeasure (Corollary 6.17); the set is not enough (Proposition D.13(iii)). $\mathrm{SOI}^{\mathrm{hist}}$ is as good as SI against Solomonoff induction's joint loss (Theorem 6.18(a)); against its conditional label loss the difference is M's loss on the inputs (Proposition 6.19), and for unnormalised M it fails both ways, conditionally on Assumptions LHG-even and LHG-odd (Proposition D.20); the normalised case is open.

6.6 What SOI can and cannot exploit

For D with distinct inputs let D_σ be D in order σ, x_σ its input list, and $K^{\mathrm{pred}}(x_{1:n}) := \min\{|p| : V(p, x_i) = \langle x_1, \dots, x_{i-1}\rangle \ \forall i \le n\}$, the complexity of recovering each input's predecessors from the input.

Theorem 6.21 (Sandwich)**.** *For every data set D: $\max_\sigma L_H(D_\sigma) - c \le L_{\mathrm{SOI}}(D) \le \min_{\sigma'}[L_H(D_{\sigma'}) + K^{\mathrm{pred}}(x_{\sigma'})] + c$.*

Corollary 6.22 (Initial segments)**.** *If $\operatorname{dom} D = \{e_1, \dots, e_k\}$ for a computable injective sequence e, then $|L_{\mathrm{SOI}}(D) - L_H(D \text{ in } e\text{-order})| \le K(e) + c$, and the e-order is the worst for $\mathrm{SOI}^{\mathrm{hist}}$ up to $K(e) + c$.*

This covers the skipped strings and the full cyclic permutation in the lexicographic order of $\{0,1\}^d$ ($K(e) \le K(d) + c$); in its reveal order, $\mathrm{SOI}^{\mathrm{hist}}$ pays $O(1)$ (Proposition 7.8). The K^{pred} term can be needed beyond order information. For $a \ne b$ and r in $\{0,1\}^q$, read as elements of $\mathrm{GF}(2^q)$, and $D_{abr} := \{(a, r), (b, r + ab)\}$, $\mathrm{SOI}^{\mathrm{hist}}$ pays at most $q + c$ in both orders, as it predicts $y_2 = y_1 + x_1 x_2$, while SOI, whose hypotheses never see the first input, pays at least $3q/2 - 2$ bits in expectation over uniform triples with $a \ne b$ and at least $3q/2 - j - c$ bits on all but a 2^{1-j} fraction of them (Theorem D.21).

In expectation, for f drawn from a computable law ν, inputs drawn by any kernel that may see f and finite alphabets, the gap $\mathbb{E}L_{\mathrm{SOI}}(D_n) - \mathbb{E}L_H$ lies between $-c$ and $I(f \to x_{1:n}) + K(\nu) + c$, and is at most $K(\nu) + c$ if the input choice does not see f; the upper bound is attained up to the last input's entropy in the randomised chain, but not in general (Theorem D.23).

On individual sequences, $L_{\mathrm{SOI}}(D) \le L_H(D_\sigma) + I(x_\sigma : D) + c$, with $I(a : b) := K(b) - K(b \mid a)$; the natural strengthening with $I(x_\sigma : y_\sigma)$ in place of $I(x_\sigma : D)$ is false, by a margin linear in n (Proposition D.24). Informally, shared randomness lets a label of the one-tape version correlate with a *guess* about the other inputs (as the brooms and the optimal law for Theorem D.21 do), but not depend on which inputs are present.

Question 6.23 (history-aware prediction)**.** (i) In Formulation C (Remark D.7): can $K(n) + K(A)$ be dropped, and do defective samplers, infinite alphabets and, in case (b) of Theorem 6.12, reveal laws that see f admit the same bound? (ii) Do Assumptions LHG-even and LHG-odd hold,

and how does L_H compare with the label loss of normalised M? (iii) Do the lower bounds of Propositions D.13 and D.14 and Theorem D.21 hold for normalised SOI? (iv) Is $L_M^{\text{joint}} \geq L_H + L_{\text{in}}^x - O(1)$ (the converse is Theorem 6.18(b))? (v) How does $\max_G N(G)$ in Lemma D.22 grow with q? (vi) Does Conjecture D.10 hold? (vii) Which individual-sequence quantity measures $L_{\text{SOI}}(D) - L_H(D_\sigma)$ up to $O(\log n)$? (viii) What is the exact growth rate of the excess in Proposition D.16(d)?

7 Worked examples

We work out two examples in which SOI does much worse than SI: a random cyclic permutation revealed backwards along its cycle, and a coin example due to Sam Eisenstat (personal communication). We then give examples in which SOI does well (examples concerning out-of-distribution behaviour are in Section 8). Losses are in the semimeasure convention, $L_{\text{SOI}}(D_n) = -\log M_F(D_n)$ in every order (Proposition 3.16), unless stated otherwise; upper bounds also hold for the normalised predictor $\bar{M}_F$, lower bounds only where we say so (Remark 3.18). c is a machine constant, $j \geq 0$ a confidence parameter, and a finite function is identified with its graph. Proofs are in Appendix E.

7.1 Tools

Lower bounds come from the coding theorem given the domain, $L_{\text{SOI}}(D) \geq K(D|_T \mid T) - c$ for $T \subseteq \operatorname{dom} D$ (Theorem 3.21(a)), and from a counting fact:

Lemma 7.1 (complexity is at least surprisal)**.** *Let Z take values in a countable set, with law P (computable or not), and let w be fixed. For every $j \geq 0$, $\mathbb{P}[K(Z \mid w) < -\log P(Z) - j] < 2^{-j}$, and $\mathbb{E}K(Z \mid w) \geq H(Z)$.*

Proof. The probability is $\sum_z P(z)$ over the z with $P(z) < 2^{-K(z|w)-j}$, hence less than $\sum_z 2^{-K(z|w)-j} \leq 2^{-j}$. The expectation bound is Gibbs' inequality for the sub-probability $2^{-K(\cdot|w)}$. □

SI. SI sees the history coded as $\text{code}(h_t) = \bar{x}_1\bar{y}_1 \cdots \bar{x}_t\bar{y}_t$ with $\bar{s} := 1^{|s|}0s$ (Theorem 6.18). For $\rho \in \{M, M_{\text{norm}}\}$, with M_{norm} the normalised Solomonoff measure ($M_{\text{norm}}(\epsilon) := 1$, $M_{\text{norm}}(zb) := M_{\text{norm}}(z)M(zb)/(M(z0) + M(z1))$; $M_{\text{norm}} \geq M$), SI's label loss is

$$L_\rho^y(k) := \textstyle\sum_{t \leq k} -\log\left[\rho(\text{code}(h_{t-1})\bar{x}_t\bar{y}_t)/\rho(\text{code}(h_{t-1})\bar{x}_t)\right],$$

for $\rho = M$ this is the label loss L_M^y of Section 6.5. If $\rho(\text{code}(h)) \geq 2^{-a}\mathbb{P}(h)$ for the true law $\mathbb{P}$ of the pairs, L_ρ^y exceeds the Bayes-optimal history-aware label loss by at most a in expectation, and by at most $a + j$ at all times with probability at least $1 - 2^{-j}$, even if the inputs are incompressible (Lemma E.1).

7.2 The cyclic permutation

Definition 7.2 (the cyclic permutation)**.** Fix $d \geq 1$, $N := 2^d$ and a cyclic permutation f of $\{0,1\}^d$ (a single N-cycle). Put $x_0 := 0^d$ and $x_k := f^{-1}(x_{k-1})$ for $1 \leq k \leq N$. The k-th revealed pair is (x_k, x_{k-1}), and $D_k := \{(x_i, x_{i-1}) : i \leq k\}$. We write $(a)_k := a(a-1)\cdots(a-k+1)$, $(a)_0 := 1$.

The map $f \mapsto (x_1, \ldots, x_{N-1})$ is a bijection onto the orderings of the nonzero strings (its inverse closes the cycle $0^d \mapsto x_{N-1} \mapsto \cdots \mapsto x_1 \mapsto 0^d$), so for uniform f the ordering is uniform. For $k < N$, D_k is a chain $x_k \to \cdots \to x_1 \to 0^d$ of distinct nonzero strings. The labels add nothing to the inputs ($y_k = x_{k-1}$): all of f is in the order.

Theorem 7.3 (the full cycle)**.** *Semimeasure convention. (a) For every total $f : \{0,1\}^d \to \{0,1\}^d$, $|L_{\mathrm{SOI}}(f) - K(f \mid d)| \le c$. (b) For every cyclic permutation f of $\{0,1\}^d$, $L_{\mathrm{SOI}}(f) \le \log((N-1)!) + c$. (c) For uniform cyclic f, $\mathbb{P}[L_{\mathrm{SOI}}(f) < \log((N-1)!) - j - c] < 2^{-j}$, and $\mathbb{E}L_{\mathrm{SOI}}(f) \ge \log((N-1)!) - c$. (d) $\log((N-1)!) = (d - \log e)N - \frac{d}{2} + \frac{1}{2}\log(2\pi) + \epsilon_N$ with $\frac{1}{(12N+1)\ln 2} < \epsilon_N < \frac{1}{12N \ln 2}$. The same bounds hold for $L_{\mathrm{SOI}}(D_{N-1})$.*

So SOI pays $d - \log e \approx d - 1.443$ bits per pair over the cycle (Table 2). In (a) one function machine reads a single program for the whole table from the shared tape; in (b) one machine draws a uniform cyclic permutation of each length; (d) is Stirling's formula with Robbins' remainder bounds (Robbins, 1955).

Theorem 7.4 (k steps)**.** *Semimeasure convention; $1 \le k \le N-1$. (a) On every chain, $L_{\mathrm{SOI}}(D_k) \le \log(N-1)_{k-1} + \min\{K(k), \log(N-k)\} + c$. (b) For uniform cyclic f, $\mathbb{P}[L_{\mathrm{SOI}}(D_k) < \log(N-1)_{k-1} - j - c] < 2^{-j}$. (c) For uniform cyclic f, with probability at least $1 - 2^{-j}$, $L_{\mathrm{SOI}}(D_k) \ge \log(N-1)_{k-1} - d - j - c$ for all $k < N$ at once.*

So $L_{\mathrm{SOI}}(D_k) = \log(N-1)_{k-1} \pm O(K(k))$ with high probability: SOI pays about $\log(N-t+1)$ bits for the label at step $t \ge 2$, and the newest input is free up to $K(k)$. The upper bound uses a *broom* B_k, a function machine that draws, for each length ℓ, a uniform injective sequence $(v_1, \ldots, v_{k-1})$ of nonzero ℓ-bit strings and maps $v_1 \mapsto 0^\ell$, $v_i \mapsto v_{i-1}$, and every other input to v_{k-1}; it extends D_k with probability $1/(N-1)_{k-1}$. It randomises the upper-bound program of the chain example (Theorem 6.3). The lower bounds rest on a depth count (Lemma E.3). The cost is intrinsic to order-free prediction:

Proposition 7.5 (order-free predictors)**.** *Let f be uniform cyclic and $k < N$. (a) For every law ν of a random partial function, computable or not (so every machine law, with shared or private randomness), $\mathbb{E}[-\log \nu(D_k)] \ge \log(N-1)_{k-1}$. (b) For every weak function semimeasure ν (every order-independent predictor, Proposition 3.3), $\mathbb{E}[-\log \nu(D_k)] \ge \log(N-1)_k - \log(N-1)$ and $\mathbb{E}[-\log \nu(f)] \ge \log((N-1)!)$.*

For $k \le N/2$ the two bounds differ by at most one bit. So no random-function law, and for $k \le N/2$ no order-independent predictor, does better than SOI by more than $O(K(k))$ bits in expectation. Part (a) covers Solomonoff's per-query solution: on the distinct inputs of D_k it equals $M_{\mathrm{prod}}(D_k)$, a mixture of laws of random partial functions with private randomness (Remark 4.2). Whether (a) holds for every weak ν is open:

Question 7.6 (depth count for weak semimeasures)**.** Is $\sum_{x,z} \nu(D(x,z)) \le N - k$ for every weak function semimeasure ν and $1 \le k \le N-1$, the sum over $x \in \{0,1\}^d \setminus \{0^d\}$ and $z \in (\{0,1\}^d \setminus \{0^d\})^{k-1}$ (notation of Lemma E.2)? If so, Proposition 7.5(a) holds for every weak ν. For $N = 4$ a direct computation (a linear program; code in the accompanying repository) confirms the inequality.

The normalised predictor is not order-free, and a set-based history-aware predictor has loss 0 here (Remark E.7), so information-theoretic arguments alone cannot bound it below. The structure of M_F can:

Proposition 7.7 (normalised SOI)**.** *Let f be uniform cyclic and $\beta_t := -\log \bar{M}_F(y_t \mid D_{t-1}, x_t)$. For $2 \le t \le N-1$, $\mathbb{E}\beta_t \ge \log(N-t+1) - K(t-1) - c$, and $\beta_t \ge \log(N-t+1) - K(t-1) - j - c$ with probability at least $1-2^{-j}$. Hence $\mathbb{E}\sum_{t\le k}\beta_t \ge (k-1)(d-1-\log k - 2\log\log(k+1) - c)$ for $k \le N/2$, which is at least $(k-1)(\delta d - O(\log d))$ for $k \le 2^{(1-\delta)d}$.*

Proof idea. $\beta_t = L_{\mathrm{SOI}}(D_t) + \log \mathcal{Z}(D_{t-1}, x_t)$, and $\mathbb{E}L_{\mathrm{SOI}}(D_t) \ge \log(N-1)_{t-1}$ by Proposition 7.5(a). The broom B_{t-1} is total, consistent with the past, and sends x_t to the wrong label x_{t-2}, so it keeps the normaliser at least $2^{-K(t-1)-c}/(N-1)_{t-2}$ (Lemma D.2), as in Proposition D.3. Over the whole cycle these bounds sum to at most about $2N$ bits (Question 7.23).

Proposition 7.8 ($\mathrm{SOI}^{\mathrm{hist}}$)**.** *For every d, every cyclic f and every $k \le N$, $L_H \le c$ on $(x_1, x_0), \dots, (x_k, x_{k-1})$, with c independent of d.*

Proof. "Answer $0^{|x|}$ first, then the previous input" is computable and has loss 0; apply Corollary 6.17. □

For seeing the previous inputs to make SOI as good as SI (Section 6.4), the hypotheses must see the *list* of previous inputs; their set is not enough:

Proposition 7.9 ($\mathrm{SOI}^{\mathrm{set}}$)**.** *Let L_{set} be the loss of* $\mathrm{SOI}^{\mathrm{set}}$*, whose hypotheses see the set of previous inputs (Proposition D.12). For uniform cyclic f on the full cycle, $\mathbb{P}[L_{\mathrm{set}} < \log((N-1)!) - j - c] < 2^{-j}$, $\mathbb{E}L_{\mathrm{set}} \ge \log((N-1)!)$, and always $L_{\mathrm{set}} \le \log((N-1)!) + c$.*

Proposition 7.10 (SI)**.** *Let $\rho \in \{M, M_{\mathrm{norm}}\}$ and f uniform cyclic. There is c, independent of d, with $\mathbb{E}L^y_\rho(k) \le c$ for every $k \le N$ and $\mathbb{P}[\exists k \le N : \ L^y_\rho(k) > j + c] \le 2^{-j}$.*

SI pays about $K(d)$ bits on the header 1^d0 of the first input, which tells it d; then the labels are free. With raw d-bit blocks the bound becomes $K(d)+c$ (Proposition E.4); whether this is the right order is Question E.5. So the example separates SOI from SI: SOI pays about $(d - \log e)2^d$ bits over the cycle (semimeasure convention) and at least $\delta d - O(\log d)$ bits per step for $t \le 2^{(1-\delta)d}$ (normalised), while SI and $\mathrm{SOI}^{\mathrm{hist}}$ pay $O(1)$, uniformly in d. The cost does not depend on the version: on the full cycle $K(\mathrm{dom}\, f) \le K(d) + O(1)$, so the versions of Section 4, Solomonoff's per-query solution among them (it equals M_{prod} on distinct inputs), agree up to $O(\log d)$ bits (Theorem 4.4). It is the cost of order-free prediction: only a predictor whose loss depends on the order of the data can avoid it (Proposition 7.5(b)), as SI on the interleaved sequence and $\mathrm{SOI}^{\mathrm{hist}}$ do and $\mathrm{SOI}^{\mathrm{set}}$ does not (Proposition 7.9).

Remark E.7 locates the information and gives a separation in the other direction.

7.3 Eisenstat's example

Definition 7.11 (Eisenstat's example)**.** Let $b_0, b_1, \dots$ be i.i.d. Bern($\frac{1}{3}$) (index i is a *head* if $b_i = 1$, else a *tail*) and $\zeta_1, \zeta_2, \dots$ i.i.d. Bern($\frac{2}{3}$), independent of b. At step t, reveal the smallest unrevealed head i as $(i, 1)$ if $\zeta_t = 1$, else the smallest unrevealed tail as $(i, 0)$. After n steps: N^1_n, N^0_n count the revealed heads and tails; τ_n (η_n) is one more than the last revealed tail (head), or 0; g_n is the smallest unrevealed index; and $UB_n := -\log P_{1/3}(b_{<\tau_n}) = \tau_n \log 3 - N^0_n$, with $P_{1/3}$ the Bern($\frac{1}{3}$) product measure.

A heuristic argument (Remark 7.18) suggests that SOI's loss is that of a threshold function up to $o(n)$. We take as threshold function the *threshold law* ν^{thr}_s: the product law with $\mathbb{P}[y = 1 \mid i] = \frac{1}{3}$ for $i < s$ and $= 1$ for $i \ge s$, a machine law with index computable from s (Proposition 3.9(a)).

Lemma 7.12 (shape of the revealed set)**.** *The revealed tails (heads) are exactly the tails (heads) below τ_n (η_n). No unrevealed index lies below $\min(\tau_n, \eta_n)$; an unrevealed index below η_n is a tail, one below τ_n a head. If $\tau_n \le \eta_n$, then $g_n \ge \tau_n$ and $[\tau_n, g_n)$ consists of revealed heads.*

Proof. The process reveals the first unrevealed head (tail), so the revealed heads (tails) are the first N_n^1 heads (N_n^0 tails) of b. □

So the revealed set is about $[0, n/2)$ together with the heads in $[n/2, 2n)$.

Proposition 7.13 (upper bound)**.** *For every b, every reveal sequence of this kind (each choice of head or tail may be arbitrary) and every n: $L_{\mathrm{SOI}}(D_n) \le UB_n + K(\tau_n) + c \le UB_n + \log(\tau_n + 1) + 2\log\log(\tau_n + 2) + c$, also for $\bar{L}_{\mathrm{SOI}}$.*

Indeed $M_F \ge 2^{-K(\tau_n)-c}\nu_{\tau_n}^{\mathrm{thr}}$ (Theorem 3.8); under $\nu_{\tau_n}^{\mathrm{thr}}$ the revealed indices at or above τ_n are heads (Lemma 7.12) and cost nothing, and those below τ_n cost at most UB_n; finally $K(i) \le \log(i+1) + 2\log\log(i+2) + c$.

Theorem 7.14 (pathwise sandwich)**.** *Semimeasure convention. For every b, every reveal sequence and every n, $L_{\mathrm{SOI}}(D_n) \ge K(b_{<g_n} \mid g_n) - c$. If b is Martin-Löf random for $P_{1/3}$, then at every n with $\tau_n \le \eta_n$,*

$$UB_n - K(g_n) - c_b \;\le\; L_{\mathrm{SOI}}(D_n) \;\le\; UB_n + K(\tau_n) + c.$$

In Eisenstat's process, almost surely $|L_{\mathrm{SOI}}(D_n) - UB_n| \le \log n + O(\log\log n) + c_b$ for all large n.

Proof idea. $[0, g_n)$ is revealed with its true labels; monotonicity and the coding theorem given the domain give the first bound, and the Levin–Schnorr theorem turns it into the surprisal $-\log P_{1/3}(b_{<g_n}) \ge UB_n$. Almost surely the heads are eventually ahead (Lemma E.8).

Proposition 7.15 (the threshold loss)**.** *Let $a := \frac{3}{2}\log 3 - 1$ and $\sigma^2 := \frac{2}{9}a^2 + \frac{1}{4}(\log 3)^2 = 1.0496602$. For every n, $\mathbb{E}UB_n = \frac{H(1/3)}{2}n = 0.4591479\,n$ and $\mathrm{Var}(UB_n) = \sigma^2 n$ exactly; $UB_n/n \to \frac{H(1/3)}{2}$ almost surely, and $(UB_n - \frac{H(1/3)}{2}n)/\sqrt{n} \to \mathcal{N}(0, \sigma^2)$ in law.*

Theorem 7.16 (expected loss)**.** *Semimeasure convention. For all $n \ge 2$, $\frac{H(1/3)}{2}n - \log n - c \le \mathbb{E}L_{\mathrm{SOI}}(D_n) \le \frac{H(1/3)}{2}n + \log n + 2\log\log n + c$, and the upper bound holds for $\mathbb{E}\bar{L}_{\mathrm{SOI}}$. The product law with $\mathbb{P}[y=1] = \frac{1}{3}$ at every input pays $(\log 3 - \frac{1}{3})n = 1.2516\,n$.*

Proposition 7.17 (exact law; order-free predictors)**.** *Let D be a possible value of D_n, $S := \operatorname{dom} D$, $s^+ := 1 + \max S$, N^1 the number of 1-labels of D, and $r := \frac{2}{3}$ if the label at $\max S$ is 1, $r := \frac{1}{3}$ otherwise. (a) $\mathbb{P}[D_n = D] = \binom{n}{N^1}(\frac{2}{9})^n r^{s^+-n}$; so the labelling given $\operatorname{dom} D_n = S$ depends on the label at $\max S$, not only on N^1. (b) Every weak function semimeasure ν, computable or not, has $\mathbb{E}[-\log \nu(D_n)] \ge \frac{H(1/3)}{2}n - \log n - O(1)$.*

So SOI is optimal among order-independent predictors up to $O(\log n)$ in expectation; the lower bound of Proposition 7.17(b) also holds for Solomonoff's per-query solution, which equals the weak semimeasure M_{prod} on these distinct inputs, and, as in the cyclic example, the gap to SI is a property of order-free prediction.

Remark 7.18 (A heuristic argument for Eisenstat's example). A heuristic argument (from correspondence with S. Eisenstat) bounds SOI's loss by that of the threshold hypothesis, up to about $\log n$ bits for naming the threshold, and argues the converse by incompressibility: a much smaller loss would give a short description of the coin flips in the initial segment, which a random sequence of Bern($\frac{1}{3}$) flips does not have. The claim that the two losses agree up to $o(n)$ holds in

the semimeasure convention, almost surely with $O(\log n)$ in place of $o(n)$ and the threshold at the tail frontier (Theorem 7.14); the upper bound is Proposition 7.13. The converse needs two changes: the step from a lower bound on the semimeasure to a short description is the coding theorem given the domain of the fully revealed segment $[0, g_n)$, which costs $K(g_n) = O(\log n)$ bits; and incompressibility must be measured by the empirical surprisal UB_n, not by $\tau_n H(1/3)$, which differs from it by $\Theta(\sqrt{n})$ (enough for $o(n)$, not for $O(\log n)$; Appendix E.3). For the normalised predictor see Question 7.23.

History-aware prediction and SI. Call a history *touching* if its revealed set is the initial segment $[0, \max(\tau, \eta))$. The Bayes-optimal history-aware predictor P^* is explicit (Lemma E.9): at a touching history it gives the input $\max(\tau, \eta) + k$ the label 1 with probability $2^k/(2^k + 1)$; at every other history it is certain and correct.

Proposition 7.19 (the burn-in)**.** *The number N_{touch} of touching steps in the infinite run has $\mathbb{P}[N_{\text{touch}} = k] = (\frac{2}{3})^{k-1}\frac{1}{3}$. Each costs P^* an expected $\vartheta = \sum_{k\geq 0} \frac{2}{9}[(\frac{2}{3})^k \log(1 + 2^{-k}) + (\frac{1}{3})^k \log(1 + 2^k)] = 0.7995167$ bits, so $\mathbb{E}L_{P^*}(n) \leq \mathbb{E}L_{P^*}(\infty) = 3\vartheta = 2.39855$ bits.*

Theorem 7.20 (SOI^{hist} and SI)**.** *(a) For every b, every coin sequence and every n, $L_H(n) \leq L_{P^*}(n) + c$; so $\mathbb{E}L_H(n) \leq 3\vartheta + c$. (b) Code indices as strings and labels as one-bit strings. For $\rho \in \{M, M_{\text{norm}}\}$, $\mathbb{E}L^y_\rho(n) \leq 3\vartheta + c$ for all n, and $\mathbb{P}[\exists n :\ L^y_\rho(n) > L_{P^*}(n) + j + c] \leq 2^{-j}$.*

Proof. (a) Corollary 6.17 for the computable P^*. (b) The pair process has a computable law $\mathbb{P}$ whose label conditionals are those of P^*, so $M(\text{code}(h)) \geq 2^{-c}\mathbb{P}(h)$ (Lemma D.17); apply Lemma E.1 and Proposition 7.19. □

A natural rule reads the labels off the history: *the new input is a tail if and only if it is the smallest unrevealed index.*

Corollary 7.21 (the rule)**.** *Almost surely there is a last touching step, and the rule is exact at every later step.*

Proposition 7.22 (errors of the rule)**.** *Over the infinite run the rule errs twice in expectation, and it makes no error with probability exactly $\frac{1}{2}$.*

In the simulations of Table 3, a finite order-free mixture pays about 0.46 bits per step, while P^* and a finite joint mixture pay a bounded total.

Question 7.23 (normalised SOI)**.** Is $\mathbb{E}\bar{L}_{\text{SOI}} = \Omega(N \log N)$ over the whole cycle of Definition 7.2, as for L_{SOI} (Theorem 7.3), and $\mathbb{E}\bar{L}_{\text{SOI}} \geq \frac{H(1/3)}{2}n - o(n)$ in Eisenstat's example? The method of Proposition 7.7 pays $O(\log t)$ bits per step, more than the information per step near the end of the cycle and in Eisenstat's example (about 0.46 bits), and set-based history-aware predictors have loss $O(1)$ (on every path for the cycle, in expectation in Eisenstat's example). In finite mixtures with most prior weight on partial chains, the full-cycle normalised loss is far below $\log((N-1)!)$ (a direct computation gives 19.2 against 40.25 bits at $d = 4$).

7.4 Positive examples

The bounds below hold in both conventions.

Proposition 7.24 (deterministic targets)**.** *Let f be partial computable, $x_1, x_2, \cdots \in \text{dom}\, f$ any inputs (adversarial, adaptive or non-computable; repeats allowed), $y_t = f(x_t)$. Then $L_{\text{SOI}}(D_n) \leq K(f)+c$, and at most $K(f)+c$ steps give $f(x_t)$ predictive probability at most $\frac{1}{2}$, raw or normalised.*

Proof. Theorem 5.1(b), Corollary 5.4, and $\mathcal{Z}(D,x) \le M_F(D)$. □

Example 7.25 (sparse parity)**.** For $S \subseteq \{0,\dots,d-1\}$ with $|S| = k$ and $\chi_S(x) := \bigoplus_{i\in S} x_i$ (bits beyond $|x|$ read as 0): $L_{\mathrm{SOI}} \le \log\binom{d}{k} + K(d) + K(k) + c$ on every input sequence. If S is uniform and the inputs form a fixed sequence whose labels determine S (e.g. containing the unit vectors), every learner, history-aware or not, has expected loss at least $\log\binom{d}{k}$.

Example 7.26 (thresholds)**.** For $f_\theta(x) := \mathbf{1}[x \ge \theta]$ on $\mathbb{N}$, $L_{\mathrm{SOI}}(D) \le -\log \sum_{\theta' : f_{\theta'} \supseteq D} 2^{-K(\theta')} + c \le K(\theta) + c$. For θ uniform on $[0, 2^\ell)$, ℓ bisection queries cost every learner at least ℓ bits in expectation, and SOI at most $\ell + K(\ell) + c$.

Example 7.27 (sorting)**.** Sorting has complexity $O(1)$, so over any number of pairs (list, sorted list), in any order, SOI's total loss and its number of steps that give the sorted list probability at most $\frac{1}{2}$ are bounded by a constant.

Example 7.28 (integer polynomials)**.** For $g(x) = \sum_{i\le k} a_i x^i$ with all $a_i \in \{0,\dots,A-1\}$, $K(g) \le (k+1)\log A + K(k) + K(A) + c$; for uniform coefficients and $k+1$ distinct inputs every learner pays at least $(k+1)\log A$ in expectation. For arbitrary integer coefficients, $K(g) \le \sum_i K(a_i) + K(k) + c$.

So SOI is optimal in these examples up to $O(\log d)$, $O(\log \ell)$ and $O(\log k + \log A)$.

Proposition 7.29 (frozen label noise)**.** *Let $g : X \to \{0,1\}$ be partial computable and $y_x := g(x) \oplus z_x$ with z_x i.i.d.* Bern(ε)*, drawn once per input. (a) For every data set D with $\operatorname{dom} D \subseteq \operatorname{dom} g$, $|\operatorname{dom} D| = n$, and e labels that disagree with g: $L_{\mathrm{SOI}}(D) \le \log\binom{n}{e} + \log(n+1) + K(g) + c$. (b) If the n distinct inputs are chosen without access to the noise at unrevealed inputs, $nH(\varepsilon) \le \mathbb{E}L_{\mathrm{SOI}}(D_n) \le nH(\varepsilon) + \log(n+1) + K(g) + c$; the lower bound holds for every learner.*

The machine behind (a) shares a flip rate across inputs, so it is a mixture of product laws (Appendix E.4). Fresh label noise is different: the one-tape version, a random function, fails as soon as an input can repeat with another label, while Solomonoff's per-query solution allows such noise (Remark E.10).

8 Out-of-distribution behaviour

If some simple property holds for no training input, SOI keeps a constant probability of a simple behaviour, such as outputting 0, on all inputs with that property, with no logarithmic term in prefix complexity K, of the kind a bound in plain complexity would carry (Section 8.1). We give a two-sided version, study converses and work out examples where SOI plausibly generalises as intended.[6]

Conventions. ν ranges over $\{W, M_F\}$ (Definitions 3.15 and 4.1); for W the normalised conditional $\bar\nu$ is defined as $\bar M_F$ is for M_F. For a target f and inputs $x_1, x_2, \dots$, $D_k := \{(x_j, f(x_j)) : j \le k\}$ and the *posterior error* $e_k := 1 - \nu(f(x_k) \mid D_{k-1}, x_k)$ is the posterior mass that is wrong *or undefined* at x_k. At a test point x^* with training labels $f|_A$: $e(x^*) := 1 - \nu(f(x^*) \mid f|_A, x^*)$;

[6] An informal version of Proposition 8.1, the question whether it loses a logarithmic term, the observation behind Proposition 8.2, properties simple given a fitted program (Proposition F.2), a two-sided statement, the question of a converse and the examples of Section 8.5 first appeared in the author's working notes (Hänni, 2026c).

$e^{\mathrm{wr}}(x^*) := \sum_{y \neq f(x^*)} \nu(f|_A + (x^*, y))/\nu(f|_A)$ (defined and wrong); $e_{\mathrm{norm}}(x^*) := 1 - \bar{\nu}(f(x^*) \mid f|_A, x^*)$. Since $\sum_y \nu(f|_A + (x^*, y)) \leq \nu(f|_A)$,

$$e^{\mathrm{wr}}(x^*) \;\leq\; e_{\mathrm{norm}}(x^*) \;\leq\; e(x^*), \tag{3}$$

so every lower bound below on e^{wr}, and every upper bound on e, holds in both conventions. A *property* $Q \subseteq X$ *separates* x^* from A if $x^* \in Q$ and $Q \cap A = \emptyset$. For decidable Q, $K(Q)$ is the least $K(e)$ over total deciders φ_e of Q; for partial computable b, $K(Q, b)$ is the least complexity of a pair $\langle e, e' \rangle$ with φ_e a total decider of Q and $\varphi_{e'}$ computing b on $Q \cap \operatorname{dom} b$, and $K(Q, b) \leq K(Q) + K(b) + c$.

8.1 Unseen simple properties

Proposition 8.1 (Unseen simple properties)**.** *Let Q be decidable, b partial computable,* $\operatorname{dom} D \cap Q = \emptyset$*, $S \subseteq Q \cap \operatorname{dom} b$ finite and $E := b|_S$. Then (a) $\nu(D \cup E) \geq 2^{-K(Q,b)-c}\, \nu(D)$, and the hypotheses (programs for W, tapes for M_F) consistent with D that agree with b on all of $Q \cap \operatorname{dom} b$ have mass at least $2^{-K(Q,b)-c}\nu(D)$; (b) $K_F(D \cup E) \leq K_F(D) + K(Q, b) + c$.*

Proof idea (Appendix F). Prefix each hypothesis consistent with D by a description of (Q, b) and let it answer b on Q; it stays consistent with D, at a cost of $K(Q, b) + c$ bits. Summing over all hypotheses avoids a logarithmic term (Remark F.1).

So, however large D is, SOI behaves like b on all unseen Q-inputs with posterior probability at least $2^{-K(Q,b)-c}$; for $|S| = 1$ and b wrong there, (a) bounds e^{wr}, so it holds in both conventions. A property may also be simple only given a program p that fits the training data, such as a test on an internal variable of p that never fires on training inputs; the same bound holds for such properties with conditional complexity, relative to the posterior weight of p (Proposition F.2).

Proposition 8.2 (Dependence on the universal machine; cf. Leike and Hutter, 2015a for AIXI)**.** *Fix V, a decidable Q and a partial computable b. (a) Some universal prefix machine V' has $2^{-K(Q,b)-c}W' \leq W \leq 2W'$ and, for every D with* $\operatorname{dom} D \cap Q = \emptyset$ *and finite $S \subseteq Q \cap \operatorname{dom} b$, $W'(D) = W(D)$ and $W'(D \cup b|_S) \geq \frac{1}{2}W'(D)$. (b) Some universal prefix machine V'' yields a one-tape version M'_F with $2^{-K(Q,b)-c}M'_F \leq M_F \leq 2M'_F$ on nonempty data sets such that, for every nonempty D with* $\operatorname{dom} D \cap Q = \emptyset$*, $M'_F(D) = M_F(D)$ and the M'_F-posterior probability that $f_\omega = b$ on all of $Q \cap \operatorname{dom} b$ is at least $\frac{1}{2}$.*

Proof idea (Appendix F). V' runs V on programs $0u$ and the wrapper of Proposition 8.1 on programs $1u$; the halves agree off Q. So for some programming language good behaviour on Q has no better constant than bad behaviour; the language is tailored to one property and one behaviour and lies $K(Q, b) + O(1)$ bits from the reference machine.

8.2 A two-sided bound

Often any output with a property $P \subseteq X \times Y$ of input–output pairs will do ("y is a correct proof of x"). Whatever the amount of data, P-abiding generalisation has at least a constant probability of happening and of not happening when no training input lies in a simple property Q of the test inputs:

Theorem 8.3 (Two-sided bound on P-abiding generalisation)**.** *Let Q be decidable,* $\operatorname{dom} D \cap Q = \emptyset$*, $S \subseteq Q$ finite and nonempty, and $K(P, Q)$ the complexity of a decider of Q paired with an enumerator of P (in (a)) or of its complement (in (b)). (a) If P is c.e. and every $x \in Q$ has*

some y with $(x,y) \in P$, then ν(every $x \in S$ gets a P-abiding output $\mid D$) $\geq 2^{-K(P,Q)-c}$. (b) If the complement of P is c.e. and every $x \in Q$ has some y with $(x,y) \notin P$, then ν(some $x \in S$ gets a P-violating output $\mid D$) $\geq 2^{-K(P,Q)-c}$, so ν(all of S is P-abiding $\mid D$) $\leq 1 - 2^{-K(P,Q)-c}$. (c) For $x \in Q$ and $(x,z) \notin P$: $M_F(D + (x,z)) \geq 2^{-K(Q)-K(z|x)-c} M_F(D)$ and $W(D + (x,z)) \geq 2^{-K(Q)-K^{\mathcal{P}}(z|x)-c} W(D)$, where $K^{\mathcal{P}}(z \mid x)$ is the length of a shortest program $h \in \operatorname{dom} V$ with $\varphi_{V(h)}(x) = z$ (Remark B.1).

These bounds concern defined outputs, so they hold in both conventions and for every amount of data. *Proof idea* (Appendix F). (a), (b) apply Proposition 8.1 to the first abiding (violating) output in an enumeration; (c) runs a conditional program on half of the tape on Q and T_U elsewhere. Two natural pointwise bounds of this kind fail (Proposition F.3); in place of the second, an upper bound on the probability of abiding, (c) bounds that of violation from below.

8.3 Out-of-distribution failure is a bounded resource

Theorem 8.4 (Cumulative bound)**.** *Let f be computable and $x_1, x_2, \ldots$ any distinct inputs (adversarial, adaptive or non-computable). For every n, with $c = 0$ for W,*

$$\sum_{k \leq n} -\log(1-e_k) \leq K(f)+c, \quad \sum_{k \leq n} e_k \leq (\ln 2)(K(f)+c), \quad \#\{k \leq n : e_k \geq \delta\} \leq \frac{(\ln 2)(K(f)+c)}{\delta}.$$

The same holds normalised, and with $K(f) + c$ replaced by $L_\nu(f|_B) \leq K_F(f|_B) + c$, $B = \{x_1, \ldots, x_n\}$.

Proof. By Proposition 3.16 the first sum is $-\log(\nu(D_n)/\nu(\emptyset)) \leq L_\nu(D_n) \leq K_F(f|_B) + c \leq K(f) + c$; use $-\ln(1-e) \geq e$, and count. Normalised errors are smaller by (3). □

This is the error form of the mistake bound (Corollary 5.4). Separating properties force errors, but only a bounded total:

Proposition 8.5 (Novelty budget)**.** *Let $Y = \{0,1\}$ and $\operatorname{sep}_k := \min\{K(Q) : Q$ decidable, $x_k \in Q$, $Q \cap \{x_1, \ldots, x_{k-1}\} = \emptyset\}$ ($\operatorname{sep}_k := \infty$ if no decidable Q separates x_k, e.g. if x_k repeats). Then $e_k \geq e_k^{\mathrm{wr}} \geq 2^{-\operatorname{sep}_k - c}$ for every k, and $\sum_k 2^{-\operatorname{sep}_k} \leq 1$.*

Proof. Proposition 8.1 with $b \equiv 1 - f(x_k)$. A property separating x_k from its predecessors separates no later input, so distinct k have distinct minimisers; apply Kraft's inequality. □

Remark 8.6 (The novelty bound is lopsided). The novelty lower bounds sum to at most 2^{-c}: separation forces constant error at single steps but only $O(1)$ in total; each simple property surprises only once. The binding bound is the upper bound of Theorem 8.4, and the positive statement is relative (Theorem 5.1(a)): if a computable x-only predictor generalises correctly across all the shifts, SOI's total excess log loss over it is at most its complexity plus c.

8.4 Converses

Propositions 8.1 and F.2 suggest a natural converse: that SOI works well if no property simple given a program doing well on the training data separates the test point. For complex targets it fails:

Proposition 8.7 (The natural converse fails for complex targets)**.** *Let $f : \{0,1\}^d \to \{0,1\}$ be uniformly random and $A = \{x_1, \dots, x_n\}$, x^* i.i.d. uniform, all independent. (a) Every learner has $\mathbb{E}[e_{\mathrm{norm}}(x^*) \mid x^* \notin A] \geq \frac{1}{2}$. (b) For every string z determined by f (e.g. a shortest program for f) and $s \geq 0$, the probability that some decidable Q with $K(Q \mid z) \leq s$ separates x^* from A is at most $2^{s+1}/(n+1)$. (c) (*A program fitted to the training data.*) Let $K(x^* \mid d) \geq d$, $A := \{0,1\}^d \setminus \{x^*\}$, $D := \{(x,1) : x \in A\}$, $f := D + (x^*, 0)$, and p_1 a shortest program for the constant 1 (consistent with D). Every decidable Q separating x^* from A has $K(Q \mid p_1) \geq d - c$, yet $e_{\mathrm{norm}}(x^*) \geq 1 - 2^{-d+c}$; here $K(f) \geq d - c$.*

So this converse fails, both for a program determined by the target (a, b) and for one fitted to the training data (c); the obstruction is $K(f)$. With a factor depending on $K(f)$, a converse holds once the property may depend on the labelled sample. For one test point, the probability of a wrong label y is at least $2^{-K(Q,b)-c}$ for every separating Q and every b with $b(x^*) = y$, and at most $2^{K_F(f|_A) - K(f|_A + (x^*,y) | A \cup \{x^*\}) + c}$ (Proposition F.4): SOI errs at x^* only if a wrong completion is, given all inputs, not much more complex than the training labels are for a program that does not know the inputs. In terms of separating properties:

Proposition 8.8 (Converse given the labelled sample)**.** *Let f be a labelling of a finite B, $x^* \in B$, $A := B \setminus \{x^*\}$, and $\mathrm{sep}(x^*; A \mid f)$, $\mathrm{sep}(x^*; A)$ the least $K(Q \mid f)$, $K(Q)$ of a decidable Q separating x^* from A. Then*

$$e^{\mathrm{wr}}(x^*) \leq e_{\mathrm{norm}}(x^*) \leq 2^{L_\nu(f) - K(x^*|f) + c} \leq 2^{K_F(f) - K(x^*|f) + c'}, \qquad |K(x^* \mid f) - \mathrm{sep}(x^*; A \mid f)| \leq c,$$

and $e^{\mathrm{wr}}(x^*) \geq 2^{-\mathrm{sep}(x^*;A) - K(y) - c}$ for every $y \neq f(x^*)$. Also $e_{\mathrm{norm}}(x^*) \leq 2^{L_\nu(f) + K_F(f) - K(x^*|B) + c}$, and $e_{\mathrm{norm}}(x^*) \leq 2^{|p| - K(x^*|B,p) + c}$ for every $p \in \mathrm{dom}\, V$ that computes f on B.

Proof idea (Appendix F). The hypotheses that agree with f on $B \setminus \{x\}$ and are defined and wrong at x form disjoint sets for distinct x, with masses lsc uniformly in (f, x); the coding theorem bounds the mass for x^* by $2^{-K(x^*|f)+c}$, and the normaliser at x^* is at least $\nu(f)$.

So if f is simple and no property simple *given the labelled sample* separates the test point, the error there is small; the gap between the two separating complexities is the value of knowing the other inputs. The argument bounds only the mass that is defined and wrong at x^*; Corollary 5.10 bounds the semimeasure-convention error $e(x^*)$ in distribution.

Remark 8.9 (Summary of the positive statements). (i) Over any shifts, SOI's total loss exceeds that of every computable x-only predictor P by at most $K(P) + c$, and its total error is at most $(\ln 2)(K(f) + c)$ (Theorems 5.1 and 8.4). (ii) A random train/test split of any input set B costs at most $L_\nu(f|_B)/|B|$ in expectation (Proposition 5.8). (iii) For simple f and binary labels, $e^{\mathrm{wr}}(x^*) \geq 2^{-\mathrm{sep}(x^*;A) - O(1)}$ and $e_{\mathrm{norm}}(x^*) \leq 2^{L_\nu(f) - \mathrm{sep}(x^*;A|f) + O(1)}$ (Proposition 8.8); for random f no learner has a converse. So the implication from no simple separating property to small error fails only for bounds independent of $K(f)$, or for properties simple unconditionally. (iv) In distribution the per-step error is at most $\min\{((K(f) + c)\ln 2 + \ln n + 1)/n,\ 2^{K(f)+c}/(n+1)\}$, and for computable binary f and uniform inputs on $\{0,1\}^d$ with $d \geq \lceil \log n \rceil + 1$ at least of order $2^{-K(f)}/(n \log^2 n)$ (Proposition 5.9 and Corollary 5.10).

Question 8.10. Is there a positive statement for properties that are simple given a program fitted to the training data only (as in Proposition 8.7(c)), with a factor depending on $K(f)$?

8.5 Three examples

Three examples show what these bounds give where SOI plausibly generalises as intended: sentences of a formal theory labelled by provability (for instance a decidable Riemann hypothesis),

sentences labelled by their shortest proofs, and string lengths, learned on odd lengths and tested on even ones.

Proposition 8.11 (The intended answer)**.** *If the labels follow a partial computable g and $x \in \operatorname{dom} g$ is new, SOI gives $g(x)$ predictive probability at least $2^{L_{\mathrm{SOI}}(D)-K(g)-c}$, in both conventions.*

Proof. $M_F(D + (x, g(x))) \geq 2^{-K(g)-c}$ (Theorem 5.1(b)); divide by $M_F(D)$; the normalised conditional is larger. □

This is near 1 only if $L_{\mathrm{SOI}}(D) \approx K(g)$, i.e. if g-like programs carry most of the posterior.

Example 8.12 (Three examples)**.** Let $\mathcal{T}$ be a consistent theory with decidable proof checking, and $p_{\mathcal{T}}$ its proof search, of length $K(\mathcal{T}) + c$. (a) *Provability labels.* $p_{\mathcal{T}}$ fits every correctly labelled training set, so $L_{\mathrm{SOI}}(D) \leq K(\mathcal{T}) + c$, Proposition 8.11 applies, and over any sequence of sentences the correct label gets probability at most $\frac{1}{2}$ at most $K(\mathcal{T}) + c$ times. If every training sentence has a proof or refutation of length at most L and a test sentence decided by $\mathcal{T}$ has none of length at most $L' := 2^{\lceil \log L \rceil}$, its wrong label keeps probability at least $2^{-K(\mathcal{T})-2\log\log L-c}$ however many training sentences were seen (Example F.5); this floor decays slowly as L grows. So easy-to-hard generalisation is limited in one precise sense: no number of easy sentences removes a (small) floor on the wrong-label probability of a hard one. Relative to proof search in the form "fixed searcher, then a description of $\mathcal{T}$" (Proposition F.2), the floor is $2^{-K(L'|p_{\mathcal{T}})-c} \geq 2^{-c'}/(\log L(\log\log L)^2)$ times the posterior weight of $p_{\mathcal{T}}$. (b) *Shortest proofs as labels.* The same bounds hold, with the floor of Example F.5. (c) *Lengths of odd strings.* The correct answer has probability at least 2^{-c} on every even-length input, and SOI makes at most c online errors (steps at which the correct label gets probability at most $\frac{1}{2}$; Theorem 8.4). "$|x|$ even" and "output 1" cost $O(1)$ bits, so a wrong answer keeps probability at least $2^{-c'}$ on every even-length input. Both constants depend on the machine. An exact computation over a small enumerated grammar of programs (Appendix F; code in the accompanying repository) gives the intended answer probability at least 0.979, 0.997 or 0.9997 on every even-length test string, depending on one prior weight, and about 0.11 when a single extra program with ten times the prior mass of the intended one is added.

Table 4 (Appendix F) summarises what is proved, what is only plausible, and natural strengthenings that fail.

Question 8.13 (Natural machines)**.** Is there a natural universal machine, for instance a standard prefix-free encoding of a programming language, for which the intended answers of Example 8.12 get posterior probability close to 1, say at least 0.99 in (c) when the training data are all strings of odd length up to 7? By Proposition 8.2 no such statement holds for every universal machine, and by Proposition 8.1 a wrong answer keeps a probability bounded below however many odd-length strings are seen; what is machine-invariant is the bounded number of errors (Theorem 8.4).

9 Computable versions

SOI is not computable (Proposition A.10). Two computable relatives are natural: a time-bounded Solomonoff predictor with constant regret against all predictors of a given running time, and induction over Boolean circuits, for which, by analogy with Theorem 3.21, regret and circuit

complexity should be polynomially related.[7] This section states what holds and shows that Occam (simplest-consistent) learning, unlike Bayes, fails exponentially out of distribution.

9.1 Time-bounded SOI

Definition 9.1 (Time-bounded predictors; length-activated mixture)**.** Let Y be finite, $|Y| \geq 2$. A *probability predictor* is a RAM program A that on input x halts with rationals $(A(y \mid x))_{y \in Y}$, nonnegative and summing to 1, in time $T_A(x)$. For a nondecreasing $t : \mathbb{N} \to \mathbb{N}_{\geq 1}$, $\mathcal{C}_t := \{A : \exists c_A \, \forall x \; T_A(x) \leq c_A t(|x|)\}$. Fix RAM programs $A_1, A_2, \ldots$ with computable weights $w_i > 0$, $\sum_i w_i \leq 1$, activation lengths $\ell_1 < \ell_2 < \cdots$, and a nondecreasing unbounded clock factor $g \geq 1$, with t and $g \cdot t$ time-constructible. Let $q_i(\cdot \mid x) := A_i(\cdot \mid x)$ if $|x| \geq \ell_i$ and A_i halts on x within $g(|x|)t(|x|)$ steps with a valid vector, and the uniform vector u otherwise. The *length-activated mixture* is the mixture of product laws $M_{\mathrm{len}}(D) := \sum_i w_i \prod_{(x,y) \in D} q_i(y \mid x)$.

Theorem 9.2 (Time-bounded SOI)**.** *Let $A = A_i \in \mathcal{C}_t$ and $\ell_A := \max(\ell_i, \min\{n : g(n) \geq c_A\})$. On every sequence of* distinct *inputs, with any labels, the regret of M_{len} against A after k rounds is*

$$\sum_{j \leq k} \log \frac{A(y_j \mid x_j)}{M_{\mathrm{len}}(y_j \mid D_{j-1}, x_j)} \;\leq\; \log \tfrac{1}{w_i} + \#\{j \leq k : |x_j| < \ell_A\} \log |Y| \;\leq\; \log \tfrac{1}{w_i} + (2^{\ell_A} - 1) \log |Y|.$$

Proof idea (Appendix G). The loss of M_{len} is at most $\log(1/w_i)$ plus that of $\prod q_i$; $q_i = A$ on inputs of length at least ℓ_A, and on shorter ones $q_i \in \{A, u\}$ costs at most $\log |Y|$ more than A. The constant is large, triple exponential in the index i (Remark G.1); on a toy instance a direct computation (code in the accompanying repository) gives regret 132.4 bits against the proved 184.

Remark 9.3 (Running time). We assume the RAM model and do not analyse the cost of running the clocks. With $\ell_i = 2^{2^i}$ and $g(n) = \lceil \log \log(n+4) \rceil$, round k of M_{len} costs at most $\log \log |x_k| + 1$ simulations of $g(|x_k|)t(|x_k|)$ steps plus as many arithmetic operations on stored weights, and weights stored with relative precision $\varepsilon_k = (k+1)^{-2}$ change the cumulative loss by at most $\sum_j \log \frac{1+\varepsilon_j}{1-\varepsilon_j} \leq 1.9$ bits. Round k of M_{rd} (Proposition 9.4) and of the learner of Corollary 9.5 costs $\#\{i : k_i \leq k\}$ simulations of $g(k)t(|x_k|)$, respectively $(|x_k| + 2)^{g(k)}$, steps plus as many arithmetic operations; the learner of Proposition 9.6 costs $O(\#\{i : k_i \leq k\} \, g(k)t(\tilde{n}_k))$ steps in round k plus the memory for replaying the history. *Proof sketch* (Appendix G). Only active machines change their weights, and rounding errors telescope; for Proposition 9.6 add the replay cost.

Theorem 9.2 needs distinct inputs: with noisy labels a repeated input with a new label has probability 0 under M_{len}, and length clocking can leave a comparator over budget on short inputs forever (Remark G.2). Activation and clocks by round remove the restriction:

Proposition 9.4 (Round-activated mixture)**.** *Fix activation rounds $k_1 < k_2 < \cdots$. Let $q_i^{(k)} := A_i(\cdot \mid x_k)$ if $k \geq k_i$ and A_i halts on x_k within $g(k)t(|x_k|)$ steps with a valid vector, and u otherwise; let M_{rd} be the Bayes mixture, with weights w_i, of the per-round products $\prod_{j \leq k} q_i^{(j)}(y_j)$. For every $A = A_i \in \mathcal{C}_t$, every stream (repeated inputs and conflicting labels allowed) and every k, the regret of M_{rd} against A is at most $\log(1/w_i) + (J_A - 1) \log |Y|$, $J_A := \max(k_i, \min\{k : g(k) \geq c_A\})$.*

[7] Both were first sketched in the author's working notes (Hänni, 2026c), the first for sequences, with activation, clocking, amortised replay, a like-for-like caveat (Remark G.3) and an extension to all polynomial-time predictors; so were in-distribution bounds for Occam learning and the question whether it makes few online mistakes on every ordering.

Proof. For $j \geq J_A$, machine i is active and $T_A(x_j) \leq c_A t(|x_j|) \leq g(j)t(|x_j|)$, so $q_i^{(j)} = A(\cdot \mid x_j)$; for $j < J_A$, $q_i^{(j)} \in \{A(\cdot \mid x_j), u\}$ and $-\log q_i^{(j)}(y_j) \leq \log|Y| - \log A(y_j \mid x_j)$. The mixture loses at most $\log(1/w_i)$ to $\prod_j q_i^{(j)}$. □

This is the time-bounded analogue of the one-tape version on time-stamped inputs (Theorem 5.5(b)) and, like Solomonoff's per-query solution, a mixture of products over queries (Remark 4.2). On the stream of Remark G.2, where the length-activated mixture has regret 469 bits, a direct computation (code in the accompanying repository) gives regret 10.0 bits, against the bound 20.

Corollary 9.5 (All polynomial-time predictors)**.** *Let g be computable, nondecreasing and unbounded, and the clock $\tau_k(x) := (|x|+2)^{g(k)}$ time-constructible. With $\tau_k(x_k)$ in place of $g(k)t(|x_k|)$ in Proposition 9.4, the regret against every x-only predictor with $T_A(x) \leq c_A(|x|+2)^{e_A}$ is bounded by a constant depending on A, on every stream.*

Proof. $c_A(|x|+2)^{e_A} \leq (|x|+2)^{e_A + \log c_A}$, so A is within budget once $g(k) \geq e_A + \log c_A$; argue as for Proposition 9.4. □

So one learner has constant regret against all polynomial-time predictors, in time $(|x_k|+2)^{g(k)}$ per round for an arbitrarily slowly growing g (Remark 9.3); with the clock $(|x|+2)^{g(|x|)}$ the same holds on distinct inputs, in time $|x|^{\omega(1)}$ per round. For x-only comparators, which have nothing to reuse, these results compare like with like; history-aware comparators need a chronological learner (Remark G.3):

Proposition 9.6 (Time-bounded history-aware predictors)**.** *Let $\mathcal{C}_t^{\text{hist}}$ be the RAM programs A that keep a state, receive x_k, output $A(\cdot \mid h_{k-1}, x_k)$, then receive y_k, with round k taking at most $c_A t(\tilde{n}_k)$ steps, where $\tilde{n}_k$ is the length of the encoded h_{k-1} and x_k. Let the learner start machine i at round k_i and give it $2g(k)t(\tilde{n}_k)$ steps in round k, spent replaying the stored history and then on round k; it uses A_i's output if this finishes in time and u otherwise, and mixes the per-round products. Then the regret against every $A = A_i \in \mathcal{C}_t^{\text{hist}}$ is at most $\log(1/w_i) + 2k_0 \log|Y|$ on every stream, $k_0 := \max(k_i, \min\{k : g(k) \geq c_A\})$.*

Proof idea (Appendix G). As $\tilde{n}_k$ grows, every past round of A fits into the current budget once $k \geq k_0$, so the replay catches up by round $2k_0 - 2$ and stays caught up; a budget depending only on $|x_k|$ could stall. For $t(n) = n$ the class contains the chain and skipped-string predictors; it is the time-bounded counterpart of SOI^{hist} (Corollary 6.17).

A mixture that reads each program both as a predictor and as a sampler has the same regret against time-bounded samplers, at a cost per round exponential in the clock (Proposition G.4); whether polynomial time per round suffices is open (Question G.5).

9.2 Circuit SOI

Definition 9.7 (Circuits and the circuit mixture)**.** Fix $d \geq 1$ and the basis of all 16 two-input Boolean gates. A circuit of size s is a list of s gates, gate j with a type and two inputs from $\{x_1, \ldots, x_d, 0, 1, \text{gate}_1, \ldots, \text{gate}_{j-1}\}$; gate s is the output. There are $N_s = \prod_{j=1}^{s} 16(d+1+j)^2$ of them. The prior $w(C) := 2^{-s}/N_s$ is $2^{-\ell}$ for the code length ℓ of "size in unary, then the gate list"; $\sum_C w(C) = 1$. Let $W_{\text{circ}}(D) := \sum\{w(C) : C \text{ consistent with } D\}$, $\text{size}(f)$ the least size of a circuit for f, and for $B \subseteq \{0,1\}^d$, $\text{size}_B(f)$ the least size of one agreeing with f on B, and $L_B := -\log W_{\text{circ}}(f|_B)$. The *exception gadget* appends $\gamma \leq d+1$ gates in all, $\max(d-1, 1)$ gates

computing $[x = a]$ and a new output gate, so that C' equals C off a and outputs a chosen bit at a; $C \mapsto C'$ is injective and $w(C')/w(C) \geq 2^{-\Gamma(s)}$, $\Gamma(s) := \gamma(5 + 2\log(d + 1 + s + \gamma))$.

For any ordering of B, let P_i be the posterior probability of the wrong label at step i (circuits are total, so both conventions agree) and $\mathcal{E} := \sum_i P_i$ the expected number of mistakes of posterior sampling (Gibbs mistakes).

Theorem 9.8 (Circuit Bayes: mistakes and circuit size)**.** *For every nonempty $B \subseteq \{0,1\}^d$, every f and every ordering of B: (a) some circuit of size at most $\lceil L_B \rceil$ agrees with f on B, and $L_B \leq \mathrm{size}_B(f) + \log N_{\mathrm{size}_B(f)} = O(\mathrm{size}_B(f)\log(\mathrm{size}_B(f)+d))$; (b) $\mathcal{E} \leq (\ln 2) L_B$ and $L_B \leq 2\mathcal{E}(\Gamma(\lceil L_B \rceil) + 3)$; explicitly, for all $\mathcal{E} > 0$ and $d \geq 1$, $\mathrm{size}_B(f) \leq \lceil L_B \rceil$ with $L_B \leq 16\,\mathcal{E}\,(d+1)\log\big(2(\mathcal{E}+1)(d+2)\big)$.*

Proof idea (Appendix G). (a) Circuits larger than S have total weight 2^{-S}. (b) The first inequality is $P_i \leq -\ln(1 - P_i)$ with Proposition 3.16. For the second, the gadget maps the small wrong circuits at step i injectively into the right ones at weight cost $2^{-\Gamma}$, so $-\log(1 - P_i) \leq \Gamma + 2$; steps with $P_i \leq \frac{1}{2}$ cost at most $2P_i$, and at most $2\mathcal{E}$ steps have $P_i > \frac{1}{2}$. Both directions hold on every ordering; the step from small log loss to a small circuit is a circuit coding theorem, which (a) provides for size priors with an exponential tail (Remark G.7).

Occam (simplest-consistent) learning, which predicts with the first consistent function in order of description length ℓ, has the classical in-distribution guarantees (Proposition G.6). Out of distribution the picture differs:

Remark 9.9 (A natural argument, which holds for weighted majority)*.* A natural argument that Occam makes $O(\ell(f))$ online mistakes on every ordering assumes that when the current guess is wrong, a constant fraction of the competing weight is wrong too. That is the guarantee of the *posterior-majority* (weighted-majority) predictor: when it errs, at least half of the posterior mass errs, so $W_{\mathrm{circ}}(D_i) \leq W_{\mathrm{circ}}(D_{i-1})/2$, and it makes at most $L_B \leq \ell(f)$ mistakes on every ordering (Littlestone, 1988; Littlestone and Warmuth, 1994). Under the circuit prior the maximum-a-posteriori *circuit* is a smallest consistent one, so the MAP-hypothesis learner is an Occam learner. For simplest-consistent learning the assumption can fail at every step (Theorem 9.10).

9.3 Occam learning fails out of distribution

The construction below is from Appendix K of the slides of Vaintrob and Hänni (2026b); we prove it for two languages and every tie rule. *Formulas* have leaves $x_j, \neg x_j, 0, 1$ and binary nodes AND, OR, XOR; size is the number of leaves; the prior is "size in unary, then uniform within the size". *Circuits* are as in Definition 9.7.

Theorem 9.10 (The conjunction ladder; construction of Vaintrob and Hänni, 2026b, Appendix K)**.** *Let $f = x_1 \wedge \cdots \wedge x_d$, and let Occam use any fixed tie rule among functions of equal least size, for formulas ($d \geq 2$) or circuits ($d \geq 3$). Present 1^d, then 0^d, then the inputs of Hamming weight $1, 2, \ldots, d-1$ level by level, inside a level an input on which the current hypothesis errs if there is one. Then Occam makes at least $2^d - 2$ mistakes (formulas) or $2^d - d - 2$ (circuits). Since $\ell(f) = \Theta(d \log d)$ in both languages, its worst-case number of mistakes is $2^{\Omega(\ell/\log \ell)}$; the trivial upper bound is $2^{\ell+1}$. For this target the bound is tight up to the constant in the exponent, since there are only 2^d inputs.*

Proof idea (Appendix G). At the start of stage k the data say $h(1^d) = 1$ and $h(x) = 0$ for $|x| < k$. A Möbius-inversion lemma (Lemma G.8) shows that the least-size consistent functions are then exactly the $\binom{d}{k}$ conjunctions of k variables (for circuits from $k = 2$). Among the inputs

of weight k, each is wrong only at its own point 1_S, so every input of weight k is a mistake; a second adversary gives the circuit count without the lemma (Remark G.9). The ordering is computable from d and the tie rule, so a simply described ordering suffices, as the slides note. A direct computation (code in the accompanying repository) for $d = 3, 4, 5$ and seven tie rules confirms that every input of every covered level is a mistake.

With the same prior, the posterior-majority predictor makes at most $\ell(f)$ mistakes on every ordering, so it is the selection rule, not the prior, that fails (Remark G.10); with uniform tie-breaking among the least-size formulas, the limit of steeper posteriors, the ladder costs only $O(d^2)$ Gibbs mistakes (Remark G.11).

Question 9.11. (a) Is there a circuit coding theorem for size priors with polynomial tails, that is, do large random circuits rarely compute functions without small circuits? (b) Do few posterior-majority mistakes imply a small circuit (steps with P_i just below $\frac{1}{2}$ cost log loss but no majority mistakes)? (c) Is there a natural language and target with $2^{\Omega(\ell)}$ worst-case Occam mistakes? (d) Can a steeper posterior over a universal machine fail exponentially on some ordering?

10 Neural-network Bayes and circuit SOI

Bayesian learning with a neural network can be viewed as a circuit version of SOI (Vaintrob and Hänni, 2026b). This section states what is proved about this link, what the slides of Vaintrob and Hänni (2026b) claim, and what is open. An *architecture* is a parameter space Θ with a prior ϖ and a map $\theta \mapsto f_\theta : \{0,1\}^d \to Y$.

Proposition 10.1 (NN Bayes is function-space Bayes)**.** *If the likelihood depends on θ only through the values of f_θ on the data (a 0–1 likelihood, a margin rule, any label-noise model), the network's posterior predictive equals the function-space Bayes predictive with prior $P_A(f) := \varpi\{\theta : f_\theta = f\}$.*

Proof. Group the parameter integral by the value of f_θ. □

So NN Bayes is Bayesian prediction with the prior P_A on functions, and the question is how P_A compares with 2^{-K}, with M_F, or with a circuit prior.

Proposition 10.2 (Coding-theorem upper bound)**.** *If P_A, as a semimeasure on truth tables, is lower-semicomputed by a program a (encoding the architecture, the prior and d), then $P_A(f) \le 2^{-K(f|a)+c} \le 2^{-K(f)+K(a)+c'}$, with c independent of the architecture.*

This is the conditional coding theorem for the family indexed by a. P_A is lsc for finite parameter sets with rational probabilities. We only sketch this for Gaussian priors on real weights with sign outputs: up to a null boundary, $\{\theta : f_\theta = f\}$ is an open semialgebraic set, approximable from below by certified rational boxes. The bound is an exact, uncomputable form of the simplicity-bias bound of Dingle et al. (2018). It owes nothing to the networks: every prior lower-semicomputed by a short program satisfies it, and nearly saturates it on its own typical set (Proposition H.1), which can omit most simple functions.[8]

Proposition 10.3 (No general lower bound)**.** *(a) Let F_d be the set of functions an architecture A_d can compute, with $|F_d| < 2^{2^d}$ and membership decidable given d (any finite parameter set; any real-weight threshold architecture, by Tarski's decision procedure). The first truth table $f_d \notin F_d$*

[8] A version of Proposition 10.2 with this observation, and the questions of comparing uniform and simplicity priors on circuits, of the Gaussian limits and of whether few mistakes imply a small circuit for NN Bayes, first appeared in the author's working notes (Hänni, 2026c).

has $K(f_d) \le K(d) + c$ and $P_{A_d}(f_d) = 0$. So no bound $P_A(f) \ge 2^{-aK(f)-b}$ with a, b independent of d holds for such a family. (b) The parameter–function map alone implies no simplicity bias: a lookup-table architecture with uniform parameters gives every f probability 2^{-2^d}.

The slides make the point of (a) with functions that diagonalise against all small circuits. So the right comparison object for networks is *circuit* SOI, not program SOI. The circuit analogue of a fixed architecture with random parameters is the uniform prior over circuits of one size; one half of its comparison with the circuit simplicity prior holds (Definition 9.7 for N_s and size):

Proposition 10.4 (Uniform circuit priors contain the circuit simplicity prior)**.** *Let $P_{\mathrm{unif},S}$ be uniform over circuits of size exactly S. For every f with $\mathrm{size}(f) < S$, $P_{\mathrm{unif},S}(f) \ge 1/(N_{\mathrm{size}(f)} \cdot 16(d{+}1{+}S)^2)$. Hence Bayes with this prior makes at most $(\ln 2)(\log N_{\mathrm{size}(f)} + 2\log(d{+}1{+}S){+}4) = O(\mathrm{size}(f)\log(\mathrm{size}(f)+d) + \log S)$ Gibbs mistakes on every ordering.*

Proof idea (Appendix H). Pad a minimal circuit with arbitrary dead gates and a final gate copying its output. The other half of the comparison, that the uniform prior is not much *larger* than the simplicity prior on functions without small circuits, is open (Question 10.9(a)). A two-sided bound of this kind would transfer the mistake bounds of circuit Bayes to NN Bayes (Remark H.2).

The Gaussian-process limits. In the infinite-width Gaussian limits (NNGP and NTK) learning reduces to kernel methods, and one may ask whether a network then behaves like circuit SOI or more like a polynomial fit. On the cube $\{\pm 1\}^d$ with uniform measure, let $\chi_S(x) = \prod_{j\in S} x_j$ be the Walsh characters and, for a kernel $\mathcal{K}$, $(T_{\mathcal{K}}g)(x) := \mathbb{E}_{x'}\mathcal{K}(x,x')g(x')$.

Theorem 10.5 (Walsh diagonalisation; dimension bound)**.** *(a) (cf. Yang and Salman, 2019; also in Appendix J of the slides.) If $\mathcal{K}(x,x') = \Psi(\langle x,x'\rangle/d)$, every χ_S is an eigenfunction of $T_{\mathcal{K}}$, with eigenvalue $\Lambda_{|S|}$ depending only on $|S|$. This covers the NNGP and NTK kernels of fully connected networks with i.i.d. Gaussian weights and biases, which are functions of $\langle x,x'\rangle$ on the cube. (b) (The standard dimension argument; cf. Kamath et al., 2020.) If $\mathcal{V}$ is any space of functions on the cube of dimension at most n (for instance $\mathrm{span}\{\mathcal{K}(x_i,\cdot)\}_{i\le n}$, which contains the GP posterior mean and the kernel ridge regression solution), then $\binom{d}{k}^{-1}\sum_{|S|=k}\min_{g\in\mathcal{V}}\|\chi_S - g\|^2 \ge 1 - n/\binom{d}{k}$.*

Proof idea (Appendix H). (a) $\mathcal{K}(x,x')$ depends only on $x\odot x'$, so characters are eigenfunctions; permutation symmetry makes the eigenvalue depend on $|S|$ only. (b) The squared projections of all characters onto $\mathcal{V}$ sum to $\dim\mathcal{V}$.

Corollary 10.6 (Kernels versus SOI on sparse parities)**.** *For $|S| = k$ and i.i.d. uniform inputs, SOI ($\nu \in \{W, M_F\}$) has cumulative posterior error at most $(\ln 2)(K(\chi_S)+c)$ on every sequence of inputs, with $K(\chi_S) \le \log\binom{d}{k} + 2\log(k+1) + 2\log(d+1) + c$, and the squared error of its posterior-mean predictor averaged over the first n steps is at most $4(\ln 2)(K(\chi_S)+c)/n$. Every kernel method or GP posterior mean needs $n \ge (1-\varepsilon)\binom{d}{k}$ samples for average squared error at most ε over random k-sparse targets, while SOI needs $O(k\log d/\varepsilon)$ samples for Cesàro-averaged squared error ε, and $O((k\log d + \log(1/\varepsilon))/\varepsilon)$ for the error at a single step (Corollary 5.10(a)).*

The separation is in squared error of the real-valued predictor; we do not claim a 0–1 version in general. For one kernel family a 0–1 failure is exact:

Remark 10.7 (Bias-free two-layer ReLU kernels and odd parities). (a) For the bias-free two-layer ReLU NNGP, $\Psi_1(\rho) = (\sqrt{1-\rho^2} + \rho(\pi - \arccos\rho))/\pi$, and for the NTK, $\Psi_1(\rho) + \rho\Psi_0(\rho)$ with

$\Psi_0(\rho) = (\pi - \arccos\rho)/\pi$, the function $\Psi(\rho) - a\rho$ is even ($a = \frac{1}{2}$, resp. $a = 1$). Hence $\Lambda_{|S|} = 0$ for every odd $|S| \geq 3$, $\mathcal{V} := \operatorname{span}\{\mathcal{K}(x_i, \cdot)\} \perp \chi_S$, and $\|\chi_S - \hat{g}\|^2 = 1 + \|\hat{g}\|^2 \geq 1$ for every n: the kernel predictor never learns these parities. This is the Boolean-cube analogue of the finding of Basri et al. (2019) that a network without bias cannot represent or learn simple, low-frequency functions with odd frequencies (on the sphere). Biases or a third layer make the eigenvalues nonzero. (b) Let $z \sim \mathrm{GP}(0, \mathcal{K})$ for one of these kernels and $|S| \geq 3$ odd. Then $\mathbb{E}_x[z(x)\chi_S(x)]$ is Gaussian with variance $\Lambda_{|S|} = 0$, so it vanishes almost surely, while $\operatorname{sign} z = \chi_S$ would force it to equal $\mathbb{E}|z| > 0$. So the GP prior, and every posterior obtained by conditioning on an event of positive probability (sign or margin likelihoods), gives probability exactly 0 to the classifier χ_S. (c) For these two kernels and odd $|S| \geq 3$, let $x_1, \ldots, x_n$ be distinct training inputs with labels $\chi_S(x_i)$ and U the set of the other $2^d - n$ points. Every $g \in \mathcal{V}$ has $\sum_{x \in U} g(x)\chi_S(x) = -\sum_{i \leq n} g(x_i)\chi_S(x_i)$. For kernel ridge regression (the GP mean with noise variance $\beta > 0$) the training sum is $y^\top G(G + \beta I)^{-1} y \geq 0$, with G the training Gram matrix, so the predictor is weakly anti-correlated with the target on the unseen points. A direct computation (code in the accompanying repository) gives kernel ridge regression a sign accuracy on unseen points of 0.05 to 0.07 at $n = 900$, $d = 10$, far below chance; we do not prove this.

Remark 10.8 (Claims of the slides). The slides of Vaintrob and Hänni (2026b) note that Bayesian learning and SGD are "importantly different". They state (i) an $O(C)$ sample bound for programs with prior $\propto 3^{-|p|}$ (C the length of a shortest program for f) and a relevance-based sample bound for countable classes; (ii) a $\mathrm{poly}(d, C)$ sample bound for MLPs of depth L and any width $N \geq C := C_L(f)$, the least width and weight bound of a depth-L MLP implementing f, with Gaussian weights, for each width, with the prior scale small enough and outputs conditioned with margin $\frac{1}{2}$; (iii) that the orderings by C_L and by prior mass are "scrambled only polynomially"; (iv) an in-distribution mistake bound and a circuit bound polynomial in d and the number of mistakes; (v) a mean-field analogue (Vaintrob and Hänni, 2026a), while the NNGP "does not have a good circuit prior"; (vi) that out of distribution the in-distribution guarantee "is simply false"; an online mistake bound might still hold for NN Bayes (the slides give a heuristic argument and list it as open), while the analogous claim for simplest-consistent learning over circuits fails (their Appendix K; Theorem 9.10). The slides give proofs or proof sketches for several of these claims; we use only what is proved here. Of these claims, this paper proves the Markov half of the relevance-based bound in (i) (Corollary 5.10(a)), the character argument of their Appendix J (Theorem 10.5(a)) and the ladder of their Appendix K (Theorem 9.10); we do not prove the other claims.

Question 10.9. (a) Is the uniform prior over circuits of size S also bounded above by the circuit simplicity prior, $P_{\mathrm{unif},S}(f) \leq \mathrm{poly}(d, S)\, 2^{\mathrm{size}(f)}\, W_{\mathrm{circ}}(f)$ (the converse of Proposition 10.4)? (b) Is there a lower bound on the log prior of a function under NN Bayes at a fixed (not vanishing) prior scale? (c) Does the main theorem of the slides hold as stated, and do few mistakes imply a small circuit for NN Bayes? (d) What is the per-point test error of GP classification on odd parities (Remark 10.7(b))? (e) Does the sign accuracy of kernel ridge regression on unseen points fall below $\frac{1}{2}$ (Remark 10.7(c))?

11 Steeper simplicity priors (summary of the companion paper)

The companion paper (Hänni, 2026a) studies the following proposal: penalise program length more steeply than likelihood, by a factor $\lambda > 1$, for hypotheses that use private randomness at each

input.[9] These hypotheses form the class of Solomonoff's per-query solution M_{pq} (Remark 4.2), so the companion paper studies that solution with the prior $2^{-\lambda\ell}$ on programs of length ℓ, and the λ-MAP over the same class. A standard objection is that, under a steeper penalty, a hypothesis that is itself a universal mixture absorbs all structure of the data. The companion paper shows that this happens in every class that contains its own mixture as a member, such as Solomonoff induction and the one-tape version of SOI, whose randomness is shared by all inputs, and in the history-aware class, whose mixture is M_H (Corollary 6.17); no hypothesis with private randomness is universal. The restriction to private randomness has a price on latent causes shared by several inputs; at $\lambda = 1$ it is the gap between M_{prod}, which equals the per-query solution on distinct inputs, and M_F (Proposition 4.7 and Remark 4.8). The companion paper gives two reasons for such a penalty. The first, and the main one, is the split of data into a model and noise. At $\lambda = 1$, with private randomness and in Solomonoff induction, a computable sampler and the programs that fill in its sampled bits get similar posterior weight on a data set; with i.i.d. inputs and a sampler that draws fresh coins at every query, every λ-MAP is eventually a shortest program for the sampler, for fixed $\lambda > 1$ at a rate that depends only on λ, the program's length, a margin and the confidence; with shared randomness, or in Solomonoff induction, a universal mixture beats every sufficiently long sampler. The second reason is not learning a teacher's systematic errors. For a random parity whose labels a teacher flips, at a rate below 0.05596, on an error set of product form that is incompressible relative to the parity, with uniform inputs and λ proportional to the sample size n, the companion paper proves that for λ/n in an explicit window every hypothesis with private randomness that is near-optimal for the population objective knows the parity and, when the error rule is large, gives the erroneous labels far less probability than a memoriser does; λ must grow linearly in n, because each bit of an error rule is reused at many inputs, and the density and description length of the error rule alone do not suffice.

12 Discussion

SFI as ideal supervised learning. Remark 5.12 states in what sense SFI is ideal supervised learning and where the analogy stops: its guarantees compare it with computable labellers, in log loss, and need no assumption on how the inputs are chosen. The out-of-distribution results have the same character (Remarks 8.6 and 8.9): on inputs that a simple property separates from the training data, SFI keeps a probability bounded below, depending on the universal machine, for every simple behaviour, whatever the amount of data (Propositions 8.1 and 8.2); and for a computable target f its total posterior error is at most $(\ln 2)(K(f) + c)$ (Theorem 8.4).

What the hypotheses may see. The large gaps between SFI and Solomonoff induction in Sections 6 and 7 are properties of order-free prediction rather than of SFI. Every order-independent predictor pays them in expectation (Propositions 6.5, 7.5 and 7.17), and hypotheses that see the list of previous inputs remove them (Corollary 6.17, Proposition 7.8, and Theorem 7.20). Which of SFI and $\mathrm{SOI}^{\mathrm{hist}}$ is the right ideal therefore depends on whether the order of the data carries information about the labels. With i.i.d. inputs it does not, and SFI's guarantees apply as stated. When a teacher chooses the order, it can carry almost all of the information, as in the cyclic permutation, where the labels repeat the inputs. The set of previous inputs is not enough to recover this information (Propositions 7.9 and D.13), and even with the list, the loss of $\mathrm{SOI}^{\mathrm{hist}}$ is at most SI's joint loss plus a constant, not automatically at most SI's conditional label loss plus

[9] The proposal first appeared in the author's note (Hänni, 2026b).

a constant (Theorem 6.18 and Proposition 6.19).

Which version to call SFI. The versions of Section 4 agree up to $K(\operatorname{dom} D) + O(1)$ bits, and their gaps are unbounded. Between the program-based versions and SFI they are small on the known witnesses: about $\log d$ for the single shortest program on two points with d-bit inputs, and a padded tape-length cost for the program mixture, SFI's coin-free version (Corollary 4.10 and Proposition 4.6); the weak semimeasure, which has no posterior over functions, can exceed M_F by linearly many bits (Theorem 3.12). For deterministic computable targets all of them learn at the same rate (Remark 4.12 and Proposition 5.11), so the choice matters mostly for random labels and for code lengths. We call M_F SFI because it is the law of the one-tape definition (one universal machine, one sequence of coin flips), it has a posterior over functions, it dominates every machine law with constant $2^{-K(\nu)}$, it is invariant, and its semimeasure-convention loss does not depend on the order of the data (Remark 3.19). Where the randomness lives also matters for noise: a random function cannot produce fresh noise on a repeated input, while a per-query mixture can (Propositions 5.6 and 5.7).

Computable versions and networks. Time-bounded mixtures keep constant regret against all predictors that see only the input and run within a time bound (Proposition 9.4 and Corollary 9.5). For circuits, Bayes mistakes and circuit size are polynomially related on every ordering (Theorem 9.8), while simplest-consistent learning on the same prior makes $2^{\Omega(\ell/\log \ell)}$ mistakes, exponentially many in the input dimension, on some ordering for the conjunction of d variables, whose description length is $\ell = \Theta(d \log d)$ (Theorem 9.10); with the same prior the posterior-majority predictor makes at most ℓ mistakes on every ordering, so it is the selection rule, not the prior, that fails (Remarks 9.9 and G.10). For neural networks, the parameter–function map alone implies no simplicity bias (Proposition 10.3); any bridge to circuit SFI must come from the architecture and the prior, which is what the slides of Vaintrob and Hänni (2026b) claim for small prior scales (Remark 10.8). In the Gaussian-process limits, kernel regression and Gaussian-process posterior means need about $\binom{d}{k}$ samples, in squared error, on random k-sparse parities where SFI needs $O(k \log d)$ for any fixed accuracy, and bias-free two-layer ReLU kernels never learn odd parities of degree at least 3 (Corollary 10.6 and Remark 10.7); whether NN Bayes generalises like SGD-trained networks is not addressed here.

The companion paper. Where the randomness lives also matters for steeper simplicity priors (Section 11), and the companion paper gives two reasons for them (Hänni, 2026a). With private randomness, a penalty steeper than likelihood by a fixed factor splits per-query samples into the sampler and its noise, exactly in the limit and at a rate that depends only on the factor, the program's length, a margin and the confidence. A penalty steeper by a factor proportional to the sample size, in an explicit window, keeps hypotheses that are near-optimal for the population objective from modelling a teacher's errors on a parity task when the errors are made at a rate below 0.05596 on an error set of product form that is incompressible relative to the parity. With shared randomness, or in sequence prediction, a universal mixture is itself a hypothesis and every selected hypothesis is shorter than a bound that does not depend on the data. Whether structured error rules leave a window for steeper selection is open there.

12.1 Open problems

(1) Is every lower-semicomputable law of a random partial function dominated by M_F up to a constant factor (Question 3.14)? Is there a two-sided characterisation of $L_{\mathrm{SOI}}(D)$ by standard complexities (Question 3.23)? What is the arithmetical level of SFI's conditionals (Question A.11)?

(2) Is the normalised loss of M_F order-dependent, and which lower bounds proved here only in the semimeasure convention hold for it (Question A.8), in particular over the full cyclic permutation and in Eisenstat's example (Question 7.23)?

(3) Is $\log(M_F(D)/W(D)) = O(K(d, |D|))$ for inputs of length at most d, and what are the exact rates of the gaps of Table 1 (Question 4.13)? Does some gap tend to infinity along a computable family of data sets (Question B.5)?

(4) Does the last iterate, not only the average, have expected risk $O(K(f)/n)$ (Remark C.2)? Is the leave-one-out error of SFI $O(K_F(f|_B)/|B|)$ up to logarithmic factors (Conjecture C.4)?

(5) For computable reveal laws, can the term $K(n) + K(A)$ of Corollary 6.14 be dropped, and do defective samplers, infinite alphabets and reveal laws that see f admit the same bound (Remarks 6.13 and D.7)? Can $K(\Delta_f)$ be replaced by the algorithmic mutual information of Δ_f and f (Conjecture D.10)? How does $\mathrm{SOI}^{\mathrm{hist}}$ compare with normalised Solomonoff induction, and do the separations of Proposition D.13 and Theorem D.21 hold for normalised SFI (Question 6.23)?

(6) Does the depth count for the cyclic permutation hold for every weak function semimeasure (Question 7.6)? With raw, undelimited blocks, is SI's expected label loss on the cyclic permutation at least $K(d) - K^{\mathrm{pad}}(d) - O(1)$ (Question E.5)?

(7) Is there a positive out-of-distribution statement for properties that are simple given a program fitted to the training data (Question 8.10)? Is there a natural universal machine for which the intended answers of Example 8.12 get posterior probability close to 1 (Question 8.13)?

(8) Is there a learner with time per round polynomial in the time bound and constant regret against all time-bounded samplers (Question G.5)? Is there a circuit coding theorem for size priors with polynomial tails, and are there Occam failures with $2^{\Omega(\ell)}$ mistakes (Question 9.11)?

(9) Is the uniform prior over circuits of a given size bounded above by the circuit simplicity prior up to polynomial factors, is there a lower bound on the log prior under NN Bayes at a fixed prior scale, and does the main theorem of the slides hold as stated (Question 10.9)?

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revised in response. No human has yet independently checked all of the mathematics, and readers should weigh the results accordingly. The author takes responsibility for the content of this paper.

Code and data

All computations reported here were done with seeded or deterministic Python scripts (Python 3.11 with NumPy, SciPy and mpmath). The scripts, their saved outputs and independent re-implementations of the main computations are in the folder `solomonoff-function-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each reported number and how to re-run them. Re-running the scripts reproduces the saved outputs, apart from timing information and one time-limited optimisation.

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A Proofs for Section 3 (function semimeasures and SOI)

Notation is that of Section 3. Tapes are $\omega = \omega_1\omega_2\cdots$; a finite union of cylinders $[w]$ is identified with the corresponding finite union of half-open dyadic intervals of $[0,1)$, which changes nothing up to a $\mathbb{U}$-null set.

A.1 Function semimeasures

Proof of Proposition 3.3. (a) Along any order the conditionals telescope to $\nu(D)/\nu(\emptyset) = \nu(D)$; if an intermediate value vanishes, so does $\nu(D)$, by monotonicity. They sum to at most 1 by the weak inequality. (b) The empty product is 1. Revealing D first and x last gives $\nu_q(D + (x,y)) = \nu_q(D)\, q(y \mid D, x)$; summing over y gives the weak inequality, and dividing gives the formula for q. □

Proposition A.1 (Normalised predictors are random total functions)**.** *If every conditional of an order-independent predictor q (Proposition 3.3) sums to* 1*, then ν_q is the containment functional of a unique law of a random total function $F \in Y^X$.*

Proof of Proposition A.1. Now $\sum_y \nu_q(D + (x,y)) = \nu_q(D)$ for all D and $x \notin \operatorname{dom} D$ (also where $\nu_q(D) = 0$, by monotonicity). By induction, for each finite $A \subseteq X$ the map $l \mapsto \nu_q(l)$ on labellings $l \in Y^A$ is a probability distribution, and these distributions are consistent under marginalisation. Y is countable, so Kolmogorov's extension theorem gives a law on Y^X; it is unique because its cylinder probabilities are determined. □

For a weak ν and finite $T \subseteq X \setminus \operatorname{dom} D$ put $\nabla_T \nu(D) := \sum_{E:\operatorname{dom} E \subseteq T} (-1)^{|E|} \nu(D \cup E)$; the weak inequality is the case $|T| = 1$ of $\nabla_T \nu(D) \geq 0$.

Proposition A.2 (Möbius characterisation)**.** *A weak ν with $\nu(\emptyset) = 1$ is the containment functional of a random partial function if and only if $\nabla_T \nu(D) \geq 0$ for all D and all finite $T \subseteq X \setminus \operatorname{dom} D$. The law is then unique, and $\nabla_T \nu(D) = P(F \supseteq D,\ F|_T \equiv \perp)$.*

Proof of Proposition A.2. Convergence. For each $R \subseteq T$, $\sum_{E:\operatorname{dom} E = R} \nu(D \cup E) \leq \nu(D)$, by (W2) after Definition 3.2, applied to the weak semimeasure $E \mapsto \nu(D \cup E)$ on data sets with domain disjoint from $\operatorname{dom} D$. So $\sum_{E:\operatorname{dom} E \subseteq T} \nu(D \cup E) \leq 2^{|T|} \nu(D)$, and $\nabla_T \nu(D)$ converges absolutely.

Necessity. Let $\nu = \nu_P$ and $A_t := \{F(t) \neq \perp\}$ for $t \in T$. By inclusion–exclusion,

$$P\Big(F \supseteq D,\ \textstyle\bigcap_{t \in T} A_t^c\Big) = \sum_{R \subseteq T} (-1)^{|R|} P\Big(F \supseteq D,\ \bigcap_{t \in R} A_t\Big),$$

and $P(F \supseteq D, \bigcap_{t \in R} A_t) = \sum_{E:\operatorname{dom} E = R} \nu(D \cup E)$, a countable disjoint union over the labellings of R. Collecting terms gives $\nabla_T \nu(D)$, which is therefore a probability.

Sufficiency. For finite $S \subseteq X$ and $v \in Y_\perp^S$ let v^{def} be the data set of the defined entries of v and $S_\perp(v)$ the set where $v = \perp$, and put $q_S(v) := \nabla_{S_\perp(v)} \nu(v^{\mathrm{def}}) \geq 0$. For $x \notin S$, split the sum defining $q_{S \cup \{x\}}(v, \perp)$ by whether $x \in \operatorname{dom} E$; absolute convergence allows the rearrangement, and

$$q_{S \cup \{x\}}(v, \perp) = q_S(v) - \textstyle\sum_{a \in Y} q_{S \cup \{x\}}(v, a).$$

So $\sum_{a \in Y_\perp} q_{S \cup \{x\}}(v, a) = q_S(v)$, and $q_\emptyset = \nu(\emptyset) = 1$. By induction each q_S is a probability distribution on the countable set $Y_\perp^S$, and the family is consistent under marginalisation. Kolmogorov's extension theorem gives a law P on $Y_\perp^X$ with these marginals, and $P(F \supseteq D) = q_{\operatorname{dom} D}(D) = \nabla_\emptyset \nu(D) = \nu(D)$. The cylinder probabilities of any such P are the Möbius sums, by necessity; so P is unique. □

A.2 Machine laws and filling-in processes

Proof of Proposition 3.7. Each coordinate of f_ω^T is determined by countably many cylinders, so $\omega \mapsto f_\omega^T$ is measurable into $Y_\perp^X$. The set $\{\omega : f_\omega^T(x) = y\} = \bigcup_{(x,w,y) \in T} [w]$ is effectively open, uniformly in (x, y) and an index of T; a finite intersection of effectively open sets is effectively open uniformly; and the measure of an effectively open set is lsc uniformly in its description. $\{f_\omega \supseteq \emptyset\}$ is the whole space.

Enumeration. Let (W_i) enumerate all c.e. sets of triples. Let T_i be obtained by enumerating W_i and accepting each triple unless it conflicts (same input, comparable strings, different labels) with a triple already accepted. Each T_i is c.e. and consistent, and $T_i = W_i$ whenever W_i is consistent. So (T_i) lists exactly the function machines. □

Proof of Theorem 3.8. T_U is c.e. If $(x, pw, y), (x, p'w', y') \in T_U$ with $pw, p'w'$ comparable, then p, p' are comparable elements of the prefix-free set $\operatorname{dom} V$, so $p = p'$, and $(x, w, y), (x, w', y')$ are triples of the consistent $T_{V(p)}$ with w, w' comparable; so $y = y'$. For $\omega = p\omega'$ with $p \in \operatorname{dom} V$,

$f_\omega^{T_U} = f_{\omega'}^{T_{V(p)}}$, and ω' is uniform and independent of p. A tape with no prefix in $\operatorname{dom} V$ gives the empty function, which extends only $\emptyset$. Summing over p gives (1), and $\mathbf{m}(i) \geq 2^{-K(i)}$ gives the bound. □

Lemma A.3 (Laws of random total functions)**.** *For countable Y, every lsc law ν of a random total function ($\nu(\emptyset) = 1$, $\nu(D) = \sum_y \nu(D + (x, y))$) is computable, uniformly in an index. So the machine laws of random total functions are exactly the computable laws of random total functions.*

Proof of Lemma A.3. Induction on $|D|$. $\nu(\emptyset) = 1$ is computable. If $\nu(D)$ is computable, then $\nu(D + (x, y)) = \nu(D) - \sum_{y' \neq y} \nu(D + (x, y'))$ is a computable real minus an lsc (possibly infinite) sum, hence upper semicomputable; it is also lsc, hence computable. The induction is uniform. The converse is Proposition 3.9(c), last clause. □

Corollary A.4 (Universality among normalised order-independent predictors)**.** *For countable Y: if ν is lsc, $\nu(\emptyset) = 1$ and $\sum_y \nu(D + (x, y)) = \nu(D)$ for all D and $x \notin \operatorname{dom} D$, then $M_F \geq 2^{-K(\nu)-O(1)}\nu$.*

Proof. As in the proof of Proposition A.1, ν is the law of a random total function, hence computable (Lemma A.3), hence a machine law whose index is computable from that of ν (Proposition 3.9(c)); apply Theorem 3.8. □

So order-independence alone gives the weak semimeasures, where M^w is universal; normalised predictions give the random total functions, where M_F is.

Remark A.5 (Computable filling-in processes)*.* A *computable filling-in process* is a Markov chain $F_0 = \emptyset \subseteq F_1 \subseteq F_2 \subseteq \cdots$ of finite partial functions whose transition kernel $\Pr[F_{t+1} = g \mid F_t = f]$ is supported on finitely many extensions $g \supseteq f$, given as an explicit list with rational probabilities computable from (t, f). The law of its limit $F_\infty := \bigcup_t F_t$ is a machine law, with an index computable from the kernel. Conversely, every machine law is the law of the limit of the nested process $F_t :=$ "the part of f_ω certified by triples enumerated within t steps with $|w| \leq t$", which depends on $\omega_{1:t}$; this process is Markov in $\omega_{1:t}$, not in F_t. So machine laws are exactly the limits of computable nested processes driven by a hidden finite state.

Proof. The machine reads the tape bit by bit and samples $F_1, F_2, \ldots$ in turn: to sample from a finitely supported law with rational probabilities it reads bits until the dyadic interval named by the bits read so far lies inside one of the consecutive half-open intervals whose lengths are those probabilities, and takes that outcome. This fails to terminate only when the tape's real is an endpoint, a null set. On query x it runs the process until the first t with $x \in \operatorname{dom} F_t$ and outputs $F_t(x)$; it diverges if there is none. So T contains (x, w, y) when the process, run on the bits of w, produces $F_1, \ldots, F_t$ with t the first index such that $x \in \operatorname{dom} F_t$, and $F_t(x) = y$. The process reads its bits in order, so on comparable strings it produces the same initial segment of the chain; with $F_t \subseteq F_{t+1}$ this makes T consistent. Off the null set the sampled chain has the law of the process, and $f_\omega = F_\infty$. The converse is a restatement of Definition 3.6. □

A.3 Realising lower-semicomputable tree semimeasures

Lemma A.6 (Realisation; for binary alphabets this is classical, see Zvonkin and Levin, 1970)**.** *Let A be a countable alphabet with a fixed enumeration and ψ an lsc semimeasure on A^*: $\psi(\epsilon) \leq 1$ and $\psi(s) \geq \sum_{a \in A} \psi(sa)$. There are finite unions of cylinders $I_t(s)$, increasing in t and computable from (t, s) and an index of ψ, such that $I(s) := \bigcup_t I_t(s)$ satisfies $I(sa) \subseteq I(s)$, $I(sa) \cap I(sb) = \emptyset$*

for $a \neq b$, and $\mathbb{U}(I(s)) = \psi(s)$ for every s; the cylinders enumerated for strings of one length are pairwise disjoint. Consequently $\omega \mapsto$ "the longest s with $\omega \in I(s)$" is a monotone machine whose output extends s with probability exactly $\psi(s)$, and the event "the output extends s" is certified by a finite prefix of ω.

Proof. The proof covers countable alphabets. Let ψ_t be computable dyadic lower approximations, nondecreasing in t, with $\psi_t \to \psi$ and $\psi_t(s) = 0$ unless $|s| \leq t$ and s uses only the first t letters (round a rational approximation down to a multiple of 2^{-t}; this keeps monotonicity in t). On this finite tree define ψ'_t from the leaves up by $\psi'_t(s) := \max\{\psi_t(s),\ \sum_a \psi'_t(sa)\}$. Then:

- $\psi'_t \leq \psi$, by induction from the leaves, since $\sum_a \psi'_t(sa) \leq \sum_a \psi(sa) \leq \psi(s)$;
- ψ'_t is nondecreasing in t (by induction from the leaves of the tree of stage t, using $\psi_{t+1} \geq \psi_t$), and $\psi'_t \geq \psi_t \to \psi$, so $\psi'_t \to \psi$;
- $\psi'_t(s) \geq \sum_a \psi'_t(sa)$, and $\psi'_t(\epsilon) \leq \psi(\epsilon) \leq 1$.

Build I_t from the root down. Let $I_t(\epsilon) \supseteq I_{t-1}(\epsilon)$ be a finite union of cylinders of measure $\psi'_t(\epsilon)$. Given $I_t(s) \supseteq I_{t-1}(s)$ of measure $\psi'_t(s)$, the room $I_t(s) \setminus \bigcup_a I_{t-1}(sa)$ has measure $\psi'_t(s) - \sum_a \psi'_{t-1}(sa) \geq \sum_a (\psi'_t(sa) - \psi'_{t-1}(sa))$. All quantities are dyadic, so after refining to a common cylinder length we can add to each $I_{t-1}(sa)$ fresh cylinders inside the room, pairwise disjoint and of total measure $\psi'_t(sa) - \psi'_{t-1}(sa)$. The invariants are preserved, and $\mathbb{U}(I(s)) = \lim_t \psi'_t(s) = \psi(s)$. Siblings stay disjoint because the I_t increase in t. Each $I_t(s)$ is a finite union of cylinders, so $\omega \in I(s)$ is witnessed by a finite prefix. □

A.4 What machine laws include

Proof of Proposition 3.9. (a) Apply Lemma A.6 with alphabet Y, depth one and $\psi(\epsilon) := 1$, $\psi(y) := P(y \mid x)$, uniformly in x. This gives c.e. unions of cylinders $G_{x,y}$, disjoint in y, with $\mathbb{U}(G_{x,y}) = P(y \mid x)$. The private stream of x is $\omega_i^{(x)} := \omega_{\langle j(x),i\rangle}$. Let T contain (x, w, y) when w contains the first L bits u of $\omega^{(x)}$, for some L, and $[u] \subseteq G_{x,y}$ is certified. If $(x, w, y), (x, w', y') \in T$ with $w \sqsubseteq w'$, the two stream prefixes u, u' are comparable, so $[u] \cap [u'] \neq \emptyset$ and $G_{x,y} \cap G_{x,y'} \neq \emptyset$; hence $y = y'$. For distinct inputs the streams use disjoint tape coordinates, so they are independent and uniform, and $\nu_T(D) = \prod_{(x,y)\in D} \mathbb{U}(G_{x,y}) = \nu_P(D)$. The index of T is computable from one of P. For the claim on samplers: the output probabilities of a probabilistic machine are lsc (enumerate halting computations); the construction just given is the converse.

(b) $T := \{(x, \epsilon, \varphi(x)) : x \in \operatorname{dom}\varphi\}$.

(c) Enumerate X as $x_1, x_2, \ldots$ and $Y_\perp$ with $\perp$ last. Build nested partitions of $[0, 1)$: the interval $J(v)$ of $v \in Y_\perp^k$ is split, from left to right, into the intervals $J(va)$ of lengths $q_{k+1}(va)$, $a \in Y$ in order, and then $J(v\perp)$. Each left endpoint is $\operatorname{left}(v) + \sum_{a'<a} q_{k+1}(va')$, and $J(v\perp)$ starts at $\operatorname{left}(v) + q_k(v) - q_{k+1}(v\perp)$; all are computable reals. Let T contain (x_k, w, y) when the dyadic interval of w is certified, by rational approximations of the endpoints, to lie in the interior of some $J(v)$ with $v \in Y_\perp^k$ and $v_k = y \neq \perp$. Intervals of one level are disjoint and the levels are nested, so T is consistent. Off the null set of endpoints a tape lies in the interior of exactly one $J(v)$ at each level, so $\mathbb{U}\{f_\omega(x_i) = v_i,\ i \leq k\} = \mathbb{U}(J(v)) = q_k(v)$ for every v, and the law of f_ω is P. If Y is finite, each $q_S(v) = \nabla_{S_\perp(v)} \nu_P(v^{\mathrm{def}})$ is a finite alternating sum of computable reals. For a random total function $q_S(v) = \nu_P(v)$ if v has no $\perp$, and 0 otherwise.

(d) Realise w by Lemma A.6 (depth one, alphabet $\mathbb{N}$). This enumerates pairwise disjoint cylinders $[u]$, each labelled by some i, those labelled i having total measure w_i; disjointness makes the set of strings u prefix-free. Let $T := \{(x, uw', y) : [u] \text{ enumerated with label } i,\ (x, w', y) \in T_i\}$.

As in Theorem 3.8, T is consistent, the tape after u is uniform and independent of u, and $\nu_T(D) = \sum_i w_i \nu_{T_i}(D)$ for $D \neq \emptyset$; tapes outside all enumerated cylinders give the empty function. □

A.5 Weak universality and its gap

Proof of Theorem 3.11. Let (ϕ_i) enumerate all lsc maps from data sets to $[0,1]$, with computable approximations $\phi_{i,t}$ nondecreasing in t. At stage t let U_t be the finite set of data sets whose inputs and labels are among the first t inputs and labels. Define $\phi'_{i,t}$ on U_t (and 0 outside) by decreasing $|\operatorname{dom} D|$:

$$\phi'_{i,t}(D) := \max\Big\{\phi_{i,t}(D),\ \max_{x\le t,\ x\notin \operatorname{dom} D}\ \textstyle\sum_{y\le t} \phi'_{i,t}(D+(x,y))\Big\}.$$

If $\phi'_{i,t}(\emptyset) > 1$, freeze: $\phi'_{i,s} := \phi'_{i,t-1}$ for all $s \ge t$. Then $\phi'_{i,t}$ is nondecreasing in t (by induction from the largest domains) and satisfies the weak inequality everywhere: inside U_t by construction; outside U_t both sides vanish, since a data set outside U_t has no extension inside it; a frozen $\phi'_{i,t-1}$ is weak for the same reason. Let $\nu_i := \lim_t \phi'_{i,t}$. For $x \notin \operatorname{dom} D$, monotone convergence gives $\sum_y \nu_i(D+(x,y)) = \lim_t \sum_y \phi'_{i,t}(D+(x,y)) \le \lim_t \phi'_{i,t}(D) = \nu_i(D)$, and $\nu_i(\emptyset) \le 1$; so ν_i is an lsc weak semimeasure. If ϕ_i is already weak, induction from the largest domains gives $\phi'_{i,t} \le \phi_i$; so there is no freezing, and $\phi_{i,t} \le \phi'_{i,t} \le \phi_i$ gives $\nu_i = \phi_i$. Put $M^w := \sum_i \mathbf{m}(i)\nu_i$. The weak inequalities are linear and $\sum_i \mathbf{m}(i) \le 1$, so M^w is an lsc weak semimeasure, and it dominates each ν_i with factor $\mathbf{m}(i) \ge 2^{-K(i)}$. □

Proof of Theorem 3.12. (i) M_F is an lsc weak semimeasure (Proposition 3.7); apply Theorem 3.11.

(ii) Fix $S = \{s_1 < \cdots < s_k\}$ and put $\psi_S(y_1 \cdots y_j) := \nu(\{(s_1,y_1),\dots,(s_j,y_j)\})$ for $j \le k$ and $\psi_S := 0$ on longer strings. The weak inequality at s_{j+1} is exactly the semimeasure inequality for ψ_S, and ψ_S is lsc uniformly in (ν, S). Let $I(\cdot)$ realise ψ_S (Lemma A.6), and let $T_{\nu,S}$ contain (s_j, w, y) when $[w]$ lies in an enumerated cylinder of some $I(u)$ with $|u| = j$ and $u_j = y$. The sets $I(u)$ with $|u| = j$ are pairwise disjoint, so $T_{\nu,S}$ is consistent; and a tape lies in $I(u^{(j)})$ for each $j \le k$ exactly when it lies in $I(u^{(k)})$ and $u^{(j)}$ is the length-j prefix of $u^{(k)}$. So for $\operatorname{dom} D = S$, $\nu_{T_{\nu,S}}(D) = \psi_S(y_{s_1} \cdots y_{s_k}) = \nu(D)$. (For proper subsets of S only "$\le$" holds.) The index of $T_{\nu,S}$ is computable from (ν, S), so Theorem 3.8 gives $M_F(D) \ge 2^{-K(\nu,S)-O(1)}\nu(D)$, and $K(\nu,S) \le K(\nu) + K(S) + O(1)$. Take $\nu = M^w$, of complexity $O(1)$, for the second claim.

(iii) Use $\nu_\triangle$ of Example 3.5 on the blocks $B_j := \{3j, 3j+1, 3j+2\}$, $j < k$ (inputs as natural numbers). Put $\nu_\triangle^{\otimes k}(D) := \prod_{j<k} \nu_\triangle(D|_{B_j})$ if $\operatorname{dom} D \subseteq \bigcup_{j<k} B_j$, and 0 otherwise. A one-point inequality at $x \in B_j$ involves only the j-th factor, and at x outside the blocks both sides vanish; so $\nu_\triangle^{\otimes k}$ is weak. $\tilde\nu := \sum_k \mathbf{m}(k)\nu_\triangle^{\otimes k}$ is an lsc weak semimeasure, so $M^w \ge 2^{-O(1)}\tilde\nu$. For $\vec\imath \in \{1,2,3\}^k$ let $E_{\vec\imath}$ be the union over j of the copy of E_{i_j} in block j. These 3^k data sets are pairwise inconsistent, each has $|\operatorname{dom} E_{\vec\imath}| = 2k$ and $\nu_\triangle^{\otimes k}(E_{\vec\imath}) = 2^{-k}$. M_F is the containment functional of a random partial function, so $\sum_{\vec\imath} M_F(E_{\vec\imath}) \le 1$, and some $D_k := E_{\vec\imath}$ has $M_F(D_k) \le 3^{-k}$. But $M^w(D_k) \ge 2^{-O(1)}\mathbf{m}(k)2^{-k} \ge 2^{-k-K(k)-O(1)}$.

(iv) Each $-\log M^w(E_{\vec\imath}) \le k + K(k) + O(1)$ as in (iii). By concavity of log,

$$3^{-k}\textstyle\sum_{\vec\imath} -\log M_F(E_{\vec\imath}) \ \ge\ -\log\left(3^{-k}\sum_{\vec\imath} M_F(E_{\vec\imath})\right) \ \ge\ k\log 3.$$

Subtract.

(v) The argument of (iii) shows $\min_{\vec\imath} P(F \supseteq E_{\vec\imath}) \le 3^{-k}$ for every law P of a random partial function, so the minimum of $P(F \supseteq D)/\nu_\triangle^{\otimes k}(D)$ over the support of $\nu_\triangle^{\otimes k}$ is at most $3^{-k}/2^{-k} = (2/3)^k$. Let P_1 be the uniform law on the three total functions of Example 3.5(b); it satisfies $P_1 \ge \frac{2}{3}\nu_\triangle$ on all ten data sets of the support of $\nu_\triangle$. Its k-fold product over independent blocks

has $P_1^{\otimes k}(F \supseteq D) = \prod_j P_1(F|_{B_j} \supseteq D|_{B_j}) \geq (2/3)^k \nu_\triangle^{\otimes k}(D)$, with equality on the $E_{\vec{\imath}}$; so the supremum is attained. □

On the family of (iii), $\operatorname{dom} E_{\vec{\imath}}$ is determined by $(\vec{\imath}, k)$, so $K(\operatorname{dom} D_k) \leq k \log 3 + K(k) + O(1)$. Whether other constructions give gaps closer to $K(\operatorname{dom} D)$ is part of Question 4.13.

A.6 A sufficient condition for Question 3.14

A sufficient condition (proof sketch). Suppose an lsc ν_P has approximations $\nu_{P_t} \uparrow \nu_P$, where each P_t is a law of a random partial function on a finite domain with finitely many labels and rational probabilities, computable from t, and consecutive laws admit a coupling with $F_t \subseteq F_{t+1}$. The couplings form a polytope given by rational linear constraints, so a rational coupling can be found by search, and Remark A.5 then realises ν_P exactly as a machine law. By Strassen's theorem on monotone couplings for finite partially ordered sets (Strassen, 1965), such a coupling exists if and only if $P_t(U) \leq P_{t+1}(U)$ for every up-set U of the extension order. The containment events $\{F \supseteq D\}$ are only the principal up-sets, so $\nu_{P_t} \leq \nu_{P_{t+1}}$ does not give this.

We do not carry out the reduction from general lower approximations to finite rational laws.

A.7 The normalised convention

Proposition A.7 (Two-point criterion)**.** *Let P be the law of a random partial function on inputs a, b, with $A := \{F(a) \neq \bot\}$, $B := \{F(b) \neq \bot\}$ and $P(A), P(B) > 0$. Consider*

$$(\star) \qquad P(B \mid F(a) = \alpha) = P(B \mid A) \quad \textit{and} \quad P(A \mid F(b) = \beta) = P(A \mid B)$$

for all α, β of positive probability. (i) If $(\star)$ holds, the normalised probability of $\{(a, \alpha), (b, \beta)\}$ is the same in both reveal orders, for every (α, β). (ii) Conversely, if every two-point probability $P(F(a) = \alpha, F(b) = \beta)$, $\alpha, \beta \in Y$, is positive (as for M_F) and the two orders agree for every (α, β), then $(\star)$ holds. (iii) Without positivity (ii) can fail.

Proof. Put $p(\alpha, \beta) := P(F(a) = \alpha, F(b) = \beta)$, $u(\alpha) := P(B \mid F(a) = \alpha)/P(B)$ and $v(\beta) := P(A \mid F(b) = \beta)/P(A)$. Revealing a first gives the normalised product

$$\frac{P(F(a) = \alpha)}{P(A)} \cdot \frac{p(\alpha, \beta)}{P(F(a) = \alpha,\ B)} = \frac{p(\alpha, \beta)}{P(A)P(B)\, u(\alpha)},$$

and revealing b first gives $p(\alpha, \beta)/(P(A)P(B)\, v(\beta))$. When $p(\alpha, \beta) > 0$ the two agree exactly when $u(\alpha) = v(\beta)$; when $p(\alpha, \beta) = 0$ both are 0. (i) $(\star)$ gives $u \equiv P(A \cap B)/(P(A)P(B)) \equiv v$. (ii) Agreement on all pairs forces $u(\alpha) = v(\beta)$ for all α, β, so $u \equiv v \equiv \gamma$ for a constant γ. Averaging u over α with weights $P(F(a) = \alpha)/P(A)$ gives $\gamma = P(A \cap B)/(P(A)P(B))$, which is $(\star)$. (iii) The law with $(F(a), F(b))$ equal to $(0, 0)$ with probability $\frac{1}{4}$, $(1, 1)$, $(1, \bot)$, $(\bot, 1)$ with probability $\frac{1}{8}$ each, and $(\bot, \bot)$ with probability $\frac{3}{8}$ has $P(B \mid F(a) = 0) = 1$, $P(B \mid F(a) = 1) = \frac{1}{2}$, $P(B \mid A) = \frac{3}{4}$; it puts positive mass only on the diagonal defined pairs and is symmetric in a, b, so its normalised probabilities are order-independent. □

The law of Proposition 3.17(b) violates $(\star)$: $P(B \mid F(a) = 0) = \frac{1}{2} \neq \frac{2}{3} = P(B \mid A)$. For a randomly generated law on six inputs a direct computation (code in the accompanying repository) gives the semimeasure loss 10.92 bits in all 720 orders and normalised losses between 7.33 and 10.64 bits.

Question A.8. Is the normalised loss of M_F order-dependent, and which of the lower bounds of Remark 3.18 that are proved only in the semimeasure convention hold for $\bar{M}_F$? On two inputs the question is whether M_F satisfies ($\star$): the normaliser leaks order information exactly when whether one input is defined says something about another input's label. We expect ($\star$) to fail for M_F but have no proof.

Total laws always satisfy ($\star$). A direct computation (code in the accompanying repository) confirms both directions: random laws with positive two-point cells violate ($\star$) and are order-dependent (400 of 400), and laws constructed to satisfy ($\star$) are order-independent (400 of 400).

Proposition A.9 (What transfers to the normalised convention)**.** *For each step t let Z_t be the posterior probability that $f_\omega(x_t)$ is defined, $Z_t := \mathcal{Z}(D_{t-1}, x_t)/M_F(D_{t-1})$ ($Z_t := 1$ at a repeated input). (a) $\bar{L}_{\mathrm{SOI}} = L_{\mathrm{SOI}}(D_n) - \sum_t \log(1/Z_t)$. As $Z_1 \geq 2^{-c}$, single points transfer: $-\log \bar{M}_F(y \mid \emptyset, x) = K(y \mid x) \pm O(1)$. (b) $\bar{L}_{\mathrm{SOI}} \geq K^{\emptyset'}(y_{1:n} \mid x_{1:n}) - O(1)$ for all inputs and labels, $K^{\emptyset'}$ relative to the halting problem. (c) For any joint law of inputs and labels (inputs may depend on past labels), $\mathbb{E}\bar{L}_{\mathrm{SOI}} \geq \sum_t \mathbb{E}H(y_t \mid h_{t-1}, x_t)$, the loss of the chronological Bayes predictor.*

Proof of Proposition A.9. (a) Each normalised factor is the raw factor divided by Z_t. At the first step $Z_1 = \mathbb{U}\{f_\omega(x_1) \neq \perp\} \geq 2^{-c}$, because a constant function is a machine law of complexity $O(1)$; combine with Theorem 3.21(c).

(b) Each normalised conditional is a ratio of two positive lsc reals, hence computable relative to $\emptyset'$, uniformly. For a fixed input list, $y_{1:n} \mapsto \prod_t \bar{M}_F(y_t \mid D_{t-1}, x_t)$ is a probability distribution on Y^n: sum out y_n, then y_{n-1}, and so on; each factor sums to 1, and a repeated input contributes a point mass on its stored label. Apply the conditional coding theorem relative to $\emptyset'$ (it relativises; Section 3.1).

(c) The normalised predictor is a history-aware predictor issuing probability vectors. At each step, conditionally on (h_{t-1}, x_t), Gibbs' inequality gives $\mathbb{E}[-\log \bar{M}_F(y_t \mid D_{t-1}, x_t) \mid h_{t-1}, x_t] \geq H(y_t \mid h_{t-1}, x_t)$; take expectations and sum. □

A.8 Invariance, coding bounds, sequence induction, computability

Proof of Theorem 3.20. (a) Each of M_F, M'_F is a machine law in the other's enumeration, so each dominates the other up to a constant factor by Theorem 3.8; the same argument applies to any two function machines whose laws dominate all machine laws up to constants.

(b) α^{-1} and β^{-1} are partial computable (by search). The set of triples $(\alpha(x), w, \beta(y))$ with $(x, w, y) \in T_U$ is a function machine over X', Y' (consistent because α is injective), whose random function is $\beta \circ f_\omega \circ \alpha^{-1}$, undefined off the range of α. Its law ν' satisfies $\nu'(D^{\alpha\beta}) = \mathbb{U}\{\beta(f_\omega(x)) = \beta(y)\ \forall (x, y) \in D\} = M_F(D)$, since β is injective. Its index is computable from indices of α, β, so $M'_F(D^{\alpha\beta}) \geq 2^{-K(\alpha,\beta)-O(1)} M_F(D)$. Conversely, the machine with random function $\beta^{-1} \circ f'_\omega \circ \alpha$ on X has law ν with $\nu(D) = M'_F(D^{\alpha\beta})$, so $M_F(D) \geq 2^{-K(\alpha,\beta)-O(1)} M'_F(D^{\alpha\beta})$.

(c) For W: a program for φ becomes a program for $\beta \circ \varphi \circ \alpha^{-1}$ by a prefix of length $K(\alpha, \beta) + O(1)$, and conversely. For M_{prod}: transform each conditional semimeasure in the same way. For M^w: given a weak ν on $X \times Y$, put $\nu'(D') := \nu(D)$ if $D' = D^{\alpha\beta}$ and $\nu'(D') := 0$ otherwise; ν' is lsc (search for preimages). If $D' = D^{\alpha\beta}$ and $x' \notin \operatorname{dom} D'$, the extensions of D' at x' of the form $E^{\alpha\beta}$ have $x' = \alpha(x)$ and labels $\beta(y)$, so the one-point inequality transfers; if D' is not an image, neither is any extension of it. The converse transformation is symmetric. □

Proof of Theorem 3.21. (a) $P(E \mid S) := M_F(E)$ for $\operatorname{dom} E = S$ is lsc uniformly in (E, S) and sums to at most 1 over E ((W2) after Definition 3.2). The conditional coding theorem gives $K(E \mid S) \leq -\log M_F(E) + O(1)$. Take $E = D|_T$, $S = T$, and use $M_F(D) \leq M_F(D|_T)$.

(b) First term: the shortest consistent program p gives the machine law $\mathbf{1}[\varphi_{V(p)} \supseteq D]$, whose complexity is at most $|p| + O(1)$, so $M_F(D) \geq 2^{-K_F(D)-O(1)}$ (equivalently, $M_F \geq 2^{-O(1)}W$ by Theorem 4.4). Second term: $\mathbf{m}(y \mid x) := \sum_{q:V(q,x)=y} 2^{-|q|}$, with $V(\cdot, x)$ the conditional universal prefix machine, is an lsc conditional semimeasure with $\mathbf{m}(y \mid x) \geq 2^{-K(y|x)}$; its product law is a machine law of complexity $O(1)$ (Proposition 3.9(a)). Last inequality: $K_F(D) \leq K(D) + O(1)$ (a program computes the lookup table of D from a shortest description of D) and $K(D) \leq K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + O(1)$.

(c) Combine (a) with $T = \{x\}$ and (b). □

Proof of Proposition 3.24. ($\geq$) Let T_1 contain (i, w, b) when the universal monotone machine, run on w, has printed more than i bits and its bit $i + 1$ is b. Outputs on comparable strings are comparable, so T_1 is a function machine, and $\nu_{T_1}(D_x) = M(x)$; apply Theorem 3.8. ($\leq$) The monotone machine that prints $f_\omega(0), f_\omega(1), \ldots$ in order, as long as each value is certified along ω and lies in $\{0, 1\}$, prints a string extending x exactly when $f_\omega \supseteq D_x$. So $x \mapsto M_F(D_x)$ is the output semimeasure of a monotone machine, and M dominates it (Zvonkin and Levin, 1970). □

Proposition A.10 (Computability)**.** *(a) $M_F(D)$ is lsc uniformly in D, and positive. (b) M_F is not computable. (c) Neither $M_F(y \mid D, x)$ nor $\bar{M}_F(y \mid D, x)$ is computable as a function of (D, x, y). (d) Both are computable relative to the halting problem $\emptyset'$.*

Proof of Proposition A.10. (a) Proposition 3.7 and the remark after Definition 3.15.

(c) Take $X = \mathbb{N}$ and suppose the raw conditional is computable. Define f by recursion: with $D_t := \{(s, f(s)) : s < t\}$, let $f(t)$ be the first $y \in \{0, 1\}$ for which approximation certifies $M_F(y \mid D_t, t) < 0.6$. The two conditionals sum to at most 1, so one of them is at most $\frac{1}{2}$, and the search halts. f is total computable, and $M_F(D_n) = \prod_{t<n} M_F(f(t) \mid D_t, t) < 0.6^n$, which contradicts $M_F(D_n) \geq 2^{-K(f)-O(1)}$ for large n. For the normalised conditional the same construction gives normalised loss at least $n \log(1/0.6)$, while it is at most $L_{\mathrm{SOI}}(D_n) \leq K(f)+O(1)$ (Proposition 3.17(a)).

(b) If M_F were computable, the raw conditional would be a ratio of positive computable reals, hence computable.

(d) Lsc reals are computable relative to $\emptyset'$; the conditionals are ratios of positive lsc reals ($M_F(D)$ and, for the normalised one, $\mathcal{Z}(D, x)$). □

Question A.11. What is the exact arithmetical level of the conditionals $M_F(y \mid D, x)$ and $\bar{M}_F(y \mid D, x)$? For Solomonoff's M see Leike and Hutter (2015b).

B Proofs for Section 4 (Kolmogorov and semimeasure versions)

Notation is that of Sections 3.1 and 4.

B.1 The variants and the chain

Remark B.1 (Why W's programs delimit themselves before reading the input). With this choice W is the containment functional of the random partial function "draw p with probability $2^{-|p|}$, output $\varphi_{V(p)}$": a mixture of fixed functions with input-independent weights. Summing $2^{-|p|}$ over the programs of a conditional prefix machine $V(p, x)$ is a different object: for each x the halting programs form a prefix-free set, but their union over x need not, so the weights define no prior over functions. Reading a conditional program lazily from one shared tape does give a random

function, the function machine $x \mapsto V(\omega, x)$; it is a machine law of complexity $O(1)$, but not a countable mixture of partial computable functions with input-independent weights, and on single points it gives $2^{-K(y|x)\pm O(1)}$ (Theorem 3.21(c)). The two notions of conditional complexity are close: with $K^{\mathcal{P}}(z \mid x) := \min\{|h| : h \in \operatorname{dom} V,\ \varphi_{V(h)}(x) = z\}$,

$$K(z \mid x) - c \ \le\ K^{\mathcal{P}}(z \mid x) \ \le\ K(z \mid x) + K(K(z \mid x)) + c,$$

since every $h \in \operatorname{dom} V$ is a conditional program, and a conditional program prefixed with a self-delimiting code of its length delimits itself without reading x.

Proof of Theorem 4.4. (1) The shortest consistent program is one term of W. (2) $\delta_e(y \mid x) := \mathbf{1}[\varphi_e(x) = y]$ is a conditional semimeasure, lsc uniformly in e; let g be computable with $P_{g(e)} = \delta_e$. Then $i \mapsto \sum_{e:g(e)=i} \mathbf{m}(e)$ is an lsc discrete semimeasure, so $\mathbf{m}(i) \ge 2^{-O(1)} \sum_{e:g(e)=i} \mathbf{m}(e)$, and $M_{\mathrm{prod}}(D) \ge 2^{-O(1)} \sum_e \mathbf{m}(e)\mathbf{1}[\varphi_e \supseteq D] = 2^{-O(1)} W(D)$. (3) M_{prod} is an lsc mixture of product laws, a machine law (Proposition 3.9(a,d)); apply Theorem 3.8. (4) Theorem 3.12(i). (5) $K_F(D) \le K(D) + O(1)$ (lookup table), $K(D) \le K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + O(1)$, and $K(D \mid \operatorname{dom} D) \le -\log M^w(D) + O(1)$ by the conditional coding theorem applied to $E \mapsto M^w(E)$ on data sets with domain $\operatorname{dom} D$, whose total is at most 1 ((W2) after Definition 3.2). □

B.2 Programs that cannot separate inputs

Proof of Proposition 4.5. $\Omega_V^{\ge \ell} \to 0$ because $\sum_{p \in \operatorname{dom} V} 2^{-|p|} \le 1$.

(a) Let E_ℓ be the set of indices $V(p)$ with $|p| < \ell$; it has fewer than 2^ℓ elements. Colour $x \in \{0,1\}^d$ by $(\chi_e(x))_{e \in E_\ell}$, where $\chi_e(x) = \varphi_e(x)$ if this is 0 or 1 and $\chi_e(x) = *$ otherwise. There are at most 3^{2^ℓ} colours, so if $2^d > 3^{2^\ell}$ two distinct strings x_1, x_2 share a colour. No $e \in E_\ell$ has $\varphi_e(x_1) = 0$ and $\varphi_e(x_2) = 1$, so $W(\{(x_1, 0), (x_2, 1)\}) \le \Omega_V^{\ge \ell}$. Choose ℓ with $\Omega_V^{\ge \ell} \le \varepsilon$, then d with $2^d > 3^{2^\ell}$. For the lower bounds, the uniform product law $P(0 \mid x) = P(1 \mid x) = \frac{1}{2}$ has complexity $O(1)$ and gives every two-point binary data set probability $\frac{1}{4}$; so $M_{\mathrm{prod}}(D) \ge 2^{-O(1)}/4$, and $M_F \ge 2^{-O(1)} M_{\mathrm{prod}}$ by Theorem 4.4.

(b) Fewer than 2^{d-1} programs have length at most $d-2$, and each gives at most one value at 0^d. So at least 2^{d-1} strings $y \in \{0,1\}^d$ receive W-mass only from programs of length at least $d-1$. Each program contributes to at most one y, so these $W(\{(0^d, y)\})$ sum to at most $\Omega_V^{\ge d-1}$, and one of them is at most $\Omega_V^{\ge d-1}/2^{d-1}$. For the lower bound, $P(y \mid x) := 2^{-|x|}\mathbf{1}[|y| = |x|]$ is a computable conditional probability of complexity $O(1)$, so $M_{\mathrm{prod}}(\{(0^d, y)\}) \ge 2^{-O(1)} 2^{-d}$. □

Remark B.2 (How fast the gap of W grows). Let $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, the largest ℓ with $3^{2^\ell} < 2^d$. By Proposition 4.5(a), some two-point binary D on d-bit inputs has $-\log W(D) \ge \log(1/\Omega_V^{\ge \ell_d})$. By Proposition 4.6(b), every such D has $-\log W(D) \le 2 + K^{\mathrm{pad}}(2^{d+1}) + O(1)$. Both bounds are unbounded and both are of padded type, since $\log(1/\Omega_V^{\ge \ell}) \le K^{\mathrm{pad}}(\ell) + O(1)$. By Proposition 4.6(c) the gap is at most $2\log(h(d)+2) + O_h(1)$ for every computable, nondecreasing, unbounded h. Its exact rate is open (Question 4.13).

Proof of the claims in Remark B.2. ℓ_d is the largest ℓ with $2^\ell \log 3 < d$, i.e. with $3^{2^\ell} < 2^d$. $\log(1/\Omega_V^{\ge \ell}) \le K^{\mathrm{pad}}(\ell) + O(1)$: choose $L \ge \ell$ with $K(L) = K^{\mathrm{pad}}(\ell)$. The prefix machine that reads a V-program for L, then L more bits, and halts, is simulated by V with a fixed prefix r. The resulting elements of $\operatorname{dom} V$ have length at least $L \ge \ell$ and total weight $2^L \cdot 2^{-|r|-K(L)-L}$, so $\Omega_V^{\ge \ell} \ge 2^{-K^{\mathrm{pad}}(\ell)-|r|}$. *Rate*: apply Proposition 4.6(c) with $g(N) := h(\max(0, \lfloor \log N \rfloor - 1))$, which is computable, nondecreasing and unbounded; then $K^{\mathrm{pad}}(2^{d+1}) \le 2\log(h(d)+2) + O_h(1)$. □

B.3 The tape-length bound and the hashing bound

Proof of Proposition 4.6. (a) *Monotonicity and limit.* A prefix of $\omega_{1:L}$ is a prefix of $\omega_{1:L'}$ for $L' \geq L$. The event $f_\omega \supseteq D$ holds exactly when each of the finitely many pairs of D has a certifying triple along ω, and then all of them lie within the longest one; so $\{f_\omega \supseteq D\} = \bigcup_L \{\text{certified within } L \text{ bits}\}$, an increasing union.

The bound. Fix $L' \geq L$ with $K(L') = K^{\mathrm{pad}}(L)$. For $w \in \{0,1\}^{L'}$ let $\varphi_{T,w}(x) := y$ if some $(x, w', y) \in T$ has w' a prefix of w, and undefined otherwise. This is well defined because T is consistent and any two prefixes of w are comparable, and it is partial computable (enumerate T). Let V' be the prefix machine that reads a V-program for an index i of T, then a V-program for L', then L' further bits w, and outputs an index of $\varphi_{T_i,w}$; its domain is prefix-free. By universality $V(rs) = V'(s)$ for a fixed string r. With shortest programs $q_i, q_{L'}$, the strings $r\, q_i\, q_{L'}\, w$, $w \in \{0,1\}^{L'}$, are distinct elements of $\operatorname{dom} V$, so

$$W(D) \;\geq \sum_{w \in \{0,1\}^{L'}} 2^{-|r|-K(T)-K(L')-L'}\, \mathbf{1}[\varphi_{T,w} \supseteq D] \;=\; 2^{-|r|-K(T)-K^{\mathrm{pad}}(L)}\, \nu^{(L')}(D),$$

and $\nu^{(L')}(D) \geq \nu^{(L)}(D)$. The consequence for $L_{1/2}$ is immediate, and $K(T_U) = O(1)$.

(b) Apply (a) to the machine $F(x) := \omega_{j(x)+1}$, of complexity $O(1)$, whose law gives every binary-labelled D probability $2^{-|D|}$. For $|x| \leq d$ its certificates lie within the first $2^{d+1} - 1$ tape bits, and $K^{\mathrm{pad}}(2^{d+1} - 1) \leq K^{\mathrm{pad}}(2^{d+1})$ since K^{pad} is nondecreasing. For the second inequality put $s := \lceil \log(d+1) \rceil$ and $L'' := 2^{2^s} \geq 2^{d+1}$; then $K^{\mathrm{pad}}(2^{d+1}) \leq K(L'') \leq K(s) + O(1)$.

(c) K^{pad} is nondecreasing because the minimum runs over a shrinking set, and unbounded because only finitely many L have $K(L) \leq c$ for each c. $K^{\mathrm{pad}}(2^{2^s}) \leq K(2^{2^s}) \leq K(s) + O(1)$. For g: let $N^* := \min\{N' : g(N') \geq g(N) + 1\}$. It exists because g is unbounded, it exceeds N because g is nondecreasing, and it is computable from $g(N) + 1$. So $K^{\mathrm{pad}}(N) \leq K(N^*) \leq K(g(N) + 1) + c_g \leq 2\log(g(N) + 2) + c'_g$, using $K(m) \leq 2\log(m+1) + O(1)$. □

Remark B.3 (A hashing bound). Proposition 4.6(b) is stronger; this bound uses a different construction. For every D with labels in $\{0,1\}$, $|D| \leq k$ and $\operatorname{dom} D \subseteq \{0,1\}^{\leq d}$: $W(D) \geq 2^{-|D|-K(d,k)-c}$. Here $K(d,k) \geq K(d) - O(1) \geq K^{\mathrm{pad}}(2^{d+1}) - O(1)$, and Proposition 4.6(b) does not depend on $|D|$. One might expect a logarithmic factor from k-wise independent hashing; that is the behaviour of K_F, not of W: for K_F a logarithmic cost in d occurs already on two points (Theorem 4.9), and Proposition 4.11 bounds it by a covering argument rather than by hashing.

Proof. We may assume $k < 2^{d+1}$, since $|\{0,1\}^{\leq d}| < 2^{d+1}$ (replacing k by $\min(k, 2^{d+1} - 1)$ changes $K(d,k)$ by $O(1)$). Put $q := d + 1$ and fix, computably from q, an irreducible polynomial of degree q over GF(2). Map $x \in \{0,1\}^{\leq d}$ injectively to the element z_x of $\mathrm{GF}(2^q)$ whose coefficient vector is the binary string $1x$. For a seed $s \in \mathrm{GF}(2^q)^k$ let $h_s(z)$ be the lowest bit of $\sum_{i<k} s_i z^i$. For distinct $z_1, \ldots, z_k$ the Vandermonde map $s \mapsto (\sum_i s_i z_l^i)_{l \leq k}$ is a bijection of $\mathrm{GF}(2^q)^k$, so for uniform s the values at any $j \leq k$ distinct points are uniform and independent, and so are their lowest bits. Hence $\Pr_s[h_s(z_x) = y\ \forall (x,y) \in D] = 2^{-|D|}$. Let V' read a V-program for (d,k), then kq further bits s, and output an index of $x \mapsto h_s(z_x)$ on $\{0,1\}^{\leq d}$; its domain is prefix-free. With $V(rs') = V'(s')$, summing over the 2^{kq} seeds gives $W(D) \geq 2^{-|r|-K(d,k)}\ \Pr_s[\cdots] = 2^{-|r|-K(d,k)-|D|}$. □

B.4 Shared randomness

Proof of Proposition 4.7. ν_{len} is the law of the function machine $\{(x,w,b) : |w| = |x|+1,\ w_{|x|+1} = b\}$, of complexity $O(1)$. Suppose $M_{\mathrm{prod}} \geq \gamma\, \nu_{\mathrm{len}}$ for some $\gamma \in (0,1]$. Put $\varepsilon := \gamma/8$, write

$M_{\mathrm{prod}} = \sum_i w_i \nu_{P_i}$ with $w_i = \mathbf{m}(i)$, and choose J with $\sum_{i>J} w_i \le \varepsilon$. For $n \ge 1$ let $a_n := 0^n$, $b_n := 1^n$, and $s_i(n,y) := P_i(y \mid a_n) P_i(y \mid b_n)$ for $y \in \{0,1\}$.

Compactness. Choose an infinite set of lengths along which $P_i(y \mid a_n) \to \alpha_i(y)$ and $P_i(y \mid b_n) \to \beta_i(y)$ for all $i \le J$ and $y \in \{0,1\}$. Let $S_i := \alpha_i(0)\beta_i(0) + \alpha_i(1)\beta_i(1) \le \max_y \alpha_i(y) \cdot (\beta_i(0) + \beta_i(1)) \le 1$. If $S_i = 1$, then $\max_y \alpha_i(y) = 1$, so $\alpha_i = \delta_{y_i}$ for some y_i, and then $\beta_i(y_i) = 1$, so $\beta_i = \delta_{y_i}$. Let $I_1 := \{i \le J : S_i = 1\}$, $I_0 := \{i \le J : S_i < 1\}$ and $\varrho := \max_{i \in I_0}(1 + S_i)/2 < 1$ ($\varrho := 0$ if I_0 is empty).

Subsequence. Choose lengths $n_1 < n_2 < \cdots$ from that set such that, for every j, $s_i(n_j, 0) + s_i(n_j, 1) \le \varrho$ for $i \in I_0$, and $s_i(n_j, 1 - y_i) \le \varepsilon' 2^{-j}$ for $i \in I_1$, where $\varepsilon' := \varepsilon/2$.

Data. For $b \in \{0,1\}^k$ let $D_b := \{(a_{n_j}, b_j), (b_{n_j}, b_j) : j \le k\}$. Then $\nu_{\mathrm{len}}(D_b) = 2^{-k}$. All D_b have the same domain, so $\sum_b \nu_{P_i}(D_b) = \prod_j \big(s_i(n_j,0) + s_i(n_j,1)\big) \le 1$ for every i.

Count. Let B_k be $\{0,1\}^k$ without the two constant vectors. For $i \in I_1$ and $b \in B_k$ the set $T_b := \{j : b_j \ne y_i\}$ is nonempty, so $\prod_j s_i(n_j, b_j) \le \prod_{j \in T_b} \varepsilon' 2^{-j}$, and $b \mapsto T_b$ is injective; hence $\sum_{b \in B_k} \prod_j s_i(n_j, b_j) \le \prod_j (1 + \varepsilon' 2^{-j}) - 1 \le e^{\varepsilon'} - 1 \le 2\varepsilon'$. Therefore

$$\textstyle\sum_{b \in B_k} M_{\mathrm{prod}}(D_b) \;\le\; \sum_{i>J} w_i + \sum_{i \in I_0} w_i \varrho^k + \sum_{i \in I_1} w_i \, 2\varepsilon' \;\le\; \varepsilon + \varrho^k + \varepsilon \;\le\; 3\varepsilon$$

once $\varrho^k \le \varepsilon$. But $\sum_{b \in B_k} \gamma \, \nu_{\mathrm{len}}(D_b) = \gamma(1 - 2^{1-k}) \ge \gamma/2 = 4\varepsilon$ for $k \ge 2$, a contradiction. ($M_F \ge 2^{-O(1)} \nu_{\mathrm{len}}$ by Theorem 3.8, so M_F / M_{prod} is unbounded.) □

For Remark 4.8: the machine above certifies the labels of all inputs of length at most d within $\omega_{1:d+1}$, so $\nu_{\mathrm{len}}^{(d+1)} = \nu_{\mathrm{len}}$ on such data, and Proposition 4.6(a) with Theorem 4.4 gives $M_{\mathrm{prod}}(D) \ge 2^{-O(1)} W(D) \ge 2^{-K^{\mathrm{pad}}(d+1) - O(1)} \nu_{\mathrm{len}}(D)$.

B.5 No coding theorem; the covering bound

Proof of Theorem 4.9. (i) Let $g := \ell_d = \lceil \log(d/\log 3) \rceil - 1$, so that $3^{2^g} < 2^d$. Fewer than 2^g programs have length less than g; let E be the set of their indices $V(p)$. Colour $x \in \{0,1\}^d$ by $(\chi_e(x))_{e \in E} \in \{0,1,*\}^E$ as in Proposition 4.5. There are at most $3^{|E|} < 2^d$ colours, so two distinct x_1, x_2 share a colour, and no $e \in E$ has $\varphi_e(x_1) = 0$, $\varphi_e(x_2) = 1$. So $K_F(D) \ge g \ge \log d - \log\log 3 - 1 > \log d - 2$. (ii) Proposition 4.6(b) with $|D| = 2$. (iii) The uniform product law, as in Proposition 4.5. For the last claim: by (i), $\log d \le K_F(D) + 2$, so $\lceil \log(d+1) \rceil \le K_F(D) + 4$; and $K(m) \le \log m + 2\log\log m + O(1)$ for $m \ge 2$. □

In terms of sets of programs: with $S_D := \{e : \varphi_e \supseteq D\}$, $W(D) = \mathbf{m}(S_D)$ and $K_F(D) = \min_{e \in S_D} K(e)$; on the lowest complexity of the elements of a set cf. Levin (2016); Epstein and Levin (2011). For binary sequences the analogous comparison is between monotone complexity and $-\log M$ (cf. Gács, 1983; Day, 2011).

Proof of Corollary 4.10. Lower bound: Theorem 4.9(i),(iii). Upper bound: $L_{\mathrm{SOI}}(D) \ge 0$, and Proposition 4.11 with $k = 2$ gives $K_F(D) \le 2 + \log(2(d+2)) + K(\lceil \log(d+2) \rceil, 2) + O(1)$. □

Proof of Proposition 4.11. Put $s := \lceil \log(d+2) \rceil$ and $d' := 2^s - 2 \ge d$; then $d' + 2 < 2(d+2)$. There are at most $\binom{2^{d'+1}-1}{k} 2^k \le 2^{k(d'+2)}$ binary-labelled data sets of size k on $\{0,1\}^{\le d'}$. Put $N_{\mathrm{cov}} := \lfloor 2^k k (d'+2) \ln 2 \rfloor + 1$. N_{cov} independent uniform truth tables on $\{0,1\}^{\le d'}$ all miss a fixed such D with probability $(1 - 2^{-k})^{N_{\mathrm{cov}}} < e^{-N_{\mathrm{cov}} 2^{-k}} \le 2^{-k(d'+2)}$. By the union bound the expected number of uncovered data sets is below 1, so some family of N_{cov} truth tables covers all of them. The lexicographically first such family is computable from (d', k), hence

from (s,k), by exhaustive search. A program consists of a V-program for (s,k) followed by a $\lceil \log N_{\mathrm{cov}} \rceil$-bit index (its length is computable from (s,k), so the program is self-delimiting); it outputs an index of the chosen truth table, undefined off $\{0,1\}^{\le d'}$. Its length is at most $K(s,k) + \log N_{\mathrm{cov}} + 1 + O(1)$, and $\log N_{\mathrm{cov}} \le k + \log(k(d'+2)) \le k + \log(k(d+2)) + 1$. For the consequence, $L_{\mathrm{SOI}}(D) \ge K(D \mid \mathrm{dom}\, D) - O(1) = k - \delta_D - O(1)$ by Theorem 3.21(a). □

Explicit two-point coverings also exist (cf. Kleitman and Spencer, 1973): the $\lfloor N/2 \rfloor$-subsets of $\{0,\dots,N-1\}$ that contain 0, read as N functions on $\binom{N-1}{\lfloor N/2 \rfloor - 1}$ inputs, show all four label patterns on every pair of inputs.

Remark B.4 (Witnesses of large gaps cannot be computable). If $k \mapsto D_k$ is computable, then $K_F(D_k) \le K(k) + O(1)$, so the five quantities of Theorem 4.4, all between $2^{-K_F(D_k)-O(1)}$ and 1, agree on D_k up to $K(k) + O(1)$ bits; if D_d has d-bit inputs and $d \mapsto D_d$ is computable, they agree up to $O(1)$. So the witnesses of Proposition 4.5 and Theorem 4.9 and Theorem 3.12(iii) are non-constructive by necessity; Theorem 3.12(iv) is an explicit family with a linear gap on average.

Question B.5. Is there a computable family $k \mapsto D_k$ along which some gap of Table 1 between the five quantities of Theorem 4.4 tends to infinity (necessarily no faster than $K(k)$)?

B.6 Structured outputs and the history-aware version

Proposition B.6 (Single queries and structured outputs)**.** *Let $M(\cdot \mid x)$ be the universal monotone semimeasure given x, over the alphabet $\{0,1,\$\}$: Solomonoff induction predicting an output y symbol by symbol after x. Then $2^{-K(y|x)} \le 2^c M(y \mid x)$; but $M(0^d \mid \epsilon) \ge 2^{-c}$, while $2^{-K(0^d)} = 2^{-K(d) \pm O(1)}$. With an end marker, $M(y\$ \mid x) = 2^{-K(y|x) \pm O(1)}$.*

Since $L_{\mathrm{SOI}}(\{(x,y)\}) = K(y \mid x) \pm O(1)$ (Theorem 3.21(c)), on a single query the one-tape version treats an output as a complete object: it commits to where the output ends and pays for its length, while sequence induction on an undelimited output does not. With delimiters the two agree up to constants; across many examples they differ in how information is shared (Section 6).

Proof of Proposition B.6. A conditional prefix program for y given x is a monotone program, given x, that prints y; so $M(y \mid x) \ge 2^{-c} \sum_{q : V(q,x) = y} 2^{-|q|} \ge 2^{-K(y|x)-c}$. A constant program prints 0^∞, so $M(0^d \mid \epsilon) \ge 2^{-c}$, while $K(0^d) = K(d) \pm O(1)$. For the end marker, "$\ge$" is the first claim applied to $y\$$, with $K(y\$ \mid x) = K(y \mid x) \pm O(1)$. For "$\le$": the strings $y\$$, $y \in \{0,1\}^*$, are pairwise incomparable, so their cylinders are disjoint and $\sum_y M(y\$ \mid x) \le 1$; $y \mapsto M(y\$ \mid x)$ is lsc uniformly in x, and the conditional coding theorem gives $M(y\$ \mid x) \le 2^{-K(y|x)+c}$. □

Remark B.7 (The same gap for the history-aware version). For $\mathrm{SOI}^{\mathrm{hist}}$ the two versions also differ unboundedly: on fair-coin labels W_{hist} pays at most $K^{\mathrm{tail}}(n) + c \le g(n) + K(g) + c$ extra bits for every computable, nondecreasing, unbounded g, while for the worst labels the single shortest program pays more: along every input sequence, $\max_{y \in \{0,1\}^n} K_{\mathrm{hist}}(y \| x_{1:n}) - n > \log n + \log\log n$ for infinitely many n (Proposition D.16(d)).

C Proofs for Section 5 (learning theorems)

Notation is that of Section 5; KL is in bits. For a probability p and a sub-probability q on Y: $\mathrm{KL}(p\|q) \ge -\log \sum_y q(y) \ge 0$ (Jensen); and adding to q one atom that carries its defect $1 - \sum_y q(y)$ leaves $\mathrm{KL}(p\|q)$ unchanged and does not decrease $\|p - q\|_1$, so Pinsker's inequality gives

$$\|p - q\|_1^2 \le 2 \ln 2 \cdot \mathrm{KL}(p \,\|\, q). \tag{4}$$

In proofs about $\nu \in \{W, M_F\}$ we treat each hypothesis as a deterministic partial function: for W a program p with weight $2^{-|p|}$ and function $\varphi_{V(p)}$; for M_F a tape ω with measure $\mathbb{U}$ and function f_ω. Then $\nu(D)$ is the mass of the hypotheses that extend D.

C.1 Dominance

Proof of Theorem 5.1. Theorem 3.8 gives $M_F \geq 2^{-K(\nu)}\nu$, and Proposition 3.16 gives $L_{\mathrm{SOI}}(D_n) = -\log M_F(D_n)$ in every order. (a) The product law $\nu_P(D) = \prod_{(x,y)\in D} P(y \mid x)$ is a machine law with $K(\nu_P) \leq K(P) + c$ (Proposition 3.9(a)); it counts each input once, and the extra factors of repeats are at most 1. (b) Proposition 3.9(b). (c) Theorem 3.12(ii),(iii); each domain of that family is determined by k and a vector in $\{1,2,3\}^k$. □

C.2 Merging

Proof of Theorem 5.3. Posterior given the past. Fix t and a history h of length $t-1$ with positive probability (histories take countably many values). The event $\{h_{t-1} = h\}$ is $\{F \supseteq D_h\}$ intersected with an event that depends only on $(\xi_s)_{s<t}$: that the rule, fed the labels of h, produces the inputs of h. As (ξ_s) is independent of F, given $h_{t-1} = h$ and $(\xi_s)_{s\leq t}$ the law of F is $\nu(\cdot \mid F \supseteq D_h)$. Since $x_t = s_t(h, \xi_t)$, $\mathbb{P}[y_t = y \mid h_{t-1}, (\xi_s)_{s\leq t}] = \nu(y \mid D_{t-1}, x_t)$.

One step. Let $p := \nu(\cdot \mid D_{t-1}, x_t)$, a probability since ν is the law of a random total function, $q := M_F(\cdot \mid D_{t-1}, x_t)$, and $Z_t := \log(p(y_t)/q(y_t))$. At a repeated input both are point masses at the stored label and $Z_t = d_t = 0$. Otherwise the conditional expectation of Z_t given the past is $d_t \geq 0$.

Integrability. With $\log u \leq (u-1)\log e$, the conditional expectation of Z_t^- is

$$\textstyle\sum_y p(y) \max\left(0, \log \frac{q(y)}{p(y)}\right) \leq \log e \sum_y (q(y) - p(y))^+ \leq \log e.$$

On every path the conditionals telescope, since $\nu(\emptyset) = M_F(\emptyset) = 1$, and Theorem 5.1 gives

$$\textstyle\sum_{t\leq n} Z_t = \log\left(\nu(D_n)/M_F(D_n)\right) \leq K(\nu).$$

Hence $\sum_t Z_t^+ \leq K(\nu) + \sum_t Z_t^-$, each $\mathbb{E}Z_t^+$ is finite, and linearity with the tower property gives $\sum_{t\leq n} \mathbb{E}d_t = \mathbb{E}\log(\nu(D_n)/M_F(D_n)) \leq K(\nu)$.

Consequences. $\sum_t d_t$ has finite expectation, so it is finite almost surely and $d_t \to 0$ almost surely. Summing (4) and taking expectations gives the ℓ_1 bound. □

C.3 Online-to-batch and noise

Proof of Theorem 5.5. (a) Labels are defined almost surely since $\mu(\operatorname{dom}\varphi) = 1$. As $x_{t+1} \sim \mu$ is independent of D_t, $\mathbb{E}R(Q_t) = \mathbb{E}[-\log M_F(y_{t+1} \mid D_t, x_{t+1})]$ (a repeated input contributes 0). Summing over $t < n$ gives $\mathbb{E}L_{\mathrm{SOI}}(D_n) \leq K(\varphi) + c$ (Theorem 5.1(b)). By convexity of $-\log$, $R(\bar{Q}_n) \leq \frac{1}{n}\sum_{t<n} R(Q_t)$. If $\bar{Q}_n(\varphi(x) \mid x) > \frac{1}{2}$, then $\varphi(x)$ is the unique maximiser; elsewhere the loss is at least one bit, so Markov's inequality bounds that set's μ-mass by $R(\bar{Q}_n)$.

(b) Stamped inputs are distinct, so $M_F(\tilde{D}_n) = \prod_{t\leq n} q_t(y_t \mid x_t)$. Given $\tilde{D}_{t-1}$ and x_t, $y_t \sim P(\cdot \mid x_t)$; so $Z_t := \log(P(y_t \mid x_t)/q_t(y_t \mid x_t))$ has conditional expectation $\mathrm{KL}(P(\cdot \mid x_t)\|q_t(\cdot \mid x_t))$, and, $x_t \sim \mu$ being independent of the past, $\mathbb{E}Z_t = \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t)$. The stamp-ignoring product law $\tilde{P}'(y \mid \langle x, t\rangle) := P'(y \mid x)$ is a machine law of complexity at most $K(P') + c$ (Proposition 3.9(a)), so on every path $\sum_{t\leq n} Z_t \leq K(P') + c + \sum_{t\leq n} W_t$ with $W_t := \log(P(y_t \mid x_t)/P'(y_t \mid x_t))$. *Integrability.* If $\mathbb{E}_\mu \mathrm{KL}(P\|P') = \infty$ there is nothing to prove. Otherwise, as for Z_t^- above,

$\mathbb{E}Z_t^- \le \log e$ and $\mathbb{E}W_t^- \le \log e$; with $\mathbb{E}W_t = \mathbb{E}_\mu \mathrm{KL}(P\|P') < \infty$ this gives $\mathbb{E}W_t^+ < \infty$ and then $\mathbb{E}Z_t^+ < \infty$. Linearity gives $\sum_{t\le n} \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t) \le K(P') + c + n\,\mathbb{E}_\mu \mathrm{KL}(P\|P')$, and convexity of KL in its second argument (also for sub-probabilities) gives the bound for $\bar{q}_n$. An lsc conditional probability is computable, since $P(y \mid x) = 1 - \sum_{y'\ne y} P(y' \mid x)$ is also upper semicomputable (as in Lemma A.3); take $P' = P$. For 0–1 loss, let y^* be a Bayes label and $\hat{y}$ the maximiser of $\bar{q}_n(\cdot \mid x)$. As $\bar{q}_n(y^* \mid x) \le \bar{q}_n(\hat{y} \mid x)$, $P(y^* \mid x) - P(\hat{y} \mid x) \le [P(y^* \mid x) - \bar{q}_n(y^* \mid x)] + [\bar{q}_n(\hat{y} \mid x) - P(\hat{y} \mid x)] \le \|P(\cdot \mid x) - \bar{q}_n(\cdot \mid x)\|_1$. Apply (4) and Jensen's inequality for the square root.

(c) The product law of a computable conditional probability is a machine law of complexity at most $K(P) + c$ and the law of a random total function; apply Theorem 5.3. □

Proof of Proposition 5.6. (a) Two draws of x_0 with different labels have positive probability, and the second gets probability 0, because the stored label gets probability 1 in both conventions. (b) The labels at x_0 agree with probability $(1-\alpha)^n + \sum_{j\ge1} \binom{n}{j}\alpha^j(1-\alpha)^{n-j}\sum_y P(y \mid x_0)^j$. For $j \ge 1$, $\sum_y P(y \mid x_0)^j \le p_{\max}^{j-1}$, so this is at most $(1-\alpha)^n + p_{\max}^{-1}(1-\alpha+\alpha p_{\max})^n$. (c) If some x with $\mu(x) > 0$ has two labels of positive probability, (a) gives infinite expected loss for $n \ge 2$, while P's is finite. Otherwise the label $\varphi(x)$ with $P(\varphi(x) \mid x) > \frac{1}{2}$ can be found by search, so φ is partial computable with $K(\varphi) \le K(P) + c$ and $\mu(\mathrm{dom}\,\varphi) = 1$, and Theorem 5.5(a) applies. □

C.4 Solomonoff's per-query solution

Proof of Proposition 5.7. (a) Each term $\prod_t P_i(y_t \mid x_t)$ satisfies the two inequalities because $\sum_y P_i(y \mid x) \le 1$; mixing with $\sum_i \mathbf{m}(i) \le 1$ preserves them; lower semicomputability is clear. The product is symmetric in the pairs, and on distinct inputs it is the definition of $M_{\mathrm{prod}}(D_n)$. On consistent sequences the factors at repeats are at most 1, which gives $M_{\mathrm{pq}} \le M_{\mathrm{prod}}(D_n)$, strictly when some term that is positive on D_n has a factor below 1 at a repeat. A positive computable conditional probability (for $Y = \{0,1\}^*$, for instance $P(y \mid x) = 2^{-2|y|-1}$) is a term with positive weight, so every sequence gets positive mass. (b) $\mathbf{m}(i) \ge 2^{-K(i)}$ for an index i of P. (c) The product of the predictions telescopes to $M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})/M_{\mathrm{pq}}(\epsilon\|\epsilon) \ge M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$. Given the past, $y_t \sim P(\cdot \mid x_t)$ and $x_t \sim \mu$ is independent of it, so the expected log ratio $\log(P(y_t \mid x_t)/q_t(y_t \mid x_t))$ is $\mathbb{E}_\mu \mathrm{KL}(P\|q_t) \ge 0$. Summing and applying (b) to P' bounds $\sum_t \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t)$ by $K(P') + \sum_t \mathbb{E}\log\frac{P(y_t|x_t)}{P'(y_t|x_t)} = K(P') + n\,\mathbb{E}_\mu \mathrm{KL}(P\|P')$, with integrability as in Theorem 5.5(b). Convexity of KL gives the averaged bound. (d) Stamping makes all inputs distinct, so on every sequence the set mixture of the stamp-ignoring laws $\tilde{P}_i(y \mid \langle x,t\rangle) := P_i(y \mid x)$ equals $\sum_i \mathbf{m}(i)\prod_t P_i(y_t \mid x_t) = M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$. The $\tilde{P}_i$ have indices computable from i, so this mixture is a machine law of complexity $O(1)$ (Proposition 3.9(a),(d)); apply Theorem 3.8. □

C.5 Learning curves, splits and leave-one-out

Proposition C.1 (the expected risk of a Bayes predictor need not decrease)**.** *Let $X = \{a,b\}$, μ uniform, target $\varphi \equiv 0$, and a Bayes mixture of three total functions: φ with prior weight w, and $(1,0)$ and $(0,1)$ with weight $\frac{1-w}{2}$ each. The expected log risk r_t of the posterior predictive after t i.i.d. samples is $r_0 = -\log\frac{1+w}{2}$, $r_1 = \frac{1}{2}\log\frac{1+w}{2w}$ and $r_t = 2^{1-t}r_1$ for $t \ge 1$. So $r_1 > r_0$ whenever $(1+w)^3 > 8w$, in particular for all $w \le \frac{1}{5}$. The sum $\sum_t r_t$ increases to $\log(1/w)$.*

Proof of Proposition C.1. Before data, the predictive probability of 0 at either input is $w + \frac{1-w}{2} = \frac{1+w}{2}$. After $(a,0)$ the consistent hypotheses are φ and $(0,1)$, and the predictive probability of 0 at b is $\frac{w}{w+(1-w)/2} = \frac{2w}{1+w}$; symmetrically after $(b,0)$. After $t \ge 1$ samples both inputs have been seen with probability $1-2^{1-t}$ (risk 0) and one with probability 2^{1-t} (risk $\frac{1}{2}\log\frac{1+w}{2w}$, as the unseen

input has probability $\frac{1}{2}$). $r_1 > r_0$ is equivalent to $(1+w)^3 > 8w$; the left side minus the right is decreasing on $[0, \frac{1}{5}]$ and equals 0.128 at $\frac{1}{5}$. Finally $\sum_t r_t = r_0 + 2r_1 = \log \frac{1}{w}$. □

For $w = \frac{1}{16}$, $r_0 = 0.913$ and $r_1 = 1.544$ bits. The mixture is a random total function, so both conventions agree, but it is not M_F: "more data never hurts in expectation" fails for Bayes predictors with a fixed target; for M_F this is not established. In a direct computation (code in the accompanying repository), 63 of 300 random finite instances had non-monotone curves; see also Example C.6.

Remark C.2 (last iterate and leave-one-out). In the setting of Theorem 5.5(a), exchangeability gives $\mathbb{E}R(Q_n) = \mathbb{E}[\frac{1}{n+1}\sum_{i \le n+1} - \log Q^{(-i)}(\varphi(x_i) \mid x_i)]$, where $Q^{(-i)}$ is the predictive of M_F given the data set of the samples $j \neq i$ (a held-out input that occurs among the others costs 0). Whether $\mathbb{E}R(Q_n) \le (K(\varphi)+c)/(n+1)$ is thus a leave-one-out question (cf. Conjecture C.4); it is open. A direct computation on 300 random finite instances (code in the accompanying repository) gives $\max(n+1)r_n/K = 0.873$.

Proof of the identity in Remark C.2. Let x_{n+1} be an independent sample; then $\mathbb{E}R(Q_n)$ is the expected loss $\mathbb{E}[-\log Q_n(\varphi(x_{n+1}) \mid x_{n+1})]$. M_F sees its data only as a set and $(x_1, \dots, x_{n+1})$ is exchangeable, so the same expectation results with any sample i held out; average over i. □

Proof of Proposition 5.8. In every order the step losses sum to $-\log(\nu(f|_B)/\nu(\emptyset)) \le L_\nu(f|_B)$ (Proposition 3.16, $\nu(\emptyset) \le 1$). The split point is uniform and independent of the order; divide by $|B|$. A shortest program consistent with $f|_B$ is a term of $W(f|_B)$, so $L_W \le K_F$, and $L_{M_F} \le L_W + c$ (Theorem 4.4). For the error use $e \le -\ln(1-e)$. □

Proposition C.3 (leave-one-out). *For $\nu \in \{W, M_F\}$, a labelling f of a finite B, and $e_{\mathrm{LOO}}(x) := 1 - \nu(f|_B)/\nu(f|_{B\setminus\{x\}})$: $\sum_{x\in B} e_{\mathrm{LOO}}(x) \le (\nu(\emptyset) - \nu(f|_B))/\nu(f|_B) < 2^{L_\nu(f|_B)} \le 2^{K_F(f|_B)+c}$.*

Proof of Proposition C.3. $e_{\mathrm{LOO}}(x)\,\nu(f|_{B\setminus\{x\}}) = \nu(f|_{B\setminus\{x\}}) - \nu(f|_B)$ is the mass of hypotheses correct on $B \setminus \{x\}$ and wrong or undefined at x. A hypothesis in the sets of two points $x \neq x'$ would be correct and not correct at x', so the sets are disjoint, and none contains a hypothesis correct on all of B. So the masses sum to at most $\nu(\emptyset) - \nu(f|_B)$. Divide by $\nu(f|_{B\setminus\{x\}}) \ge \nu(f|_B)$; the last inequality is as in Proposition 5.8. □

Conjecture C.4 (leave-one-out for the one-tape version). *For $\nu = M_F$, $\frac{1}{|B|}\sum_{x\in B} e_{\mathrm{LOO}}(x) = O(K_F(f|_B)/|B|)$, up to logarithmic factors.*

We have neither a proof nor a counterexample. *Heuristic for Conjecture C.4 (not a proof).* The natural one-exception hypotheses "f except at x" have weight about $2^{-K(f)-K(x|f)}$, and $\sum_x 2^{-K(x|f)} \le 1$. A counterexample needs many more than $K(f)$ points x whose one-point flips f^x have $K(f^x) \lesssim K(f)$; natural attempts (halting-set indicators, pseudo-random seeds, thresholds) give only $O(1)$ or $O(K(f))$ such points. For general prior weights Proposition C.3 is tight within a factor of about 2 (Example C.6).

C.6 Typical separating properties and per-step error

Separation step (the special case of Proposition 8.1 needed here, proved so as not to rely on a later section). Let G be decidable, b total computable, $\operatorname{dom} D \cap G = \emptyset$ and $x^* \in G$. Then for $\nu \in \{W, M_F\}$, with $K(G)$ the least complexity of an index of a total decider of G,

$$\nu(D + (x^*, b(x^*))) \;\ge\; 2^{-K(G)-K(b)-c}\,\nu(D). \tag{5}$$

For M_F: $\{(x, \epsilon, b(x)) : x \in G\} \cup \{(x, w, y) \in T_U : x \notin G\}$ is c.e. and consistent, hence a function machine with an index computable from those of the decider and of b; every tape whose f_ω extends D makes it extend $D + (x^*, b(x^*))$. Apply Theorem 3.8. For W: let V' read a V-program z for the two indices, then a V-program p, and output an index of "$b(x)$ if $x \in G$, else $\varphi_{V(p)}(x)$". Then $V(wu) = V'(u)$ for a fixed w; the map $p \mapsto wzp$ is injective and sends programs consistent with D to programs consistent with $D + (x^*, b(x^*))$.

Proof of Proposition 5.9. (1) The prefix u depends only on x^*, which is independent of the training inputs, and $x^* \in G_u$. Each training input lies in G_u with probability $2^{-\ell} \le 1/(2n)$, independently, so the probability is $(1 - 2^{-\ell})^n \ge 1 - n2^{-\ell} \ge \frac{1}{2}$. (2) A decider needs u: a self-delimiting code of ℓ, then ℓ bits. (3) On the event of (1), (5) with $G = G_u$ and $b \equiv 1$ gives $e^{\mathrm{wr}}(x^*) \ge 2^{-K(G_u)-c}$. As $\ell \le \log n + 2$, $2^{-\ell} \ge 1/(4n)$ and $(\ell + 1)^2 \le (\log n + 3)^2$. For computable binary f take $b = 1 - f$, with $K(b) \le K(f) + c$. □

Proof of Corollary 5.10. (a) Let $\varepsilon_h := \mu\{x : h(x) \ne f(x) \text{ or undefined}\}$. For a fresh x^*, $e(x^*)$ is the posterior mass of hypotheses not correct at x^* (zero at a training input, where all survivors are correct). By Fubini,

$$\mathbb{E}[e(x^*) \mid D_n] = \frac{\int_{h \supseteq D_n} \varepsilon_h \, d\nu(h)}{\nu(D_n)} \le \varepsilon + \frac{\nu\{h \supseteq D_n : \varepsilon_h > \varepsilon\}}{\nu(D_n)}.$$

The denominator is at least $2^{-K(f)-c}$ on every sample, as a shortest program for f is consistent with D_n. A fixed hypothesis survives the n independent inputs with probability $(1 - \varepsilon_h)^n$, so $\mathbb{E}\,\nu\{h \supseteq D_n : \varepsilon_h > \varepsilon\} \le \exp(-\varepsilon n)\nu(\emptyset) \le \exp(-\varepsilon n)$. With $\varepsilon := ((K(f) + c)\ln 2 + \ln n)/n$ the second term is $1/n$ (for $\varepsilon \ge 1$ the bound is trivial). Markov's inequality on the surviving bad mass gives the high-probability form. The posterior-majority predictor errs at x only if $e(x) \ge \frac{1}{2}$. (b) Condition on the multiset of the $n + 1$ sampled values; by exchangeability x^* is a uniformly random one of the $n + 1$ positions. If its value occurs more than once it is a training input and $e(x^*) = 0$. Otherwise the training set is $S \setminus \{x^*\}$ and $e(x^*) = e_{\mathrm{LOO}}(x^*; S)$, since ν sees its data as a set. So the conditional expectation is at most $\frac{1}{n+1}\sum_{x \in S} e_{\mathrm{LOO}}(x; S)$; apply Proposition C.3. For $f \equiv 0$ the lower bound is Proposition 5.9, as $e \ge e^{\mathrm{wr}}$, and the upper bound is (b) with $K(f) = O(1)$. □

C.7 Occam bound and chosen inputs

Proof of Proposition 5.11. Put $\varepsilon_p := (|p| \ln 2 + \ln(1/\delta))/n$. A fixed p whose disagreement with φ exceeds ε_p is consistent with D_n with probability at most $(1 - \varepsilon_p)^n \le e^{-n\varepsilon_p} = 2^{-|p|}\delta$. Sum over $p \in \operatorname{dom} V$ and use $\sum_p 2^{-|p|} \le 1$. A shortest program for φ is consistent, so $K_F(D_n) \le K(\varphi)$. □

We state one further result on chosen inputs.

Proposition C.5 (chosen inputs)**.** *(1) For every weak semimeasure ν and every query set A, chosen adaptively or not, the total semimeasure-convention log loss on the labels of A is $-\log(\nu(f|_A)/\nu(\emptyset))$ in every order: for log loss, choosing the order is worthless. (2) Small total loss does not give efficient identification. For $f_a(x) := \mathbf{1}[x = a]$ on $\{0,1\}^d$, the total loss of M_F on any query sequence, even all of $\{0,1\}^d$, is at most $K(f_a) + c \le d + 2\log d + c$, yet every query strategy, adaptive or randomised, needs at least $(2^d - 1)/2$ queries in expectation over a uniform a to identify a.*

Proof. (1) is Proposition 3.16. (2) f_a is computable from a and $K(a) \le d + 2\log d + c$; apply Theorem 5.1(b). Fix the strategy's coins. Until a is queried every answer is 0, so the queries are those made on $f \equiv 0$. Identifying a requires querying a or all other $2^d - 1$ points, so at least $\min(\mathrm{pos}(a), 2^d - 1)$ queries, where $\mathrm{pos}(a)$ is the time of the first query of a on $f \equiv 0$. These times are distinct, so their mean over uniform a is at least $2^{-d}\big(\sum_{k=1}^{2^d-1} k + (2^d - 1)\big) \ge \frac{2^d-1}{2}$. Average over the coins. □

So for active learning the log-loss guarantee must be paired with another objective, such as expected information gain (cf. Orseau et al., 2013).

C.8 Further remarks

Example C.6 (learning curves can increase)**.** Take a Bayes mixture of the truth f (weight w_f) and $|B|$ hypotheses each wrong at exactly one point of B (weight w each). Sampling without replacement, the loss at step k is $\log \frac{w_f + w(|B|-k+1)}{w_f + w(|B|-k)}$ in every order, and it increases with k. With $w = w_f = 1/(|B| + 1)$ and $|B| = 16$, the last step (leave-one-out) costs 1 bit, while for the random-split average a direct calculation gives 0.255 bits. Each leave-one-out error is $\frac{1}{2}$, within a factor of about 2 of Proposition C.3. By Kraft's inequality such weights $w = w_f = 2^{-K}$ need $|B| \le 2^K - 1$.

Remark C.7 (per-step versus averaged bounds)*.* Proposition 5.9 is a per-step lower bound and cannot be compared with the Cesàro bound of Theorem 5.5(a): since $\sum_k \mathbb{E}e_k \le (\ln 2)(K(f) + c)$, $\liminf_k k\,\mathbb{E}e_k = 0$. The per-step rate is given by Corollary 5.10.

Remark C.8 (relations among the in-distribution bounds)*.* (i) For computable f and distinct inputs, $\sum_{k \le n} e_k \le (\ln 2)(K(f) + c)$ on every sequence (Theorem 5.1(b) with $e \le -\ln(1 - e)$; this is Theorem 8.4). Corollary 5.10(a) gives a per-step rate in distribution at the price of an additive $\ln n$; Theorem 5.5(a) is the Cesàro statement for the log loss of the averaged predictor. (ii) Corollary 5.10(a) is the Markov half of the relevance-based bound in the slides of Vaintrob and Hänni (2026b, Appendix A), specialised to W and M_F. The union-bound half and the slides' geometric length prior are not needed: long programs enter only multiplied by $\exp(-\varepsilon n)$. (iii) Leave-one-out is different. For a fixed finite B and a random held-out point, the training set eliminates every hypothesis with two or more errors on B; what remains is the mass of one-exception hypotheses (Conjecture C.4). If Conjecture C.4 holds, the proof of Corollary 5.10(b) gives $\mathbb{E}e(x^*) = O(K(f)/(n + 1))$ up to logarithmic factors, without the additive $\ln n$.

Comparison with other bounds. *PAC-Bayes* (McAllester, 1999): risk bounds with a $\sqrt{\mathrm{KL}(Q\|\mathrm{prior})/n}$ term, or a KL/n term in the realisable case; with prior 2^{-K} and a posterior concentrated on the target the term is about $K(\varphi)\ln 2/n$, the same rate. The difference is in kind: PAC-Bayes bounds hold with high probability for a randomised classifier, Theorem 5.5 in expectation for the averaged Bayes predictive. *MDL and the index of resolvability* (Barron and Cover, 1991): Theorem 5.5(b) and Proposition 5.7(c) are Cesàro-average resolvability bounds $\min_{P'}[K(P')/n + \mathrm{KL}(P\|P')]$ in the conditional setting, for every input distribution.

Remark C.9 (positive-example induction (informal))*.* Positive-example induction, learning a set or a distribution from samples, is a natural relative of SOI. In SOI terms positive labels $(x, 1)$ alone are useless, because the constant-1 program explains them. A natural formalisation is a sampling likelihood, whose mixture is $\sum_P w_P \prod_i P(x_i)$ over computable distributions P on X: the private-randomness analogue of the universal semimeasure on the sample sequence (cf. the ideal learner from positive evidence of Chater and Vitányi, 2007). *Proof sketch.* We do not fix the

encoding in detail. The samples are the labels of the index inputs $i \mapsto x_i$, under index-ignoring hypotheses with private randomness per sample; each such hypothesis draws every sample from one distribution P on X, and mixing over them gives the displayed mixture.

Computations. The numbers quoted in Section 5 and here come from seeded or deterministic scripts in the accompanying repository: exact finite analogues of the dominance, merging, online-to-batch and last-iterate statements, the fresh-noise consistency probabilities, the per-query version on a finite class of laws, the learning curves, random splits, separating prefixes and the leave-one-out bounds.

D Proofs for Section 6 (history-aware prediction)

Notation is that of Section 6. We use: (F1) Theorem 3.8 for any countable decidable input set, $M_F(D) \geq 2^{-K(T)}\nu_T(D)$ for $D \neq \emptyset$, and $M_F(D) \geq 2^{-K(g)-c}$ if a partial computable g extends D (Proposition 3.9(b)); (F2) the conditional coding theorem of Section 3.1: if $\nu(z \mid w)$ is lsc uniformly in w and $\sum_z \nu(z \mid w) \leq 1$, then $K(z \mid w) \leq -\log \nu(z \mid w) + K(\nu) + c$; (F3) Levin–Schnorr: ω is Martin-Löf random iff $K(\omega_{1:\ell}) \geq \ell - c_\omega$ for all ℓ (Li and Vitányi, 2008); (F4) $K(a,b) \leq K(b) + K(a \mid b) + c$ and $K(s) \leq |s| + K(|s|) + c$ (Li and Vitányi, 2008); (F5) $M \geq 2^{-K(\nu)-c}\nu$ for every lsc semimeasure ν on $\{0,1\}^*$ (Zvonkin and Levin, 1970; Li and Vitányi, 2008); (F6) Sion's minimax theorem (Sion, 1958), for compact convex sets in $\mathbb{R}^m$ and a continuous payoff convex in one variable and affine in the other.

D.1 The counterexamples

Proof of Lemma 6.2. The events for different next labels are disjoint subsets of the event for $y_{1:n}$, and at a repeated input only the stored label survives; lower semicomputability is as for M_F. The second claim follows from Corollary 6.17(a) with $K(\rho_F) = O(1)$. For the third, at a new input the chronological inequality is the weak inequality, and at a repeated input only one term is nonzero. □

Lemma D.1 (Chain semimeasure)**.** *For a string x and a sequence $z = (z_1, \dots, z_k)$ of nonempty strings ($k \geq 0$) let $\nu(z \mid x) := \mathbb{U}\{\omega : f_\omega(x) = z_1,\ f_\omega(z_1) = z_2, \dots, f_\omega(z_k) = \epsilon\}$. Then $\sum_z \nu(z \mid x) \leq 1$, and ν is lsc uniformly in x.*

Proof. Fix ω, x, and $z \neq z'$ that both satisfy the condition; put $z_0 = z'_0 := x$. If they first differ at $j \geq 1$, then $z_j = f_\omega(z_{j-1}) = z'_j$. If z is a proper prefix of z', of length k, then $\epsilon = f_\omega(z_k) = z'_{k+1} \neq \epsilon$. Both are contradictions, so the events are disjoint; they are effectively open, uniformly. □

Proof of Theorem 6.3. (a) Every ω with $f_\omega \supseteq D_n$ satisfies the condition of Lemma D.1 with $x = x_n$, $z = (x_{n-1}, \dots, x_1)$, so $M_F(D_n) \leq 2^{-K(z|x_n)+c}$ by (F2). The program with $x_1, \dots, x_{n-1}$ built in that outputs ϵ at x_1, x_{i-1} at x_i ($2 \leq i \leq n-1$) and x_{n-1} elsewhere extends D_n; apply (F1). (b) $x_{1:n}$ determines $\omega_{1:\bar{\ell}_n}$, so $K(x_{1:n}) \geq \bar{\ell}_n - c_\omega$ by (F3); by (F4), $K(x_{1:n}) \leq \ell_n + K(\ell_n) + K(x_{1:n-1} \mid x_n) + c$. For the upper bound, $x_{1:n-1}$ is determined by n and $\omega_{1:\bar{\ell}_{n-1}}$, and $\bar{\ell}_{n-1}$ is computable from n. For $\ell_i = i$ use $K(n) \leq 2\log n + c$. (c) At step $i \geq 2$ the history's inputs are $x_1, \dots, x_{i-1}$ and its labels $\epsilon, x_1, \dots, x_{i-2}$. □

Proof of Theorem 6.4. (a) In block ℓ the strings other than s_ℓ come first, so s_ℓ is unseen when they are queried, and all its length-ℓ siblings are seen when it is. (b) $\operatorname{dom} D^{(\ell)}$ and ℓ are computable from each other, and given the domain the labelling determines $s_{1:\ell}$; apply Theorem 3.21(a). A

function machine that reads, for each length j, its own block of j tape bits as a guess s'_j and answers 1 at x iff $x = s'_{|x|}$ extends $D^{(\ell)}$ with probability $2^{-\ell(\ell+1)/2}$; apply (F1). (c) $K(s_{1:\ell} \mid \ell) \geq K(s_{1:\ell}) - K(\ell) - c \geq \ell(\ell+1)/2 - K(\ell) - c_\omega$ by (F3); and fewer than 2^{m-j} tuples have $K(\cdot \mid \ell) < m - j$, $m = \ell(\ell+1)/2$. □

Proof of Proposition 6.5. (a) For distinct nonempty $z_1, \ldots, z_j \neq x$ let

$$O_j(z_{1:j}) := \{(x, z_1), (z_1, z_2), \ldots, (z_{j-1}, z_j)\}, \qquad O_0 := \emptyset.$$

As z_j (or x, if $j = 0$) is not in $\operatorname{dom} O_j$, the weak inequality gives $\sum_y \nu(O_j \cup \{(z_j, y)\}) \leq \nu(O_j)$; the term $y = \epsilon$ is $\nu(D(x, z_{1:j}))$, each nonempty $y \notin \{x, z_1, \ldots, z_j\}$ gives $\nu(O_{j+1}(z_{1:j}, y))$, and the other terms are dropped. Induction on J gives $\sum_{|z| \leq J} \nu(D(x, z)) \leq \nu(\emptyset) \leq 1$. (b) $D_n = D(x_n, (x_{n-1}, \ldots, x_1))$, and $z \mapsto \nu(D(x, z))$ is a semimeasure by (a), lsc uniformly in x; apply (F2) and continue as for Theorem 6.3. For the skipped strings, the labellings of a finite set have total ν-mass at most 1, so $-\log \nu(D) \geq K(D \mid \operatorname{dom} D) - K(\nu) - c$. (c) Gibbs' inequality given x_n against the sub-probability $z \mapsto \nu(D(x_n, z))$; given x_n, $(x_{n-1}, \ldots, x_1)$ is uniform on $\bar{\ell}_{n-1}$ bits. For the skipped strings, against the labellings of the domain. □

Lemma D.2 (Normaliser lower bound)**.** *For every D, $x \notin \operatorname{dom} D$ and function machine B: $\mathcal{Z}(D, x) \geq 2^{-K(B)-c}\, \mathbb{U}\{\omega : f^B_\omega \supseteq D,\ f^B_\omega(x) \neq \perp\}$, which is $2^{-K(B)-c}\nu_B(D)$ if f^B_ω is always total.*

Proof. Apply (F1) to each $D + (x, y)$ and sum over y; the events $\{f^B_\omega \supseteq D,\ f^B_\omega(x) = y\}$ are disjoint. □

For the chain, the *broom* B_j reads blocks $v_1, \ldots, v_j$ of lengths $\ell_1, \ldots, \ell_j$, sets $f(v_1) := \epsilon$, $f(v_i) := v_{i-1}$, and sends every other input to v_j; it extends D_{j+1} with probability $2^{-\bar{\ell}_j}$ at a cost of $K(j) + c$ bits (as in Proposition 7.7).

Proposition D.3 (Normalised SOI on both examples)**.** *(a) In the setting of Theorem 6.3(b), $-\log \bar{M}_F(y_t \mid D_{t-1}, x_t) \geq \ell_{t-1} - K(\ell_t) - K(t-2) - c_\omega$ for $t \geq 2$; for $\ell_i = i$, $\bar{L}_{\mathrm{SOI}} \geq n(n-1)/2 - n(4 \log n + c_\omega)$. With independent uniform blocks the expected normalised loss at step t is at least $\ell_{t-1} - K(t-2) - c$, so $\mathbb{E}\bar{L}_{\mathrm{SOI}} \geq n(n-1)/2 - n(2 \log n + c)$ for $\ell_i = i$. (b) In the setting of Theorem 6.4(c), the normalised loss at s_ℓ is at least $\ell - 2K(\ell) - c_\omega$, so $\bar{L}_{\mathrm{SOI}} \geq \ell(\ell+1)/2 - \ell(4 \log \ell + c_\omega)$ after block $\ell \geq 2$; with independent uniform s_j, at least $\ell - K(\ell) - c$ in expectation.*

Proof of Proposition D.3. Let $\beta_t := -\log \bar{M}_F(y_t \mid D_{t-1}, x_t) = -\log M_F(D_t) + \log \mathcal{Z}(D_{t-1}, x_t) \geq 0$. (a) The broom B_j above is well defined (the v_i have distinct lengths), total, and $K(B_j) \leq K(j) + c$. On $\{v_i = x_i\ \forall i \leq j\}$, of probability $2^{-\bar{\ell}_j}$, it extends D_{j+1}: $f(x_{j+1}) = v_j = x_j$ since $\ell_{j+1} \notin \{\ell_1, \ldots, \ell_j\}$. At step $t \geq 2$, Lemma D.2 with B_{t-2} gives $\mathcal{Z}(D_{t-1}, x_t) \geq 2^{-K(t-2)-c-\bar{\ell}_{t-2}}$, so $\beta_t \geq -\log M_F(D_t) - \bar{\ell}_{t-2} - K(t-2) - c$. Insert Theorem 6.3(b) at horizon t, or, for uniform blocks, Proposition 6.5(c) in expectation. Sum the positive parts; for $\ell_i = i$ use $K(t), K(t-2) \leq 2 \log n + c$ for $t \leq n$. (b) T_ℓ reads blocks g_j of length j for $j < \ell$ and outputs 1 at x iff $|x| < \ell$ and $x = g_{|x|}$, and 0 elsewhere; it is total and $K(T_\ell) \leq K(\ell) + c$. Before the query s_ℓ the data are $D' := D^{(\ell-1)} \cup \{(x, 0) : |x| = \ell,\ x \neq s_\ell\}$, which T_ℓ extends iff $g_j = s_j$ for all $j < \ell$, with probability $2^{-\ell(\ell-1)/2}$. So $\mathcal{Z}(D', s_\ell) \geq 2^{-K(\ell)-c-\ell(\ell-1)/2}$; insert Theorem 6.4(c), or Proposition 6.5(c) in expectation, and use $2K(j) \leq 4 \log \ell + c$ for $j \leq \ell$. □

The method needs more information per step than the $O(\log t)$ bits of the broom, so it gives nothing for Proposition D.13 and Theorem D.21.

D.2 A dominance theorem for computable reveal laws

Lemma D.4 (Private memory)**.** *Let a reveal mechanism that does not see f consist of a random element Γ (private memory and coins) independent of f and maps $x_i = \mathrm{sel}_i(\Gamma, h_{i-1})$, with reveal-sequence law $\Delta(x_{1:n}\|y_{1:n-1}) := \mathbb{P}[\mathrm{sel}_i(\Gamma, h_{i-1}) = x_i \ \forall i \le n]$. (a) Given h_{i-1}, f and x_i are conditionally independent; so $I(f \to x_{1:n}) = 0$ for finite-valued inputs and labels. (b) Given f, the input sequence has probability $\Delta(x_{1:n}\|f(x_1), \dots, f(x_{n-1}))$, and the effective kernel $\Delta(x_{1:i}\|y_{1:i-1})/\Delta(x_{1:i-1}\|y_{1:i-2})$ depends neither on f nor on its law. (c) If the mechanism is a computable sampler with private memory that halts with probability 1 at each step whatever labels it is fed, and its inputs up to time n lie in a finite listed set A_n, then $\Delta(x_{1:n}\|y_{1:n-1})$ is a computable real, uniformly in its arguments.*

Proof. The joint law of (f, Γ, h_n) factorises (on countable spaces; in general via densities with respect to $\mathrm{law}(f) \otimes \mathrm{law}(\Gamma)$) as $[\mathrm{law}(f) \prod_i \mathbf{1}[y_i = f(x_i)]] \cdot [\mathrm{law}(\Gamma) \prod_i \mathbf{1}[x_i = \mathrm{sel}_i(\Gamma, h_{i-1})]]$. (a) Given (h_{i-1}, x_i) the law of (f, Γ) is proportional to a function of f times a function of Γ, and the f-factor does not involve x_i. (b) Sum the Γ-factor over Γ. (c) $\Delta(x\|y)$ is lsc (enumerate halting computations fed the labels y). Feed the sampler the labels y_j while its inputs agree with $x_{1:j}$, and a fixed label otherwise; its input sequences in A_n^n then have total probability 1, and $x_{1:n}$ has probability $\Delta(x\|y)$. So $\Delta(x\|y) = 1 - \sum_{x' \neq x} \mathbb{P}[x']$, a finite sum of lsc terms subtracted from 1. □

Proof of Proposition 6.7. P pays 1 bit for y_1 and none for y_2. The simulator at a outputs P's uniform y_1; at b it simulates a then b on fresh bits and outputs $y_2' = y_1'$, a uniform bit independent of the run at a; so (β, β) gets probability $\frac{1}{4}$. The mixture of the two constant functions gives it $\frac{1}{2}$. □

Proof of Theorem 6.8. The function machine T uses ω for one run of the alternating process: $x_1' \sim \Delta(\cdot \mid \emptyset)$, $y_1' \sim P(\cdot \mid \emptyset, x_1')$, $x_2' \sim \Delta(\cdot \mid h_1')$, …, each call reading fresh consecutive bits (continuing from the stored state if there is private memory). On query x it runs until the first j with $x_j' = x$ and outputs y_j', diverging if there is none. Its answer depends on a finite prefix of ω, so T is a function machine, with $K(T) \le K(\Delta, P) + c$. If the trajectory begins with $x_1 y_1 \cdots x_n y_n$, each $x \in \mathrm{dom}\, D_n$ is first visited with its label in D_n (the sequence is consistent), so $f_\omega^T \supseteq D_n$. The tapes producing this beginning are the disjoint union of the cylinders $[w_1 v_1 \cdots w_n v_n]$, w_i ranging over the bits read by halting runs of Δ on h_{i-1} with output x_i and v_i over those of P on (h_{i-1}, x_i) with output y_i; each set is prefix-free, so the union has measure $\prod_i \Delta(x_i \mid h_{i-1}) P(y_i \mid h_{i-1}, x_i)$. For distinct inputs, different orders give different trajectory prefixes of equal length, hence disjoint events. Apply (F1). □

Corollary D.5 (Deterministic reveal policies)**.** *If $x_i = \delta(h_{i-1})$ for a computable δ, then for every f, every sampler P and every n, $L_{\mathrm{SOI}}(D_n) \le L_P + K(\delta) + K(P) + c$ on every path.*

Corollary D.6 (Against Solomonoff induction on the joint sequence)**.** *With M and* code *as in Section 6.5: $L_{\mathrm{SOI}}(D_n) \le -\log M(\mathrm{code}(h_n)) + c$ on every consistent sequence; for distinct inputs, $L_{\mathrm{SOI}}(D_n) \le -\log \sum_\sigma M(\mathrm{code}(D_n \text{ in order } \sigma)) + c$.*

Proof of Corollaries D.5 and D.6. For Corollary D.5, take the input sampler δ in Theorem 6.8; $L_\Delta = 0$ on the realised path. For Corollary D.6, run the universal monotone machine on ω, parse its output (uniquely) into blocks $\bar{x}_1' \bar{y}_1' \bar{x}_2' \cdots$, and answer x with the label after the first input block equal to $\bar{x}$. If the output extends $\mathrm{code}(h_n)$ then $f_\omega \supseteq D_n$, and that event has probability $M(\mathrm{code}(h_n))$. For distinct inputs the codes of different orders have equal length and differ, so the events are disjoint. □

Proof of Theorem 6.9. The joint law of (f, Γ, h_n) is $\nu(df)\mathbb{P}(d\Gamma)\prod_i \mathbf{1}[x_i = \mathrm{sel}_i(\Gamma, h_{i-1})]\mathbf{1}[f \supseteq D_n]$, a function of f times a function of Γ for fixed h_{i-1}. So given h_{i-1} and x_i the law of f is $\nu(\cdot \mid f \supseteq D_{i-1})$, whose prediction for y_i is the Bayes predictor; the product telescopes to $\nu(D_n)$, and (F1) gives $M_F \geq 2^{-K(\nu)}\nu$. Gibbs' inequality, valid for sub-probabilities, gives $\mathbb{E}L_P \geq \mathbb{E}[-\log \nu(D_n)]$ step by step. □

Proof of Theorem 6.10. (a) Let Q_f be a regular conditional law of $x_{1:n}$ given f. Then $\mathbb{P}(h_n \mid f) = Q_f(x_{1:n})\mathbf{1}[f \supseteq D_n]$ and $\mathbb{P}(h_n) = \nu(D_n)\bar{Q}_D(x_{1:n})$ with $\bar{Q}_D := \mathbb{E}_{f'\sim\nu}[Q_{f'} \mid f' \supseteq D_n]$; the posterior of f has density $Q_f/\bar{Q}_D$ with respect to $\nu(\cdot \mid f \supseteq D_n)$. Hence $I(f; h_n) = \mathbb{E}\,\mathrm{KL}(\mathbb{P}(f \in \cdot \mid h_n) \| \nu(\cdot \mid f \supseteq D_n)) + \mathbb{E}[-\log \nu(D_n)]$. By the chain rule $I(f; h_n) = \sum_i [I(f; x_i \mid h_{i-1}) + I(f; y_i \mid h_{i-1}, x_i)] = I(f \to x) + \mathbb{E}L_{\mathbb{P}}$, because y_i is a function of (f, x_i). All terms are finite ($I(f; h_n) \leq H(h_n)$). (b) (F1), then (a), then $\mathbb{E}L_P \geq \mathbb{E}L_{\mathbb{P}}$ by Gibbs' inequality. □

Proof of Theorem 6.11. The machine compares a query of length i with tape block i and returns block $i-1$ if they agree and $i \geq 2$, and ϵ otherwise. Given h_{i-1}, x_i is uniform on $\{0,1\}^i$; given f it is determined; so $I(f; x_i \mid h_{i-1}) = i$. The losses are Proposition 6.5(c) and Proposition D.3(a). □

Proof of Theorem 6.12. Write $Y^A := Y^{A_n}$, $\mathcal{U}$ for the uniform law on it, and $\mathcal{G} := \{\frac{1}{2}G_0 + \frac{1}{2}\mathcal{U}\}$ over laws G_0 on Y^A; $\mathcal{G}$ is compact and convex, and $G(D) \geq \frac{1}{2}|Y|^{-n}$ when $|D| \leq n$. Let $\mathcal{F}_{\mathrm{fin}} := \{f : \mathbb{E}_{x|f}L_P < \infty\}$ (other f are trivial), $\Phi_f(G) := \mathbb{E}_{x|f}[-\log G(D_n)] - \mathbb{E}_{x|f}L_P$, and $\Phi(G, \nu) := \sum_f \nu(f)\Phi_f(G)$ for laws ν on $\mathcal{F}_{\mathrm{fin}}$. *Step 1.* Φ is continuous, convex in G, affine in ν; by (F6), $\min_G \max_\nu \Phi = \max_\nu \min_G \Phi$. For $G = \frac{1}{2}\nu + \frac{1}{2}\mathcal{U}$, $-\log G(D) \leq 1 - \log \nu(D)$, so Theorem 6.10(a) and Gibbs' inequality give $\Phi(G, \nu) \leq 1 + I_\nu(f \to x)$. Hence $v^* := \min_G \max_{f\in\mathcal{F}_{\mathrm{fin}}} \Phi_f(G) \leq 1 + C_{\mathrm{dir}}$. *Step 2.* For $G, G' \in \mathcal{G}$ and $|D| \leq n$, $|\log G(D) - \log G'(D)| \leq |G(D) - G'(D)|/(\ln 2 \min(G(D), G'(D))) \leq |Y|^n \|G_0 - G_0'\|_1 / \ln 2$. So a rational G_0' within ℓ^1-distance $|Y|^{-n}\ln 2$ of a minimiser has $\max_{f\in\mathcal{F}_{\mathrm{fin}}} \Phi_f(G') \leq v^* + 1 \leq 2 + C_{\mathrm{dir}}$. *Step 3, case (a).* Path probabilities given f are computable rationals, $\mathcal{F}_{\mathrm{fin}}$ is decidable ($f \notin \mathcal{F}_{\mathrm{fin}}$ iff some path of positive probability has $P(y_i \mid h, x) = 0$), and each Φ_f is computable on rational points. With the Lipschitz constant of Step 2, evaluating $\max_f \Phi_f$ to within $\frac{1}{4}$ on a fine rational grid yields a grid point G' with $\max_f \Phi_f(G') \leq v^* + 1$. *Step 3, case (b).* Let γ be rational with $C_{\mathrm{dir}} \leq \gamma \leq C_{\mathrm{dir}} + 1$ ($\gamma = 0$ if Δ does not see f). Enumerate rational G_0' and stop at the first G' certified to satisfy ($\star$): $\mathbb{E}_{x|f}[-\log G'(D_n)] < \mathbb{E}_{x|f}L_P + 3 + \gamma$ for every $f \in Y^A$. The left side is a computable real. The right side is a finite sum of products of a computable nonnegative real $\mathbb{P}[\mathrm{path} \mid f]$ and an lsc value $L_P(\mathrm{path}) \in [0, \infty]$, hence lsc in $[0, \infty]$ (with $0 \cdot \infty = 0$). A true strict inequality "computable $<$ lsc" is certified in finite time, also when the right side is $+\infty$; dovetail over candidates. By Step 2 some rational G' satisfies ($\star$), so the search halts. If Δ sees f, γ must be computed; this case is only sketched (Remark 6.13). *Step 4.* The procedure is uniform in (P, Δ, n, A), so G' has rational probabilities computable from these, and rejection sampling (undefined outside A_n) makes it a function machine with $K(G') \leq K(P) + K(\Delta) + K(n) + K(A) + c$. (F1) gives $\mathbb{E}_{x|f}L_{\mathrm{SOI}} \leq \mathbb{E}_{x|f}[-\log G'(D_n)] + K(G') + c \leq \mathbb{E}_{x|f}L_P + 4 + C_{\mathrm{dir}} + K(G') + c$; average over f for the last claim. An almost surely halting reveal sampler with private memory has $K(\Delta)$ at most its complexity plus $K(A) + c$ (Lemma D.4(c) uses the list A_n). □

Proof of Corollary 6.14. (a) Theorem 6.12(b) with $\Delta := \Delta_f$, which does not see f; then average. (b) Corollary D.5. (c) Take expectations in Theorem 6.8; given f the law of $x_{1:n}$ is $\Delta_f(x\|y)$, and $\mathbb{E}_{x|f}[-\log \Delta_f(x\|y)] = H(x_{1:n} \mid f)$, as Δ_f halts with probability 1. □

Remark D.7 (Open cases in Formulation C). (i) Whether $K(n) + K(A)$ can be dropped: the minimax law depends on the horizon; in one small instance a direct computation (code in the

accompanying repository) certifies that one law for horizons 1–4 needs at least 0.026 bits, while each single horizon needs at most 0.005, and it finds no growth up to horizon 6. (ii) Defective samplers with $K(P)$ alone ($\mathbb{E}L_P$ is then only upper semicomputable, so ($\star$) cannot be certified). (iii) Infinite alphabets at a fixed horizon. (iv) Reveal laws that see f, in case (b) (Remark 6.13).

Proposition D.8 (No pathwise version of Formulation A)**.** *No bound $L_{\mathrm{SOI}}(D_n) \le L_P + K(P) + K(\Delta) + c$ holds on every path for every fixed f and every reveal law Δ that does not see f.*

Proof. Let ω be Martin-Löf random with blocks b_i of length i, $f(b_{i+1}) := b_i$ and $f := \epsilon$ elsewhere; let Δ draw independent uniform strings of lengths $1, 2, \dots$, and P be the chain predictor. The path $x_i = b_i$ has probability $2^{-n(n+1)/2} > 0$; on it $L_P = 0$ and $L_{\mathrm{SOI}}(D_n) \ge n(n-1)/2 - 2\log n - c_\omega$ (Theorem 6.3(b)). □

Proposition D.9 (Shannon form of the mutual-information bound)**.** *Let (f, Θ) be jointly random and, given (f, Θ), let $x_i \sim \Delta_{\Theta,\Gamma}(\cdot \mid h_{i-1})$, a kernel that does not see f, with private memory Γ independent of f given Θ; finite alphabets up to time n. (a) If $I(f;\Theta) < \infty$, then $I(f \to x_{1:n}) = I(f;\Theta) - I(f;\Theta \mid h_n) - \sum_i I(y_i;\Theta \mid h_{i-1}, x_i) \le I(f;\Theta)$. (b) For $f \sim \nu$ computable, $\mathbb{E}L_{\mathrm{SOI}}(D_n) \le \mathbb{E}L_P + I(f;\Theta) + K(\nu) + c$ for every P. (c) In Formulation C, $\Theta := \Delta_f^{(n)}$, the restriction of Δ_f to n steps, is a function of f, $I(f;\Theta) = H(\Theta)$, and in Theorem 6.11 $I(f \to x) = H(\Theta) = n(n+1)/2$.*

Proof. (a) Given $\Theta = k$, the law of (f, Γ, h_{i-1}) is $p(df \mid k)p(d\Gamma \mid k)\prod_{j<i}\Delta_{k,\Gamma}(x_j \mid h_{j-1})\mathbf{1}[f(x_j) = y_j]$, so $f \perp \Gamma \mid (\Theta, h_{i-1})$, and the law of x_i given (f, Θ, h_{i-1}) does not involve f: $I(f; x_i \mid \Theta, h_{i-1}) = 0$. The chain rule gives $I(f;\Theta,h_n) = I(f;\Theta) + \sum_i H(y_i \mid \Theta, h_{i-1}, x_i)$, and also $I(f;\Theta,h_n) = I(f;h_n) + I(f;\Theta \mid h_n)$ with $I(f;h_n) = I(f \to x) + \sum_i H(y_i \mid h_{i-1}, x_i)$ (proof of Theorem 6.10). Equate, using $H(y_i \mid h_{i-1}, x_i) - H(y_i \mid \Theta, h_{i-1}, x_i) = I(y_i;\Theta \mid h_{i-1}, x_i)$. (b) Theorem 6.10(b) and (a). (c) $I(f;\Theta) = H(\Theta)$ when Θ is a function of f; in the randomised chain, $\Delta_f^{(n)}$ determines the chain, uniform on $n(n+1)/2$ bits. □

Conjecture D.10 (Algorithmic form in Formulation C)**.** *With $I(a:b) := K(b) - K(b \mid a)$ and the hypotheses of Corollary 6.14(a): $\mathbb{E}_{x|f}L_{\mathrm{SOI}} \le \mathbb{E}_{x|f}L_P + I(\Delta_f : f|_{A_n}) + K(P) + O(\log K(\Delta_f)) + K(n) + K(A) + c$.*

We have neither a proof nor a counterexample.

D.3 $\mathrm{SOI}^{\mathrm{hist}}$

Lemma D.11. *M_T^{hist} is an lsc chronological semimeasure, uniformly in an index of T.*

Proof. The events $\{T^\omega(x_{1:i}) = y_i \ \forall i \le n,\ T^\omega(x_{1:n}x) = y\}$ for different y are disjoint subsets of the event for $(y_{1:n}, x_{1:n})$ and are effectively open, uniformly (Proposition 3.7); the label at time i depends only on $x_{1:i}$. □

Proof of Theorem 6.16. Nodes are $z = (x_{1:k}, y_{1:k})$ with parent $p(z)$; the *family* $F(z)$ of a non-root node consists of the nodes with the same parent and the same last input. Take dyadic, finitely supported $\rho_t \uparrow \rho$, computable and nondecreasing in t, and a computable enumeration $z_1, z_2, \dots$ of the non-root nodes with every node after its parent. Each node carries a finite union $S(z)$ of dyadic intervals (cylinders), initially empty; $S(\text{root}) := [0,1)$. At stage t process $z_1, \dots, z_t$ in order: with $\text{free}(z) := S(p(z)) \setminus \bigcup_{z' \in F(z)} S(z')$ (current sets, z included), if $a := \min(\rho_t(z) - \mathbb{U}(S(z)), \mathbb{U}(\text{free}(z))) > 0$, add to $S(z)$ a dyadic part of $\text{free}(z)$ of measure a. By construction: (I1) $S(z) \subseteq S(p(z))$; (I2) family members are disjoint; (I3) $\mathbb{U}(S_t(z)) \le \rho_t(z)$ after stage t; (I4)

$S_t(z)$ grows with t. *Limit.* We show $\mathbb{U}(S(z)) = \rho(z)$, $S(z) := \bigcup_t S_t(z)$, by induction along the enumeration. With $p := p(z)$, the hypothesis (or $\mathbb{U}(S(\text{root})) = 1$) and the chronological inequality give $\mathbb{U}(S(p)) \geq \sum_{z' \in F(z)} \rho(z')$. When z is processed at stage t, $S(p) = S_t(p)$ (p is processed earlier), and each sibling has measure at most $\rho(z')$ by (I3). So afterwards $\mathbb{U}(S_t(z)) \geq \min(\rho_t(z), \mathbb{U}(S_t(p)) - \sum_{z' \in F(z) \setminus \{z\}} \rho(z'))$; only "$\geq$" holds, since a later sibling may still grow. Letting $t \to \infty$ gives $\liminf_t \mathbb{U}(S_t(z)) \geq \rho(z)$, and (I3) gives equality. *Machine.* $T^\omega_\rho(x_{1:k})$ recomputes $y_i := T^\omega_\rho(x_{1:i})$ for $i < k$, then searches for t, y with $\omega \in S_t((x_{1:k}, y_{1:k-1}y))$ and outputs the first y found; by (I2) at most one y is ever found, and membership depends on a finite prefix of ω. By induction with (I1), (I2), $\{\omega : T^\omega_\rho(x_{1:i}) = y_i \ \forall i \leq n\} = S((x_{1:n}, y_{1:n}))$, of measure $\rho(y_{1:n} \| x_{1:n})$. The construction is uniform in an index of (ρ_t). □

Proof of Corollary 6.17. (a) Theorem 6.16 and (F1) over X_H at non-root nodes; at the root $M_H = 1 \geq \rho(\epsilon \| \epsilon)$; two universal elements dominate each other. (b) $P(y \| x)$ is an lsc chronological semimeasure of complexity at most $K(P) + c$. (c) Lemma 6.2. □

Proposition D.12 (Other oracles)**.** *Let the inputs be distinct. (a) M_{pairs} and M_H agree up to constant factors. (b) $\text{SOI}^{\text{count}}$ is SOI on stamped inputs (i, x_i) (as in Theorem 5.5(b)), and $L_{\text{count}} \leq -\log M(\bar{y}_1 \cdots \bar{y}_n) + c$. (c) SOI $\preceq$ $\text{SOI}^{\text{count}}$ $\preceq$ SOI^{set} $\preceq$ SOI^{hist}, where $\nu \preceq \nu'$ means $\nu \leq 2^c \nu'$.*

Proof. (a) A pairs-machine T gives the history-oracle machine

$$T'^\omega(x_{1:i}) := T^\omega(x_1, T'^\omega(x_1), \ldots, x_{i-1}, T'^\omega(x_{<i}), x_i);$$

on the event that T' outputs $y_{1:n}$ the recursion reproduces the true history, and conversely by induction on i. (b) The hypothesis that outputs the i-th $\bar{y}$-block of the universal monotone machine's output. (c) Each oracle is computable from the next (the count is the size of the set plus one). □

Proposition D.13 (The hierarchy is strict)**.** *Semimeasure convention. (i) Every set S of $n \geq 2$ strings has an order such that, with labels $y_i = \mathbf{1}[i \text{ even}]$, $L_{\text{SOI}} \geq n - \log(n+1) - c$ while $L_{\text{count}} \leq c$. (ii) On the chain of Theorem 6.3(b), $L_{\text{count}} \geq \bar{\ell}_{n-1} - K(\ell_n) - c_\omega$ while $L_{\text{set}} \leq c$. (iii) Every such S has an order such that, with labels $y_1 = \epsilon$, $y_i = x_{i-1}$, $L_{\text{set}} \geq \log n! - \log n - c$ while $L_H \leq c$. Single infinite sequences with linearly growing gaps also exist (Proposition D.14).*

Proof of Proposition D.13. (i) $A := \{x_i : i \text{ even}\}$ and D are computable from each other given S, so $L_{\text{SOI}} \geq K(A \mid S) - c$ (Theorem 3.21(a)); choose A of size $\lfloor n/2 \rfloor$ with $K(A \mid S) \geq \log \binom{n}{\lfloor n/2 \rfloor} - 1 \geq n - \log(n+1) - 1$. The count hypothesis $\mathbf{1}[i \text{ even}]$ gives $L_{\text{count}} \leq c$. (ii) With U the universal count-machine, $\nu(z \mid x, n) := \mathbb{U}\{\omega : U^\omega(n, x) = z_1, U^\omega(n-1, z_1) = z_2, \ldots, U^\omega(1, z_{n-1}) = \epsilon\}$ sums to at most 1 over z (iteration determines z), so $L_{\text{count}} \geq K(x_{1:n-1} \mid x_n, n) - c$; n is computable from $|x_n|$, and the proof of Theorem 6.3(b) finishes. "Output the longest element of the set, or ϵ if it is empty" gives $L_{\text{set}} \leq c$. (iii) For a tape ω of the universal set-machine U, an order is consistent iff $U^\omega(\emptyset, x_1) = \epsilon$ and $U^\omega(\{x_{<i}\}, x_i) = x_{i-1}$. Then x_n determines $x_{n-1} = U^\omega(S \setminus \{x_n\}, x_n)$, and so on backwards, so at most n orders are consistent with ω and $\sum_\sigma M_{\text{set}}(\sigma) \leq n$. (F2) for this semimeasure divided by n, lsc uniformly in S, gives $L_{\text{set}} \geq K(\sigma \mid S) - \log n - c$; choose $K(\sigma \mid S) \geq \log n! - 1$. "$y_1 = \epsilon$, $y_i = x_{i-1}$" gives $L_H \leq c$. □

Proposition D.14 (Single infinite sequences)**.** *Let ω be Martin-Löf random and u_j, v_j the strings with indices $2j-1, 2j$ in length-lexicographic order; block j reveals u_j, v_j in this order if $\omega_j = 0$, reversed otherwise. (i) With labels $\mathbf{1}[i \text{ even}]$, $L_{\text{SOI}}(D_{2k}) \geq k - K(k) - c_\omega$ for all k, while $L_{\text{count}} \leq c$.*

(ii) The chain of Theorem 6.3(b) is a single sequence for Proposition D.13(ii). (iii) With labels $y_1 = \epsilon$, $y_i = x_{i-1}$, $L_{\mathrm{set}} \geq k - K(k) - c_\omega$ at time $2k$, while $L_H \leq c$. So each gap grows linearly along one sequence (semimeasure convention).

Proof. (i) $\operatorname{dom} D_{2k}$ is computable from k, and the labelling determines $\omega_{1:k}$; use Theorem 3.21(a), (F4) and (F3). (iii) By the backward reconstruction, a tape fixes the order given its last input (u_k or v_k), so at most two of the 2^k block orders σ_b fit one tape; apply (F2) to $b \mapsto M_{\mathrm{set}}(\sigma_b)/2$ and take $b = \omega_{1:k}$. □

M_{pq} is Solomonoff's per-query solution, his eq. (10) for the operator-induction problem (Remark 4.2).

Proposition D.15 (Solomonoff's per-query solution ignores the history)**.** *M_{pq} is an lsc chronological semimeasure that ignores the history, so $L_H \leq -\log M_{\mathrm{pq}} + c$ on every sequence. On distinct inputs $M_{\mathrm{pq}} = M_{\mathrm{prod}}(D_n)$, a weak semimeasure, so it pays the lower bounds of Theorems 6.3 and 6.4 (semimeasure convention).*

Proof. Proposition 5.7(a), Corollary 6.17(a), and $M_{\mathrm{prod}} \leq 2^c M_F$ (Theorem 4.4); M_{prod} is weak, so Proposition 6.5(b) applies. □

Proposition D.16 (Deterministic programs for $\mathrm{SOI}^{\mathrm{hist}}$)**.** *With (φ_e) the partial computable functions on X_H, let $W_{\mathrm{hist}}(y\|x) := \sum_{p \in \operatorname{dom} V} 2^{-|p|} \mathbf{1}[\varphi_{V(p)}(x_{1:i}) = y_i \ \forall i \leq n]$ and $K_{\mathrm{hist}}(y\|x)$ the least such $|p|$; let $K^{\mathrm{tail}}(n) := -\log \sum_{N \geq n} 2^{-K(N)}$ and $\{0,1\} \subseteq Y$. For every input list and $y \in \{0,1\}^n$, $W_{\mathrm{hist}}(y\|x) \geq 2^{-n-K^{\mathrm{tail}}(n)-c}$, where (a) $K^{\mathrm{tail}}(n) \leq K^{\mathrm{pad}}(n) \leq K(n)$, (b) $K^{\mathrm{tail}}(n) \leq g(n) + K(g) + c$ for every computable nondecreasing unbounded g, and (c) $K^{\mathrm{tail}}(n) \to \infty$. (d) But $n - c \leq \max_y K_{\mathrm{hist}}(y\|x_{1:n}) \leq n + K(n) + c$, and along every input sequence $\max_y K_{\mathrm{hist}}(y\|x_{1:n}) - n > \log n + \log\log n$ for infinitely many n.*

Proof. W_{hist} is an lsc chronological semimeasure (programs consistent with different next labels differ). Let $\varphi_{N,w}(x_{1:i}) := w_i$ for $i \leq N$, $w \in \{0,1\}^N$. A prefix machine reads a V-program for N and then N bits w and outputs an index of $\varphi_{N,w}$; by universality, with a fixed string r and q_N a shortest program for N, the strings rq_Nw are distinct elements of $\operatorname{dom} V$ ($q_N, q_{N'}$ are incomparable for $N \neq N'$), of length $|r| + K(N) + N$. For $N \geq n$, $\varphi_{N,w}$ is consistent iff $w_{1:n} = y$, for 2^{N-n} strings w, so $W_{\mathrm{hist}}(y\|x) \geq \sum_{N \geq n} 2^{N-n-|r|-K(N)-N} = 2^{-n-|r|-K^{\mathrm{tail}}(n)}$. (a) One term of the sum. (b) $q(N) := 2^{-g(N)} - 2^{-g(N+1)}$ is a computable discrete measure with $\sum_{N \geq n} q(N) = 2^{-g(n)}$, and (F2) gives $2^{-K(N)} \geq 2^{-K(g)-c} q(N)$. (c) The tails of $\sum_N 2^{-K(N)} \leq 1$ tend to 0. (d) The upper bound is the program rq_ny; the lower bound holds since a consistent program and $x_{1:n}$ determine y. Fix the input sequence, let $e(n) := \max_y K_{\mathrm{hist}}(y\|x_{1:n}) - n$, $e^*(n) := \max_{m \leq n} e(m)$, and let $C(L)$ and k_L count the $p \in \operatorname{dom} V$ with $|p| \leq L$ and $|p| = L$. Distinct labellings need distinct programs, so $C(n + e^*(n)) \geq 2^n$. By summation by parts and Kraft, $\sum_{L \leq m} C(L) 2^{-L-1} \leq \sum_L k_L 2^{-L} \leq 1$; as $n + e^*(n)$ is strictly increasing, $\sum_n 2^{-e^*(n)} \leq 2$. If $e(n) \leq \log n + \log\log n$ for all $n \geq n_0$, then $\sum_n 2^{-e^*(n)} \geq 2^{-O(1)} \sum_n 1/(n \log n) = \infty$. □

D.4 Solomonoff induction on the joint sequence

Block sequences are $b = (x_1, y_1, x_2, \dots)$, possibly ending after an input; b^- is b without its last block.

Lemma D.17 (Block codes)**.** *Let ν be an lsc semimeasure on block sequences ($\nu(\emptyset) \le 1$, $\sum_\beta \nu(b\beta) \le \nu(b)$), and $s \mapsto \bar{s}$ a prefix-free code with nonempty codewords. Let $B(z)$ be the set of b with $\mathrm{code}(b) \sqsupseteq z$ and ($b = \emptyset$ or $\mathrm{code}(b^-) \not\sqsupseteq z$). Then $\nu_{\mathrm{bin}}(z) := \sum_{b \in B(z)} \nu(b)$ is an lsc semimeasure on $\{0,1\}^*$ with $\nu_{\mathrm{bin}}(\mathrm{code}(b)) = \nu(b)$, so $M(\mathrm{code}(b)) \ge 2^{-K(\nu)-c}\nu(b)$.*

Proof. $B(\mathrm{code}(b)) = \{b\}$: codes are prefix-free and nonempty, so $\mathrm{code}(b') \sqsupseteq \mathrm{code}(b)$ forces b' to extend b blockwise, and if $b' \ne b$ then $\mathrm{code}(b'^-) \sqsupseteq \mathrm{code}(b)$. $B(\epsilon) = \{\emptyset\}$. Let $b \in B(za)$. Either $\mathrm{code}(b^-) \not\sqsupseteq z$, and $b \in B(z)$ with $\mathrm{code}(b) \ne z$; or $\mathrm{code}(b^-) \sqsupseteq z$, and since $\mathrm{code}(b^-)$ and za are prefixes of $\mathrm{code}(b)$ with $\mathrm{code}(b^-) \not\sqsupseteq za$, $\mathrm{code}(b^-) = z$, so b extends by one block the element b^- of $B(z)$ with code z. Conversely each $b \in B(z)$ with $\mathrm{code}(b) \ne z$ lies in exactly one $B(za)$. Hence $\nu_{\mathrm{bin}}(z0) + \nu_{\mathrm{bin}}(z1) \le \sum_{b \in B(z), \mathrm{code}(b) \ne z} \nu(b) + \sum_{b' \in B(z), \mathrm{code}(b') = z} \sum_\beta \nu(b'\beta) \le \nu_{\mathrm{bin}}(z)$. Apply (F5). □

Proof of Theorem 6.18. (a) $M(\mathrm{code}(h_n))$ is an lsc chronological semimeasure: with $z := \mathrm{code}(h_n)$, $\sum_y M(z\bar{x}\bar{y}) \le M(z\bar{x}) \le M(z)$, as $\{\bar{y}\}$ is prefix-free. Apply Corollary 6.17(a). Likewise $M(\mathrm{code}(h_{n-1})\bar{x}_n) \ge M(\mathrm{code}(h_n))$ is one for inputs given labels. (b) $\nu(h_n) := M_H^{\mathrm{in}}(x_{1:n}\|y_{1:n-1})M_H(y_{1:n}\|x_{1:n})$ and $\nu(h_{n-1}x_n) := M_H^{\mathrm{in}}(x_{1:n}\|y_{1:n-1})M_H(y_{1:n-1}\|x_{1:n-1})$ define an lsc semimeasure on block sequences (sum over y_n with M_H's chronological inequality, over x_n with M_H^{in}'s); apply Lemma D.17. (c) Combine with $L_M^{\mathrm{joint}} = L_M^x + L_M^y$. (d) Proposition 6.19. □

Proof of Proposition 6.19. $\nu(h_n) := \mathbf{1}[x_i = \delta(h_{i-1})\ \forall i \le n]M_H(y_{1:n}\|x_{1:n})$ and $\nu(h_{n-1}x_n) := \mathbf{1}[x_i = \delta(h_{i-1})\ \forall i \le n]M_H(y_{1:n-1}\|x_{1:n-1})$ define an lsc semimeasure on block sequences with $K(\nu) \le K(\delta) + c$: the indicator is c.e., summing over the next input leaves only $\delta(h_{n-1})$, and summing over the next label uses M_H's chronological inequality. Lemma D.17 gives $L_M^{\mathrm{joint}} \le L_H + K(\delta) + c$; with $L_M^{\mathrm{joint}} = L_M^x + L_M^y$ this is the left inequality. The right one is Theorem 6.18(c). □

On δ-consistent sequences also $L_{\mathrm{in}}^x \le K(\delta) + c$, so $L_H + L_{\mathrm{in}}^x - K(\delta) - c \le L_M^{\mathrm{joint}} \le L_H + L_{\mathrm{in}}^x + c$.

Lattimore, Hutter and Gavane (LHG below) (Lattimore et al., 2011) show that unnormalised Solomonoff induction can fail to predict simple recursive sub-patterns of a sequence. We do not show that their construction gives the following assumptions, which are not proved here.

Assumption D.18 (LHG-even)**.** With $X = Y = \{0,1\}$ and the identity code, for some ω the loss of unnormalised M on positions $2, 4, \dots$ of $\omega_1\omega_1\omega_2\omega_2\cdots$ is unbounded.

Assumption D.19 (LHG-odd)**.** The same for positions $3, 5, \dots$ of $0\omega_1\omega_1\omega_2\omega_2\cdots$.

Proposition D.20 (The conditional label loss)**.** *With $X = Y = \{0,1\}$ and the identity code: under Assumption LHG-even, $L_M^y \le L_H + O(1)$ fails on some sequence (labels copy inputs); under Assumption LHG-odd, $L_H \le L_M^y + O(1)$ fails on some sequence (inputs copy the previous label, Proposition 6.19). For the code $\bar{s} = 1^{|s|}0s$ the assumptions must also hold for framed blocks.*

Proof of Proposition D.20. (Even.) $x_i := \omega_i$, $y_i := x_i$: $L_H \le c$ by the copy predictor (Corollary 6.17(b)), while L_M^y is M's loss on the copied bits. (Odd.) $\delta(\emptyset) := 0$, $\delta(h) :=$ the last label of h, $y_i := \omega_i$: the joint sequence is $0\omega_1\omega_1\omega_2\omega_2\cdots$, the inputs $x_2, x_3, \dots$ are the copied bits, so L_M^x is unbounded, and Proposition 6.19 with $K(\delta) = O(1)$ gives $L_H - L_M^y \to \infty$. Non-convergence of the copied bits' conditional probabilities to 1 already makes the cumulative loss unbounded. □

D.5 What SOI can and cannot exploit

Proof of Theorem 6.21 and Corollary 6.22. The left inequality is Lemma 6.2. For the right one fix σ' and p attaining $K^{\mathrm{pred}}(x_{\sigma'})$. The function machine T, which does not depend on p, on query x enumerates the prefix-free set $\operatorname{dom} V(\cdot, x)$ and takes the first element q found that is a prefix of the tape; checking a candidate reads the tape only up to the first mismatch, which comes within $|q|$ bits, so T reads exactly q. It computes $V(q, x) = \langle x'_1, \dots, x'_k \rangle$ and runs T_U^H on the rest of the tape at $(x'_1, \dots, x'_k, x)$. On tapes $p\omega'$, every $x \in \operatorname{dom} D$ finds p (as $p \in \operatorname{dom} V(\cdot, x)$, a prefix-free set), so all queries share ω', and $\nu_T(D) \geq 2^{-|p|} M_H(y_{\sigma'} \| x_{\sigma'})$; apply (F1). For Corollary 6.22, $K^{\mathrm{pred}}(e\text{-order}) \leq K(e) + c$: given x, enumerate e until x appears. □

To make the example of Section 6.6 precise, identify $\{0,1\}^q$ with $\mathrm{GF}(2^q)$ (polynomial basis modulo the lexicographically least irreducible polynomial) and let $D_{abr} := \{(a, r), (b, r + ab)\}$, $a \neq b$.

Theorem D.21 (A $\mathrm{GF}(2^q)$ example)**.** *(a) In both orders $L_H(D_{abr}) \leq q + c$, with c independent of q. (b) $L_{\mathrm{SOI}}(D_{abr}) \geq K(a, b, r \mid q) - 3q/2 - 1 - c$; so for all but a 2^{1-j} fraction of triples, $L_{\mathrm{SOI}}(D_{abr}) \geq 3q/2 - j - c \geq \max_\sigma L_H(D_\sigma) + q/2 - j - c'$. (c) For uniform (a, b, r), $a \neq b$, $\mathbb{E} L_{\mathrm{SOI}}(D_{abr}) \geq 3q/2 - 2$ for every $q \geq 1$. These lower bounds are in the semimeasure convention; for $\bar{L}_{\mathrm{SOI}}$ they are open.*

The key step of the proof is a character-sum bound.

Lemma D.22 (Character-sum bound)**.** *Let $F = \mathrm{GF}(2^q)$. Every partial $G : F \to F$ has $N(G) := \#\{(a, b) : a \neq b,\ G(a) + G(b) = ab\} < 2^q + 2^{3q/2}$ (pairs where G is defined).*

Proof. Extending G to a total function and including $a = b$ does not decrease the count. With $\chi_u(z) := (-1)^{\mathrm{Tr}(uz)}$, $\sum_u \chi_u(z) = 2^q \mathbf{1}[z = 0]$, so in characteristic 2

$$N \leq 2^{-q} \textstyle\sum_u \sum_{a,b} \chi_u(G(a)) \chi_u(G(b)) \chi_u(ab).$$

The term $u = 0$ is 2^{2q}. For $u \neq 0$, $(\mathbf{C}_u)_{ab} := \chi_u(ab)$ has $\mathbf{C}_u \mathbf{C}_u^{\mathsf{T}} = 2^q I$, so $|v^{\mathsf{T}} \mathbf{C}_u v| \leq 2^{q/2} |v|^2 = 2^{3q/2}$ for $v_a := \chi_u(G(a))$. Hence $N \leq 2^{-q}(2^{2q} + (2^q - 1) 2^{3q/2}) < 2^q + 2^{3q/2}$. □

Proof of Theorem D.21. (a) P predicts y_1 uniformly on $\{0,1\}^{|x_1|}$ and then $y_2 = y_1 + x_1 x_2$; in the order (b, a) it predicts $(r + ab) + ba = r$. Apply Corollary 6.17(b). (b) For fixed ω, $f_\omega \supseteq D_{abr}$ iff $r = f_\omega(a)$ and $f_\omega(a) + f_\omega(b) = ab$, so $\sum_{a \neq b, r} M_F(D_{abr}) = \mathbb{E}_\omega N(f_\omega|_F) < 2^{3q/2+1}$ by Lemma D.22 (both sides count ordered pairs). So $M_F(D_{abr}) 2^{-3q/2-1}$ is a semimeasure on triples, lsc uniformly in q; apply (F2). Fewer than 2^{3q-j} of the at least 2^{3q-1} triples have $K(a, b, r \mid q) < 3q - j$. (c) Jensen: $\mathbb{E} L_{\mathrm{SOI}} \geq -\log \mathbb{E} M_F(D_{abr}) \geq \log(2^{2q}(2^q - 1)) - 3q/2 - 1 \geq 3q/2 - 2$. □

A direct computation (code in the accompanying repository) gives the optimum over *all* laws of random functions, $\min_w \mathbb{E}[-\log w(D_{abr})] = 2.708$ bits for $q = 2$ and 4.348 for $q = 3$, which is a lower bound on $\mathbb{E} L_{\mathrm{SOI}}$ (the one-tape version restricted to the field is a random partial function), against the Bayes history-aware loss q ($L_H \leq q + c$); and $\max_G N(G) = 8$ for $q = 2$ and 22 for $q = 3$, against the bounds 12 and 30.6 of Lemma D.22, while for $q = 4$ it lies in $[56, 80]$.

Theorem D.23 (Expected gap)**.** *Let $f \sim \nu$ computable and the inputs drawn by any kernel $\Delta(\cdot \mid f, h_{i-1})$, with finite alphabets up to time n. Then $-c \leq \mathbb{E} L_{\mathrm{SOI}}(D_n) - \mathbb{E} L_H \leq I(f \to x_{1:n}) + K(\nu) + c$, and the gap is at most $K(\nu) + c$ if the input choice does not see f.*

The upper bound is attained up to the last input's entropy in the randomised chain, but not in general. Let (a, b, r) be uniform, let $f(a) = r$, $f(b) = r + ab$ and $f = \epsilon$ elsewhere, and reveal a and b in a uniformly random order. Then $I(f \to x) = (q-1) + \log(2^q - 1)$, while the gap $\mathbb{E}L_{\mathrm{SOI}} - \mathbb{E}L_H$ is at most $q + c$. The predictor of Theorem D.21(a) has $\mathbb{E}L_P = q$ and its reveal law has an $O(1)$ program relative to an oracle for f, and $\mathbb{E}L_{\mathrm{SOI}} \geq 3q/2 - 2$: a second process on which the oracle form fails.

Proof of Theorem D.23. The lower bound is Lemma 6.2, pathwise; the upper bound is Theorem 6.10(b) with P the predictor of M_H (sub-probability conditionals), and Lemma D.4(a). In the GF process after the theorem, uniform labels give $\mathbb{E}L_{\mathrm{SOI}} \leq 2q + c$ and $\mathbb{E}L_H$ is at least the Bayes loss q, so the gap is at most $q + c$. As f determines $\{a, b\}$, x_1 has entropy q and entropy 1 given f, and x_2 is uniform on $2^q - 1$ points given h_1 but determined by f and x_1; so $I(f \to x) = (q-1) + \log(2^q - 1)$. □

Proposition D.24 (An individual-sequence information bound)**.** *With $I(a : b) := K(b) - K(b \mid a)$: (a) $L_{\mathrm{SOI}}(D) \leq L_H(D_\sigma) + I(x_\sigma : D) + c$ for every D and order σ. (b) The natural strengthening with $I(x_\sigma : y_\sigma)$ in place of $I(x_\sigma : D)$ is false, even up to $O(\log n)$: on the sequences of Proposition D.13(i), $L_{\mathrm{SOI}}(D) - L_H(D_\sigma) - I(x_\sigma : y_\sigma) \geq n - \log(n+1) - K(n) - c$ (semimeasure convention).*

Proof. (a) The lookup table gives $L_{\mathrm{SOI}}(D) \leq K(D) + c$ (F1). For fixed $x_{1:n}$, $y_{1:n} \mapsto M_H(y_{1:n} \| x_{1:n})$ is a semimeasure on Y^n, lsc uniformly in $x_{1:n}$, so (F2) gives $K(D \mid x_\sigma) \leq L_H(D_\sigma) + c$; and $K(D) = I(x_\sigma : D) + K(D \mid x_\sigma)$. (b) $L_{\mathrm{SOI}}(D) \geq n - \log(n+1) - 1 - c$ (proof of Proposition D.13(i)); $L_H(D_\sigma) \leq c$, as $\mathbf{1}[i \text{ even}]$ knows i; and $I(x_\sigma : y_\sigma) \leq K(y_\sigma) + c \leq K(n) + c$, as $y_\sigma = 0101\cdots$. With A uniform, the events $\{f_\omega \supseteq D_A\}$ are disjoint, so Gibbs' inequality gives $\mathbb{E}_A L_{\mathrm{SOI}}(D_A) \geq \log \binom{n}{\lfloor n/2 \rfloor} \geq n - \log(n+1)$ exactly. □

E Proofs for Section 7 (worked examples)

Notation is that of Section 7. The coding theorem is used in the form of Section 3.1: if $q(a \mid w)$ is lsc uniformly in (a, w) and $\sum_a q(a \mid w) \leq 1$, then $K(a \mid w) \leq -\log q(a \mid w) + K(q) + c$. "Gibbs" is $\mathbb{E}[-\log q(Z)] \geq H(Z)$ for every q with $\sum_z q(z) \leq 1$. Numbers obtained by direct computation come from seeded or deterministic scripts in the accompanying repository; the finite mixtures used there are illustrations, not SOI or SI.

E.1 SI's label loss

Lemma E.1 (SI's label loss)**.** *Let $\mathbb{P}$ be a probability law on infinite sequences of pairs in countable sets, coded by a prefix-free code with nonempty codewords, and ρ a semimeasure on $\{0,1\}^*$ with $\rho(\mathrm{code}(h_t)) \geq 2^{-a}\mathbb{P}(h_t)$ for all t and histories. Let $L^y_{\mathbb{P}}(n) := \sum_{t \leq n} -\log \mathbb{P}(y_t \mid h_{t-1}, x_t)$. Then (i) $\mathbb{E}L^y_\rho(n) \leq \mathbb{E}L^y_{\mathbb{P}}(n) + a$ for every n, and (ii) $\mathbb{P}[\exists n : \; L^y_\rho(n) > L^y_{\mathbb{P}}(n) + a + j] \leq 2^{-j}$.*

Proof. Put $Q_t := \rho(\mathrm{code}(h_{t-1})\bar{x}_t)/\rho(\mathrm{code}(h_{t-1}))$. Telescoping, $\rho(\mathrm{code}(h_n)) = \rho(\epsilon)\prod_{t \leq n} Q_t \cdot 2^{-L^y_\rho(n)}$ with $\rho(\epsilon) \leq 1$, and $-\log \mathbb{P}(h_n) = L^y_{\mathbb{P}}(n) + \sum_t -\log \mathbb{P}(x_t \mid h_{t-1})$. Hence, $\mathbb{P}$-almost surely,

$$L^y_\rho(n) \leq -\log \rho(\mathrm{code}(h_n)) + \textstyle\sum_t \log Q_t \leq a + L^y_{\mathbb{P}}(n) + \log \Lambda_n, \qquad \Lambda_n := \prod_{t \leq n} Q_t / \mathbb{P}(x_t \mid h_{t-1}).$$

Given h_{t-1}, $\mathbb{E}[Q_t / \mathbb{P}(x_t \mid h_{t-1})] \leq \sum_x \rho(\mathrm{code}(h_{t-1})\bar{x})/\rho(\mathrm{code}(h_{t-1})) \leq 1$, since $\sum_{w \in A} \rho(sw) \leq \rho(s)$ for prefix-free A. So Λ is a nonnegative supermartingale with $\Lambda_0 = 1$. Jensen gives (i), and Ville's inequality $\mathbb{P}[\sup_n \Lambda_n \geq 2^j] \leq 2^{-j}$ (Ville, 1939) gives (ii). □

The normalisation, $M_{\mathrm{norm}}(\epsilon) = 1$ and $M_{\mathrm{norm}}(zb) = M_{\mathrm{norm}}(z)M(zb)/(M(z0) + M(z1))$, is a probability measure with $M_{\mathrm{norm}} \geq M$, since

$$M_{\mathrm{norm}}(z) = \tfrac{M(z)}{M(\epsilon)} \prod_{i<|z|} \tfrac{M(z_{1:i})}{M(z_{1:i}0)+M(z_{1:i}1)}$$

and each factor is at least 1. For computable $\mathbb{P}$, $M(\mathrm{code}(h)) \geq 2^{-K(\mathbb{P})-c}\mathbb{P}(h)$ (Lemma D.17), so the lemma applies to M and M_{norm} with $a = K(\mathbb{P}) + c$.

E.2 The cyclic permutation

For a string x and strings $z = (z_1, \ldots, z_i)$ put

$$D(x, z) := \{(x, z_1), (z_1, z_2), \ldots, (z_{i-1}, z_i), (z_i, 0^{|x|})\}, \qquad D(x, ()) := \{(x, 0^{|x|})\},$$

so that $D_k = D(x_k, (x_{k-1}, \ldots, x_1))$.

Lemma E.2 (chain semimeasure)**.** *For every law ν of a random partial function F and every x, $\sum_z \nu(D(x, z)) \leq 1$, the sum over the finite sequences z of strings different from $0^{|x|}$ for which $D(x, z)$ is a data set. Hence $K(z \mid x) \leq L_{\mathrm{SOI}}(D(x, z)) + c$ for these z.*

Proof. If $F \supseteq D(x, z)$, then z is the orbit of x up to its first visit to $0^{|x|}$; so for each F at most one z qualifies. $M_F(D(x, z))$ is lsc; apply the coding theorem. □

Lemma E.3 (depth count)**.** *For a partial function F and $1 \leq k \leq N - 1$, let $A_k(F)$ be the set of nonzero $x \in \{0, 1\}^d$ such that $F(x), \ldots, F^{k-1}(x)$ are nonzero d-bit strings and $F^k(x) = 0^d$. Then $|A_k(F)| \leq N - k$, with equality for the broom B_k.*

Proof. If $x \in A_k(F)$, the strings $F(x), \ldots, F^{k-1}(x)$ first reach 0^d at times $k-1, \ldots, 1$, so they are distinct, nonzero and not in $A_k(F)$: $|A_k(F)| \leq (N - 1) - (k - 1)$. For the broom, every nonzero input off the spine is in A_k. □

Proof of Theorem 7.3. (a) The lower bound is Theorem 3.21(a) with $T = \{0, 1\}^d$, computable from d. For the upper bound, the function machine T, on input x, runs $V(\cdot, \{0, 1\}^{|x|})$ on program bits read from the start of the tape and, if V halts with a table E of $\{0, 1\}^{|x|}$, outputs $E(x)$. For fixed x the halting prefixes are prefix-free, so T is consistent; all inputs of length d read the same program, so $\nu_T(f) \geq 2^{-K(f|d)}$; apply Theorem 3.8. (The same argument gives $L_{\mathrm{SOI}}(D) \leq K(D \mid \mathrm{dom}\, D) + K(e) + c$ whenever a partial computable e maps each $x \in \mathrm{dom}\, D$ to $\mathrm{dom}\, D$.) (b) T_{cyc} reads, for each length ℓ, its own sub-stream $(\omega_{\langle \ell, i \rangle})_i$ of the tape, draws a uniform ordering of the nonzero ℓ-bit strings by rejection sampling, closes it into a cyclic permutation $\hat{f}_\ell$ (the bijection after Definition 7.2), and outputs $\hat{f}_{|x|}(x)$. It halts with probability 1, each value is fixed by finitely many tape bits, $K(\nu_{T_{\mathrm{cyc}}}) = O(1)$, and $\nu_{T_{\mathrm{cyc}}}(f) = 1/(N - 1)!$; apply Theorem 3.8. (c) By (a), $L_{\mathrm{SOI}}(f) \geq K(f \mid d) - c$; apply Lemma 7.1 with $Z = f$, $w = d$. (d) Robbins' bounds $\ln m! = m \ln m - m + \frac{1}{2}\ln(2\pi m) + r_m$, $\frac{1}{12m+1} < r_m < \frac{1}{12m}$ (Robbins, 1955), for $(N - 1)! = N!/N$ give $\epsilon_N = r_N / \ln 2$. D_{N-1} has the computable domain $\{0, 1\}^d \setminus \{0^d\}$ and determines f, so the bounds transfer. □

Proof of Theorem 7.4. (a) If $k-1 > 2^\ell - 1$, B_k answers 0^ℓ on length ℓ; for $k = 1$ it maps everything to 0^ℓ. Each value is fixed by finitely many tape bits almost surely, B_k is total almost surely, and its index is computable from k. On the event $(v_1, \ldots, v_{k-1}) = (x_1, \ldots, x_{k-1})$, of probability $1/(N - 1)_{k-1}$, it maps $x_1 \mapsto 0^d$, $x_i \mapsto x_{i-1}$, and x_k, which is off the spine, to $v_{k-1} = x_{k-1}$.

Theorem 3.8 gives the term $K(k)$. A uniform cyclic permutation contains the k arcs $x_k \to \cdots \to x_1 \to 0^d$, which do not close the cycle, with probability $(N-k-1)!/(N-1)! = 1/(N-1)_k$ (contract the path), so T_{cyc} gives $\log(N-1)_{k-1} + \log(N-k) + c$. (b) Let F have law ν. Each $x \in A_k(F)$ has exactly one z of length $k-1$ with $F \supseteq D(x,z)$, its orbit, and other nonzero x have none. So, summing over $x \in \{0,1\}^d \setminus \{0^d\}$ and $z \in (\{0,1\}^d \setminus \{0^d\})^{k-1}$,

$$\textstyle\sum_{x,z} \nu(D(x,z)) = \mathbb{E}|A_k(F)| \le N-k. \tag{6}$$

Thus $M_F(D(x,z))/(N-k)$ is a semimeasure in (x,z), lsc uniformly in (d,k), and $K(x,z \mid d,k) \le L_{\mathrm{SOI}}(D(x,z)) + \log(N-k) + c$. As $(x_k,\dots,x_1)$ is uniform over $(N-1)_k$ values, Lemma 7.1 with $w=(d,k)$ gives (b). (c) Use $j+d$ in (b) and a union bound over the $N-1 < 2^d$ values of k. □

Proof of Proposition 7.5. (a) Gibbs with $q(x,z) := \nu(D(x,z))/(N-k)$ on the same range of (x,z) and (6); $(x_k,(x_{k-1},\dots,x_1))$ lies in this range and is uniform over $(N-1)_k$ values. (b) First, for weak ν and nonzero x, $\sum_z \nu(D(x,z)) \le 1$ over sequences z of distinct nonzero strings other than x: this is the proof of Proposition 6.5(a), with the terminal label 0^d in place of ϵ and nonzero labels in place of nonempty ones. Given x_k, $(x_{k-1},\dots,x_1)$ is uniform over $(N-2)_{k-1}$ values; Gibbs gives $\log(N-2)_{k-1} = \log(N-1)_k - \log(N-1)$. For the full cycle, $\sum_g \nu(g) \le 1$ over total g on $\{0,1\}^d$ (Definition 3.2); Gibbs. □

Proof of Proposition 7.7. $\beta_t = -\log M_F(D_t) + \log \mathcal{Z}(D_{t-1}, x_t)$. (i) M_F is the law of a random partial function (Proposition 3.7), so $\mathbb{E}[-\log M_F(D_t)] \ge \log(N-1)_{t-1}$ by Proposition 7.5(a). (ii) B_{t-1} is total almost surely and $x_t \notin \operatorname{dom} D_{t-1}$, so Lemma D.2 and the proof of Theorem 7.4(a) give $\mathcal{Z}(D_{t-1},x_t) \ge 2^{-K(t-1)-c}\nu_{B_{t-1}}(D_{t-1}) \ge 2^{-K(t-1)-c}/(N-1)_{t-2}$ on every path. (iii) $\log(N-1)_{t-1} - \log(N-1)_{t-2} = \log(N-t+1)$. The high-probability form uses Theorem 7.4(b) for (i). For the sum use $\beta_t \ge 0$, $K(t-1) \le \log t + 2\log\log(t+1) + c$ and $N-t+1 \ge N/2$. *Why the method stops.* Over the cycle these bounds sum to at most $\sum_{t<N}(d-K(t))^+ \le 2N$ bits, since fewer than 2^{d-i+1} integers have $K(t) \le d-i$. □

Proof of Proposition 7.9. The labels are determined by the order $\sigma = (x_1,\dots,x_{N-1})$. Fix a tape ω of the universal set-machine, with answer $U^\omega(A,x)$ at x given the set A. If σ is consistent with ω, then $x_{N-1} = U^\omega(\{0,1\}^d \setminus \{0^d\}, 0^d)$, $x_{N-2} = U^\omega(\{0,1\}^d \setminus \{0^d, x_{N-1}\}, x_{N-1})$, and so on. So $\sum_\sigma M_{\mathrm{set}}(y(\sigma)\|(\sigma,0^d)) \le 1$, lsc uniformly in d. Gibbs gives the expectation bound; the coding theorem and Lemma 7.1 give the probability bound. The upper bound is $L_{\mathrm{set}} \le L_{\mathrm{SOI}} + c$ on distinct inputs (Proposition D.12) and Theorem 7.3(b). □

Proof of Proposition 7.10. Let $\mathbb{P}_d$ be the law of the pairs for bit length d (x_t uniform over unused nonzero strings, $x_N = 0^d$, $y_t = x_{t-1}$, continued after step N by a fixed computable rule). It is computable uniformly in d, so by Lemma D.17 applied to $\sum_d \mathbf{m}(d)\mathbb{P}_d$, $\rho(\mathrm{code}(h)) \ge M(\mathrm{code}(h)) \ge 2^{-c}\mathbf{m}(d)\mathbb{P}_d(h)$. Telescoping as in Lemma E.1, with label conditionals 1, $L^y_\rho(k) \le c - \log \mathbf{m}(d) + \log\rho(\epsilon) + \log\Lambda_k$. At the first step $\mathbb{E}\Lambda_1 = \sum_x \rho(1^d0x)/\rho(\epsilon) \le \rho(1^d0)/\rho(\epsilon)$, so $V_k := \Lambda_k\rho(\epsilon)/\rho(1^d0)$ is a nonnegative supermartingale with $V_0 := 1$, and $L^y_\rho(k) \le c + K(d) + \log\rho(1^d0) + \log V_k$. Finally $\log\rho(1^d0) \le -K(d)+c$: for M by the coding theorem, as $\sum_d M(1^d0) \le 1$ and $d \mapsto M(1^d0)$ is lsc; for M_{norm} because $M_{\mathrm{norm}}(1^d0) \le M_{\mathrm{norm}}(0 \mid 1^d) \le M(1^d0)/M(1^{d+1})$ and $M(1^{d+1}) \ge 2^{-c}$. Jensen and Ville finish. □

Proposition E.4 (SI on raw blocks)**.** *Code the history as $x_1y_1x_2y_2\cdots$ in raw d-bit blocks without delimiters. For $\rho \in \{M, M_{\mathrm{norm}}\}$ and uniform cyclic f: $\mathbb{P}[\exists k \le N :\ L^y_\rho(k) > K(d) + j + c] \le 2^{-j}$, and $\mathbb{E}L^y_\rho(k) \le K(d) + c$.*

d	$\log((N-1)!)$	per pair	$d - \log e$	order-free	history-aware	joint
4	40.250	2.516	2.557	42.748	1.585	1.544
8	1675.996	6.547	6.557	1678.576	1.585	1.583
12	43238.047	10.556	10.557	43240.632	1.585	1.585

Table 2: The cyclic permutation: constants and full-cycle losses (bits) of exact finite mixtures, computed exactly on permutations drawn with a fixed seed (code in the accompanying repository). Order-free: weight $\frac{1}{6}$ each on the uniform cyclic-permutation, permutation and function laws and the maps $x \mapsto 0^d$, $x \mapsto x$, $x \mapsto x+1 \bmod N$. History-aware: $\frac{1}{3}$ each on "copy the previous input", uniform labels, uniform unused labels. Joint: label loss of a mixture of the true process, true inputs with uniform labels, and uniform blocks.

Proof. For fixed d the raw blocks are a prefix-free code and Lemma D.17 applies uniformly in d, so $M \geq 2^{-K(d)-c}\mathbb{P}_d$ on coded histories; the label loss of $\mathbb{P}_d$ is 0. Apply Lemma E.1. □

For illustration, for an exact Bayes mixture of the true processes for $d' \leq 10$ and a uniform-bits model, a direct computation (code in the accompanying repository) gives full-cycle label losses of 0.13–0.26 bits with the prefix code and 1.3–7.8 bits with raw blocks ($d = 2, \ldots, 6$).

Question E.5 (raw blocks)**.** With raw d-bit blocks, is SI's expected label loss at least $K(d) - K^{\mathrm{pad}}(d) - O(1)$? (The upper bound $K(d) + c$ is Proposition E.4; a heuristic: SI does not know where the first block ends, and the first label costs about $K(d) - \min_{L \geq d} K(L)$ bits.)

Remark E.6 (a prior over d). If d is drawn from a computable prior w, then with raw blocks $\mathbb{E}L_M^y \leq H(w) + K(w) + c$.

Proof. $M \geq 2^{-K(w)-c} w(d) \mathbb{P}_d^{\mathrm{bin}}$, with $\mathbb{P}_d^{\mathrm{bin}}$ the law of the raw coding. Telescoping gives $L_M^y(k) \leq K(w) + c - \log w(d) + \log M(\epsilon) + \log \Lambda_k$, and $\mathbb{E}[\Lambda_1 \mid d] \leq \sum_{|x|=d} M(x)/M(\epsilon) \leq 1$, since strings of one length are prefix-free. (Raw blocks of different lengths are not, so Lemma E.1 does not apply to the mixture directly.) □

Remark E.7 (where the information is). A history-aware predictor that sees the previous *pairs* as a set predicts perfectly: the most recent input is the unique input of D_{t-1} that is not a label. So the separation is between scoring a data set as a whole and predicting with the past in view, not between sets and sequences; the previous *inputs* as a set are not enough (Proposition 7.9). In the other direction, take $X = Y = \{0, 1\}$ and $y_i = x_i$: $L_{\mathrm{SOI}} \leq c$ on every input sequence, while, under Assumption LHG-even (Assumption D.18), SI's unnormalised label loss with the identity code is unbounded on the inputs $\omega_1\omega_2\cdots$ given there (Proposition D.20).

E.3 Eisenstat's example

In Remark 7.18, $UB_n - \tau_n H(\frac{1}{3}) = \frac{2}{3}(G_{N_n^0} - \frac{1}{2}N_n^0)$, with G as in the proof of Proposition 7.15.

Lemma E.8 (the heads frontier is rarely behind)**.** $\mathbb{P}[\eta_n < \tau_n] \leq 20e^{-n/18}$.

Proof. If $\eta_n < \tau_n$, all revealed indices lie below τ_n, so $\tau_n \geq n$ and $b_{<\tau_n}$ has exactly N_n^0 tails (Lemma 7.12). So $N_n^0 \geq n/2$, or some $k \geq n$ has fewer than $k/2$ tails in $b_{<k}$. By Hoeffding's inequality (Hoeffding, 1963) these have probabilities at most $e^{-n/18}$ and $e^{-k/18}$; the sum is at most $e^{-n/18}(1 + 1/(1 - e^{-1/18})) \leq 19.6\, e^{-n/18}$. □

n	runs	L_{of}/n	UB_n/n	iid$/n$	sd$(UB_n)/\sqrt{n}$	$L_{\mathrm{of}} - UB_n$	L_{P^*}	L_{joint}	touching
100	400	0.5785	0.4498	1.2532	0.989	12.9	2.18	3.32	2.83
1000	200	0.4840	0.4630	1.2502	1.113	21.0	2.30	3.43	2.88
10000	40	0.4627	0.4597	1.2510	1.113	29.4	2.17	3.29	2.85
limit			0.4591	1.2516	1.0245		2.3986		3

Table 3: Eisenstat's example: means over the runs of a simulation with a fixed seed (code in the accompanying repository), in bits. L_{of}: loss of the exact order-free Bayes mixture of an i.i.d. law and two families of threshold laws, with Laplace biases; iid: loss of the product law with $\mathbb{P}[y=1] = \frac{1}{3}$; L_{P^*}: loss of the Bayes predictor of Lemma E.9; L_{joint}: label loss of the exact joint mixture of the true process, the true inputs with Laplace labels, and a null model; touching: number of touching steps (Proposition 7.19). $L_{\mathrm{of}} - UB_n$ grows like $2.5 \log n$ (cf. Proposition 7.13). Last row: the limits of Propositions 7.15 and 7.19.

A direct computation (code in the accompanying repository; a dynamic program over the frontier process of Proposition 7.19) gives $\mathbb{P}[\eta_n < \tau_n] = 6.48 \cdot 10^{-2}$, $1.55 \cdot 10^{-2}$, $1.11 \cdot 10^{-3}$, $7.31 \cdot 10^{-6}$, $4.27 \cdot 10^{-10}$ at $n = 10, 20, 40, 80, 160$ (three digits).

Proof of Theorem 7.14. $[0, g_n) \subseteq \operatorname{dom} D_n$ with labels $b_{<g_n}$, and $[0, g_n)$ and g_n are computable from each other; monotonicity and Theorem 3.21(a) give $L_{\mathrm{SOI}}(D_n) \geq K(b_{<g_n} \mid g_n) - c$. For b Martin-Löf random for the computable $P_{1/3}$, the Levin–Schnorr theorem (see Li and Vitányi, 2008) gives $K(b_{<g}) \geq -\log P_{1/3}(b_{<g}) - c_b$, so $K(b_{<g_n} \mid g_n) \geq -\log P_{1/3}(b_{<g_n}) - K(g_n) - c_b$; if $\tau_n \leq \eta_n$, then $g_n \geq \tau_n$ and $-\log P_{1/3}(b_{<g_n}) \geq UB_n$. The upper bound is Proposition 7.13. By Lemma E.8 and Borel–Cantelli the heads are eventually ahead; by the strong law $\eta_n/n \to 2$ and $\tau_n/n \to \frac{1}{2}$, so $g_n \leq \max(\tau_n, \eta_n) \leq 3n$ for large n. Martin-Löf random b have probability 1. □

Proof of Proposition 7.15. Let G_i be the number of heads before the i-th tail. Then $\tau_n = N_n^0 + G_{N_n^0}$ and, using $H(\frac{1}{3}) = \log 3 - \frac{2}{3}$,

$$UB_n - \tfrac{H(1/3)}{2} n = a(N_n^0 - \tfrac{n}{3}) + \log 3\,(G_{N_n^0} - \tfrac{1}{2} N_n^0),$$

where $G_i = \sum_{l \leq i} \gamma_l$ with γ_l i.i.d. geometric ($\mathbb{P}[\gamma = m] = \frac{2}{3}(\frac{1}{3})^m$, mean $\frac{1}{2}$, variance $\frac{3}{4}$), independent of $N_n^0 \sim \mathrm{Bin}(n, \frac{1}{3})$. The second term has conditional mean 0 given N_n^0, so the terms are uncorrelated, with variances $a^2 \frac{2n}{9}$ and $(\log 3)^2 \frac{3}{4} \frac{n}{3}$. The strong law follows from $N_n^0/n \to \frac{1}{3}$, $G_i/i \to \frac{1}{2}$. For the limit law, let $\psi(s) := \mathbb{E} e^{is(\gamma - 1/2)} = 1 - \frac{3}{8} s^2 + o(s^2)$; by independence the characteristic function at θ is $\mathbb{E}[e^{i\theta a (N_n^0 - n/3)/\sqrt{n}} \psi(\theta \log 3/\sqrt{n})^{N_n^0}]$, the second factor tends to $e^{-\theta^2 (\log 3)^2/8}$ in probability with modulus at most 1, and bounded convergence with the de Moivre–Laplace and Lévy continuity theorems gives $\mathcal{N}(0, \sigma^2)$. □

Proof of Theorem 7.16. Upper. Take expectations in Proposition 7.13; $t \mapsto \log(t+1) + 2 \log\log(t+2)$ is concave and $\mathbb{E}\tau_n = \frac{n}{3} + \frac{n}{6} = \frac{n}{2}$. *Lower.* By Theorem 7.14 and Kraft, conditionally on each value of g_n, $\mathbb{E} L_{\mathrm{SOI}}(D_n) \geq H(b_{<g_n} \mid g_n) - c = H(b_{<g_n}) - H(g_n) - c$. Since $\mathbb{P}[b_{<g_n} = s] \leq P_{1/3}(s)$, $H(b_{<g_n}) \geq \mathbb{E}[-\log P_{1/3}(b_{<g_n})] \geq \mathbb{E}[UB_n; \tau_n \leq \eta_n]$, and $\mathbb{E}[UB_n; \eta_n < \tau_n] \leq \log 3\,(\mathbb{E}\tau_n^2)^{1/2} \mathbb{P}[\eta_n < \tau_n]^{1/2} = O(n) e^{-n/36}$. Next, g_n is at most the first tail at or after τ_n; $\{\tau_n = k\}$ depends only on the coins and $b_{<k}$, so $\mathbb{E} g_n \leq \frac{n}{2} + \frac{1}{2}$, and a variable on $\mathbb{N}$ with mean m has entropy at most $\log(m+1) + \log e$. The i.i.d. law pays $\mathbb{E}[N_n^1 \log 3 + N_n^0 \log \frac{3}{2}] = (\log 3 - \frac{1}{3}) n$. □

Proof of Proposition 7.17. (a) $D_n = D$ exactly when the coins contain N^1 ones (probability $\binom{n}{N^1}(\frac{2}{3})^{N^1}(\frac{1}{3})^{n-N^1}$) and the first N^1 heads and $n-N^1$ tails of b are the inputs of D labelled 1 and 0. By Lemma 7.12 this fixes b on $[0, s^+)$: each of the $s^+ - n$ unrevealed indices below s^+ is a tail if the top label is 1, a head otherwise; probability $(\frac{1}{3})^{N^1}(\frac{2}{3})^{n-N^1} r^{s^+-n}$. Given $\operatorname{dom} D_n = S$, with $g := \min(\mathbb{N}\setminus S)$, the labelling has probability $\binom{n}{N^1} r^{s^+-n}/Z(S)$, where $Z(S) = ((\frac{2}{3})^{s^+-n} + (\frac{1}{3})^{s^+-n})\binom{n+g}{g}$ if $g < s^+$ and $Z(S) = \binom{2n}{n}$ if $S = [0, n)$. Indeed, a possible labelling with top label 1 is fixed by its tail frontier $\tau' \in [0, g]$ (arbitrary labels on $[0, \tau' - 1)$, 0 at $\tau' - 1$, 1 on $S \setminus [0, \tau')$); summing $\binom{n}{N^1}$ gives $\sum_i \binom{\tau'-1}{i}\binom{n}{\tau'-i} = \binom{n+\tau'-1}{\tau'}$, and $\sum_{\tau' \le g} \binom{n-1+\tau'}{\tau'} = \binom{n+g}{g}$; top label 0 is symmetric, and $\sum_m \binom{n}{m}^2 = \binom{2n}{n}$. On $S = \{0, 1, 3\}$ the labellings $(0, 1, 1)$ and $(1, 1, 0)$ both have $N^1 = 2$ but probabilities $\frac{1}{5}$ and $\frac{1}{10}$. (b) Weak semimeasures are monotone and $\sum_l \nu(l) \le 1$ over the labellings of $[0, g)$, so Gibbs given g_n yields $\mathbb{E}[-\log \nu(D_n)] \ge H(b_{<g_n} \mid g_n)$; continue as for Theorem 7.16. □

Lemma E.9 (the Bayes predictor)**.** *Let h be a history with frontiers τ, η, $F := \max(\tau, \eta)$ and smallest unrevealed index g, and x the next input. The conditional probability of label 1 at x is 0 if $x < \eta$ and 1 if $x < \tau$ (for $x < F$ exactly one holds); for $x \ge F$ it is 1 if $g < F$ and $\tau \le \eta$, 0 if $g < F$ and $\eta < \tau$, and $2^k/(2^k + 1)$, $k := x - F$, if $g = F$ (touching).*

Proof. The first case is Lemma 7.12. Let $x \ge F$. If $\tau \le \eta$ and $g < F$, the first unrevealed index in $[\tau, \eta)$ is the next tail, so a tail reveal would give an input below η; the case $\eta < \tau$ is symmetric. If h is touching, it depends only on $\zeta_{<t}$ and $b_{<F}$, so a head at $F + k$ has probability $\frac{2}{3}(\frac{2}{3})^k\frac{1}{3}$ and a tail there $\frac{1}{3}(\frac{1}{3})^k\frac{2}{3}$; the ratio is 2^k. □

Proof of Proposition 7.19. After each step the process is touching, or in a state (A, u) (heads ahead, $u \ge 1$ unrevealed tails in $[\tau, \eta)$), or (B, u) (heads behind, $u \ge 1$ unrevealed heads in $[\eta, \tau)$); indices $\ge F$ are fresh. From touching, a head reveal skipping k tails (probability $\frac{2}{9}(\frac{2}{3})^k$) leads to (A, k), a tail reveal skipping k heads (probability $\frac{2}{9}(\frac{1}{3})^k$) to (B, k); $k = 0$ means touching. From (A, u): $u - 1$ with probability $\frac{1}{3}$, $u + m$ with probability $\frac{2}{9}(\frac{2}{3})^m$. From (B, u): $u - 1$ with probability $\frac{2}{3}$, $u + m$ with probability $\frac{2}{9}(\frac{1}{3})^m$. The A-walk moves down by exactly one and has drift $+1$; its probability β of ever going one level down satisfies $\beta = \frac{1}{3} + \frac{2}{3}\sum_m \frac{1}{3}(\frac{2}{3})^m\beta^{m+1}$, i.e. $2\beta^2 - 3\beta + 1 = 0$, and $\beta = 1$ would contradict its convergence to $+\infty$; so $\beta = \frac{1}{2}$, and (A, k) returns with probability 2^{-k}. The B-walk has drift $-\frac{1}{2}$ and returns almost surely. The return probability from touching is $\frac{2}{9} + \frac{2}{9} + \sum_{k\ge1}\frac{2}{9}(\frac{1}{3})^k + \sum_{k\ge1}\frac{2}{9}(\frac{1}{3})^k = \frac{2}{3}$, and after a touching step the process starts afresh, so N_{touch} is geometric. By Lemma E.9, a touching step costs P^* an expected ϑ and other steps cost 0, so $\mathbb{E}L_{P^*}(n) = \vartheta\mathbb{E}N_{\text{touch}}(n)$. □

Proof of Corollary 7.21. In (A, u) the unrevealed indices in $[\tau, \eta)$ are tails, so g is the next tail to be revealed and a head lands at an index $\ge \eta > g$. Almost surely $N_{\text{touch}} < \infty$; after the last touching step the process stays in A-states, since B-states return to touching almost surely and A-states are left only through touching. □

Proof of Proposition 7.22. At a touching step ($g = F$, input $F + k$) the rule errs on a head with $k = 0$ (probability $\frac{2}{9}$) and on a tail with $k \ge 1$ (probability $\frac{1}{9}$). It never errs in A-states and always in B-states, where g is the next head. A B-excursion from (B, k) moves down by exactly one with i.i.d. increments of mean $-\frac{1}{2}$; optional stopping for the martingale $u_t + t/2$ shows that its length S is integrable, and Wald's identity gives $\mathbb{E}S = 2k$. So a touching step is followed by $\sum_{k\ge1}\frac{2}{9}(\frac{1}{3})^k 2k = \frac{1}{3}$ B-steps on average, and a cycle (a touching step and its excursion) has $\frac{2}{9} + \frac{1}{9} + \frac{1}{3} = \frac{2}{3}$ expected errors. Cycles are i.i.d., N_{touch} is a stopping time for them, and Wald's identity gives $3 \cdot \frac{2}{3} = 2$. For the second claim, the probability p of no error from a touching history

satisfies $p = \frac{2}{9}p + \sum_{k\ge1} \frac{2}{9}(\frac{2}{3})^k(2^{-k}p + 1 - 2^{-k}) = \frac{1}{3}p + \frac{1}{3}$ (an error-free step is a tail at $k = 0$ or a head at $k \ge 1$), so $p = \frac{1}{2}$. □

E.4 Positive examples

Proof of Proposition 7.29. (a) T_g reads a uniform real ε' bit by bit from a shared part of the tape and, for each input x, a private uniform real u_x, and outputs $g(x) \oplus \mathbf{1}[u_x < \varepsilon']$ once the comparison is decided (after finitely many bits, with probability 1). Given ε' the flips are independent, so $\nu_{T_g}(D) = \int_0^1 \varepsilon'^e(1-\varepsilon')^{n-e}d\varepsilon' = 1/((n+1)\binom{n}{e})$, with $K(\nu_{T_g}) \le K(g) + c$. So T_g is a mixture of product laws, not a product law. Among product laws, "g flipped independently with probability e/n" has complexity $K(g) + O(\log n)$ and loss $nH(e/n) \le \log\binom{n}{e} + \log(n+1)$, which gives (a) up to $O(\log n)$. (b) Each new input's flip is Bern(ε) and independent of the past, so $e \sim \mathrm{Bin}(n, \varepsilon)$; use $\log\binom{n}{e} \le nH(e/n)$ and concavity. Given the past and a new input, the label has conditional entropy $H(\varepsilon)$, so Gibbs at each step bounds every predictor below by $H(\varepsilon)$ per step. □

Finite versions. A direct computation (code in the accompanying repository) gives the following. For sparse parity ($d = 16$, $k = 3$, prior $w(S) = 2^{-|S|-1}/\binom{d}{|S|}$ on $|S| \le 4$) the loss of the Bayes mixture saturates at $\log\binom{16}{3} + 4 = 13.13$ bits after 16 to 24 random examples, in any order. For frozen noise (parities of at most 3 of 12 bits, $\varepsilon = 0.1$, Laplace flip rate) the regret against the noise-aware predictor is 14.9–17.3 bits for $n = 100$–3200, close to $-\log w(S^*) + \frac{1}{2}\log n$ for the target parity S^*.

Remark E.10 (fresh noise). Let the pairs be i.i.d. from $\mu \times P(y \mid x)$ with fresh noise at each draw. If inputs never repeat, fresh and frozen noise coincide and Theorem 5.1(a) bounds the regret against P by $K(P) + c$ on every path. If some input has positive probability and $P(\cdot \mid x)$ is not a point mass there, the one-tape version has $\mathbb{E}L_{\mathrm{SOI}} = \infty$ in both conventions (Proposition 5.6); time stamps or Solomonoff's per-query solution repair this (Theorem 5.5(b), Proposition 5.7).

F Proofs for Section 8 (out-of-distribution behaviour)

Notation and conventions are those of Section 8; c, c' are machine constants. We use the conditional coding theorem as stated in Section 3: a discrete semimeasure $\mu(\cdot \mid z)$ that is lsc uniformly in z satisfies $\mu(a \mid z) \le 2^{-K(a|z)+c_\mu}$.

Proof of Proposition 8.1. For W: let V' read a V-program z for $\langle e, e'\rangle$, then a V-program p, and output an index of the function equal to $\varphi_{e'}$ on Q and to $\varphi_{V(p)}$ off Q; let $V(wu) = V'(u)$. The map $p \mapsto wzp$ is injective, preserves consistency with D (no input of D lies in Q), and its image agrees with b on $Q \cap \mathrm{dom}\, b$. Summing over consistent p with $|z| = K(Q, b)$ gives (a); a shortest p gives (b). For M_F: the function machine $T_{Q,b}$ that answers $b(x)$ on Q without reading the tape and runs T_U off Q has an index computable from $\langle e, e'\rangle$, and maps every tape with $f_\omega \supseteq D$ to a function extending $D \cup b|_{Q\cap\mathrm{dom}\, b}$; Theorem 3.8 gives the bound on $M_F(D \cup E)$. For the event on all of $Q \cap \mathrm{dom}\, b$: the tapes pw of T_U with p a shortest program for an index of $T_{Q,b}$ and $f_w^{T_U} \supseteq D$ have measure $2^{-|p|}M_F(D) \ge 2^{-K(Q,b)-c}M_F(D)$, and each gives a function that extends D and equals b on $Q \cap \mathrm{dom}\, b$. □

Remark F.1 (Where a logarithmic term would come from). No logarithmic term arises, for W or for M_F: the modified program is a concatenation of self-delimiting codes, and the proof sums over

all consistent programs. Plain complexity would cost $K(Q) \le C(Q)+2\log C(Q)+O(1)$. For semi-decidable Q the argument gives no bound, since the wrapper cannot decide Q; a time-truncated decidable subset of Q restores it there.

Proposition F.2 (Properties simple given a training-consistent program)**.** *Let $p \in \operatorname{dom} V$ be consistent with D, and q, r indices with $\varphi_q(\langle p, \cdot\rangle)$ total; let $Q_p := \{x : \varphi_q(\langle p, x\rangle) = 1\}$ and $b_p(x) := \varphi_r(\langle p, x\rangle)$. If $Q_p \cap \operatorname{dom} D = \emptyset$ and $S \subseteq Q_p \cap \operatorname{dom} b_p$ is finite, then $\nu(D \cup b_p|_S) \ge 2^{-K(q,r|p)-c}\, 2^{-|p|}$; so the posterior probability of b_p on S is at least $2^{-K(q,r|p)-c}$ times $2^{-|p|}/\nu(D)$, the posterior weight of p under W (up to a constant factor under M_F).*

Proof. The program *wpz* reads p first, then a conditional program z for $\langle q, r\rangle$ given p; it outputs b_p on Q_p and p's values elsewhere. Reading z first would give the complexity of the pair instead. For M_F use $M_F \ge 2^{-c}W$ (Theorem 4.4). □

Proof of Proposition 8.2. (a) Let $V'(0u) := V(u)$, and let $V'(1u)$ be an index of the function that is b on Q and $\varphi_{V(u)}$ off Q (defined when $u \in \operatorname{dom} V$). $\operatorname{dom} V'$ is prefix-free, and V' is universal because it simulates V with one extra bit. If $\operatorname{dom} D \cap Q = \emptyset$, the programs $0u$ and $1u$ agree on $\operatorname{dom} D$, so each half of $\operatorname{dom} V'$ contributes $W(D)/2$: $W'(D) = W(D)$. Every program $1u$ outputs b on S, so $W'(D \cup b|_S) \ge W(D)/2 = W'(D)/2$. For all D: $W(D) \le 2W'(D)$ by the programs $0u$, and $W(D) \ge 2^{-K(Q,b)-c}W'(D)$ because V simulates $1u$ by the wrapper wzu of Proposition 8.1 and $0u$ by u.

(b) For a function machine T_i let $T_i^{Q,b} := \{(x, \epsilon, b(x)) : x \in Q \cap \operatorname{dom} b\} \cup \{(x, w, y) \in T_i : x \notin Q\}$: it answers b on Q without reading the tape and runs T_i elsewhere; it is c.e. because Q is decidable, consistent because T_i is, and has an index computable from i and $\langle e, e'\rangle$. Let $V''(0u) := V(u)$, let $V''(1u)$ be an index of $T^{Q,b}_{V(u)}$, and build M'_F from V'' as in Theorem 3.8: for $D \ne \emptyset$, $M'_F(D) = \frac{1}{2}\sum_{u \in \operatorname{dom} V} 2^{-|u|}\big(\nu_{T_{V(u)}}(D) + \nu_{T^{Q,b}_{V(u)}}(D)\big)$. If $\operatorname{dom} D \cap Q = \emptyset$, the functions $f_\omega^{T_i^{Q,b}}$ and $f_\omega^{T_i}$ agree on $\operatorname{dom} D$ for every ω, so both halves equal $\frac{1}{2}M_F(D)$ and $M'_F(D) = M_F(D)$. Every tape in the second half gives a function equal to b on $Q \cap \operatorname{dom} b$, which gives the posterior bound $\frac{1}{2}$. The first half gives $M'_F \ge \frac{1}{2}M_F$ on nonempty sets, so M'_F dominates every machine law. Conversely M'_F is a machine law with an index computable from $\langle e, e'\rangle$, so $M_F \ge 2^{-K(Q,b)-c}M'_F$. The decomposition holds only for nonempty data sets; for $D = \emptyset$ the same computation gives $M'_F(b|_S) \ge \frac{1}{2}\Omega_V$ with $\Omega_V := \sum_{p \in \operatorname{dom} V} 2^{-|p|}$. □

Proof of Theorem 8.3. (a) Let $b_P(x)$ be the first y found in a fixed enumeration of P with first coordinate x. It is partial computable, total on Q, P-abiding, and $K(Q, b_P) \le K(P, Q)+c$. Apply Proposition 8.1 with S. (b) The same with $b_{\neg P}$ from an enumeration of the complement. For the consequence: the events "every $x \in S$ is P-abiding" and "some $x \in S$ is P-violating" concern disjoint sets of labellings of S; for W they are disjoint sets of programs and for M_F disjoint sets of tapes, so their conditional probabilities sum to at most 1. (c) For W: let $h \in \operatorname{dom} V$ be shortest with $\varphi_{V(h)}(x) = z$, so $|h| = K^{\mathcal{P}}(z \mid x)$; apply Proposition 8.1 with $b := \varphi_{V(h)}$ and $K(Q, b) \le K(Q)+|h|+c$. For M_F: let $T(x', \omega)$ split ω into its even and odd bits; if $x' \in Q$, run the conditional universal prefix machine on input x', reading program bits lazily from the odd bits; otherwise run T_U on the even bits. T is a function machine. The two streams are independent and uniform and no input of D lies in Q, so $\nu_T(D + (x, z)) = M_F(D) \cdot \sum\{2^{-|u|} : u \text{ outputs } z \text{ on } x\} \ge 2^{-K(z|x)}M_F(D)$. T has an index computable from a decider of Q, so universality costs $K(Q) + c$. □

For a property $P \subseteq X \times Y$ of input–output pairs (for example, "y is a correct proof of x"), two natural pointwise forms of the two-sided statement of Section 8.2, a lower and an upper bound

on the probability of an abiding output, are, for an input x and data D,

$$\begin{aligned} \nu(P\text{-abiding output at } x \mid D) \;&\geq\; 2^{-K(P)}, \\ \nu(P\text{-abiding output at } x \mid D) \;&\leq\; c\,2^{-K(Q)-K(z_x)} \quad \text{if } x \in Q \text{ and } \operatorname{dom} D \cap Q = \emptyset, \end{aligned} \tag{7}$$

where z_x is the simplest output with $(x, z_x) \notin P$. Both fail:

Proposition F.3 (Two natural two-sided bounds fail)**.** *Let $P = \{(x, 0) : x \in X\}$. (a) If D labels the inputs outside a decidable Q_m with 0 and $x^* \in Q_m$, then $\nu((x^*, 0) \mid D) \geq 2^{-c}$, while the second bound of (7) tends to 0 as $K(Q_m) \to \infty$ ($m \to \infty$). (b) If $x^* \in \{0,1\}^d$ has $K(x^* \mid d) \geq d$ and $D = \{(x, 1) : x \in \{0,1\}^d \setminus \{x^*\}\}$, then $K(P) = O(1)$ but $\nu((x^*, 0) \mid D) \leq 2^{-d+c}$, also normalised. With $Q = \{x^*\}$, Theorem 8.3(a) gives $2^{-d-2\log d-c}$, so its $K(Q)$ term is needed and nearly tight.*

Proof of Proposition F.3. (a) The constant-0 program p_0 is consistent with $D + (x^*, 0)$, so $\nu((x^*, 0) \mid D) \geq 2^{-|p_0|-c}/\nu(D) \geq 2^{-|p_0|-c}$; the violating output 1 has $K(1) = O(1)$, so the bound $c\,2^{-K(Q_m)-K(1)}$ tends to 0. (b) For $x \in \{0,1\}^d$ let $\mu_d(x) := \nu(\{(x', 1) : x' \neq x\} \cup \{(x, 0)\})$. These are labellings of $\{0,1\}^d$, so the events are disjoint and $\sum_x \mu_d(x) \leq 1$; μ_d is lsc uniformly in d. The conditional coding theorem gives $\mu_d(x^*) \leq 2^{-K(x^*|d)+c} \leq 2^{-d+c}$. The constant-1 program is consistent with D and with $D + (x^*, 1)$, so $\nu(D) \geq 2^{-c_1}$ and $\sum_y \nu(D + (x^*, y)) \geq 2^{-c_1}$. Dividing $\mu_d(x^*) = \nu(D + (x^*, 0))$ by either gives the bound in both conventions. Finally $K(\{x^*\}) \leq d + 2\log d + c$. □

Proof of Proposition 8.7. (a) Given $(A, f|_A, x^*)$ with $x^* \notin A$, the label $f(x^*)$ is a fair coin independent of everything the learner sees. A normalised prediction (q_0, q_1) has expected error $1 - \frac{q_0+q_1}{2} = \frac{1}{2}$. (b) Condition on f, hence on z; then A and x^* are still i.i.d. uniform. Fewer than 2^{s+1} properties have $K(Q \mid z) \leq s$. For one of them, with density δ in $\{0,1\}^d$, $\mathbb{P}[x^* \in Q,\ A \cap Q = \emptyset] = \delta(1-\delta)^n \leq \frac{1}{n+1}$. Take a union bound. (c) p_1 is a fixed string, so $K(Q \mid p_1) \geq K(Q) - c$. A separating Q meets $\{0,1\}^d$ exactly in $\{x^*\}$, so x^* is computable from d and a decider of Q: $d \leq K(x^* \mid d) \leq K(Q) + c$. The error bound is Proposition F.3(b), which holds normalised. Finally x^* is the unique 0 of f among strings of length d, so $d \leq K(x^* \mid d) \leq K(f) + c$. □

For A finite, $x^* \notin A$, $B := A \cup \{x^*\}$ and $y \neq f(x^*)$, put $D^*_y := f|_A + (x^*, y)$.

Proposition F.4 (Sandwich for one test point)**.** *For every decidable Q separating x^* from A and partial computable b with $b(x^*) = y$, $2^{-K(Q,b)-c} \leq \nu(D^*_y)/\nu(f|_A) \leq 2^{L_\nu(f|_A)-K(D^*_y|B)+c} \leq 2^{K_F(f|_A)-K(D^*_y|B)+c'}$.*

Proof. Left: Proposition 8.1. Right: for B given as a list, ν restricted to labellings of B is a semimeasure (disjoint programs or tapes), lsc uniformly in B, so the conditional coding theorem gives $\nu(D^*_y) \leq 2^{-K(D^*_y|B)+c}$; and $\nu(f|_A) = 2^{-L_\nu(f|_A)} \geq 2^{-K_F(f|_A)-c}$. □

Proof of Proposition 8.8. For $x \in B$ let $\mu_f(x) := \sum_{y \neq f(x)} \nu(f|_{B\setminus\{x\}} + (x, y))$: the mass of hypotheses (programs or tapes) that agree with f on $B \setminus \{x\}$ and give a defined wrong label at x. A hypothesis counted for x and for $x' \neq x$ would be wrong at x (first set) and right at x (second set); so the sets are disjoint and $\sum_{x \in B} \mu_f(x) \leq \nu(\emptyset) \leq 1$. $\mu_f(x)$ is lsc uniformly in the finite object f and x (enumerate programs, or tape cylinders), so $\mu_f(x^*) \leq 2^{-K(x^*|f)+c}$. The normaliser at x^* satisfies $\sum_y \nu(f|_A + (x^*, y)) \geq \nu(f) = 2^{-L_\nu(f)}$, and $L_\nu(f) \leq K_F(f) + c$. Hence

$$e_{\mathrm{norm}}(x^*) = \frac{\mu_f(x^*)}{\sum_y \nu(f|_A + (x^*, y))} \leq 2^{L_\nu(f)-K(x^*|f)+c}.$$

Separating properties. $Q = \{x^*\}$ is decidable from x^*, so $\mathrm{sep}(x^*; A \mid f) \le K(x^* \mid f) + c$. Conversely, given f (hence the list B) and a decider of a separating Q, x^* is the unique element of $B \cap Q$, so $K(x^* \mid f) \le \mathrm{sep}(x^*; A \mid f) + c$. *Lower bound.* Proposition 8.1 with $b \equiv y$ and $K(Q, b) \le K(Q) + K(y) + c$. *The other two bounds.* $K(x^* \mid B) \le K(x^* \mid f) + K(f \mid B) + c$ with $K(f \mid B) \le K_F(f) + c$ (run a shortest consistent program on the list B); and from $\langle B, p\rangle$ one computes f, so $K(x^* \mid B, p) \le K(x^* \mid f) + c$, while $L_\nu(f) \le K_F(f) + c \le |p| + c$. The last form is never sharper than the main bound by more than a constant. □

Example F.5 (Easy-to-hard generalisation; shortest proofs)**.** Let the inputs be sentences of a consistent theory $\mathcal{T}$ with decidable proof checking, labelled 1 if $\mathcal{T}$ proves them and 0 if it refutes them. Let every training sentence have a proof or refutation of length at most L, and let the test sentence x^* be decided by $\mathcal{T}$, with no proof or refutation of length at most $L' := 2^{\lceil \log L \rceil} \le 2L$. The property "no proof or refutation of length $\le L'$" is decidable, separates x^* from the training set, and has complexity at most $K(\mathcal{T}) + K(\lceil \log L \rceil) + c \le K(\mathcal{T}) + 2 \log\log L + c$ for large L. With $b \equiv 0$ and $b \equiv 1$, Proposition 8.1 gives the wrong label probability at least $2^{-K(\mathcal{T}) - 2\log\log L - c}$, however many easy sentences were seen. Proof search, of length $K(\mathcal{T}) + c$, fits every correctly labelled training set, so the correct label has probability at least $2^{-K(\mathcal{T}) - c}$. If instead each provable sentence is labelled by its shortest proof, Theorem 8.3(c) with the violating output ϵ (the empty string; $K(\epsilon \mid x^*)$ and $K^{\mathcal{P}}(\epsilon \mid x^*)$ are $O(1)$) gives probability at least $2^{-K(\mathcal{T}) - 2\log\log L - c}$ that x^* does not get its shortest proof.

The grammar computation of Example 8.12(c). A deterministic script (code in the accompanying repository) computes an exact Bayes mixture over an enumerated grammar. A plain program is an arithmetic expression of depth at most 2 over the leaves $|x|$, $\#_1(x)$, $\#_0(x)$, 0, 1, 2 with $+, -, \times$; an `if` program is "if C then E else E'" with one of 80 simple conditions C (parity of a leaf, comparisons and equalities of leaves, first and last bit). Plain expressions with the same values on all strings of length 1 to 8 are merged into 1345 classes; an `if` program is a condition and two such classes. The prior gives weight p_{plain} to plain programs and $1 - p_{\text{plain}}$ to `if` programs, with fixed grammar weights inside each. Training: all 170 strings of length $1, 3, 5, 7$, labelled by their length; test: all 340 strings of length $2, 4, 6, 8$. Only one plain class ("$|x|$") fits the training data. For $p_{\text{plain}} = 0.5$, 0.9 and 0.99 the posterior predictive of the intended answer $|x|$ at lengths $2, 4, 6, 8$ is 0.9824 to 0.9835, 0.9975 to 0.9977 and 0.99977 to 0.99978, with minima over the test strings 0.9790, 0.9971 and 0.99972. A tilted prior adds one program "if $|x|$ even then 0 else $|x|$" with prior mass equal to (or ten times) that of the intended program $|x|$, the only plain program consistent with the training data; with the factor ten it carries 44% of the prior mass (the predictive is computed after normalising). The intended answer then gets 0.557 (or 0.114) at every even length, with minima 0.555 (or 0.113). A finite grammar is not a universal machine, so this only illustrates Proposition 8.2, which proves that the answer depends on the machine.

G Proofs for Section 9 (computable versions)

Proof of Theorem 9.2. M_{len} is a weak semimeasure with $M_{\text{len}}(\emptyset) = \sum_i w_i \le 1$ (each term is a product of probability vectors). On distinct inputs Proposition 3.16 gives cumulative loss $-\log(M_{\text{len}}(D_k)/M_{\text{len}}(\emptyset)) \le \log(1/w_i) + \sum_{j \le k} -\log q_i(y_j \mid x_j)$. If $|x_j| \ge \ell_A$, machine i is active and $T_A(x_j) \le c_A t(|x_j|) \le g(|x_j|)t(|x_j|)$, so $q_i(\cdot \mid x_j) = A(\cdot \mid x_j)$. If $|x_j| < \ell_A$, either $q_i(\cdot \mid x_j) = A(\cdot \mid x_j)$ (the machine is active and happens to finish in time) or $q_i(\cdot \mid x_j) = u$; in both cases

example	proved	plausible only	false strengthening
provability labels	correct label $\geq 2^{L_{\mathrm{SOI}}(D)-K(\mathcal{T})-c}$; $\leq K(\mathcal{T})+c$ online errors; wrong label $\geq 2^{-K(\mathcal{T})-2\log\log L-c}$, and $\geq 2^{-K(L'|p_{\mathcal{T}})-c}$ times the weight of proof search	a hard sentence (e.g. the Riemann hypothesis) labelled correctly with probability near 1	at fixed difficulty L of the training sentences, the wrong-label probability tends to 0 as more are added
shortest proofs	the same form	a hard sentence gets its shortest proof w.h.p.	the same, for shortest proofs
odd lengths	correct $\geq 2^{-c}$, wrong $\geq 2^{-c'}$, $\leq c$ online errors	correct w.h.p. on natural machines (at least 0.979 on the grammar)	correct w.h.p. on every universal machine (Prop. 8.2; 0.11 with a tilted prior)

Table 4: The out-of-distribution examples of Ex. 8.12: what is proved, what is plausible, and strengthenings that fail. The floor $2^{-K(\mathcal{T})-2\log\log L-c}$ is tiny for a natural theory; the relative bound is the stronger one.

$-\log q_i(y_j \mid x_j) \leq \log|Y| - \log A(y_j \mid x_j)$, because $-\log A \geq 0$. Subtract A's loss. Distinctness bounds the number of inputs shorter than ℓ_A by $2^{\ell_A}-1$. □

Proof sketch for Remark 9.3. Store $v_i := w_i|Y|^k \prod_{j\leq k} q_i(y_j \mid x_j)$ for the machines with $\ell_i \leq \max_{j\leq k}|x_j|$; every other machine has $v_i = w_i$. Then $M_{\mathrm{len}}(D_k) = |Y|^{-k}\sum_i v_i$. A machine inactive at x_k has $q_i = u$, and $|Y|u(y) = 1$, so its v_i does not change in round k; hence

$$M_{\mathrm{len}}(y \mid D_{k-1}, x_k) = \frac{\sum_{i \text{ active at } x_k} v_i q_i(y \mid x_k) + (V - \sum_{i \text{ active at } x_k} v_i)/|Y|}{V}, \qquad V := \sum_i v_i,$$

and an update touches only the $\#\{i : \ell_i \leq |x_k|\} \leq \log\log|x_k|+1$ active weights, with V maintained by adding their changes to its start value $\sum_i w_i$. *Precision.* Let the learner store approximations $\hat{v}_i$ and use $\hat{V}_{k-1}$, computed exactly from them, with each active $\hat{v}_i$ replaced in round k by $\hat{v}_i|Y|q_i(y_k \mid x_k)$ up to relative error ε_k. Its prediction $\hat{p}_k$ is a probability vector, and $\hat{V}_k \leq (1+\varepsilon_k)|Y|\hat{p}_k(y_k)\hat{V}_{k-1}$, so the cumulative loss is at most $k\log|Y| + \log\hat{V}_0 - \log\hat{V}_k + \sum_{j\leq k}\log(1+\varepsilon_j)$. Since $\hat{V}_0 \leq 1$ and $\hat{V}_k \geq \hat{v}_i \geq w_i|Y|^k\prod_{j\leq k} q_i(y_j \mid x_j)(1-\varepsilon_j)$, the regret bound of Theorem 9.2 holds up to $\sum_j \log\frac{1+\varepsilon_j}{1-\varepsilon_j}$, which is at most 1.88 bits for $\varepsilon_j = (j+1)^{-2}$. The round-activated learners are handled in the same way, with the round in place of the input length; for Proposition 9.6 the replay adds its cost and memory.

Remark G.1 (Size of the constant). With $w_i = 2^{-i}$ and $\ell_i = 2^{2^i}$, the constant $\log(1/w_i) + (2^{\ell_A}-1)\log|Y|$ of Theorem 9.2 is at least $2^{2^{2^i}}$, triple exponential in the index i, and i can be exponential in the description length of A. Weights $w_A = 2^{-\ell(A)}$ from a prefix-free code make $\log(1/w_A)$ the description length, but the activation term remains. For comparison, the sequence analogue of Proposition 9.4, which starts machine i at the round r_1 with $\log\log r_1 \geq i$, has regret $\max(r_1, r_2)$ bits, with r_2 the round after which the comparator runs within its clock: doubly exponential in i. Length activation adds one exponential, because up to 2^{ℓ_i} distinct inputs are shorter than ℓ_i; it buys a per-round cost that depends only on the current input, as an x-only learner's should. The round-activated Proposition 9.4 returns to the doubly exponential form, in i and in c_A. Activating machine i at round k_i but clocking by input length removes the extra exponential in i, not in

c_A: on distinct inputs the regret is at most $\log(1/w_i) + (k_i - 1)\log|Y| + \#\{j : |x_j| < \ell'_A\}\log|Y|$, $\ell'_A := \min\{n : g(n) \geq c_A\}$, by the proof of Theorem 9.2.

Remark G.2 (Distinct inputs are needed). Theorem 9.2 needs distinct inputs. With noisy labels, a repeated input with a new label has probability 0 under M_{len}, as under the one-tape version of SOI (Proposition 5.6); and even if the product hypotheses are mixed round by round, length clocking leaves a comparator over budget on short inputs forever. On 4000 rounds that repeat 12 short inputs with labels drawn from a comparator in $\mathcal{C}_t$, a direct computation (code in the accompanying repository) gives regret 469 bits, growing linearly.

Remark G.3 (Like-for-like comparison). A time-bounded learner that reuses computations from earlier rounds, against comparators that recompute each round, does not compare like with like. For x-only comparators, which see one input and have nothing to reuse, Theorem 9.2 and Proposition 9.4 compare like with like. History-aware comparators need a chronological learner: every order-independent predictor that sees the past as a set, such as a mixture of x-only predictors, pays the lower bounds of the chain and skipped-string examples (Proposition 6.5), while history-aware predictors that run in linear time avoid them (Theorems 6.3 and 6.4).

Proof of Proposition 9.6. $\tilde{n}_k$ and t are nondecreasing, so for $k \geq k_0$ and $j \leq k$ one round of A at time j costs at most $c_A t(\tilde{n}_j) \leq g(k)t(\tilde{n}_k)$. Hence in every learner round $k \geq k_0$ the budget $2g(k)t(\tilde{n}_k)$ completes at least two further rounds of A (counting a partly done one), unless the replay reaches the present. If r_k is the number of rounds of A completed after learner round k, then $r_k \geq \min(k, r_{k-1}+2)$ for $k \geq k_0$, so $r_k \geq \min(k, 2(k-k_0+1)) = k$ for $k \geq 2k_0 - 2$. A caught-up replay stays caught up, since one round costs at most the budget. So $q_i^{(k)} = A(\cdot \mid h_{k-1}, x_k)$ for $k \geq 2k_0-2$; for the earlier rounds $q_i^{(k)} \in \{A(\cdot \mid h_{k-1}, x_k), u\}$ and $-\log q_i^{(k)}(y_k) \leq \log|Y| - \log A(y_k \mid h_{k-1}, x_k)$. Sum as in Proposition 9.4. If the budget depended only on $|x_k|$, a stream of short inputs after a long one would leave a budget too small to replay even that one round, and the catch-up could stall forever. □

Proposition G.4 (Samplers). *Let $A(x; r)$ be a probabilistic RAM program that halts within $c_A t(|x|)$ steps on all coin sequences r, with output law $P_A^{\text{samp}}(\cdot \mid x)$. For $|x| \geq \ell_i$ let $q_i^{\text{samp}}(y \mid x)$ be the probability that $A_i(x; r)$ halts within $g(|x|)t(|x|)$ steps with output y, plus $1/|Y|$ times the probability that it does not; for $|x| < \ell_i$ let $q_i^{\text{samp}} := u$. The mixture $\sum_i \frac{w_i}{2}(\prod_D q_i + \prod_D q_i^{\text{samp}})$ satisfies Theorem 9.2 against every such sampler (regret against P_A^{samp}) and every predictor in $\mathcal{C}_t$, with $\log(2/w_i)$ for $\log(1/w_i)$; Proposition 9.4 extends likewise. Computing q_i^{samp} exactly takes time $O(2^{g(|x|)t(|x|)} g(|x|)t(|x|))$.*

A sampler's program is not a predictor's program, so the mixture must contain both readings; the proof of Theorem 9.2 applies because $q_i^{\text{samp}}(y \mid x) \geq P_A^{\text{samp}}(y \mid x)/|Y|$ on short inputs.

Question G.5. Is there a learner with per-round time polynomial in $t(|x|)$ and constant regret against all time-$O(t)$ samplers? (Heuristically, Monte Carlo with s samples per round adds, when the sampler's probabilities are bounded away from 0 and 1, an expected excess of order $(|Y| - 1)/(2s \ln 2)$ bits per round, so constant regret would need $s_k \to \infty$ with $\sum_k 1/s_k < \infty$.) Speed priors (Schmidhuber, 2002; Filan et al., 2016) treat the sequence case; on efficient universal extrapolation and one-way functions, cf. Impagliazzo and Levin (1990).

Proposition G.6 (Occam learning in distribution). *Order functions by shortest description length in a prefix-free language, then by a fixed tie rule; $\ell(f)$ is the shortest description length of f. The* Occam learner *predicts with the first function consistent with the data, so it uses only $H_f :=$*

$\{h$ *preceding or equal to* $f\}$, $|H_f| < 2^{\ell(f)+1}$. *For i.i.d. inputs from* μ*: (a) after* $n \geq (\ln|H_f| + \ln(1/\delta))/\varepsilon$ *samples its hypothesis has* μ*-error at most* ε *with probability at least* $1-\delta$ *(Blumer et al., 1987); (b) its expected number of mistakes is at most* $6.7 + (\ln|H_f|)(1+\ln n_0) + \ln^2 n_0$, $n_0 := \lceil \ln|H_f|/p_{\min} \rceil$ *with* $p_{\min}$ *the least positive point mass of* μ*; for uniform* μ *on* $\{0,1\}^d$ *this is* $O(\ell(f)(d + \log \ell(f)) + d^2)$.

Part (a) is a union bound over H_f; with $\varepsilon = \delta = 0.01$ it needs $70\ell(f) + 531$ samples.

Proof of Proposition G.6(b). For step $n+1$ with $n \geq 1$ and every ε, $\mathbb{P}[\text{mistake}] = \mathbb{E}[\text{err}_\mu(h_n)] \leq \varepsilon + |H_f|(1-\varepsilon)^n$. With $\varepsilon_n := \ln(|H_f|(n+1)^2)/n$ this is at most $\varepsilon_n + (n+1)^{-2}$. Summing over steps $1, \ldots, n_0$ (step 1 costs at most 1), with $\ln(n+1) \leq \ln n + 1/n$ and $\sum_{n<n_0} \frac{\ln n}{n} \leq \frac{1}{2}\ln^2 n_0 + \frac{1}{e}$, gives at most $5.7 + (\ln|H_f|)(1+\ln n_0) + \ln^2 n_0$. After step n_0, each mistake is made by the current hypothesis, which lies in H_f, is consistent with the first n_0 samples, is wrong at a point of positive mass (so has error at least $p_{\min}$), and is eliminated by the mistake. The expected number of such hypotheses is at most $|H_f|(1-p_{\min})^{n_0} \leq 1$. □

Proof of Theorem 9.8. (a) $\sum_{s>S} 2^{-s} = 2^{-S}$. If no circuit of size at most S agrees with f on B, then $W_{\text{circ}}(f|_B) < 2^{-S}$, strictly because at every size some constant circuit disagrees with f on $B \neq \emptyset$; so $S < L_B$. Take $S = \lceil L_B \rceil$. Also $W_{\text{circ}}(f|_B) \geq w(C_{\min}) = 2^{-\text{size}_B(f)}/N_{\text{size}_B(f)}$, and $\log N_s \leq s(4 + 2\log(d+1+s))$.

(b) First inequality: $P_i \leq -\ln(1-P_i)$, and $\sum_i -\log(1-P_i) = L_B$ by Proposition 3.16 with $W_{\text{circ}}(\emptyset) = 1$. Second: fix i and $S := \lceil L_B \rceil$; let $\mathcal{R}$ and $\mathcal{W}$ be the circuits consistent with D_{i-1} that give x_i the right and the wrong label. The exception gadget at x_i maps $\mathcal{W}_{\leq S}$ (size at most S) injectively into $\mathcal{R}$, since earlier inputs differ from x_i, and Γ is increasing; so $w(\mathcal{W}_{\leq S}) \leq 2^{\Gamma(S)} w(\mathcal{R})$. Also $w(\mathcal{W}_{>S}) \leq 2^{-S} \leq 2^{-L_B} = W_{\text{circ}}(f|_B) \leq w(\mathcal{R})$, because $\mathcal{R}$ contains every circuit that agrees with f on B. So $P_i/(1-P_i) \leq 2^{\Gamma(S)} + 1$ and $-\log(1-P_i) \leq \Gamma(S) + 2$. In $L_B = \sum_i -\log(1-P_i)$, terms with $P_i \leq \frac{1}{2}$ contribute at most $2P_i$ (the chord from $(0,0)$ to $(\frac{1}{2}, 1)$), and at most $2\mathcal{E}$ indices have $P_i > \frac{1}{2}$. So $L_B \leq 2\mathcal{E}(\Gamma(S) + 3)$.

Explicit form. $\gamma \leq d+1$ and $d+1+S+\gamma \leq 2d+4+L_B$, so $L_B \leq R(L_B)$ with $R(L) := 2\mathcal{E}[(d+1)(5+2\log(2d+4+L))+3]$. R is concave and increasing with $R(0) > 0$, so $\{L \geq 0 : L \leq R(L)\}$ is an interval $[0, L^*]$, and it suffices to show $K > R(K)$ for $K := 16a\Lambda$, where $a := \mathcal{E}(d+1)$, $\Lambda := X + Y$, $X := \log(2(d+2)) \geq \log 6$, $Y := \log(\mathcal{E}+1)$. Since $6\mathcal{E} \leq 3a$, $R(K) \leq 13a + 4a\log(2d+4+K)$, and $2d + 4 + 16a\Lambda \leq 2(d+2)(1+8\mathcal{E}\Lambda) \leq 2(d+2)(1+8\mathcal{E})(1+X)(1+Y)$. So it suffices that

$$\big[12X - 4\log(1+X) - 13\big] + \big[16Y - 4\log(1+8\mathcal{E}) - 4\log(1+Y)\big] > 0.$$

The first bracket is increasing in X and equals 10.65 at $X = \log 6$. For the second, $\log(1+Y) \leq Y + 0.087$ for all $Y \geq 0$, and $\log(1+Y) \leq Y$ for $Y \geq 1$. If $\mathcal{E} \leq 1$, then $(1+8\mathcal{E})/(1+\mathcal{E}) \leq \frac{9}{2}$ and the bracket is at least $12Y - 4\log 4.5 - 4(Y + 0.087) \geq -9.03$. If $\mathcal{E} \geq 1$, then $(1+8\mathcal{E})/(1+\mathcal{E}) < 8$ and the bracket exceeds $12Y - 12 - 4Y \geq -4$. The sum is positive. □

Remark G.7 (Regret and circuit size). A natural form of the relation is that expected regret m when learning f implies a circuit of size $O(m \log m \cdot d)$ for f. This is Theorem 9.8(b) with (a), with $\log((\mathcal{E}+1)(d+2))$ for $\log m$; it holds for every single ordering, needs no Markov step (total log loss is order independent), and the factor $O(\log m)$ is the $\log(d+s)$ in Γ. Here the step from log loss at most poly(m) to a circuit of size at most poly(m) is a *circuit coding theorem*, which (a) gives for a size prior with exponential tail, while a polynomial tail gives only size$(f) \leq 2^{O(L)}$. The converse holds on every ordering: $\mathcal{E} \leq (\ln 2)L \leq (\ln 2)\,\text{size}(f)(5 + 2\log(d+1+\text{size}(f)))$. For formulas on $d = 3$, an exhaustive computation over all 256 functions and all orderings (code in the accompanying repository) confirms inequalities of the same form.

Lemma G.8 (Möbius lemma)**.** *Let $1 \le k \le d$ and $h : \{0,1\}^d \to \{0,1\}$ with $h(1^d) = 1$ and $h(x) = 0$ whenever $|x| < k$. Then (i) h depends on at least k variables; (ii) if it depends on exactly k variables, then $h = \mathrm{AND}_S := \bigwedge_{j \in S} x_j$ for some $|S| = k$.*

Proof. Write $h = \bigoplus_S c_S \prod_{j \in S} x_j$ over GF(2). Evaluating at the indicator 1_T of T gives $h(1_T) = \bigoplus_{S \subseteq T} c_S$, so $c_S = \bigoplus_{T \subseteq S} h(1_T)$. If $|S| < k$, every $T \subseteq S$ has $|T| < k$, so $c_S = 0$. Since $h \neq 0$, some $c_S = 1$ with $|S| \ge k$; by uniqueness of this representation, h depends on every variable of such a monomial, which gives (i). For (ii), let h depend only on the variables in R, $|R| = k$. For $a \in \{0,1\}^R$ with $a \neq 1_R$, the input equal to a on R and 0 elsewhere has weight less than k, so $h = 0$ there; and $h(1_R, \cdot) = h(1^d) = 1$. So $h = \mathrm{AND}_R$. □

Proof of Theorem 9.10. Stage k ($1 \le k \le d-1$) begins when the data are $h(1^d) = 1$ and $h(x) = 0$ for $|x| < k$, that is, after 1^d, 0^d and all inputs of weight below k have been shown with the labels of f.

Formulas. A formula with s leaves depends on at most s variables. By Lemma G.8(i) every function consistent at the start of stage k has formula size at least k; one of size exactly k depends on exactly k variables and is a k-conjunction by Lemma G.8(ii). Every AND_S with $|S| = k$ is consistent and has a formula of size k. So the consistent functions of least size are exactly the $\binom{d}{k}$ k-conjunctions, and Occam's hypothesis is one of them, since description length increases with size.

Circuits, $k \ge 2$. In a circuit with s gates, let $t \le s$ be the number of gates from which the output can be reached. Each of the $t-1$ reachable non-output gates occupies at least one of the $2t$ input slots of reachable gates, so at most $t+1$ slots hold variables, and the function depends on at most $s+1$ variables. By Lemma G.8(i) a consistent circuit has $s \ge k-1$. If $s = k-1$, then $t = s$ and exactly k slots hold variables, which must be k distinct ones; so the function depends on exactly k variables and is a k-conjunction. AND_S has a circuit of $k-1$ gates. So again the least-size consistent functions are the k-conjunctions. (For $k = 1$ this fails: the one-gate circuit $x_1 \wedge x_2$ is consistent at the start of stage 1.)

Every input of weight k is a mistake. For $|S| = |S'| = k$, $\mathrm{AND}_{S'}(1_S) = 1$ iff $S' \subseteq S$ iff $S' = S$. If Occam's hypothesis in stage k is AND_S, then 1_S is unrevealed (else AND_S would be inconsistent), and the ordering shows it: Occam predicts 1, the label is $f(1_S) = 0$, and exactly AND_S among the k-conjunctions is eliminated. So all $\binom{d}{k}$ inputs of weight k are mistakes, after which stage $k+1$ begins.

Count and description length. Formulas: $\sum_{k=1}^{d-1} \binom{d}{k} = 2^d - 2$; circuits: $\sum_{k=2}^{d-1} \binom{d}{k} = 2^d - d - 2$. At $k = d$ the argument shows that f has least size d (formulas) or $d-1$ (circuits), so $\ell(f) = d + \log(\mathrm{Cat}(d-1)\, 3^{d-1}(2d+2)^d)$ for formulas, with Cat the Catalan numbers counting tree shapes, and $\ell(f) = (d-1) + \log N_{d-1}$ for circuits. Both are $\Theta(d \log d)$: $\log((2d+2)^d)$ dominates $\mathrm{Cat}(d-1) \le 4^d$, and $\log N_{d-1} = \sum_{j<d}(4 + 2\log(d+1+j))$. So $d = \Theta(\ell / \log \ell)$ and the number of mistakes is at least $2^{d-1} = 2^{\Omega(\ell/\log \ell)}$ for $d \ge 4$. The trivial upper bound is $\min(2^d, |H_f| - 1) \le 2^{\ell+1}$, and there are only 2^d inputs, which gives the tightness for this target. □

Remark G.9 (A second adversary)*.* For circuits the count $2^d - d - 2$ also follows without the stage lemma, for every tie rule. Reveal 1^d; then, while Occam's hypothesis h has $\mathrm{size}(h) < d-1$, let V_h be the set of variables h depends on and reveal 1_{V_h}. Each step is a mistake: by the slot count, $|V_h| \le \mathrm{size}(h) + 1 < d$; h depends only on V_h, so $h(1_{V_h}) = h(1^d) = 1$, while $f(1_{V_h}) = 0$; and the point is new, or h would be inconsistent. The phase reveals a new point at each step, so it ends, and then $\mathrm{size}(h) \ge d-1$. Fix $2 \le k \le d-1$ and $|T| = k$, so $\mathrm{size}(\mathrm{AND}_T) = k-1$. While AND_T is consistent, Occam's hypothesis precedes it, so $\mathrm{size}(h) \le k-1 < d-1$: the phase continues and

reveals points of weight $|V_h| \le k$. When the phase ends AND_T has been killed by a revealed point x with $\mathrm{AND}_T(x) = 1 \ne f(x)$, revealed while AND_T was consistent; so the ones of x contain T, $x \ne 1^d$ and $|x| \le k$, hence $x = 1_T$, revealed as a mistake. So every point of weight $2, \dots, d-1$ is a mistake. (For formulas, $|V_h|$ is at most the formula size of h, and every point of weight $1, \dots, d-1$ is a mistake.) A direct computation under seven tie rules ($d = 3, 4, 5$; code in the accompanying repository) confirms this.

Remark G.10 (It is the selection rule, not the prior). With the same formula prior the posterior-majority predictor makes at most $\ell(f)$ mistakes on every ordering (Remark 9.9). For $d = 3$ and $d = 4$ an exact computation (code in the accompanying repository; ties count $\frac{1}{2}$) gives 7 and 15 mistakes for Occam on the ladder against 2.5 and 3.5 for posterior majority. The slides of Vaintrob and Hänni (2026b) (Appendix K) tabulate four selection rules on one prior: deterministic argmax, $2^{\Theta(C/\log C)}$ worst-case mistakes; argmax with randomised tie-breaking, $\tilde{\Theta}(C^2)$; a uniformly random consistent hypothesis with a doubling budget, $O(C)$; weighted majority, at most C. The first row (for the conjunction target) and the last row are proved here (Theorem 9.10 and Remark 9.9); the slides also state the other rows, and that a random ordering gives $O((C+d)d)$ expected mistakes in any language and the descending-weight ordering exactly $d+1$, which we do not prove.

Remark G.11 (Uniform tie-breaking on the ladder). All formulas of one size have the same prior weight, so the limit $\lambda \to \infty$ of the power posterior $\propto w(\text{formula})^\lambda \mathbf{1}[\text{consistent}]$ (the direction of the companion paper on steeper priors, Hänni 2026a) is uniform over the least-size consistent formulas, and by symmetry uniform over the surviving k-conjunctions at stage k. On the ladder it makes at most $1 + \sum_{k=1}^{d-1} H_{\binom{d}{k}} \le d + d^2 \ln 2$ Gibbs mistakes (H_m the harmonic number) and at most $\frac{3}{2}(d-1)+1$ majority mistakes. So Theorem 9.10 is a failure of deterministic tie-breaking; it does not show that Solomonoff's per-query solution with a steeper prior fails on this ordering. The slides make this observation for their strong-prior learner (Appendix K); we give the formula-prior version with explicit constants.

H Proofs for Section 10 (neural-network Bayes)

Proposition H.1 (Every prior nearly saturates the bound on its typical set (standard))**.** *For every semimeasure P on truth tables, every string a and $j \ge 0$: $P\{f : P(f) \le 2^{-K(f|a)-j}\} \le 2^{-j}$. Hence if P_A is lower-semicomputed by a and ϖ is a probability measure, then with probability at least $1 - 2^{-j}$ over $\theta \sim \varpi$, $K(f_\theta \mid a) - c \le -\log P_A(f_\theta) < K(f_\theta \mid a) + j$.*

Proof. The left side is at most $\sum_f 2^{-K(f|a)-j} \le 2^{-j}$ by Kraft's inequality; the lower bound is Proposition 10.2. □

This is the standard fact that $-\log P \approx K$ on P-most objects (Li and Vitányi, 2008) (cf. Lemma 7.1). So P_A looks like 2^{-K} on its own typical set, which can omit most simple functions.

Proof of Proposition 10.4. Let C be a circuit for f of size $s = \mathrm{size}(f) < S$. The circuits of size S whose first s gates equal C, whose gates $s+1, \dots, S-1$ are arbitrary, and whose last gate is $\mathrm{AND}(\mathrm{gate}_s, \mathrm{gate}_s)$ all compute f. Their fraction among all circuits of size S is $\prod_{j=s+1}^{S-1} 16(d+1+j)^2 / \prod_{j=1}^{S} 16(d+1+j)^2 = 1/(N_s \cdot 16(d+1+S)^2)$. For the mistakes: a law of a random total function has order-independent cumulative log loss $-\log P_{\mathrm{unif},S}(f)$ on every ordering (Proposition 3.16), and the expected number of Gibbs mistakes is at most $\ln 2$ times the log loss, as in Theorem 9.8(b). □

Remark H.2 (What a bridge would give). If $P_A(f) \geq 2^{-a\ell(f)-b}$ for a description length ℓ, then on every ordering NN Bayes makes at most $(\ln 2)(a\ell(f)+b)$ Gibbs mistakes and at most $a\ell(f)+b$ posterior-majority mistakes (as in Theorem 8.4 and Remark 9.9); if also $P_A(f) \leq 2^{-a'\ell(f)+b'}$, the log loss on the full domain is at least $a'\ell(f) - b'$. Proposition 10.4 is the lower half for uniform circuit priors. The converse, that few mistakes imply a small circuit, is stated for NN Bayes by the slides in their setting and in distribution (Remark 10.8), and holds for circuit Bayes on every ordering (Theorem 9.8).

Proof of Theorem 10.5. (a) Write $\mathcal{K}(x,x') = h(x \odot x')$ with $h(z) := \Psi(\sum_j z_j/d)$ and $\odot$ the coordinatewise product, using $\langle x, x'\rangle = \sum_j x_j x'_j$. For fixed x, $x' = x \odot z$ with z uniform, and $\chi_S(x \odot z) = \chi_S(x)\chi_S(z)$, so $(T_{\mathcal{K}}\chi_S)(x) = \mathbb{E}_z[h(z)\chi_S(z)]\,\chi_S(x)$. The eigenvalue $\Lambda_{|S|} := \mathbb{E}_z[h(z)\chi_S(z)]$ depends only on $|S|$, because h is invariant under permutations of the coordinates. For the NNGP and NTK recursions with i.i.d. Gaussian weights and biases, every layer's kernel depends on the inputs only through $\langle x,x\rangle$, $\langle x',x'\rangle$ and $\langle x,x'\rangle$, and the first two equal d on the cube. (b) The characters form an orthonormal basis of L^2 of the uniform measure. For each S, $\min_{g\in\mathcal{V}} \|\chi_S - g\|^2 = 1 - \|P_{\mathcal{V}}\chi_S\|^2$ with $P_{\mathcal{V}}$ the orthogonal projection, and $\sum_{\text{all } T} \|P_{\mathcal{V}}\chi_T\|^2 = \operatorname{tr} P_{\mathcal{V}} = \dim \mathcal{V} \leq n$. So $\sum_{|S|=k} \|P_{\mathcal{V}}\chi_S\|^2 \leq n$. □

Proof of Corollary 10.6. χ_S is total and computable, so Theorem 8.4 gives $\sum_t e_t \leq (\ln 2)(K(\chi_S)+c)$ on every sequence of distinct inputs; repeated inputs have $e_t = 0$, so the bound holds for i.i.d. inputs too (Proposition 3.16). An enumerative code of S given d and k gives the bound on $K(\chi_S)$. Let $y^* = \chi_S(x) \in \{\pm 1\}$ and let $w \leq e$ be the posterior mass of the wrong label at x; the posterior mean is $\hat{g} = y^*(1-e-w)$, so $|y^* - \hat{g}| = e + w \leq 2e$ and $(y^* - \hat{g})^2 \leq 4e^2 \leq 4e$. For i.i.d. inputs the expected test error after $t-1$ samples equals the expected online error at step t, which gives the averaged bound. The kernel side is Theorem 10.5(b) with $\mathcal{V} = \operatorname{span}\{\mathcal{K}(x_i,\cdot)\}$. Since $\log \binom{d}{k} \leq k \log d$, the SOI side needs $n = O(k \log d/\varepsilon)$ for the averaged error, while $\binom{d}{k} \geq (d/k)^k$. For the error at a single step, Corollary 5.10(a) bounds the expected error by $((K(\chi_S)+c)\ln 2 + \ln n + 1)/n$, and the squared error by four times this. □

Proof of Remark 10.7. (a) With $\arccos(-\rho) = \pi - \arccos\rho$ one gets $\Psi_1(\rho) - \Psi_1(-\rho) = \rho$ and $\rho\Psi_0(\rho) - (-\rho)\Psi_0(-\rho) = \rho$. So $\Psi_1(\rho) - \frac{1}{2}\rho$ and $\Psi_1(\rho) + \rho\Psi_0(\rho) - \rho$ are even. The function $z \mapsto \Psi(\rho(z)) - a\rho(z)$, $\rho(z) = \sum_j z_j/d$, is then invariant under $z \mapsto -z$, so its Walsh coefficients at odd $|S|$ vanish, while $a\rho(z)$ has Walsh coefficients only at $|S| = 1$. Hence $\Lambda_{|S|} = 0$ for odd $|S| \geq 3$. Then $\langle \mathcal{K}(x_i,\cdot), \chi_S\rangle = (T_{\mathcal{K}}\chi_S)(x_i) = 0$, so $\mathcal{V} \perp \chi_S$ and $\|\chi_S - \hat{g}\|^2 = 1 + \|\hat{g}\|^2$. (b) $\langle z, \chi_S\rangle := \mathbb{E}_x[z(x)\chi_S(x)]$ is a linear functional of a centred Gaussian process, hence centred Gaussian, with variance $\mathbb{E}_{x,x'}\mathcal{K}(x,x')\chi_S(x)\chi_S(x') = \langle \chi_S, T_{\mathcal{K}}\chi_S\rangle = \Lambda_{|S|} = 0$. So $\langle z, \chi_S\rangle = 0$ almost surely. If $\operatorname{sign} z(x) = \chi_S(x)$ for all x, then $z(x)\chi_S(x) = |z(x)|$ and $\langle z, \chi_S\rangle = \mathbb{E}|z| > 0$. So this event has prior probability 0, and conditioning on an event of positive probability keeps it at 0. (c) From Theorem 10.5(a), $\mathcal{K}(x,x') = \sum_T \Lambda_{|T|}\chi_T(x)\chi_T(x')$, so $\sum_x \mathcal{K}(x_i,x)\chi_S(x) = 2^d\Lambda_{|S|}\chi_S(x_i) = 0$, and $\sum_x g(x)\chi_S(x) = 0$ for every $g \in \mathcal{V}$; split the sum into training and unseen points. For kernel ridge regression $g = \sum_i \alpha_i \mathcal{K}(x_i,\cdot)$ with $\alpha = (G+\beta I)^{-1}y$ and G the training Gram matrix, so $\sum_i g(x_i)y_i = y^\top G(G+\beta I)^{-1}y \geq 0$: G and $(G+\beta I)^{-1}$ are commuting positive semidefinite matrices. If some $g \in \mathcal{V}$ interpolates the training labels, its unseen sum is exactly $-n$. □